

SINAMICS S120

Safety Integrated
Function Manual · 05/2010

SINAMICS

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S120 Safety Integrated

Function Manual

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Legal information

Warning notice system

This manual contains notices you have to observe in order to ensure your personal safety, as well as to prevent damage to property. The notices referring to your personal safety are highlighted in the manual by a safety alert symbol, notices referring only to property damage have no safety alert symbol. These notices shown below are graded according to the degree of danger.

⚠ DANGER
indicates that death or severe personal injury will result if proper precautions are not taken.
⚠ WARNING
indicates that death or severe personal injury may result if proper precautions are not taken.
⚠ CAUTION
with a safety alert symbol, indicates that minor personal injury can result if proper precautions are not taken.
CAUTION
without a safety alert symbol, indicates that property damage can result if proper precautions are not taken.
NOTICE
indicates that an unintended result or situation can occur if the corresponding information is not taken into account.

If more than one degree of danger is present, the warning notice representing the highest degree of danger will be used. A notice warning of injury to persons with a safety alert symbol may also include a warning relating to property damage.

Qualified Personnel

The product/system described in this documentation may be operated only by **personnel qualified** for the specific task in accordance with the relevant documentation for the specific task, in particular its warning notices and safety instructions. Qualified personnel are those who, based on their training and experience, are capable of identifying risks and avoiding potential hazards when working with these products/systems.

Proper use of Siemens products

Note the following:

⚠ WARNING
Siemens products may only be used for the applications described in the catalog and in the relevant technical documentation. If products and components from other manufacturers are used, these must be recommended or approved by Siemens. Proper transport, storage, installation, assembly, commissioning, operation and maintenance are required to ensure that the products operate safely and without any problems. The permissible ambient conditions must be adhered to. The information in the relevant documentation must be observed.

Trademarks

All names identified by ® are registered trademarks of the Siemens AG. The remaining trademarks in this publication may be trademarks whose use by third parties for their own purposes could violate the rights of the owner.

Disclaimer of Liability

We have reviewed the contents of this publication to ensure consistency with the hardware and software described. Since variance cannot be precluded entirely, we cannot guarantee full consistency. However, the information in this publication is reviewed regularly and any necessary corrections are included in subsequent editions.

Preface

SINAMICS documentation

The SINAMICS documentation is organized in 2 parts:

- General documentation / Catalogs
- Manufacturer/service documentation

A current overview of the documentation in the available languages is provided in the Internet:

<http://www.siemens.com/motioncontrol>

Select the menu items "Support" --> "Technical Documentation" --> "Overview of Publications."

The Internet version of DOConCD (DOConWEB) is available on the Internet:

<http://www.automation.siemens.com/doconweb>

Information on the range of training courses and FAQs (Frequently Asked Questions) is available on the Internet:

<http://www.siemens.com/motioncontrol>

Follow the menu item "Support".

Usage phases and their tools/documents (as an example)

Table 1 Usage phases and the available documents/tools

Usage phase	Document/tool
Orientation	SINAMICS S Sales Documentation
Planning/configuration	• SIZER configuration tool • Configuration Manuals, Motors
Decision/ordering	SINAMICS S Catalogs
Installation/assembly	• SINAMICS S120 Equipment Manual for Control Units and Additional System Components • SINAMICS S120 Equipment Manual for Booksize / Booksize Compact Power Units • SINAMICS S120 Equipment Manual for Chassis Power Units • SINAMICS S120 Equipment Manual for AC Drives
Commissioning	• STARTER parameterization and commissioning tool • SINAMICS S120 Getting Started • SINAMICS S120 Commissioning Manual • SINAMICS S120 CANopen Commissioning Manual • SINAMICS S120 Function Manual • SINAMICS S120/S150 List Manual

Usage phase	Document/tool
Usage/operation	• SINAMICS S120 Commissioning Manual • SINAMICS S120/S150 List Manual
Maintenance/servicing	• SINAMICS S120 Commissioning Manual • SINAMICS S120/S150 List Manual
References	• SINAMICS S120/S150 List Manual

Target group

This documentation is intended for machine manufacturers, plant manufacturers, commissioning engineers, and service personnel who use the SINAMICS S drive system.

Benefits

The Safety Integrated Function Manual covers all information, procedures and operations required for commissioning safety functions and servicing of SINAMICS S120.

Search guides

The following guides are provided to help you locate information in this manual:

1. Contents
2. List of abbreviations
3. Index

Standard scope

The scope of the functionality described in this document can differ from the scope of the functionality of the drive system that is actually supplied.

- Other functions not described in this documentation might be able to be executed in the drive system. However, no claim can be made regarding the availability of these functions when the equipment is first supplied or in the event of servicing.
- Functions can be described in the documentation that are not available in a particular product version of the drive system. The functionality of the supplied drive system should only be taken from the ordering documentation.
- Extensions or changes made by the machine manufacturer must be documented by the machine manufacturer.

For reasons of clarity, this documentation does not contain all of the detailed information on all of the product types. This documentation cannot take into consideration every conceivable type of installation, operation and service/maintenance.

Technical Support

In case of questions, please contact us through the following hotline:

	Europe / Africa
Phone	+49 180 5050 - 222
Fax	+49 180 5050 - 223
0.14 €/min. from German landlines, max. 0.42 €/min for calls from a mobile phone	
Internet	http://www.siemens.de/automation/support-request

	America
Phone	+1 423 262 2522
Fax	+1 423 262 2200
E-mail	mailto:techsupport.sea@siemens.com

	Asia/Pacific
Phone	+86 1064 757 575
Fax	+86 1064 747 474
E-mail	mailto:support.asia.automation@siemens.com

Note

Country-specific telephone numbers for technical support are provided under the following Internet address:

<http://www.siemens.com/automation/service&support>

Questions on the manual

If you have any questions (suggestions, corrections) regarding this documentation, please fax or e-mail us at:

Fax	+49 9131 98 2176
E-mail	E-mail to: docu.motioncontrol@siemens.com

A fax form is available in the appendix of this document.

Internet address for SINAMICS

<http://www.siemens.com/sinamics>.

Internet address for Safety Integrated

<http://www.siemens.com/safety>

This address contains detailed application examples for Safety Integrated.

Notation

The following notation and abbreviations are used in this documentation:

Notation for parameters (examples):

- p0918 Adjustable parameter 918
- r1024 Display parameter 1024
- p1070[1] Adjustable parameter 1070, index 1
- p2098[1].3 Adjustable parameter 2098, index 1, bit 3
- p0099[0...3] Adjustable parameter 99 indices 0 to 3
- r0945[2](3) Display parameter 945 index 2 of drive object 3
- p0795.4 Adjustable parameter 795 bit 4

Notation for faults and alarms (examples):

- F12345 Fault 12345
- A67890 Alarm 67890

ESD Notes

CAUTION

Electrostatic sensitive devices (ESD) are single components, integrated circuits or devices that can be damaged by electrostatic fields or electrostatic discharges.

Regulations for the ESD handling:

During the handling of electronic components, pay attention to the grounding of the person, workplace and packaging!

Electronic components may be touched by persons only when

- these persons are grounded using an ESD bracelet, or
- these persons in ESD areas with a conducting floor wear ESD shoes or ESD grounding straps.

Electronic components should be touched only when this is unavoidable. The touching is permitted only on the front panel or on the circuit board edge.

Electronic components must not be brought into contact with plastics or clothing made of artificial fibers.

Electronic components may only be placed on conducting surfaces (table with ESD coating, conducting ESD foamed material, ESD packing bag, ESD transport container).

Electronic components may not be placed near display units, monitors or televisions (minimum distance from the screen > 10 cm).

Measurements must only be taken on boards when the measuring instrument is grounded (via protective conductors, for example) or the measuring probe is briefly discharged before measurements are taken with an isolated measuring device (for example, touching a bare metal housing).

Safety notices

DANGER

- Commissioning is absolutely prohibited until it has been completely ensured that the machine, in which the components described here are to be installed, is in full compliance with the provisions of the EC Machinery Directive.
- SINAMICS devices and AC motors must only be commissioned by suitably qualified personnel.
- The personnel must take into account the information provided in the technical customer documentation for the product, and be familiar with and follow the specified danger and warning notices.
- When electrical equipment and motors are operated, the electrical circuits automatically conduct a dangerous voltage.
- When the machine or system is operated, hazardous axis movements can occur.
- All of the work carried-out on the electrical machine or system must be carried-out with it in a no-voltage condition.
- SINAMICS devices with three-phase motors must only be connected to the power supply via an AC-DC residual-current-operated device with selective switching once verification has been provided that the SINAMICS device is compatible with the residual-current-operated device in accordance with IEC 61800-5-1, Section 5.2.11.2.

WARNING

- The successful and safe operation of this equipment and motors is dependent on correct transport, proper storage, installation and mounting as well as careful operator control, service and maintenance.
- For special versions of the drive units and motors, information and data in the Catalogs and quotations additionally apply.
- In addition to the danger and warning information provided in the technical customer documentation, the applicable national, local, and plant-specific regulations and requirements must be taken into account.
- Only protective extra-low voltages (PELV, DVC-A) that comply with EN 60204-1:2006 can be connected to the connections and terminals between 0 V and 48 V.

CAUTION

- The motors can have surface temperatures of over +80 °C.
- This is the reason that temperature-sensitive components, e.g. cables or electronic components may neither be in contact nor be attached to the motor.
- When attaching the connecting cables, you must ensure that:
 - they are not damaged
 - they are not under tension
 - they cannot come into contact with any rotating parts

CAUTION
• As part of routine tests, SINAMICS devices are subject to a voltage test in accordance with EN 61800-5-1. Before the voltage test is performed on the electrical equipment of industrial machines to EN 60204-1:2006, Section 18.4, all connectors of SINAMICS equipment must be disconnected/unplugged to prevent the equipment from being damaged. • Motors should be connected-up according to the circuit diagram provided. otherwise they can be destroyed.

Note

When operated in dry areas, SINAMICS devices with three-phase motors conform to the Low-Voltage Directive 2006/95/EC.

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Standards and regulations

1.1 General information

1.1.1 Aims

Manufacturers and operating companies of equipment, machines, and products are responsible for ensuring the required level of safety. This means that plants, machines, and other equipment must be designed to be as safe as possible in accordance with the current state of the art. To ensure this, companies describe in the various standards the current state of the art covering all aspects relevant to safety. When the relevant Standards are observed, this ensures that state-of-the-art technology has been utilized and, in turn, the erector/builder of a plant or a manufacturer of a machine or a piece of equipment has fulfilled his appropriate responsibility.

Safety systems are designed to minimize potential hazards for both people and the environment by means of suitable technical equipment, without restricting industrial production and the use of machines more than is necessary. The protection of man and environment must be assigned equal importance in all countries, which is it is important that rules and regulations that have been internationally harmonized are applied. This is also designed to avoid distortions in the competition due to different safety requirements in different countries.

There are different concepts and requirements in the various regions and countries of the world when it comes to ensuring the appropriate degree of safety. The legislation and the requirements of how and when proof is to be given and whether there is an adequate level of safety are just as different as the assignment of responsibilities.

The most important thing for manufacturers of machines and companies that set up plants and systems is that the legislation and regulations in the country where the machine or plant is being operated apply. For example, the control system for a machine that is to be used in the US must fulfill local US requirements even if the machinery construction company (OEM) is based in the European Economic Area (EEA).

1.1.2 Functional safety

Safety, from the perspective of the object to be protected, cannot be split-up. The causes of hazards and, in turn, the technical measures to avoid them can vary significantly. This is why a differentiation is made between different types of safety (e.g. by specifying the cause of possible hazards). "Functional safety" is involved if safety depends on the correct function.

To ensure the functional safety of a machine or plant, the safety-related parts of the protection and control devices must function correctly. In addition, the systems must behave in such a way that either the plant remains in a safe state or it is brought into a safe state if a fault occurs. In this case, it is necessary to use specially qualified technology that fulfills the requirements described in the associated Standards. The requirements to achieve functional safety are based on the following basic goals:

- Avoiding systematic faults
- Controlling systematic faults
- Controlling random faults or failures

Benchmarks for establishing whether or not a sufficient level of functional safety has been achieved include the probability of hazardous failures, the fault tolerance, and the quality that is to be ensured by minimizing systematic faults. This is expressed in the Standards using different terms. In IEC/EN 61508, IEC/EN 62061, IEC/EN 61800-5-2: "Safety Integrity Level" (SIL) and EN ISO 13849-1:2006 "Categories" and "Performance Level" (PL).

1.2 Safety of machinery in Europe

The EU Directives that apply to the implementation of products are based on Article 95 of the EU contract, which regulates the free exchange of goods. These are based on a new global concept ("new approach", "global approach"):

- EU Directives only specify general safety goals and define basic safety requirements.
- Technical details can be defined by means of standards by Standards Associations that have the appropriate mandate from the commission of the European Parliament and Council (CEN, CENELEC). These standards are harmonized in line with a specific directive and listed in the official journal of the commission of the European Parliament and Council. Legislation does not specify that certain standards have to be observed. When the harmonized Standards are observed, it can be assumed that the safety requirements and specifications of the Directives involved have been fulfilled.
- EU Directives specify that the Member States must mutually recognize domestic regulations.

The EU Directives are equal. This means that if several Directives apply for a specific piece of equipment or device, the requirements of all of the relevant Directives apply (e.g. for a machine with electrical equipment, the Machinery Directive and the Low-Voltage Directive apply).

1.2.1 Machinery Directive

The basic safety and health requirements specified in Annex I of the Directive must be fulfilled for the safety of machines.

The protective goals must be implemented responsibly to ensure compliance with the Directive.

Manufacturers of a machine must verify that their machine complies with the basic requirements. This verification is facilitated by means of harmonized standards.

1.2.2 Harmonized European Standards

The two Standards Organizations CEN (Comité Européen de Normalisation) and CENELEC (Comité Européen de Normalisation Électrotechnique), mandated by the EU Commission, drew-up harmonized European standards in order to precisely specify the requirements of the EC directives for a specific product. These standards (EN standards) are published in the official journal of the commission of the European Parliament and Council and must be included without revision in domestic standards. They are designed to fulfill basic health and safety requirements as well as the protective goals specified in Annex I of the Machinery Directive.

When the harmonized standards are observed, it is "automatically assumed" that the Directive is fulfilled. As such, manufacturers can assume that they have observed the safety aspects of the Directive under the assumption that these are also covered in this standard. However, not every European Standard is harmonized in this sense. Key here is the listing in the official journal of the commission of the European Parliament and Council.

The European standards regarding the safety of machines are structured in a hierarchical manner as follows:

- A standards (basic standards)
- B standards (group standards)
- C standards (product standards)

Type A standards/basic standards

A standards include basic terminology and definitions relating to all types of machine. This includes EN ISO 12100 (previously EN 292) "Safety of Machines, Basic Terminology, General Design Principles."

A standards are aimed primarily at the bodies responsible for setting the B and C standards. The measures specified here for minimizing risk, however, may also be useful for manufacturers if no applicable C standards have been defined.

Type B standards/group standards

B standards cover all safety-related standards for various different machine types. B standards are aimed primarily at the bodies responsible for setting C standards. They can also be useful for manufacturers during the machine design and construction phases, however, if no applicable C standards have been defined.

A further sub-division has been made for B standards:

- Type B1 standards for higher-level safety aspects (e.g. ergonomic principles, safety clearances from sources of danger, minimum clearances to prevent parts of the body from being crushed).
- Type B2 standards for protective safety devices are defined for different machine types (e.g. EMERGENCY STOP devices, two-hand operating circuits, interlocking elements, contactless protective devices, safety-related parts of controls).

Type C standards/product standards

C standards are product-specific standards (e.g. for machine tools, woodworking machines, elevators, packaging machines, printing machines, etc.). Product standards list requirements for specific machines. The requirements can, under certain circumstances, deviate from the basic and group standards. Type C/product standards have the highest priority for machine manufacturers who can assume that it fulfills the basic requirements of Annex I of the Machinery Directive (automatic presumption of compliance). If no product standard has been defined for a particular machine, type B standards can be applied when the machine is constructed.

A complete list of the standards specified and the mandated draft standards are available on the Internet at the following address:

<http://www.newapproach.org/>

Recommendation: Due to the rapid pace of technical development and the associated changes in machine concepts, the standards (and C standards in particular) should be checked to ensure that they are up to date. Please note that the application of a particular standard may not be mandatory provided that all the safety requirements of the applicable EU directives are fulfilled.

1.2.3 Standards for implementing safety-related controllers

If the functional safety of a machine depends on various control functions, the controller must be implemented in such a way that the probability of the safety functions failing is sufficiently minimized. EN ISO 13849-1:2006 (formerly EN 954-1) and EN 62061 define principles for implementing safety-related machine controllers which, when properly applied, ensure that all the safety requirements of the EC Machinery Directive are fulfilled. These standards ensure that the relevant safety requirements of the Machinery Directive are fulfilled.

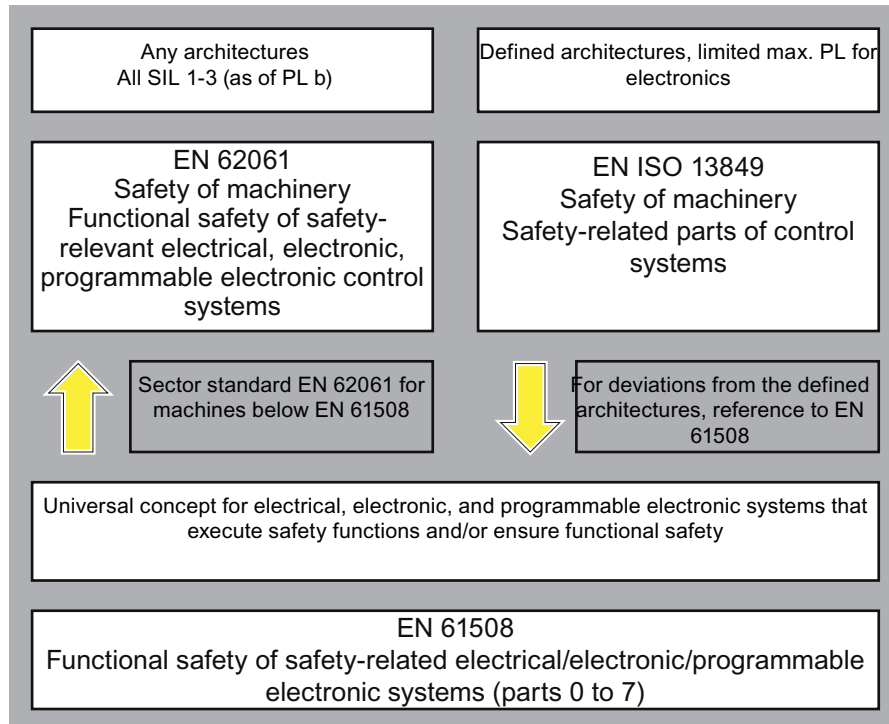


Figure 1-1 Standards for implementing safety-related controllers

The application areas of EN ISO 13849-1:2006, EN 62061 and EN 61508 are very similar. To help users make an appropriate decision, the IEC and ISO associations have specified the application areas of both standards in a joint table in the introduction to the standards. EN ISO 13849-1:2006 or EN 62061 is applied, depending on the technology (mechanical, hydraulic, pneumatic, electrical, electronic, programmable electronic), risk classification, and architecture.

	Systems for executing safety-related control functions	EN ISO 13849-1:2006	EN 62061
A	Non-electrical (e.g. hydraulic, pneumatic)	X	Not covered
B	Electromechanical (e.g. relay and/or basic electronics)	Restricted to the designated architectures (see comment 1) and max. up to PL = e	All architectures and max. up to SIL 3
C	Complex electronics (e.g. programmable electronics)	Restricted to the designated architectures (see comment 1) and max. up to PL = d	All architectures and max. up to SIL 3
D	A standards combined with B standards	Restricted to the designated architectures (see comment 1) and max. up to PL = e	X See comment 3
E	C standards combined with B standards	Restricted to the designated architectures (see comment 1) and max. up to PL = d	All architectures and max. up to SIL 3
F	C standards combined with A standards or C standards combined with A standards and B standards	X See comment 2	X See comment 3

"X" indicates that the point is covered by this standard.

Comment 1:

Designated architectures are described in Annex B of EN ISO 13849-1:2006 and provide a simplified basis for quantification.

Comment 2:

For complex electronics: Use designated architectures in compliance with EN ISO 13849-1:2006 up to PL = d or every architecture in compliance with EN 62061

Comment 3:

For non-electrical systems: Use parts that comply with EN ISO 13849-1:2006 as sub-systems.

1.2.4 EN ISO 13849-1 (previously EN 954-1)

A qualitative analysis (to EN 954-1) is not sufficient for modern controllers due to their technology. Among other things, EN 954-1 does not take into account time behavior (e.g. test interval and/or cyclic test, lifetime). This led to the probability-based approach of EN ISO 13849-1:2006 (probability of failure per time unit).

EN ISO 13849-1:2006 is based on the familiar categories used in EN 954-1. It now also takes into account complete safety functions and all the devices required to execute these. In addition to the qualitative approach of EN 954-1, EN ISO 13849-1:2006 now includes a quantitative analysis of the safety functions. Performance levels (PL), which are based on the categories, are used. The following safety-related characteristic quantities are required for devices/equipment:

- Category (structural requirement)
- PL: Performance level
- MTTF_d: Mean time to dangerous failure

- DC: Diagnostic coverage

- CCF:
Common cause failure

The standard describes how the performance level (PL) is calculated for safety-related components of the controller on the basis of designated architectures. In the event of any deviations from this, EN ISO 13849-1:2006 refers to EN 61508.

When combining several safety-related parts to form a complete system, the Standard explains how to determine the resulting PL.

Note

Since May 2007, EN ISO 13849-1:2006 has been harmonized as part of the Machinery Directive. EN 954-1 will continue to apply until 30.12.2011.

1.2.5 EN 62061

EN 62061 (identical to IEC 62061) is a sector-specific standard subordinate to IEC/EN 61508. It describes the implementation of safety-related electrical machine control systems and looks at the complete lifecycle, from the conceptual phase to decommissioning. The standard is based on the quantitative and qualitative analyses of safety functions,

whereby it systematically applies a top-down approach to implementing complex control systems (known as "functional decomposition"). The safety functions derived from the risk analysis are sub-divided into sub-safety functions, which are then assigned to real devices, sub-systems, and sub-system elements. Both the hardware and software are covered. EN 62061 also describes requirements regarding the implementation of application programs.

A safety-related control systems comprises different sub-systems. From a safety perspective, the sub-systems are described in terms of the SIL claim limit and PFH_D characteristic quantities.

Programmable electronic devices (e.g. PLCs or variable-speed drives) must fulfill EN 61508. They can then be integrated in the controller as sub-systems. The following safety-related characteristic quantities must be specified by the manufacturers of these devices.

Safety-related characteristic quantities for subsystems:

- SIL CL: SIL claim limit
- PFH_D:
Probability of dangerous failures per hour
- T1:
Lifetime

Simple sub-systems (e.g. sensors and actuators) in electromechanical components can, in turn, comprise sub-system elements (devices) interconnected in different ways with the characteristic quantities required for determining the relevant PFH_D value of the sub-system.

Safety-related characteristic quantities for subsystem elements (devices):

- λ :
Failure rate
- B10 value: For elements that are subject to wear
- T1:
Lifetime

For electromechanical devices, a manufacturer specifies a failure rate λ with reference to the number of operating cycles. The failure rate per unit time and the lifetime must be determined using the switching frequency for the particular application.

Parameters for the sub-system, which comprises sub-system elements, that must be defined during the design phase:

- T2:
Diagnostic test interval
- β :
Susceptibility to common cause failure
- DC:
Diagnostic coverage

The PFH_D value of the safety-related controller is determined by adding the individual PFH_D values for subsystems.

The user has the following options when setting up a safety-related controller:

- Use devices and sub-systems that already comply with EN ISO 13849-1:2006, IEC/EN 61508, or IEC/EN 62061. The standard provides information specifying how qualified devices can be integrated when safety functions are implemented.
- Develop own subsystems:
 - Programmable, electronic systems and complex systems: Application of EN 61508 or EN 61800-5-2.
 - Simple devices and subsystems: Application of EN 62061.

EN 62061 does not include information about non-electric systems. The standard provides detailed information on implementing safety-related electrical, electronic, and programmable electronic control systems. EN ISO 13849-1:2006 must be applied for non-electrical systems.

Note

Details of simple sub-systems that have been implemented and integrated are now available as "functional examples".

Note

IEC 62061 has been ratified as EN 62061 in Europe and harmonized as part of the Machinery Directive.

1.2.6 Series of standards EN 61508 (VDE 0803)

This series of standards describes the current state of the art.

EN 61508 is not harmonized in line with any EU directives, which means that an automatic presumption of conformity for fulfilling the protective requirements of a directive is not implied. The manufacturer of a safety-related product, however, can also use EN 61508 to fulfill basic requirements of European directives in accordance with the latest conceptual design, for example, in the following cases:

- If no harmonized standard exists for the application in question. In this case, the manufacturer can use EN 61508, although no presumption of conformity exists here.
- A harmonized European standard (e.g. EN 62061, EN ISO 13849:2006, EN 60204-1) references EN 61508. This ensures that the appropriate requirements of the directives are fulfilled ("standard that is also applicable"). When manufacturers apply EN 61508 properly and responsibly in accordance with this reference, they can use the presumption of conformity of the referencing standard.

EN 61508 covers all the aspects that must be taken into account when E/E/PES systems (electrical, electronic, and programmable electronic System) are used in order to execute safety functions and/or to ensure the appropriate level of functional safety. Other hazards (e.g. electric shock) are, as in EN ISO 13849:2006, not part of the standard.

EN 61508 has recently been declared the "International Basic Safety Publication", which makes it a framework for other, sector-specific standards (e.g. EN 62061). As a result, this standard is now accepted worldwide, particularly in North America and in the automotive industry. Today, many regulatory bodies already stipulate it (e.g. as a basis for NRTL listing).

Another recent development with respect to EN 61508 is its system approach, which extends the technical requirements to include the entire safety installation from the sensor to the actuator, the quantification of the probability of hazardous failure due to random hardware failures, and the creation of documentation covering all phases of the safety-related lifecycle of the E/E/PES.

1.2.7 Risk analysis/assessment

Risks are intrinsic in machines due to their design and functionality. For this reason, the Machinery Directive requires that a risk assessment be performed for each machine and, if necessary, the level of risk reduced until the residual risk is less than the tolerable risk. To assess these risks, the following standards must be applied:

- EN ISO 12100-1 "Safety of Machinery - basic terminology, general principles for design"
- EN ISO 13849-1:2006 (successor to EN 954-1) "Safety of machinery"
- EN ISO 14121-1 (previously EN 1050, Paragraph 5) "Safety of machinery - Risk assessment"

EN ISO 12100-1 focuses on the risks to be analyzed and the design principles for minimizing risk. EN ISO 14121-1 describes the iterative process for assessing and minimizing risk to achieve the required level of safety.

The risk assessment is a procedure that allows hazards resulting from machines to be systematically investigated. Where necessary, the risk assessment is followed by a risk reduction procedure. When the procedure is repeated, this is known as an iterative process. This can help eliminate hazards (as far as this is possible) and can act as a basis for implementing suitable protective measures.

The risk assessment involves the following:

- Risk analysis
 - Determining the limits of the machine (EN ISO 12100-1, EN ISO 14121-1 Paragraph 5)
 - Identifying the hazards (EN ISO 12100-1, EN ISO 14121-1 Paragraph 6)
 - Estimating the level of risk (EN 1050 Paragraph 7)
- Risk assessment (EN ISO 14121-1 Paragraph 8)

As part of the iterative process to achieve the required level of safety, a risk assessment is carried out after the risk estimation. A decision must be made here as to whether the residual risk needs to be reduced. If the risk is to be further reduced, suitable protective measures must be selected and applied. The risk assessment must then be repeated.

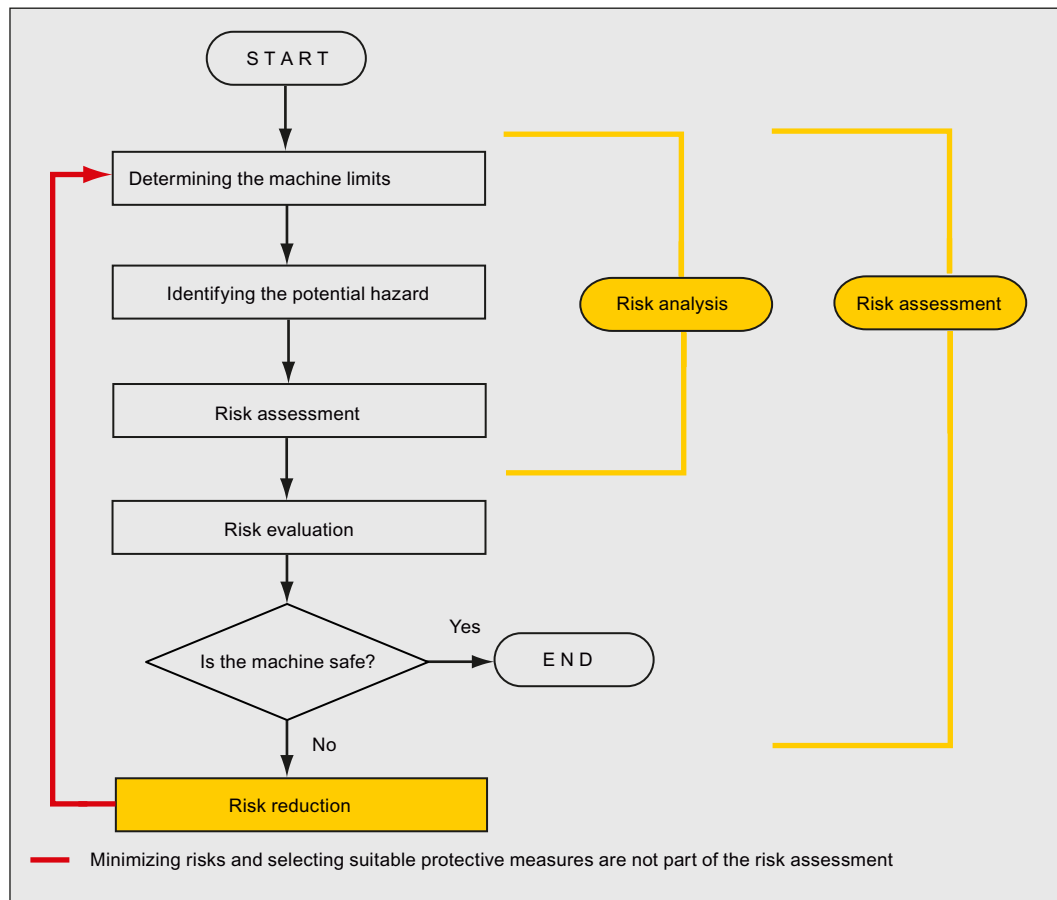


Figure 1-2 Iterative process to achieve the required level of safety to ISO 14121-1

Risks must be reduced by designing and implementing the machine accordingly (e.g. by means of controllers or protective measures suitable for the safety-related functions).

If the protective measures involve the use of interlocking or control functions, these must be designed in accordance with EN ISO 13849-1:2006. For electrical and electronic controllers, EN 62061 can be used as an alternative to EN ISO 13849-1:2006. Electronic controls and bus systems must also comply with IEC/EN 61508.

1.2.8 Risk reduction

Risk reduction measures for a machine can be implemented by means of safety-related control functions in addition to structural measures. To implement these control functions, special requirements graded according to the magnitude of the risk must be taken into account. These are described in EN ISO 13849-1:2006 or, in the case of electrical controllers (particularly programmable electronics), in EN 61508 or EN 62061. The requirements regarding safety-related controller components are graded according to the magnitude of the risk and the level to which the risk needs to be reduced.

EN ISO 13849-1:2006 defines a risk graph, which can be used instead of the categories to create hierarchical performance levels (PL).

IEC/EN 62061 uses "Safety Integrity Level" (SIL) for classification purposes. This is a quantified measure of the safety-related performance of a controller. The required SIL is also determined in accordance with the risk assessment principle to ISO 14121 (EN 1050). Annex A of the standard describes a method for determining the required Safety Integrity Level (SIL).

Regardless of which standard is applied, steps must be taken to ensure that all the machine controller components required for executing the safety-related functions fulfill these requirements.

1.2.9 Residual risk

In today's technologically advanced world, the concept of safety is relative. In practice, the ability to ensure safety to the extent that risk is permanently excluded – "zero-risk guarantee" – is impossible. The residual risk is the risk that remains once all the relevant protective measures have been implemented in accordance with the latest state of the art.

Machine/plant documentation must always refer to the residual risk (user information to EN ISO 12100-2).

1.3 Machine safety in the USA

A key difference in the legal requirements regarding safety at work between the USA and Europe is that, in the USA, no legislation exists regarding machinery safety that is applicable in all of the states and that defines the responsibility of the manufacturers/supplier. A general requirement exists stating that employers must ensure a safe workplace.

1.3.1 Minimum requirements of the OSHA

The Occupational Safety and Health Act (OSHA) from 1970 regulates the requirement that employers must offer a safe place of work. The core requirements of OSHA are specified in Section 5 "Duties".

The requirements of the OSH Act are managed by the "Occupational Safety and Health Administration" (also known as OSHA). OSHA employs regional inspectors who check whether or not workplaces comply with the applicable regulations.

The OSHA regulations are described in OSHA 29 CFR 1910.xxx ("OSHA Regulations (29 CFR) PART 1910 Occupational Safety and Health"). (CFR: Code of Federal Regulations.)

<http://www.osha.gov>

The application of standards is regulated in 29 CFR 1910.5 "Applicability of standards". The concept is similar to that used in Europe. Product-specific standards have priority over general standards insofar as they cover the relevant aspects. Once the standards are fulfilled, employers can assume that they have fulfilled the core requirements of the OSH Act with respect to the aspects covered by the standards.

In conjunction with certain applications, OSHA requires that all electrical equipment and devices that are used to protect workers be authorized by an OSHA-certified, "Nationally Recognized Testing Laboratory" (NRTL) for the specific application.

In addition to the OSHA regulations, the current standards defined by organizations such as NFPA and ANSI must be carefully observed and the extensive product liability legislation that exists in the US taken into account. Due to the product liability legislation, it is in the interests of manufacturing and operating companies that they carefully maintain the applicable regulations and are "forced" to fulfill the requirement to use state-of-the-art technology.

Third-party insurance companies generally demand that their customers fulfill the applicable standards of the standards organizations. Self-insured companies are not initially subject to this requirement but, in the event of an accident, they must provide verification that they have applied generally-recognized safety principles.

1.3.2 NRTL listing

To protect employees, all electrical equipment used in the USA must be certified for the planned application by a "Nationally Recognized Testing Laboratory" (NRTL) certified by the OSHA. NRTLs are authorized to certify equipment and material by means of listing, labeling, or similar. Domestic standards (e.g. NFPA 79) and international standards (e.g. IEC/EN 61508 for E/E/PES systems) are the basis for testing.

1.3.3 NFPA 79

NFPA 79 (Electrical Standard for Industrial Machinery) applies to the electrical equipment of industrial machines with rated voltages of less than 600 V. A group of machines that operate with one another in a coordinated fashion is also considered to be a machine.

For programmable electronics and communication buses, NFPA 79 states as a basic requirement that these must be listed if they are to be used to implement and execute safety-related functions. If this requirement is fulfilled, electronic controls and communication buses can also be used for emergency stop functions, Categories 0 and 1 (refer to NFPA 79 9.2.5.4.1.4). Like EN 60204-1, NFPA 79 no longer specifies that the electrical energy must be disconnected by electromechanical means for emergency stop functions.

The core requirements regarding programmable electronics and communication buses are: system requirements (see NFPA 79 9.4.3)

1. Control systems that contain software-based controllers must:
 - In the event of a single fault
 - (a) cause the system to switch to a safe shutdown mode
 - (b) prevent the system from restarting until the fault has been rectified
 - (c) prevent an unexpected restart
 - Offer the same level of protection as hard-wired controllers
 - Be implemented in accordance with a recognized standard that defines the requirements for such systems.
2. IEC 61508, IEC 62061, ISO 13849-1/-2:2000, and IEC 61800-5-2 are specified as suitable standards in a note.

Underwriter Laboratories Inc. (UL) has defined a special category for "Programmable Safety Controllers" for implementing this requirement (code NRGF). This category covers control devices that contain software and are designed for use in safety-related functions.

A precise description of the category and a list of devices that fulfill this requirement can be found on the Internet at the following address:

<http://www.ul.com> → certifications directory → UL Category code/ Guide information → search for category "NRGF"

TUV Rheinland of North America, Inc. is also an NRTL for these applications.

1.3.4 ANSI B11

ANSI B11 standards are joint standards developed by associations such as the Association for Manufacturing Technology (AMT) and the Robotic Industries Association (RIA).

The hazards of a machine are evaluated by means of a risk analysis/assessment. Risk analysis is an important requirement in accordance with NFPA 79, ANSI/RIA 15.06, ANSI B11.TR-3 and SEMI S10 (semiconductors). The documented findings of a risk analysis can be used to select a suitable safety system based on the safety class of the application in question.

1.4 Machine safety in Japan

The situation in Japan is different from that in Europe and the US. Legislation such as that prescribed in Europe does not exist. Similarly, product liability does not play such an important role as it does in the US.

Instead of legal requirements to apply standards have been defined, an administrative recommendation to apply JIS (Japanese Industrial Standard) is in place: Japan bases its approach on the European concept and uses basic standards as national standards (see table).

Table 1- 1 Japanese standards

ISO/IEC number	JIS number	Comment
ISO12100-1	JIS B 9700-1	Earlier designation TR B 0008
ISO12100-2	JIS B 9700-2	Earlier designation TR B 0009
ISO14121- 1 / EN1050	JIS B 9702	
ISO13849-1:2006	JIS B 9705-1	
ISO13849-2:2006	JIS B 9705-1	
IEC 60204-1	JIS B 9960-1	Without annex F or route map of the European foreword
IEC 61508-0 to -7	JIS C 0508	
IEC 62061		JIS number not yet assigned

1.5 Equipment regulations

In addition to the requirements of the guidelines and standards, company-specific requirements must be taken into account. Large corporations in particular (e.g. automobile manufacturers) make stringent demands regarding automation components, which are often listed in their own equipment specifications.

Safety-related issues (e.g. operating modes, operator actions with access to hazardous areas, EMERGENCY STOP concepts, etc.) should be clarified with customers early on so that they can be integrated in the risk assessment/risk reduction process.

1.6 Other safety-related issues

1.6.1 Information sheets issued by the Employer's Liability Insurance Association

Safety-related measures to be implemented cannot always be derived from directives, standards, or regulations. In this case, supplementary information and explanations are required.

Some regulatory bodies issue publications on an extremely wide range of subjects.

Information sheets covering the following areas are available, for example:

- Process monitoring in production environments
- Axes subject to gravitational force
- Roller pressing machines
- Lathes and turning centers - purchasing/selling

These information sheets issued by specialist committees can be obtained by all interested parties (e.g. to provide support in factories, or when regulations or safety-related measures for plants and machines are defined). These information sheets provide support for the fields of machinery construction, production systems, and steel construction.

You can download the information sheets from the following Internet address (website is in German, although some of the sheets are available in English):

<http://www.bg-metall.de/>

Click the "Downloads" quick link and select the category "Informationblätter der Fachausschüsse".

1.6.2 Additional references

- Safety Integrated: The Safety System for Industry (5th Edition and supplement), order no. 6ZB5 000-0AA01-0BA1
- Safety Integrated - Terms and Standards - Machine Safety Terminology (Edition 04/2007), order no. E86060-T1813-A101-A1

General information about SINAMICS Safety Integrated

2

2.1 Supported functions

All of the Safety Integrated functions available under SINAMICS S120 are listed in this chapter. SINAMICS makes a distinction between Safety Integrated Basic Functions and Safety Integrated Extended Functions.

The functions listed here are in conformance with the IEC 61508, SIL2 standard, in the operating mode with a high demand, Category 3 and Performance Level d (PL d) acc. to ISO 13849-1:2006, as well as IEC 61800-5-2.

The following Safety Integrated functions (SI functions) are available:

- **Safety Integrated basic functions**

These functions are part of the standard scope of the drive and can be used without requiring an additional license:

- Safe Torque Off (STO)

STO is a safety function that prevents the drive from restarting unexpectedly, in accordance with EN 60204-1:2006 Section 5.4.

- Safe Stop 1 (SS1, time controlled)

Safe Stop 1 is based on the "Safe Torque Off" function. This means that a Category 1 stop in accordance with EN 60204-1:2006 can be implemented.

- Safe Brake Control (SBC)

The SBC function permits the safe control of a holding brake.

Note regarding Power/Motor Modules in chassis format:

For the chassis format, SBC is only supported by Power/Motor Modules with order number ...3 or higher. A Safe Brake Adapter is needed in addition for this design.

Note regarding Power/Motor Modules in blocksize format:

blocksize Power Modules additionally require a Safe Brake Relay for this function.

- **Safety Integrated Extended Functions**

- Safe Torque Off (STO)

STO is a safety function that prevents the drive from restarting unexpectedly, in accordance with EN 60204-1:2006 Section 5.4.

- Safe Stop 1 (SS1, time and acceleration controlled)

The SS1 function is based on the "Safe Torque Off" function. This means that a Category 1 stop in accordance with EN 60204-1:2006 can be implemented.

- Safe Stop 2 (SS2)

The SS2 function brakes the motor safely with a subsequent transition to "Safe Operating Stop" (SOS). This means that a Category 2 stop in accordance with EN 60204-1:2006 can be implemented.

- Safe Operating Stop (SOS)

SOS protects against unintentional movement. The drive is in closed-loop control mode and is not disconnected from the power supply.

- Safely Limited Speed (SLS)

The "Safely Limited Speed" (SLS) protects against excessively high drive speeds.

- Safe Speed Monitor (SSM)

The SSM function safely monitors the speed limit and issues a safe output signal, but without initiating a response function.

- Safe Acceleration Monitor (SBR)

The Safe Acceleration Monitor function safely monitors a drive when it accelerates. It is a part of the SS1 and SS2 functions.

- Safe Brake Ramp (SBR)

The Safe Brake Ramp function safely monitors a brake ramp. It is part of the SS1 without encoder and SLS without encoder functions.

2.2 Preconditions for the Safety Extended Functions

- A license is required to use the Safety Integrated Extended Functions. The associated license key is entered in parameter p9920 in ASCII code. The license key is activated using parameter p9921 = 1. For information on how to generate the license key for the product "SINAMICS Safety Integrated Extended Functions", read the section "Licensing" in the SINAMICS S120 Function Manual. An insufficient license is indicated via the following alarm and LED:
 - A13000 --> License not sufficient
 - LED RDY --> Flashes green/red at 0.5 Hz
- Control via PROFIsafe or TM54F
- An activated speed controller in the drive
- No parallel connection
- Overview of hardware components that support the Extended Functions:
 - Control Unit CU320-2
 - Motor Modules Booksize where the order number ends as follows: ...3 or higher
 - Motor Modules Booksize Compact
 - Motor Modules Chassis where the order number ends as follows: ...3 or higher (For this design only Extended Functions **with** encoder are permitted.)
 - Motor Modules Cabinet where the order number ends with ...2 or higher
 - Power Modules Blocksize
 - Control Unit adapter CUA31 with order no.: 6SL3040-0PA00-0AA1
 - Control Unit adapter CUA32 with order no.: 6SL3040-0PA01-0AA0
 - Sensor Module SMC20, SME20/25/120/125, SMI20
 - Motors with integrated encoder and encoder evaluation with DRIVE-CLiQ interface

2.3 Controlling the Safety Integrated functions

The Safety Integrated functions can be controlled via terminals, via a PROFIsafe telegram using PROFIBUS or PROFINET or, for the Extended Functions, via the TM54F Terminal Module. In this case, control via terminals and TM54F or terminals and PROFIsafe can be simultaneously selected.

NOTICE

Safety Integrated functions with SIMOTION

PROFIsafe via PROFINET is not permitted with SIMOTION.

NOTICE

PROFIsafe or TM54F

Using a Control Unit, control is possible either via PROFIsafe or TM54F. Mixed operation is not permissible

If induction motors are used, certain Safety Integrated functions can also be used without an encoder. In operation without encoder the speed actual values are calculated from the measured electrical ACTUAL values. As a consequence, speed monitoring down to $n = 0$ rpm is also possible for operation without an encoder.

Table 2- 1 Overview of Safety Integrated functions

	Functions	Abbreviation	With encoder	Without encoder	Short description
Basic Functions	Safe Torque Off	STO	Yes	Yes	Safe Torque Off
	Safe Stop 1	SS1	Yes	Yes	Safe stopping according to stop category 1
	Safe Brake Control	SBC	Yes	Yes	Safe Brake Control
Extended Functions	Safe Torque Off	STO	Yes	Yes	Safe Torque Off
	Safe Stop 1	SS1	Yes	Yes	Safe stop according to stop category 1
	Safe Stop 2	SS2	Yes	-	Safe stop according to stop category 2
	Safe Operating Stop	SOS	Yes	-	Safe monitoring of the standstill position
	Safely Limited Speed	SLS	Yes	Yes	Safe monitoring of the maximum speed
	Safe Speed Monitor	SSM	Yes	-	Safe monitoring of the minimum speed
	Safe Acceleration Monitor	SBR	Yes	-	Safe monitoring of the drive acceleration
	Safe Brake Ramp	SBR	-	Yes	Safe Brake Ramp

The Safety Integrated functions are selected and activated and the monitoring with or without encoder is selected in the Safety screen forms of the STARTER or SCOUT tools.

2.4 Parameter, checksum, version, password

Properties of Safety Integrated parameters

The following applies to Safety Integrated parameters:

- The safety parameters are kept separate for each monitoring channel. For SINAMICS, Safety Integrated functions are controlled through two channels via a terminal at the Power Module and at the Control Unit. Both monitoring channels are tested to identify whether they are functioning correctly.
- During startup, checksum calculations (Cyclic Redundancy Check, CRC) are performed on the safety parameter data and checked. The display parameters are not contained in the CRC.
- Data storage: The parameters are stored on the non-volatile memory card.
- Factory settings for safety parameters

A reset of the safety parameters to the factory setting on a drive-specific basis using p0970 or p3900 and p0010 = 30 is only possible when the safety functions are not enabled (p9301 = p9501 = p9601 = p9801 = p10010 = 0).

A complete reset of all parameters to the factory settings (p0976 = 1 and p0009 = 30 on the Control Unit) is possible even when the safety functions are enabled (p9301 = p9501 = p9601 = p9801 = p10010 ≠ 0).

- They are password-protected against accidental or unauthorized changes.

NOTICE

The following safety parameters are not protected by the safety password:

- p9370 SI Motion acceptance test mode (Motor Module)
- p9570 SI Motion acceptance test mode (Control Unit)
- p9533 SI Motion SLS setpoint speed limitation
- p9705 BI: SI Motion Test stop signal source

Checking the checksum

For each monitoring channel, the safety parameters include one parameter for the actual checksum for the safety parameters that have undergone a checksum check.

During commissioning, the actual checksum must be transferred to the corresponding parameter for the setpoint checksum. This can be done for all checksums of a drive object at the same time with parameter p9701.

Basic functions

- r9798 SI actual checksum SI parameters (Control Unit)
- p9799 SI setpoint checksum SI parameters (Control Unit)
- r9898 SI actual checksum SI parameters (Motor Module)
- p9899 SI setpoint checksum SI parameters (Motor Module)

Extended functions

- r9398[0...1] SI Motion actual checksum SI parameters (Motor Module)
- r9399[0...1] SI Motion setpoint checksum SI parameters (Motor Module)
- r9728[0...2] SI Motion actual checksum SI parameters
- p9729[0...2] SI Motion setpoint checksum SI parameters

During each ramp-up procedure, the actual checksum is calculated via the safety parameters and then compared with the setpoint checksum.

If the actual and setpoint checksums are different, fault F01650/F30650 or F01680/F30680 is output and an acceptance test requested.

Safety Integrated versions

The safety firmware has a separate version ID for the Control Unit and Motor Module.

For the basic functions:

- r9770 SI version, drive-autonomous safety functions (Control Unit)
- r9870 SI version (Motor Module)

For the Extended Functions:

- r9590 SI Motion version safe movement monitoring (Control Unit)
- r9390 SI Motion version safe movement monitoring (Motor Module)
- r9890 SI version (Sensor Module)
- r10090 SI version TM54F

Note

For detailed requirements regarding Safety Integrated firmware, see "Safety Integrated firmware versions".

Password

The safety password protects the safety parameters against unintentional or unauthorized access.

In commissioning mode for Safety Integrated (p0010 = 95), you cannot change safety parameters until you have entered the valid safety password in p9761 for the drives or p10061 for the TM54F.

- When Safety Integrated is commissioned for the first time, the following applies:

- Safety passwords = 0
- Default setting for p10061 = 0
- Default setting for p9761 = 0

In other words:

The safety password does not need to be set during first commissioning.

- In the case of a series commissioning of Safety or in the case of spare part installation, the following applies:
 - The safety password is retained on the memory card and in the STARTER project.
 - No safety password is required in the case of spare part installation.
- Change password for the drives
 - p0010 = 95 Commissioning mode
 - p9761 = Enter "old safety password".
 - p9762 = Enter "new password".
 - p9763 = Confirm "new password".
 - The new and confirmed safety password is valid immediately.
- Change password for the TM54F
 - p0010 = 95 Commissioning mode
 - p10061 = Enter "Old TM54F Safety Password" (factory setting "0")
 - p10062 = Enter "new password"
 - p10063 = Acknowledge "new password"
 - The new and acknowledged safety password is valid immediately.

If you need to change safety parameters but you do not know the safety password, proceed as follows:

1. Set the entire drive unit (Control Unit with all connected drives/components) to the factory setting.
2. Recommission the drive unit and drives.
3. Recommission Safety Integrated.

Or contact your regional Siemens office and ask for the password to be deleted (complete drive project must be made available).

Overview of important parameters for "Password" (see SINAMICS S120/S150 List Manual)

- p9761 SI password input
- p9762 SI password new
- p9763 SI password acknowledgement
- p10061 SI password input TM54F
- p10062 SI password new TM54F
- p10063 SI password acknowledgement TM54F

2.5 DRIVE-CLiQ rules for Safety Integrated Functions

Note

For the Safety Integrated Functions (Basic and Extended Functions) the general DRIVE-CLiQ rules apply as a basic principle. You will find these rules in the chapter "Rules for connection with DRIVE-CLiQ" in the following manual:

References: /FH1/ SINAMICS S120 Drive Functions Function Manual

This specification also lists the exceptions for Safety Integrated components depending on the firmware version.

The following rules are also valid particularly for the Safety Integrated functions:

- Maximum of 6 servo axes for default cycle time settings (monitoring cycle: 12 ms; current controller cycle: = 125 μ s).
- Of which a maximum of 4 servo axes in a DRIVE-CLiQ line.
- Maximum of 6 vector axes with default cycle time settings (monitoring cycle: 12 ms; current controller cycle: = 500 μ s).
- The TM54F must be directly connected to a Control Unit via DRIVE-CLiQ. Motor Modules or infeed must not be connected to a TM54F.
- A Double Motor Module, a DMC20, or DME20, a TM54F and a CUA32 each correspond to two DRIVE-CLiQ participants.

Note: This limitation is not valid for SINUMERIK 828D-2.

3.1 Latest information

Important note for maintaining the operational safety of your system:



WARNING

Systems with safety-related characteristics are subject to special operational safety requirements on the part of the operating company. The supplier is also obliged to maintain certain measures regarding his product. For this reason, we publish a special newsletter containing information on product developments and features that are (or could be) relevant when operating safety-related systems. You should subscribe to the corresponding newsletter in order to obtain the latest information and to allow you to modify your equipment accordingly.

Go into the Internet under:

<http://automation.siemens.com>

To subscribe to the newsletter, please proceed as follows:

1. Select the desired language for the webpage.
2. Click on the menu item "Support".
3. Click on the menu item "Service & Support".
4. Click on "Subscribe to our newsletter now": You can subscribe to the individual newsletters in the following window. Under the "Product Support" heading on this page, you can see which newsletter is currently available.
5. Open the subject area "Safety Engineering - Safety Integrated".

You will now be shown which newsletter is available for this particular subject area or topic. You can subscribe to the corresponding newsletter by clicking on the box. If you require more detailed information on the newsletters then please click on these. A small supplementary window is opened from where you can take the appropriate information.

Note

You have to register and log in if you want to subscribe to any newsletters. You will be led automatically through the registration process.

6. At the very least, register for the newsletters for the following product areas:
 - Safety Integrated for SIMOTION
 - Drives

3.2 Certification

The safety functions of the SINAMICS S drive system meet the following requirements:

- Category 3 according to ISO 13849-1:2006.
- Performance Level (PL) d to EN ISO 13849-1:2006.
- Safety integrity level 2 (SIL 2) to IEC 61508

In addition, most of the safety functions of the SINAMICS S have been certified by independent institutes. An up-to-date list of certified components is available on request from your local Siemens office.

3.3 Safety instructions

Note

Additional safety information and residual risks not specified in this section are included in the relevant sections of this Function Manual.

DANGER

Safety Integrated can be used to minimize the level of risk associated with machines and plants.

Machines and plants can only be operated safely in conjunction with Safety Integrated, however, when the machine manufacturer is familiar with and observes every aspect of this technical user documentation, including the documented general conditions, safety information, and residual risks.

- Precisely knows and observes this technical user documentation - including the documented limitations, safety information and residual risks;
- Carefully constructs and configures the machine/plant. A careful and thorough acceptance test must then be performed by qualified personnel and the results documented.
- Implements and validates all the measures required in accordance with the machine/plant risk analysis by means of the programmed and configured Safety Integrated functions or by other means.

The use of Safety Integrated does not replace the machine/plant risk assessment carried out by the machine manufacturer as required by the EC machinery directive.

In addition to using Safety Integrated functions, further risk reduction measures must be implemented.

 WARNING

The Safety Integrated functions cannot be activated until the system has been completely powered up. System startup is a critical operating state with increased risk. No personnel may be present in the immediate danger zone in this phase.

The drives of vertical axes must be in torque state.

A complete forced dormant error detection cycle is required after power on (see chapter "Forced dormant error detection").

 WARNING

EN 60204-1:2006

Emergency Stop function must bring the machine to a standstill in accordance with stop category 0 or 1 (STO or SS1).

The machine must not restart automatically after EMERGENCY STOP.

When the safety functions (Basic or Extended Functions) are deactivated, an automatic restart is permitted under certain circumstances depending on the risk analysis (except when Emergency Stop is reset). An automatic start is permitted when a protective door is closed, for example.

 WARNING

After hardware and/or software components have been modified or replaced, all protective equipment must be closed prior to system startup and drive activation. Personnel shall not be present within the danger zone.

It may be necessary to carry out a partial or complete acceptance test or a simplified functional test (see chapter "Acceptance test") after having made certain changes or replacements.

Before allowing anybody to re-enter the danger zone, you should test steady control response by briefly moving the drives in forward and reverse direction (+/-).

To observe during power on:

The Safety Integrated functions are only available and can only be selected after the system has completely powered up.

 WARNING

- For a 1-encoder system, encoder faults are detected using different hardware and software monitoring functions. It is not permissible to disable these monitoring functions and they must be parameterized carefully. Depending on the fault type and responding monitoring function, stop function category 0 or 1 to EN 60204-1:2006 (fault response functions STOP A or STOP B to Safety Integrated) is selected.
- Stop function category 0 to EN 60204-1:2006 (STO or STOP A to Safety Integrated) means that the drives are not decelerate but instead coast to a standstill (the time required to coast to standstill depends on the kinetic energy). This must be included in the logic of the protective door lock, for example, by means of logic operation of SSM ($n < nx$).
- Safety Integrated functions cannot detect parameterization errors made by the machine manufacturer. The required safety level can only be reached by by means of an elaborate acceptance test.
- Motor Modules or the motor must be replaced with a device of the same type, as the parameter settings will otherwise lead to an incorrect response of the Safety Integrated functions. The corresponding drive must be re-calibrated after an encoder is replaced.

 WARNING

If an internal or external fault occurs, none or only some of the parameterized safety functions are available during the STOP-F response triggered by the fault. This must be taken into account when a delay time between STOP F and STOP B is parameterized. This applies in particular to vertical axes.

NOTICE

Changing the EDS with safe motion monitoring

An encoder which is used for Safety functions must not be switched over when a data set is switched over.

The Safety functions check the safety-relevant encoder data for changes when data sets are switched over. If a change is detected, fault F01670 is displayed with a fault value of 10, which results in a non-acknowledgeable STOP A. The safety-relevant encoder data in the various data sets must therefore be identical.

3.4 Probability of failure of the safety functions (PFH value)

Probability of failure

The probability of the failure of safety functions must be specified in the form of a PFH value (Probability of Failure per Hour) in accordance with IEC 61508, IEC 62061, and ISO 13849-1:2006. The PFH value of a safety function depends on the safety concept of the drive unit and its hardware configuration, as well as on the PFH values of other components used for this safety function.

Corresponding PFH values are provided for the SINAMICS S120 drive system, depending on the hardware configuration (number of drives, control type, number of encoders used). The various integrated safety functions are not differentiated.

The PHF values can be requested from your local sales office.

3.5 Response times

The Basic Functions are executed in the monitoring clock cycle (p9780). PROFIsafe telegrams are evaluated in the PROFIsafe scan cycle, which corresponds to twice the monitoring clock cycle (PROFIsafe scan cycle = $2 \times r9780$).

Controlling Basic Functions via terminals on the Control Unit and Motor Module

The following table lists the response times from the control via terminals until the response actually occurs.

Table 3- 1 Response times for control via terminals on the Control Unit and the Motor Module.

Function	Typical	Worst case
STO	$2 \times r9780 + p0799$	$4 \times r9780 + p0799$
SBC	$4 \times r9780 + p0799$	$8 \times r9780 + p0799$
SS1 (time controlled) Selection up until braking is initiated	$2 \times r9780 + p0799 + 2 \text{ ms}$	$4 \times r9780 + p0799 + 2 \text{ ms}$

The following tables list the response times, when selecting the STO and SS1 functions, between detecting a new selection at the Control Unit and initiating the particular brake response.

Controlling Basic Functions via Profisafe

The following table lists the response times from receiving the PROFIsafe telegram at the Control Unit up to initiating the particular response.

Table 3- 2 Response times when controlling via PROFIsafe

Function	Typical	Worst case
STO	5 x r9780	5 x r9780
SBC	4 x r9780	10 x r9780
SS1 (time controlled) Selection up until STO is initiated	5 x r9780 + p9652	5 x r9780 + p9652
SS1 (time controlled) Selection up until SBC is initiated	6 x r9780 + p9652	10 x r9780 + p9652
SS1 (time controlled) Selection up until braking is initiated	2 x r9780 + p0799 + 2 ms	4 x r9780 + p0799 + 2 ms

Control of Safety Extended Functions with encoder via PROFIsafe

The following table lists the response times from receiving the PROFIsafe telegram at the Control Unit up to initiating the particular response.

Table 3- 3 Response times when controlling via PROFIsafe

Function	Typical	Worst case
STO	4 x p9500 + r9780	4 x p9500 + 3 x r9780
SBC	4 x p9500 + 2 x r9780	4 x p9500 + 6 x r9780
SS1 (time and acceleration controlled), SS2: Selection	4 x p9500 + 2 ms	5 x p9500 + 2 ms
SBR response of safe acceleration monitoring	2 x p9500 + 2 ms	2.5 x p9500 + r9780 + t_DP ¹⁾
SOS standstill tolerance window violated	1.5 x p9500 + 2 ms	3 x p9500 + t_DP ¹⁾ + 2 ms
SLS speed limit violated ²⁾	2 x p9500 + 2 ms	3.5 x p9500 + t_DP ¹⁾ + 2 ms
SSM ³⁾	4 x p9500	4.5 x p9500 + t_DP ¹⁾

The specified response times involve internal SINAMICS response times. Program run times in the F host and the transmission time via PROFIBUS or PROFINET are not taken into account.

Control of Safety Extended Functions with encoder via TM54F

The table below shows the response times after the appearance of a signal at the terminals.

Table 3- 4 Response times with control via TM54F

Function	Typical	Worst case
STO	$2.5 \times p9500 + r9780 + 1.5 \text{ ms}$	$3 \times p9500 + 3 \times r9780 + 2 \text{ ms}$
SBC	$2.5 \times p9500 + 2 \times r9780 + 1 \text{ ms}$	$3 \times p9500 + 6 \times r9780 + 2 \text{ ms}$
SS1 (time and acceleration controlled), SS2: Selection	$2.5 \times p9500 + 3 \text{ ms}$	$4 \times p9500 + 4 \text{ ms}$
SBR response of safe acceleration monitoring	$2 \times p9500 + 2 \text{ ms}$	$2.5 \times p9500 + r9780 + t_{DP}^{1)}$
SOS standstill tolerance window violated	$1.5 \times p9500 + 2 \text{ ms}$	$3 \times p9500 + t_{DP}^{1)}) + 2 \text{ ms}$
SLS speed limit violated ²⁾	$2 \times p9500 + 2 \text{ ms}$	$3.5 \times p9500 + t_{DP}^{1)}) + 2 \text{ ms}$
SSM ⁴⁾	$3 \times p9500$	$3.5 \times p9500 + t_{DP}^{1)}$

CAUTION

Response time of Power Module PM340 for STO, controlled via terminals:
 $5 \times r9780 + p0799$

Information on the tables:

¹⁾ t_{DP} = PROFIBUS cycle with isochronous PROFIBUS master, otherwise 1 ms

²⁾ SLS: Specification of the response time required for initiation of a braking reaction in the drive, or for the output of the "SOS selected" message to the motion control system.

³⁾ SSM: The data corresponds to the times between the limit value being undershot up to sending the information via PROFIsafe.

⁴⁾ SSM: The data corresponds to the times between the limit value being undershot up to output of the information via the TM54F terminals.

Control of Safety Extended Functions without encoder via PROFIsafe

The following table lists the response times from receiving the PROFIsafe telegram at the Control Unit up to initiating the particular response.

Table 3- 5 Response times when controlling via PROFIsafe

Function	Typical	Worst case
STO	$4 \times p9500 + r9780$	$4 \times p9500 + 3 \times r9780$
SBC	$4 \times p9500 + 2 \times r9780$	$4 \times p9500 + 6 \times r9780$
SS1 (time and acceleration controlled)	$4 \times p9500 + 2 \text{ ms}$	$5 \times p9500 + 2 \text{ ms}$
SBR response of safe acceleration monitoring	$3 \times p9500 + 31 \text{ ms}$	$3.5 \times p9500 + r9780 + 57 \text{ ms}$
SLS speed limit violated ²⁾	$3 \times p9500 + 31 \text{ ms}$	$4.5 \times p9500 + r9780 + 57 \text{ ms}$

The specified response times involve internal SINAMICS response times. Program run times in the F host and the transmission time via PROFIBUS or PROFINET are not taken into account.

Control of Safety Extended Functions without encoder via TM54F

The table below shows the response times after the appearance of a signal at the terminals.

Table 3- 6 Response times with control via TM54F

Function	Typical	Worst case
STO	$2.5 \times p9500 + r9780 + 1.5 \text{ ms}$	$3 \times p9500 + 3 \times r9780 + 2 \text{ ms}$
SBC	$2.5 \times p9500 + 2 \times r9780 + 1 \text{ ms}$	$3 \times p9500 + 6 \times r9780 + 2 \text{ ms}$
SS1 (time and acceleration controlled)	$2.5 \times p9500 + 3 \text{ ms}$	$4 \times p9500 + 4 \text{ ms}$
SBR response of safe acceleration monitoring	$3 \times p9500 + 31 \text{ ms}$	$3.5 \times p9500 + r9780 + 57 \text{ ms}$
SLS speed limit violated ²⁾	$3 \times p9500 + 31 \text{ ms}$	$4.5 \times p9500 + r9780 + 57 \text{ ms}$

 CAUTION

Response time of Power Module PM340 for STO, controlled via terminals:
 $5 \times r9780 + p0799$

Information on the tables:

- ¹⁾ t_{DP} = PROFIBUS cycle with isochronous PROFIBUS master, otherwise 1 ms
- ²⁾ SLS: Specification of the response time required for initiation of a braking reaction in the drive, or for the output of the "SOS selected" message to the motion control system.
- ³⁾ SSM: The data corresponds to the times between the limit value being undershot up to sending the information via PROFIsafe.
- ⁴⁾ SSM: The data corresponds to the times between the limit value being undershot up to output of the information via the TM54F terminals.

3.6 Residual risk

The fault analysis enables the machine manufacturer to determine the residual risk at his machine with regard to the drive unit. The following residual risks are known:

WARNING

Due to the intrinsic potential of hardware faults, electrical systems are subject to additional residual risk, which can be expressed by means of the PFH value.

WARNING

- Faults in the absolute track (C-D track), cyclic interchange of the drive phases (V-W-U instead of U-V-W) and reversal of the control direction may cause acceleration of the drive. Due to the fault, however, Category 1 and 2 stop functions to EN 60204-1:2006 (fault response functions STOP B to D in accordance with Safety Integrated) are not activated.
Stop function Category 0 to EN 60204-1:2006 (fault response function STOP A in accordance with Safety Integrated) is not triggered until after the transition or delay time set in the parameter has elapsed. These faults are detected when SBR is selected (fault reaction functions STOP B/C) and stop function category 0 to EN 60204-1:2006 (fault reaction function STOP A in accordance with Safety Integrated) is triggered as early as possible regardless of this delay. Electrical faults (defective components or similar) may also lead to the response stated above.
- Simultaneous failure of two power transistors (one in the upper and the other offset in the lower inverter bridge) in the inverter may cause brief movement of the drive, depending on the number of poles of the motor.
Maximum value of this movement:
Synchronous rotary motors: Max. movement = 180° / no. of pole pairs
Synchronous linear motors: max. movement = pole width

WARNING

- Violation of limits may briefly lead to a speed higher than the speed setpoint, or the axis may pass the defined position to a certain extent, depending on the dynamic response of the drive and on parameter settings.
- Mechanical forces greater than the maximum drive torque may force a drive currently operated in position control mode out of Safe Operating Stop state (SOS) and trigger stop function category 1 to EN 60204-1:2006 (fault reaction function STOP B).

 WARNING

Within a single-encoder system:

a) a single electrical fault in the encoder

(or loose encoder shaft coupling), or a loose encoder housing will cause a static state of the encoder signals (that is, they no longer follow a movement while still returning a correct level), and prevent fault detection while the drive is in stop state (for example, drive in SOS state).

Generally, the drive is held by the active closed-loop control. Especially for drives with suspended load, from a closed-loop control perspective, it is conceivable that drives such as these move without this being detected.

The risk of an electrical fault in the encoder as described under a) is only given for few encoder types with specific function principle (for example, encoders with microprocessor controlled signal generation such as the Heidenheim EQI, Hübner HEAG 159/160, or AMO measuring systems with sin/cos signals).

The risk analysis of the machine manufacturer must include all of the faults described above. Additional safety measures have to be taken for drives with suspended/vertical or pulling loads - e.g. in order to exclude faults under a):

- Use of an encoder with analog signal generation
- Use of a two-encoder system

In order to exclude the fault described in b), for example:

- An FMEA regarding encoder shaft breakage (or slip of the encoder shaft coupling), and a solution to prevent loose encoder housings, integration of a fault exclusion process to CDV IEC 61800-5-2, or
- Implementation of a two-encoder system (the encoders must not be mounted on the same shaft).

Safety Integrated basic functions

Note

The Basic Functions are also described in the following manual:

Reference: /FH1/ SINAMICS S120 Function Manual Drive Functions.

4.1 Safe Torque Off (STO)

In conjunction with a machine function or in the event of a fault, the "Safe Torque Off" (STO) function is used to safely disconnect the torque-generating energy feed to the motor.

When the function is selected, the drive unit is in a "safe status". The switching on inhibited function prevents the drive unit from being restarted.

The two-channel pulse suppression function integrated in the Motor Modules / Power Modules is a basis for this function.

Functional features of "Safe Torque Off"

- This function is integrated in the drive; this means that a higher-level controller is not required.
- The function is drive-specific, i.e. it is available for each drive and must be individually commissioned.
- The function must be enabled using parameters.
- When the "Safe Torque Off" function is selected, the following applies:
 - The motor cannot be started accidentally.
 - The pulse suppression safely disconnects the torque-generating energy feed to the motor.
 - The power unit and motor are not electrically isolated.
- By selecting/deselecting STO, if p9307.0/p9507.0 are set to 1, in addition to the fault messages, the safety messages are also automatically withdrawn.
- A debounce function can be applied to the terminals of the Control Unit and the Motor Module in order to prevent incorrect trips due to signal disturbances. The filter times are set using parameters p9651 and p9851.

 WARNING

Appropriate measures must be taken to ensure that the motor does not undesirably move once the energy feed has been disconnected, e.g. against coasting down or for a hanging/suspended axis, the "Safe Brake Control" (SBC) function should be enabled, also refer to Chapter "Safe Brake Control".

 CAUTION

If two power transistors simultaneously fail in the power unit (one in the upper and one in the lower bridge), then this can cause brief momentary movement.

The maximum movement can be:

Synchronous rotary motors: Max. movement = $180^\circ / \text{No. of pole pairs}$

Synchronous linear motors: Max. movement = pole width

- The status of the "Safe Torque Off" function is displayed using parameters.

Enabling the "Safe Torque Off" function

The "Safe Torque Off" function is enabled via the following parameters:

- STO via terminals:
p9601.0 = 1, p9801.0 = 1
- STO via TM54F (only with "Extended Functions" option):
 - p9601.2 = 1, p9801.2 = 1
 - p9601.3 = 0, p9801.3 = 0
- STO via PROFIsafe:
 - p9601.0 = 0, p9801.0 = 0
 - Basic Functions: p9601.2 = 0, p9801.2 = 0
Extended Functions: p9601.2 = 1, p9801.2 = 1
 - p9601.3 = 1, p9801.3 = 1
- STO via PROFIsafe and terminals:
 - p9601.0 = 1, p9801.0 = 1
 - Basic Functions: p9601.2 = 0, p9801.2 = 0
Extended Functions: p9601.2 = 1, p9801.2 = 1
 - p9601.3 = 1, p9801.3 = 1

Selecting/deselecting "Safe Torque Off"

The following is executed when "Safe Torque Off" is selected:

- Each monitoring channel triggers safe pulse suppression via its switch-off signal path.
- A motor holding brake is closed (if connected and configured).

Deselecting "Safe Torque Off" represents an internal safety acknowledgement. The following is executed:

- Each monitoring channel cancels safe pulse suppression via its switch-off signal path.
- The Safety requirement "Close motor holding brake" is canceled.
- Any pending STOP F or STOP A commands are canceled (see r9772 / r9872).
- The cause of the fault must be removed.
- The messages in the fault memory must be additionally reset using the general acknowledgement mechanism.

Note

If "Safe Torque Off" is selected and de-selected through one channel within the time in p9650/p9850, the pulses are suppressed without a message being output.

However, if you want a message to be displayed, then you must reconfigure N01620/N30620 as an alarm or fault using p2118 and p2119.

Restart after the "Safe Torque Off" function has been selected

1. Deselect the function in each monitoring channel via the input terminals.
2. Issue drive enable signals.
3. Cancel the "switching on inhibited" and switch the drive back on.
 - 1/0 edge at input signal "ON/OFF1" (cancel "switching on inhibited")
 - 0/1 edge at input signal "ON/OFF1" (switch on drive)
4. Operate the drives again.

Status for "Safe Torque Off"

The status of the "Safe Torque Off" (STO) function is displayed using the parameters r9772, r9872, r9773 and r9774.

As an alternative, the status of the functions can be displayed using the configurable messages N01620 and N30620 (configured using p2118 and p2119).

Response time for the "Safe Torque Off" function

For the response times when the function is selected/deselected via input terminals, see the table in "Response times".

Examples booksize

Assumption:

Safety monitoring clock cycle CU (r9780) = 4 ms and
inputs/outputs sampling time (r0799) = 4 ms

$$t_{R_type} = 2 \times r9780 (4 \text{ ms}) + r0799 (4 \text{ ms}) = 12 \text{ ms}$$

$$t_{R_max} = 4 \times r9780 (4 \text{ ms}) + r0799 (4 \text{ ms}) = 20 \text{ ms}$$

Internal armature short-circuit with the "Safe Torque Off" function

The function "internal armature short-circuit" can be configured together with the "STO" function. However, only one of the two functions can be selected, as an OFF2 is also always triggered when STO is selected. This OFF2 disables the function "Internal armature short-circuit".

The "STO" safety function has the higher priority when simultaneously selected. If the "STO" function is initiated, then an activated "internal armature short-circuit" is disabled.

Overview of important parameters (see SINAMICS S120/S150 List Manual)

- r9720.0...10 CO/BO: SI Motion control signals integrated in the drive
- r9722.0...15 CO/BO: SI Motion status signals integrated in the drive
- r9772 CO/BO: SI Status (Control Unit)
- r9872 CO/BO: SI Status (Motor Module)
- r9773 CO/BO: SI Status (Control Unit + Motor Module)
- r9774 CO/BO: SI Status (group STO)
- p0799 CU inputs/outputs sampling time
- r9780 SI Monitoring clock cycle (Control Unit)
- r9880 SI Monitoring clock cycle (Motor Module)

4.2 Safe Stop 1 (SS1, time controlled)

General description

A Category 1 stop in accordance with EN 60204-1:2006 can be implemented with function "Safe Stop 1" (SS1). The drive decelerates with the OFF3 ramp (p1135) once "Safe Stop 1" is selected and switches to "Safe Torque Off" once the delay time set in p9652/p9852 has elapsed.

CAUTION
If the "Safe Stop 1" function (time-controlled) function has been selected by parameterizing a delay in p9652/p9852, STO can no longer be selected directly via terminals.

Functional features of Safe Stop 1

SS1 is selected by setting p9652 and p9852 (delay time) not equal to "0"

- Setting parameter p9652/p9852 has the following effect:
 - p9652/p9852 = 0: STO active
 - p9652/p9852 > 0: SS1 active
- When SS1 is selected, the drive is braked along the OFF3 ramp (p1135) and STO/SBC is automatically initiated after the delay time has expired (p9652/p9852).

After the function has been selected, the delay timer runs down - even if the function is deselected during this time. In this case, after the delay time has expired, the STO/SBC function is selected and then again de-selected immediately.
- The selection is realized through two channels - however braking along the OFF3 ramp, only through one channel.
- A debounce function can be applied to the terminals of the Control Unit and the Motor Module in order to prevent incorrect trips due to signal disturbances. The filter times are set using parameters p9651 and p9851.

Enabling Safe Stop 1 function

The "Safe Stop 1" (SS1) function is enabled via the following parameters:

- SS1 via terminals or PROFIsafe:
 - By entering the delay time in p9652 and p9852,

Prerequisite

The "Safe Torque Off" function must be enabled.

In order that the drive can brake down to a standstill even when selected through one channel, the time in p9652/p9852 must be shorter than the sum of the parameters for the data cross-check (p9650/p9850 and p9658/p9858).

The time in p9652/p9852 must be dimensioned so that after selection, the drive brakes to a standstill.

Status for Safe Stop 1

The status of the "Safe Stop 1" (SS1) function is displayed using the parameters r9772, r9872, r9773 and r9774.

Alternatively, the status of the functions can be displayed using the configurable messages N01621 and N30621 (configured using p2118 and p2119).

Overview of important parameters (see the SINAMICS S120/S150 List Manual)

- p1135[0...n] OFF3 ramp-down time
- p9652 SI Safe Stop 1 delay time (Control Unit)
- r9720.0...10 CO/BO: SI Motion drive-integrated control signals
- r9722.0...15 CO/BO: SI Motion drive-integrated status signals
- r9772 CO/BO: SI Status (Control Unit)
- r9773 CO/BO: SI Status (Control Unit + Motor Module)
- r9774 CO/BO: SI Status (group STO)
- r9872 CO/BO: SI Status (Motor Module)
- p9852 SI Safe Stop 1 delay time (Motor Module)

4.3 Safe Brake Control (SBC)

Description

The "Safe Brake Control" function (SBC) is used to control holding brakes that function according to the closed-circuit principle (e.g. motor holding brake).

The command for releasing or applying the brake is transmitted to the Motor Module/Power Module via DRIVE-CLiQ. The Motor Module/Safe Brake Relay then carries out the action and activates the outputs for the brake.

Brake activation via the brake connection on the Motor Module/Safe Brake Relay involves a safe, two-channel method.

Note

Chassis components with an order number ending from ...xxx3 and higher support this function. A Safe Brake Adapter is needed in addition for this design.

Note

To ensure that this function can be used for Blocksize Power Modules, a Safe Brake Relay must be used (for more information, see the Equipment Manual).

When the Power Module is configured automatically, the Safe Brake Relay is detected and the motor holding brake type is defaulted (p1278 = 0).

WARNING

The "Safe Brake Control" function does not detect electrical faults or mechanical defects. The system does not detect whether a brake is e.g. worn or has a mechanical defect, whether it opens or closes.

A cable break or a short-circuit in the brake winding is only detected when the state changes, i.e. when the brake either opens or closes.

Functional features of "Safe Brake Control"

- When "Safe Torque Off" (STO) is selected or when safety monitoring functions respond, SBC is executed with safe pulse suppression.
- Unlike conventional brake control, SBC is executed via p1215 through two channels.
- SBC is executed regardless of the brake control or mode set in p1215. SBC is not recommended, however, when 1215 = 0 or 3.
- The function must be enabled using parameters.
- If SBC is enabled, each time "Safe Torque Off" is selected, the holding brake is closed immediately with forced dormant error detection.
- When the state changes, electrical faults, such as e.g. a short-circuit in the brake winding or wire breakage can be detected.
- A debounce function can be applied to the terminals of the Control Unit and the Motor Module in order to prevent incorrect trips due to signal disturbances. The filter times are set using parameters p9651 and p9851.

Enabling the "Safe Brake Control" function

The "Safe Brake Control" function is enabled via the following parameters:

- p9602 SI enable safe brake control (Control Unit)
- p9802 SI enable safe brake control (Motor Module)

The "Safe Brake Control" function is not selected until at least one safety monitoring function has been enabled (i.e. $p9601 = p9801 \neq 0$).

Two-channel brake control

The brake is essentially controlled from the Control Unit. Two signal paths are available for applying the brake.

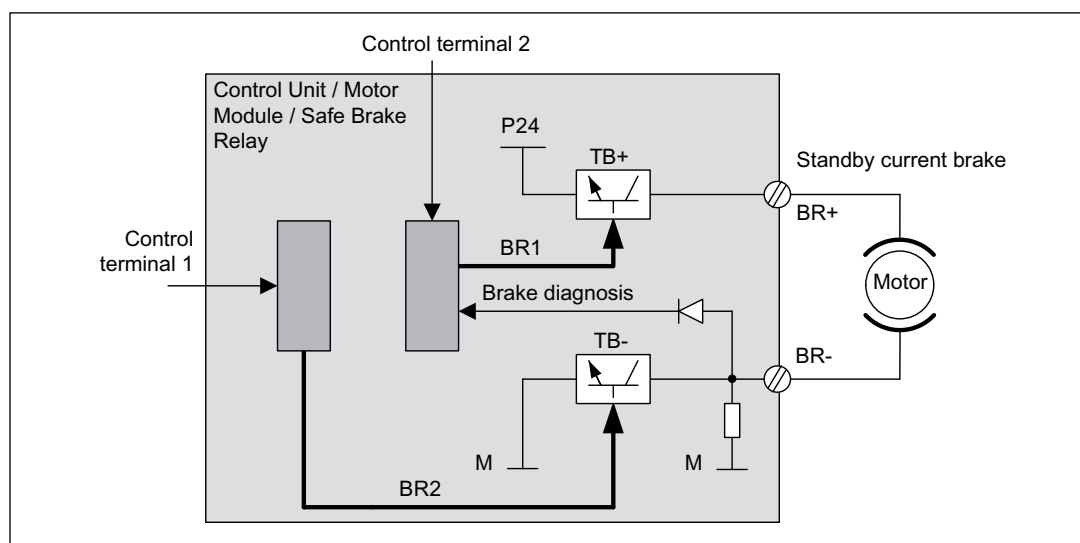


Figure 4-1 Two-channel brake control, blocksize (example)

For the "Safe Brake Control" function, the Motor/Power Module assumes a monitoring function to ensure that when the Control Unit fails or malfunctions the brake current is interrupted therefore closing the brake.

The brake diagnosis can only reliably detect a malfunction in either of the switches (TB+, TB-) when the status changes (when the brake is released or applied).

If the Motor Module or Control Unit detects a fault, the brake current is switched off and the safe status is reached.

Safe Brake Control for Motor Modules of chassis format

To be able to set higher power in the brakes of devices of this format, an additional Safe Brake Adapter (SBA) module is needed. You can find more information on the connection and wiring of the Safe Brake Adapter in the Equipment Manual.

Using parameters p9621/p9821, you can define via which digital input the Safe Brake Adapter's feedback (brake released or applied) is channeled to the Control Unit or the Motor Module.

Further functionality and the activation of the brake, i.e. reaching the safe status, are in this case the same as the above described procedure for booksize devices.

Response time with the "Safe Brake Control" function

For the response times when the function is selected/deselected via input terminals, see the table in "Response times".

Examples

Safety Integrated Basic Functions via terminals:

Safety monitoring clock cycle CU (r9780) = 4 ms and
inputs/outputs sampling time (r0799) = 4 ms

$$t_{R_typ} = 4 \times r9780 (4 \text{ ms}) + r0799 (4 \text{ ms}) = 20 \text{ ms}$$

$$t_{R_max} = 8 \times r9780 (4 \text{ ms}) + r0799 (4 \text{ ms}) = 36 \text{ ms}$$

NOTICE

When the brake is controlled via a relay with "Safe Brake Control":

If "Safe Brake Control" is used, it is not permissible to control the brake via a relay. This can result in incorrect feedback regarding a brake fault.

Overview of important parameters (see the SINAMICS S120/S150 List Manual)

- p0799 CU inputs/outputs sampling time
- p9621 BI: SI Signal source for SBA (Control Unit)
- p9622[0...1] SI SBA relay wait times (Control Unit)
- r9780 SI Monitoring clock cycle (Control Unit)
- p9821 BI: SI Signal source for SBA (Motor Module)
- p9822[0...1] SI SBA relay wait times (Motor Module)
- r9880 SI Monitoring clock cycle (Motor Module)

4.4 Safety faults

The fault messages for Safety Integrated Basic Functions are stored in the standard message buffer and can be read from there. In contrast, the fault messages for Safety Integrated Extended Functions are stored in a separate Safety message buffer (see chapter "Message buffer").

When faults associated with Safety Integrated Basic Functions occur, the following stop responses can be initiated:

Table 4- 1 Stop responses for Safety Integrated Basic Functions

Stop response	Triggered ...	Action	Effect
STOP A cannot be acknowledged	For all non-acknowledgeable Safety faults with pulse suppression.	Trigger safe pulse suppression via the switch-off signal path for the relevant monitoring channel. During operation with SBC: apply motor holding brake.	The motor coasts to a standstill or is braked by the holding brake.
STOP A	For all acknowledgeable Safety faults As a follow-up reaction of STOP F.		
STOP A is identical to stop Category 0 to EN 60204-1:2006. With STOP A, the motor is switched directly to zero torque via the "Safe Torque Off (STO)" function. A motor at standstill cannot be started again accidentally. A moving motor coasts to standstill. This can be prevented by using external braking mechanisms, e.g. holding or operating brake. When STOP A is present, "Safe Torque Off" (STO) is active.			
STOP F	If an error occurs in the data cross-check.	Transition to STOP A.	Follow-up response STOP A with adjustable delay (default setting without delay) if one of the Safety functions is selected
STOP F is permanently assigned to the data cross-check (DCC). In this way, errors are detected in the monitoring channels. After STOP F, STOP A is triggered. When STOP A is present, "Safe Torque Off" (STO) is active.			



WARNING

With a vertical axis or pulling load, there is a risk of uncontrolled axis movements when STOP A/F is triggered. This can be prevented by using "Safe Brake Control (SBC)" and a holding brake (not a safety brake!) with sufficient holding force.

Acknowledging the Safety faults

Faults associated with Safety Integrated Basic Functions must be acknowledged as follows:

1. Remove the cause of the fault.
2. Deselect "Safe Torque Off" (STO).
3. Acknowledge the fault.

If the Safety commissioning mode is exited when the Safety functions are switched off (p0010 = value not equal to 95 when p9601 = p9801 = 0), then all the Safety faults can be acknowledged.

Once Safety commissioning mode has been selected again (p0010 = 95), all the faults that were previously present reappear.

NOTICE
Safety faults can also be acknowledged (as with all other faults) by switching the drive unit off and then on again (POWER ON).
If this action has not eliminated the fault cause, the fault is displayed again immediately after power up.

Description of faults and alarms

Note

The faults and alarms for SINAMICS Safety Integrated functions are described in the following document:

Reference: /LH1/ SINAMICS S120/S150 List Manual

4.5 Forced dormant error detection

Forced dormant error detection or test of the switch-off signal paths for Safety Integrated Basic Functions

The forced dormant error detection function at the switch-off signal paths is used to detect software/hardware faults at both monitoring channels in time and is automated by means of activation/deactivation of the "Safe Torque Off" function.

To fulfill the requirements of ISO 13849-1:2006 regarding timely error detection, the two switch-off signal paths must be tested at least once within a defined time to ensure that they are functioning properly. This functionality must be implemented by means of forced dormant error detection function, triggered either in manual mode or by the automated process.

A timer ensures that forced dormant error detection is carried out as quickly as possible.

- p9659 SI timer for the forced dormant error detection.

Forced dormant error detection must be carried out at least once during the time set in this parameter.

Once this time has elapsed, an alarm is output and remains present until forced dormant error detection is carried out.

The timer returns to the set value each time the STO function is deactivated.

When the appropriate safety devices are implemented (e.g. protective doors), it can be assumed that running machinery will not pose any risk to personnel. For this reason, only an alarm is output to inform the user that a forced dormant error detection run is due and to request that this be carried out at the next available opportunity. This alarm does not affect machine operation.

The user must set the time interval for carrying out forced dormant error detection to between 0.00 and 9000.00 hours depending on the application (factory setting: 8.00 hours).

Examples of when to carry out forced dormant error detection:

- When the drives are at a standstill after the system has been switched on (POWER ON).
- When the protective door is opened.
- At defined intervals (e.g. every 8 hours).
- In automatic mode (time and event dependent)

NOTICE

The timer of the Basic Functions will be reset if the associated forced dormant error detection is executed and the Extended Functions are used simultaneously. The corresponding alarm of the Basic Functions is not triggered.

Discrepancy is not checked at the terminals used to select the Basic Functions as long as STO is set by the Extended Functions. That is, the forced dormant error detection procedure of the Basic Functions always has to be executed without simultaneous selection of STO or SS1 by the Extended Functions. It is otherwise not possible to verify the correct control through the terminals.

Safety Integrated Extended Functions

5.1 Parking note

Note

When a drive object for which Safety Integrated Extended Functions are enabled is switched to "Park" mode, the Safety Integrated software responds by selecting STO without generating a separate message. This internal STO selection is displayed in parameter r9772.19.

5.2 Safe Torque Off (STO)

5.2.1 Safe Torque Off with encoder

In addition to the control options, specified under Safety Integrated Basic Functions, "Safe Torque Off (STO) with encoder" can also be controlled via TM54F under Safety Integrated Extended Functions.

The functionality of "Safe Torque Off (STO) with encoder" is described in Chapter, "Safety Integrated basic functions".

5.2.2 Safe Torque Off without encoder

Safe Torque Off without encoder

If an induction motor is being used, Safe Torque Off (STO) can also be used without an encoder.

Function

Set p9306 = p9506 = 1 (factory setting = 0) to activate Safety Integrated functions without encoder. You can also make this setting by selecting "Without encoder" on the safety screen.

Difference between Safe Torque Off with and without an encoder

Depending on whether or not it has an encoder, STO demonstrates different restart behavior in terms of SLS after STO/OFF2 (see the "Safely Limited Speed without Encoder" chapter).

Constraint

STO without encoder is not possible for devices of chassis format.

5.3 Safe Stop 1 (SS1)

5.3.1 Safe Stop 1 with encoder (time and acceleration controlled)

Safe Stop 1 with encoder

The SS1 function with an encoder monitors whether motor acceleration reaches impermissible levels during the SS1 time.

The "Safe Stop 1" (SS1) function allows the drive to be stopped in accordance with EN 60204-1:2006, stop category 1. The drive brakes with the OFF3 ramp (p1135) once "Safe Stop 1" is selected and switches to "Safe Torque Off" (STO) once the delay time has elapsed (p9356/p9556) or when the shutdown speed is reached (p9360/p9560).

If the drive complies with acceleration monitoring limits, STO is triggered when the shutdown speed is reached. If acceleration monitoring limits are violated, messages C01706 and C30706 are output and the drive is stopped with STOP A. If the motor does not reach the shutdown speed within the set braking time, STO is still triggered and the drive coasts to a standstill. No message is issued.

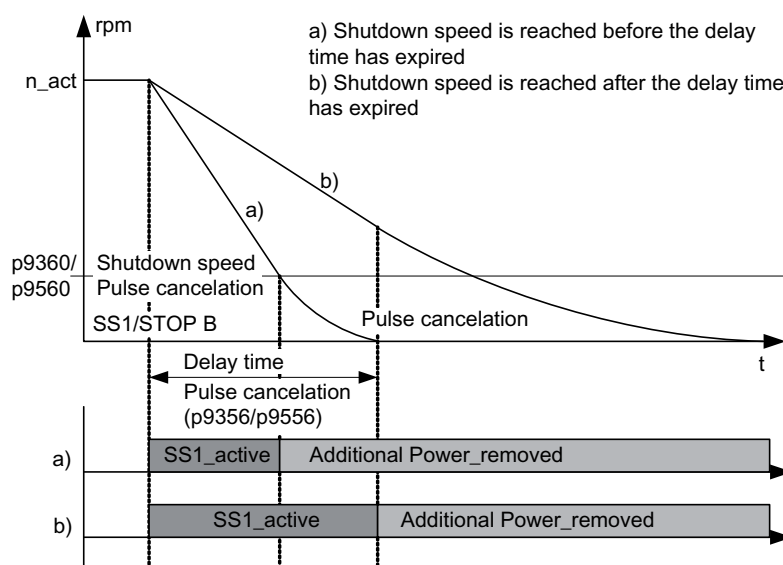


Figure 5-1 Sequence with SS1 selection

Functional features of Safe Stop 1

- The delay time starts after the function is selected. If SS1 is deselected again within this time, after the delay time has expired or after the shutdown speed has been undershot, the STO function is selected and then immediately deselected again.
- The selection is realized through two channels - however braking along the OFF3 ramp, only through one channel.
- The "Safe Acceleration Monitor" (SBR) function is activated when braking (see "Safe Acceleration Monitor").

Note

Activating SS1 can mean that the device (PLC, motion controller) which issues the speed setpoint interrupts the ramp function with OFF2.
The reason is a fault response of this device, which is initiated due to the activation of OFF3. The fault response can be prevented using suitable parameterization or wiring, which then signals the initiation of SS1 to this device.

Note

If you use SS1 under EPOS, then an OFF2 is not permitted as fault response to a following error.

Commissioning

The delay time is set by entering parameters p9356 and p9556. The wait time until the pulses are suppressed can be shortened by defining a shutdown speed in p9360 and p9560.

To enable the drive to decelerate to standstill, the time set in p9356/p9556 must be sufficient to allow the drive to decelerate to below the shutdown speed in p9360/p9560 with the OFF3 ramp (p1135).

The shutdown speed defined in p9360/p9560 must be set in such a way that personal safety or the safety of the machine is not compromised as of this speed and as a result of subsequent coasting due to the pulses being suppressed.

Responses

Speed limit violated (SBR):

- STOP A
- Safety message C01706/C30706

System errors:

1. STOP F with subsequent STOP B, followed by STOP A
2. Safety message C01711/C30711

Status for "Safe Stop 1"

The status of the "Safe Stop 1" function is displayed using the following parameters:

- r9722.1 CO/BO: SI Motion status signals, SS1 active
- r9722.0 CO/BO: SI Motion status signals, STO active (power removed)

5.3.2 Safe Stop 1 without encoder (time and speed controlled)

If an induction motor is being used, the "Safe Stop 1" (SS1) Safety Integrated function can also be activated without an encoder.

Function

The motor is immediately decelerated along the OFF3 ramp (OFF3 ramping) as soon as SS1 is triggered. Monitoring is activated once the delay time in p9582/p9382 has elapsed (SBR delay time). Monitoring ensures that the motor does not exceed the set brake ramp (envelope (monitoring ramp)) during braking. As soon as the speed drops below the shutdown speed (p9560/p9360; standstill detection), safe monitoring of the brake ramp is deactivated and safe pulse suppression (STO) is activated. If the set brake ramp is violated (exceeded), messages C01706 and C30706 are output and the drive is stopped with STO (STOP A).

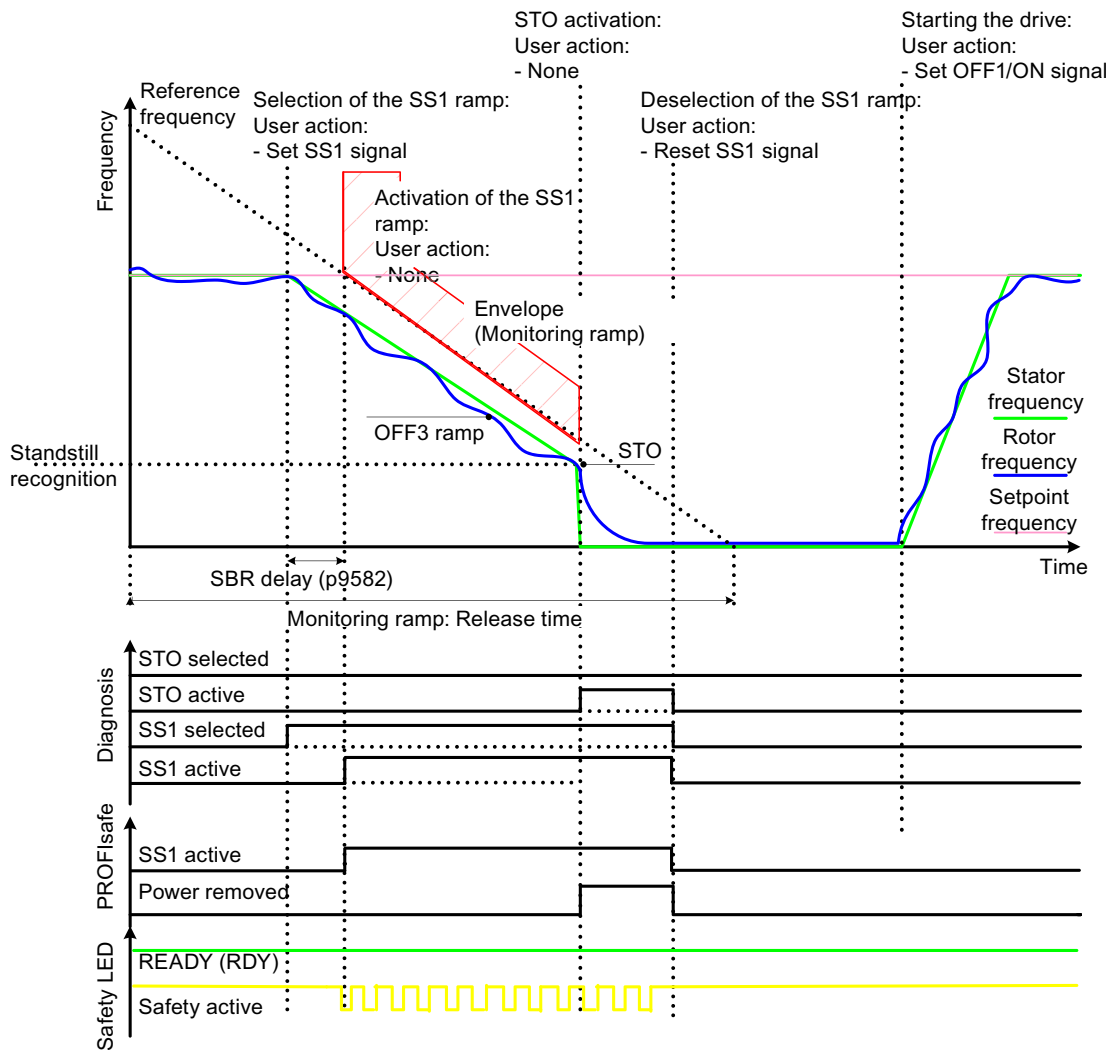


Figure 5-2 SS1 function without encoder

Constraint

STO without encoder is not possible for devices of chassis format.

Parameterization of the brake ramp "without encoder"

p9581/p9381 and p9583/p9383 are used to set the steepness of the brake ramp. Parameters p9581/p9381 determine the reference speed and parameters p9583/p9383 define the monitoring period. Parameters p9582/p9382 are used to set the time between the triggering of Safe Stop 1 and the start of brake ramp monitoring.

Restrictions

The following restrictions apply to the functions SS1 without encoder and SLS without encoder:

	Cannot be used in conjunction with
1	SINAMICS chassis format
2	Synchronous motors
3	Torque control
4	SW gating unit
	Cannot be used in conjunction with the following functions
1	Motor identification
2	Rotating measurement
3	Flying restart
4	Pole position identification
5	Vdc controller
6	DC braking
7	Measurement functions (frequency response measurement)
8	Current limitation (I_{Lim})



CAUTION

Safety functions "SS1 without encoder" and "SLS without encoder" must not be used if the motor, after it has been switched off, can still be accelerated by the mechanical elements of the connected machine component.

Whether or not a mechanical brake is installed is irrelevant here.

Examples:

1. For the hoisting gear of a crane, the suspended load can accelerate the motor as soon as the motor is switched off. In this case, safety functions SS1 and SLS are not permitted.
Even if the mechanical brake of the hoisting gear is generally applied after the motor has been switched off, the use of safety functions SS1 and SLS in this application is still forbidden.
2. A horizontal conveyor is always braked to a standstill due to friction as soon as the motor is switched off. In this case, safety functions SS1 and SLS can be used without any restriction.

5.3.3 Safe Stop 1 - Parameter

Overview of important parameters (see the SINAMICS S120/S150 List Manual)

- p1135[0...n] OFF3 ramp-down time
- p9301 SI Motion enable safety functions (Motor Module)
- p9501 SI Motion enable safety functions (Control Unit)
- p9306 SI Motion function specification (Motor Module)
- p9506 SI Motion function specification (Control Unit)
- p9348 SI Motion SBR actual velocity tolerance (Motor Module)
- p9548 SI Motion SBR actual velocity tolerance (Control Unit)
- p9356 SI Motion pulse cancelation delay time (Motor Module)
- p9556 SI Motion pulse cancelation delay time (Control Unit)
- p9360 SI Motion pulse cancelation shutdown speed (Motor Module)
- p9560 SI Motion pulse cancelation shutdown speed (Control Unit)
- p9381 SI Motion braking ramp reference value (Motor Module)
- p9581 SI Motion braking ramp reference value (Control Unit)
- p9382 SI Motion braking ramp delay time (Motor Module)
- p9582 SI Motion braking ramp delay time (Control Unit)
- p9383 SI Motion braking ramp monitoring time (Motor Module)
- p9583 SI Motion braking ramp monitoring time (Control Unit)
- r9722.0...15 CO/BO: SI Motion drive-integrated status signals

5.4 Safe Stop 2 (SS2)

5.4.1 SS2 general

The "Safe Stop 2" (SS2) function is used to brake the motor safely along the OFF3 deceleration ramp (p1135) with subsequent transition to the SOS state (see "Safe Operating Stop") after the delay time expires (p9352/p9552). The delay time set must allow for the drive to brake down to a standstill within this time. The standstill tolerance (p9330/p9530) may not be violated after this time.

After the braking operation is completed, the drives remain in the speed control mode with the speed setpoint $n = 0$.

The safety function "Safe Stop 2" (SS2) can only be used with an encoder.

WARNING

For SS2, the full rated voltage is available at the motor (V_{DCLink}) and current flows through the motor.

The default setpoint (e.g from the setpoint channel, or from a higher-level control) remains inhibited as long as SS2 is selected. The "Safe Acceleration Monitor" (SBR) function is selected during braking.

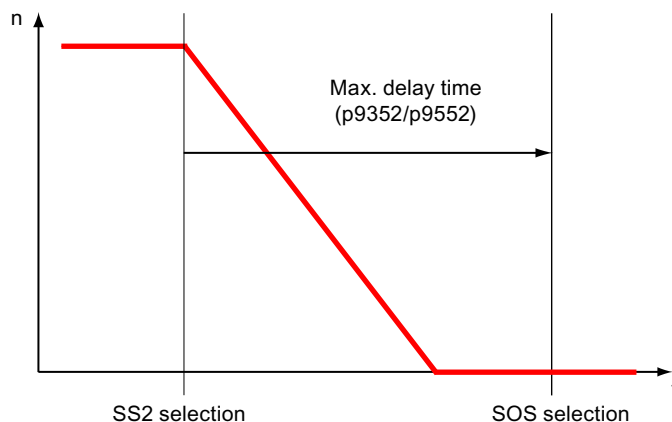


Figure 5-3 Sequence with SS2 selection

Note

If SS2 is activated, this can cause the device (PLC, motion controller) which specifies the speed setpoint to interrupt the ramp function with OFF2.

The reason is a fault response of this device, which is initiated when OFF3 is activated. The fault response can be prevented using suitable parameterization or wiring, which then signals the initiation of SS2 to this device.

Responses

Speed limit violated (SBR):

- STOP A
- Safety message C01706/C30706

Standstill tolerance violated in p9330/p9530 (SOS):

- STOP B with subsequent STOP A
- Safety message C01707/C30707

System errors:

- STOP F with subsequent STOP A
- Safety message C01711/C30711

Overview of important parameters (see the SINAMICS S120/S150 List Manual)

- p1135[0...n] OFF3 ramp-down time
- p9301 SI Motion enable safety functions (Motor Module)
- p9501 SI Motion enable safety functions (Control Unit)
- p9330 SI Motion standstill tolerance (Motor Module)
- p9530 SI Motion standstill tolerance (Control Unit)
- p9348 SI Motion SBR actual velocity tolerance (Motor Module)
- p9548 SI Motion SBR actual velocity tolerance (Control Unit)
- p9352 SI Motion transition time STOP C to SOS (Motor Module)
- p9552 SI Motion transition time STOP C to SOS (Control Unit)
- r9722.0...15 CO/BO: SI Motion drive-integrated status signals

5.4.2 EPOS and Safe Stop 2

Since the function SS2 – with its setpoint-independent braking – is not suitable for use with EPOS, the Safe Operating Stop (SOS) function can be used with delay.

By connecting the setpoint limit value (r9733) with the EPOS external speed parameter (p2594) and the EPOS function "intermediate stop" (p2640 = 0), on selection of SOS, EPOS is able to stop the drive in its tracks and then keep it under control in this state before the SOS becomes active. The maximum necessary braking time (from p2573 and p2645 from EPOS) must then be entered in the delay time for SLS/SOS (p9551/p9351) with a small safety addition: This ensures that the drive is stationary before SOS becomes active.

To do this, proceed as follows:

1. Connect the setpoint speed limit (r9733) with the external EPOS speed parameter (p2594).
2. Connect the EPOS function "intermediate stop" (p2640) with the SOS selection (p9720.3).
3. Enter the maximum necessary braking time from EPOS (p2573 und p2645) with a safety addition (+1 ...+5 %) in the SOS delay time (p9551/p9351).

Overview of important parameters (see the SINAMICS S120/S150 List Manual)

- p2645 CI: EPOS direct setpoint input/MDI, deceleration override
- p2573 EPOS maximum deceleration
- p2594 CI: EPOS maximum speed, externally limited
- p2640 BI: EPOS intermediate stop (0 signal)
- p9351 SI Motion SLS changeover delay time (Motor Module)
- p9551 SI Motion SLS(SG) changeover delay time (Control Unit)
- r9720.0...10 CO/BO: SI Motion drive-integrated control signals
- r9733[0...1] CO: SI Motion setpoint speed limit effective

5.5 Safe Operating Stop (SOS)

General description

This function serves for fail-safe monitoring of the standstill position of a drive.

Personnel can enter the protected machine areas without having to shut down the machine as long as SOS is active.

Drive standstill is monitored by means of an SOS tolerance window (p9330 and p9530). The function SOS comes into effect in the following cases:

- After SOS has been selected and after the delay time set in p9351/p9551 has expired. The drive must be braked to standstill within this delay time (e.g. by the controller).
- As a consequence of SS2.
- As a consequence of STOP C.

When this function is activated, the current actual position is saved as a comparative position, until SOS is deselected again. Any delay time is cleared after SOS is canceled and the drive can start up immediately.

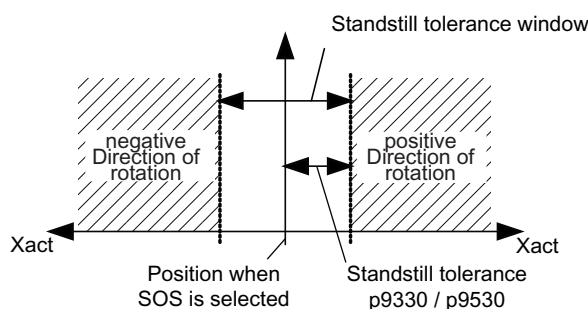


Figure 5-4 Standstill tolerance

Functional features of "Safe Operating Stop"

- The drive remains in the closed-loop control mode.
- A programmable standstill tolerance window is available.
- STOP B is the stop response after SOS has responded

Note

The size of the tolerance window should be slightly above the standard standstill monitoring limit, otherwise the standard monitoring functions will no longer be effective.

Parameter r9731 displays the safe position accuracy (load side) that can be achieved as a maximum due to the acquisition of the actual value for the safe motion monitoring functions.

Responses

Standstill tolerance violated in p9330/p9530:

- STOP B with subsequent STOP A
- Safety message C01707/C30707

System errors:

- STOP F
- Safety message C01711/C30711

Overview of important parameters (see the SINAMICS S120/S150 List Manual)

- p9301 SI Motion enable safety functions (Motor Module)
- p9501 SI Motion enable safety functions (Control Unit)
- p9330 SI Motion standstill tolerance (Motor Module)
- p9530 SI Motion standstill tolerance (Control Unit)
- p9351 SI Motion SLS changeover delay time (Motor Module)
- p9551 SI Motion SLS(SG) changeover delay time (Control Unit)
- r9722.0...15 CO/BO: SI Motion drive-integrated status signals
- r9731 SI Motion safe position accuracy

5.6 Safely-Limited Speed (SLS)

The Safely Limited Speed (SLS) function is used to protect a drive against unintentionally high speeds in both directions of rotation. This is achieved by monitoring the current drive speed up to a speed limit.

Safely Limited Speed prevents a parameterized speed limit from being exceeded. Limits must be specified based on results of the risk analysis. Up to 4 different SLS speed limits can be parameterized via parameter p9531[0..3].

5.6.1 Safely-Limited Speed with encoder

Features

- A selected speed limit is activated once SLS has been selected and after the delay time (p9351/p9551) has elapsed. When switching to a lower speed limit, the speed must be decelerated below the new maximum limit within this delay time.
- If the actual speed is higher than the new speed limit after the delay time has elapsed, a message is created with the parameterized stop response.
- Stop responses are parameterized via p9363/p9563.
- During changeover to a higher speed limit, the delay time does not take effect.
- 4 parameterizable speed limits p9331[0...3] and p9531[0...3]

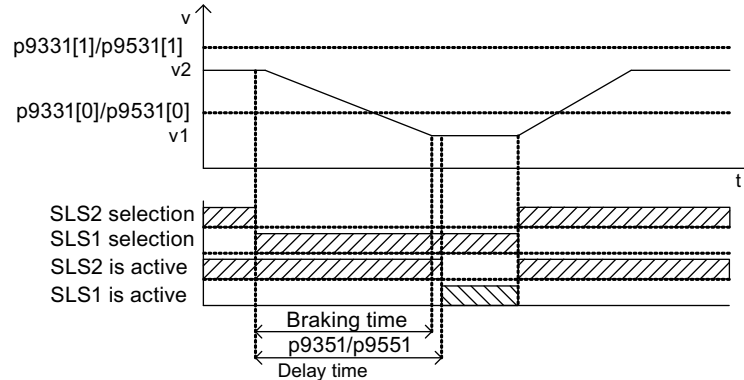


Figure 5-5 Delay time SLS speed limit value

A speed setpoint limit can be set as percentage in p9533. This value is used to calculate a speed setpoint limit r9733, depending on the selected speed limit p9531[x].

By contrast to SLS limit parameters, this parameter specifies limits on the motor side instead of limits on the load side.

- $r9733[0] = p9531[x] * p9533$; x = selected SLS stage
- $r9733[1] = - p9531[x] * p9533$; x = selected SLS stage

Changeover of speed limits

The changeover is controlled binary-coded via two F-DIs or two PROFIsafe control bits. The speed selection status can be checked using the r9720.9/r9720.10 parameters. Parameters r9722.9 and r9722.10 indicate the actual speed limit, bit r9722.4 must carry a "1" signal.

Table 5- 1 Changeover of speed limits:

F-DI for bit 0 (r9720.9)	F-DI for bit 1 (r9720.10)	Speed limit
0	0	p9331[0]/p9531[0]
1	0	p9331[1]/p9531[1]
0	1	p9331[2]/p9531[2]
1	1	p9331[3]/p9531[3]

The changeover from a lower to a higher speed limit takes effect without any delay.

The changeover from a higher to a lower limit triggers a delay time which can be set at the corresponding parameter (p9351 and p9551).

To ensure that the drive reaches the reduced speed below the new speed limit value once the delay time has elapsed, it must be decelerated accordingly within the delay time by means of the higher-level motion control/setpoint channel. The previous SLS limit value is still active during the delay time.



CAUTION

SLS level 1 must be defined as the lowest speed limit.

SLS level 1 is activated after two unacknowledged discrepancy errors; in other words, 0 is the failsafe value for the 2 F-DIs for speed level selection. The SLS levels to be switched between should, therefore, always be parameterized in ascending order, i.e. with SLS level 1 as the lowest speed and SLS level 4 as the highest.

Responses

Speed limit value exceeded:

- Configured subsequent stop STOP A / B / C / D by means of p9363/p9563
- Safety message C01714/C30714

System errors:

- STOP F
- Safety messages C01711/C30711

5.6.2 Safely Limited Speed without encoder

For induction motors, Safely Limited Speed (SLS) can also be activated without an encoder.

Features

If you are using a speed setpoint limit, after activating SLS the motor will immediately be decelerated with the OFF3-ramp from the current speed under the selected SLS [1... 4] speed limit; otherwise the motor will continue to run and violate the Safe Brake Ramp. Monitoring is activated after delay time p9582/p9382 (SI Motion brake ramp delay time Control Unit/Motor Module) has elapsed. Monitoring ensures the motor does not exceed the safe brake ramp (SBR) during braking. If the brake ramp is violated, messages C01707 and C30707 are output and the drive is stopped with STOP A or STOP B (depending on the setting made).

The new SLS speed limit is accepted as the new limit speed if either the brake ramp has reached the new SLS speed limit or the actual speed of the drive was below the new SLS speed limit for at least as long as p9582 (SI Motion brake ramp delay time Control Unit).

The SLS function then monitors whether the new actual speed remains below the selected SLS speed limit. The parameterized stop response is triggered if the limit speed is exceeded (p9563[x]).

Configuring the limits

Speed limits for SLS without encoder are configured in exactly the same way as described for SLS with an encoder.

Signal profile for SLS without encoder

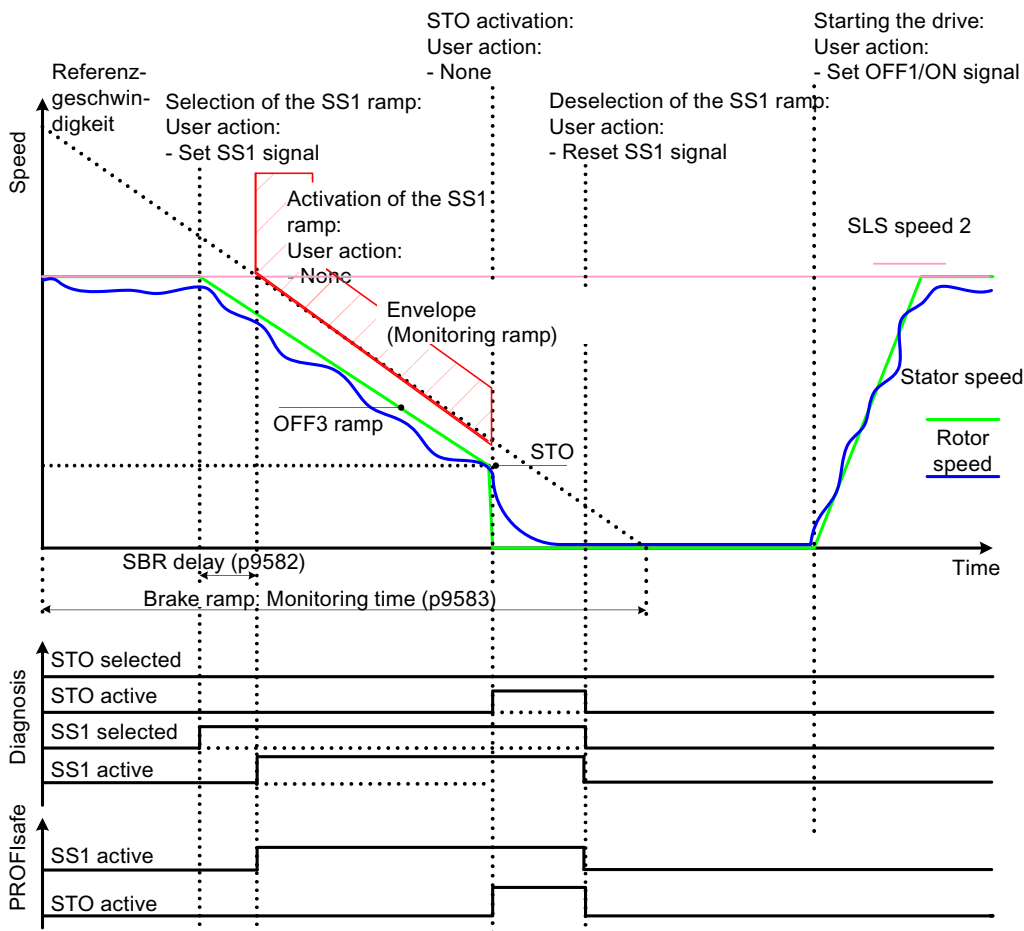


Figure 5-6 Signal profile for SLS without encoder

Constraint

STO without encoder is not possible for devices of chassis format.

Restart after OFF2

If the drive has been switched off via OFF2/STO, the following steps need to be carried out before a restart can be performed:

1st scenario:

- SLS not selected, OFF2 is active (STO active)
- Deselect STO.
- Rising edge at OFF1

2nd scenario

- Switch-on status: SLS without encoder selected, STO selected, OFF2 active
- Deselect STO.
- Unless there is a drive enable via a positive edge at OFF1 within 5 seconds, a safety fault will be output as speed monitoring will be impossible.

3rd scenario

- Switch-on status: SLS without encoder selected, STO not selected, OFF2 active
- Unless there is a drive enable via a positive edge at OFF1 within 5 seconds after ramping up is complete, a safety fault will be output as speed monitoring will be impossible.

Parameterization of the brake ramp without encoder

p9581/p9381 and p9583/p9383 are used to set the steepness of the brake ramp. Parameters p9581/p9381 determine the reference speed and parameters p9583/p9383 define the monitoring period. Parameters p9582/p9382 are used to set the time which elapses between the switchover to a lower SLS speed level and the start of brake ramp monitoring.

Restrictions

The following restrictions apply to the functions SS1 without encoder and SLS without encoder:

	Cannot be used in conjunction with
1	SINAMICS chassis format
2	Synchronous motors
3	Torque control
4	SW gating unit
	Cannot be used in conjunction with the following functions
1	Motor identification
2	Rotating measurement
3	Flying restart
4	Pole position identification
5	Vdc controller
6	DC braking
7	Measurement functions (frequency response measurement)
8	Current limitation (I_{Lim})



CAUTION

Safety functions "SS1 without encoder" and "SLS without encoder" must not be used if the motor, after it has been switched off, can still be accelerated by the mechanical elements of the connected machine component.

Whether or not a mechanical brake is installed is irrelevant here.

Examples:

1. For the hoisting gear of a crane, the suspended load can accelerate the motor as soon as the motor is switched off. In this case, safety functions SS1 and SLS are not permitted.

Even if the mechanical brake of the hoisting gear is generally applied after the motor has been switched off, the use of safety functions SS1 and SLS in this application is still forbidden.

2. A horizontal conveyor is always braked to a standstill due to friction as soon as the motor is switched off. In this case, safety functions SS1 and SLS can be used without any restriction.

5.6.3 Safely Limited Speed - Parameter

Overview of important parameters (see the SINAMICS S120/S150 List Manual)

- p9301.0 SI Motion enable safety functions (Motor Module)
- p9501.0 SI Motion enable safety functions (Control Unit)
- p9306 SI Motion function specification (Motor Module)
- p9506 SI Motion function specification (Control Unit)
- p9331[0...3] SI Motion SLS limits (Motor Module)
- p9531[0...3] SI Motion SLS (SG) limits (Control Unit)
- p9533 SI Motion speed setpoint limit (Control Unit)
- p9351 SI Motion SLS changeover delay time (Motor Module)
- p9551 SI Motion SLS(SG) changeover delay time (Control Unit)
- p9363[0...3] SI motion SLS stop response (Motor Module)
- p9563[0...3] SI motion SLS (SG)-specific stop response (Control Unit)
- r9720.0...10 CO/BO: SI Motion drive-integrated control signals
- r9722.0...15 CO/BO: SI Motion drive-integrated status signals
- r9733[0...1] CO: SI Motion setpoint speed limit effective

5.6.4 EPOS and Safely-Limited Speed

If safe speed monitoring (SLS) is also to be used at the same time as the EPOS positioning function, EPOS must be informed of the activated speed monitoring limit. Otherwise the speed monitoring limit can be violated by the EPOS setpoint input. Through the SLS monitoring, this violation leads to the drive being stopped and so abandoning the planned movement sequences. Here the relevant safety faults are output first of all and only then the sequential faults created by EPOS.

With its parameter r9733, the SLS function provides a setpoint limit value which, when taken into account, prevents the SLS limit value from being violated.

This means that the setpoint limit value in r9733 must therefore be transferred to the input for the maximum setpoint speed/velocity of EPOS (p2594), to prevent an SLS limit value violation as a result of the EPOS setpoint input. You need to set the delay time for SLS/SOS (p9551/p9351) so that the SLS only becomes active after the maximum required time for the speed to be reduced below the SLS limit. This required braking time is determined by the current speed, the jerk limit in p2574 and the maximum delay in p2573.

Overview of important parameters (see the SINAMICS S120/150 List Manual)

- p2573 EPOS maximum deceleration
- p2574 EPOS jerk limitation
- p2593 CI: EPOS LU/revolution LU/mm
- p2594 CI: EPOS maximum speed, externally limited
- p9551 SI Motion SLS(SG) changeover delay time (Control Unit)
- r9733(0,1) CO: SI Motion setpoint speed limit effective

5.7 Safe Speed Monitor (SSM)

General description

The "Safe Speed Monitor" function is used for reliably detecting when a velocity limit value has been undershot (p9346/p9546) (e.g. for standstill detection) in both directions. A fail-safe output signal is available for further processing.

The function is activated automatically as soon as the Extended Functions are enabled with p9301.0 = p9501.0 = 1.

The SSM function is only available for operation with an encoder.

NOTICE

If 0 is entered for p9368/p9568, the speed limit of the SSM function (p9346/p9546) is also used as shutdown limit for the SBR function (safe acceleration monitoring).

In this case, the effects of safe acceleration monitoring are therefore restricted if a relatively high SSM/SBR velocity limit is set when using the SS1 and SS2 stop functions.



WARNING

STOP F (indicated by safety message C01711/C30711) only results in a follow-up response (STOP B / STOP A) if at least one of the safety functions (SOS or SLS) is active or has been selected. If only the SSM function is active, a STOP F crosswise comparison error does not result in a follow-up response STOP B / STOP A.

If SSM is to be used as a safety function, at least one of the SOS or SLS functions must be active/selected (e.g. by selecting a high SLS level).

Functional features of "Safe Speed Monitor"

The parameter p9346/p9546 "SI Motion SSM (SGA n < nx) velocity limit n_x (CU)" is used to set the velocity limit. The abbreviation "SGA n < nx" indicates the safety function required for determining an output signal when a parameterizable velocity limit has been undershot.

If the velocity limit for the "Safe Speed Monitor" feedback signal (n < n_x) is undershot, the "SSM feedback signal active" signal (SGA n < n_x) is set. When the set threshold value has been undershot, the "Safe Acceleration Monitor" (SBR) function is also deactivated (see p9368/p9568). If p9368 = p9568 = 0, then p9346/p9546 (SSM feedback signal) is also used as a trip threshold for the SBR monitoring.

The hysteresis for the SSM output signal is set in parameter p9347/p9547 "SI Motion SSM Speed Hysteresis n_x". Therefore the SSM output signal can either take the state "1" or "0" – depending on which direction the belt is reached from.

If the maximum permissible velocity tolerance is overshoot (i.e. one channel displays a velocity less than $p9546 - p9547$, while the other channel displays a velocity greater than $p9546$), a STOP F is issued. As an additional function, parameters $p9347/p9547$ are used to define the maximum tolerance of the velocity actual values between the two channels.

In addition, the output signal for SSM can be smoothed by means of a PT1 filter by setting a filter time $p9345/9545$.

During safe motion monitoring, the hysteresis and filtering functions can be activated or deactivated jointly using the enable bit $p9301.16$ (Motor Modules) and $p9501.16$ (CU). In the default setting, the functions are deactivated ($p9301.16/p9501.16 = 0$).

NOTICE

Exception

The activated "hysteresis and filtering" function is evaluated as activated monitoring function and, after a STOP F, also results in a subsequent STOP B/STOP A response.

The following diagram shows the characteristic of the safe output signal SSM when the hysteresis is active:

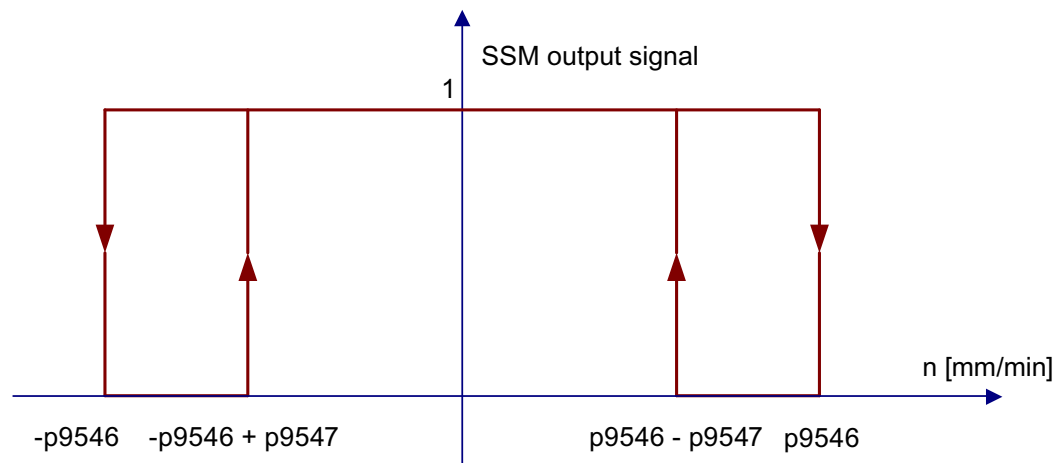


Figure 5-7 Safe output signal for SSM with hysteresis

Note

When the hysteresis and filtering are activated with output signal SSM, the axes behave in a time-delayed manner. This is a characteristic of the filter.

Features

- Safe monitoring of the speed limit specified in p9346 and p9546
- Parameterizable hysteresis via p9347 and p9547
- Variable PT1 filter via p9345 and p9545
- Fail-safe output signal
- No stop response
- This function is not available for speed monitoring without an encoder.

Overview of important parameters (see the SINAMICS S120/S150 List Manual)

- p9345 SI Motion SSM (SGA n < nx) filter time (Motor Module)
- p9545 SI Motion SSM (SGA n < nx) filter time (Control Unit)
- p9346 SI Motion SSM velocity limit (Motor Module)
- p9546 SI Motion SSM (SGA n < nx) velocity limit n_x (CU)
- p9347 SI Motion velocity hysteresis (crosswise) (Motor Module)
- p9547 SI Motion velocity hysteresis (crosswise) (Control Unit)
- p9368 SI Motion SBR velocity limit (Motor Module)
- p9568 SI Motion SBR velocity limit (CU)
- r9722.0...15 CO/BO: SI Motion drive-integrated status signals

5.8 Safe Acceleration Monitor (SBR)

Safe Acceleration Monitor with encoder

The "Safe Acceleration Monitor" (SBR) function is used to safely monitor drive acceleration. This function is part of the SS1 (time and acceleration-controlled) and SS2 (or STOP B and STOP C) safety functions.

Features

A STOP A is generated if any drive acceleration within the ramp-down phase exceeds the tolerance defined in p9348/p9548. The monitoring function is activated for SS1 (or STOP B) and SS2 (or STOP C) and is deactivated after the speed drops below the value set in p9368/p9568.

NOTICE

If 0 is entered for p9368/p9568, the speed limit of the SSM function (p9346/p9546) is also used as shutdown limit for the SBR function (safe acceleration monitoring). The SBR is deactivated if the speed is below this limit.

In this case, the effects of safe acceleration monitoring are therefore greatly restricted if a relatively high SSM/SBR velocity limit is set when using the SS1 and SS2 stop functions.

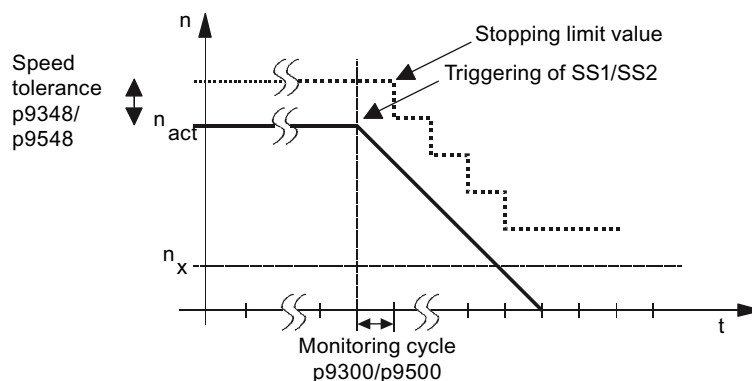


Figure 5-8 Characteristics of the shutdown limit for SBR

Calculating the SBR tolerance of the ACTUAL speed:

- The following applies when parameterizing the SBR tolerance:
 - The maximum speed increase after SS1 / SS2 is triggered is derived from the effective acceleration (a) and the duration of the acceleration phase.
 - The duration of the acceleration phase is equivalent to one monitoring clock cycle (p9300/p9500) MC (delay from detecting an SS1 / SS2 until $n_{set} = 0$):
- SBR tolerance:
Actual speed SBR = acceleration * acceleration duration
The following setup rule is derived thereof:
 - At linear axes:
$$\text{SBR tolerance [mm/min]} = a \text{ [m/s}^2\text{]} * \text{MC [s]} * 1000 \text{ [mm/m]} * 60 \text{ [s/min]}$$
 - At rotary axes:
$$\text{SBR tolerance [rev/min]} = a \text{ [rev/s}^2\text{]} * \text{MC [s]} * 60 \text{ [s/min]}$$
- Recommendation:
The SBR tolerance value entered should be approx. 20% higher than the calculated value.

Responses

Speed limit violated (SBR):

- STOP A
- Safety message C01706/C30706

System errors:

- STOP F with subsequent STOP A
- Safety message C01711/C30711

Features

- Element of the SS1 (time and acceleration controlled) and SS2 functions
- Parameterizable, minimum shutdown speed to be monitored

Overview of important parameters (see the SINAMICS S120/S150 List Manual)

- p9346 SI Motion SSM velocity limit (Motor Module)
- p9546 SI Motion SSM (SGA $n < n_x$) velocity limit n_x (CU)
- p9348 SI Motion SBR actual velocity tolerance (Motor Module)
- p9548 SI Motion SBR actual velocity tolerance (Control Unit)
- p9368 SI Motion SBR velocity limit (Motor Module)
- p9568 SI Motion SBR velocity limit (CU)

5.9 Safe Brake Ramp (SBR)

If an induction motor is being used, the "Safe Brake Ramp" (SBR) Safety Integrated function can be activated without an encoder. The Safe Brake Ramp (SBR) function provides a safe method for monitoring the brake ramp. The Safe Brake Ramp function – safe brake ramp monitoring (SBR) – is always used to monitor braking when using the "SS1 without encoder" and "SLS without encoder" functions.

Functional features of Safe Brake Ramp without encoder

The motor is immediately decelerated along the OFF3 ramp as soon as SS1 or SLS is triggered (if setpoint speed limitation is used). Monitoring of the brake ramp (envelope (monitoring ramp)) is activated after delay time p9582/p9382 (SI Motion brake ramp delay time (SBR delay time), Control Unit/Motor Module) has elapsed. Monitoring ensures that the motor does not exceed the set brake ramp (SBR) when braking. Deactivation of the safe monitoring of the brake ramp occurs

- for SS1
 - as soon as the speed drops below the shutdown speed (p9560/p9360; standstill detection).
- for SLS:
 - as soon as the new SLS level has been reached
 - or
 - as soon as the actual speed has dropped below the new level and remains below this level for the parameterized timer time.

Additional specific functions (e.g. STO, new SLS speed limit, etc.) are activated at this point, depending on the Safety Integrated function used

5.9 Safe Brake Ramp (SBR)

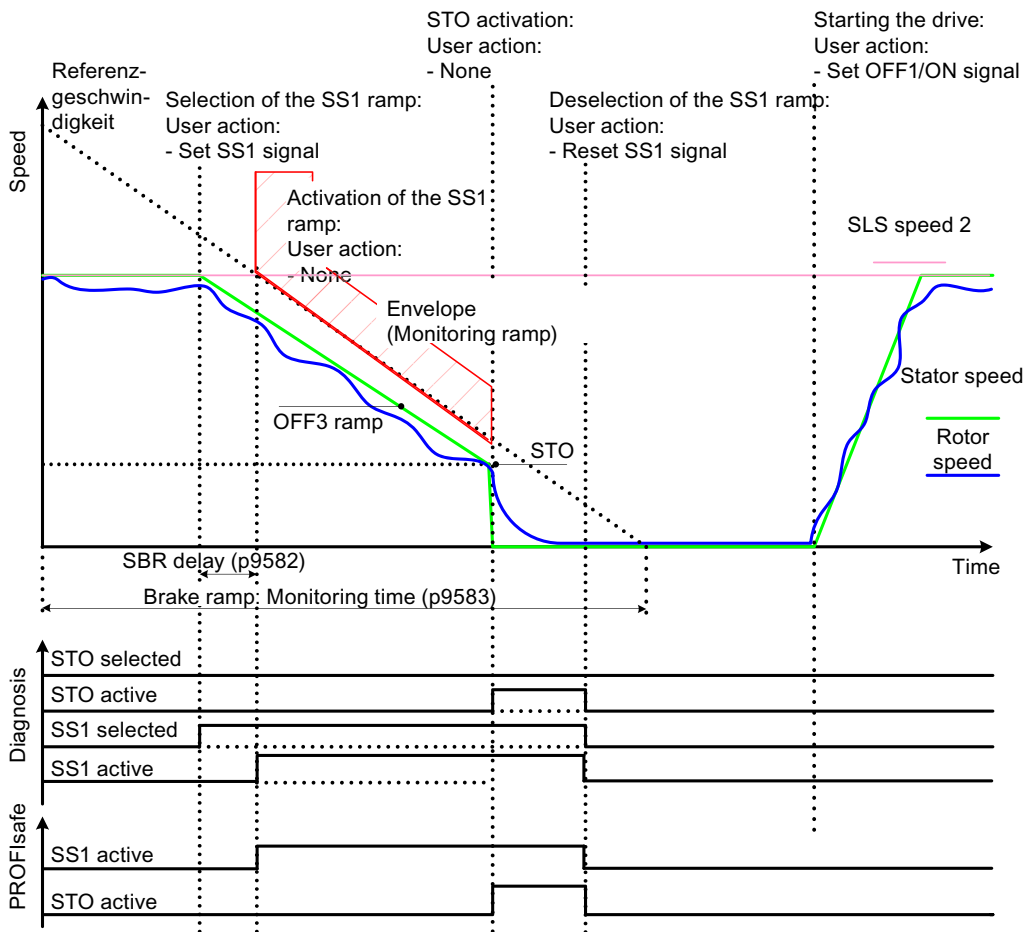


Figure 5-9 Safe Brake Ramp without encoder

Parameterization of the brake ramp "without encoder"

p9581/p9381 (SI Motion brake ramp reference value, Control Unit/Motor Module) and p9583/p9383 (SI Motion brake ramp monitoring time, Control Unit/Motor Module) are used to set the steepness of the brake ramp. Parameters p9581/p9381 determine the reference speed and parameters p9583/p9383 define the monitoring period. Parameters p9582/p9382 are used to set the time between the triggering of Safe Stop 1 and the start of brake ramp monitoring.

Responses to brake ramp violations (SBR)

- Safety messages C01706 (SI Motion CU: SBR limit exceeded) and C30706 (SI Motion MM: SBR limit exceeded)
- Drive stopped with STOP A

Features

- Part of the "SS1 without encoder" and "SLS without encoder" functions.
- Parameterizable safe brake ramp

Overview of important parameters (see the SINAMICS S120/S150 List Manual)

- p9360 SI Motion pulse suppression shutdown speed (Motor Module)
- p9560 SI Motion pulse suppression shutdown speed (Control Unit)
- p9381 SI Motion brake ramp reference value (Motor Module)
- p9581 SI Motion brake ramp reference value (Control Unit)
- p9382 SI Motion brake ramp delay time (Motor Module)
- p9582 SI Motion brake ramp delay time (Control Unit)
- p9383 SI Motion brake ramp monitoring time (Motor Module)
- p9583 SI Motion brake ramp monitoring time (Control Unit)

5.10 Safety faults

Stop responses

Faults with Safety Integrated Extended Functions and violation of limits can trigger the following stop responses:

Table 5- 2 Overview, stop responses

Stop response	Triggered ...	Action	Effect
STOP A	For all acknowledgeable safety faults with pulse disable.	Immediate pulse cancelation	Drive coasts down
STOP B	Examples: - standstill tolerance violated in p9330/p9530 (SOS). - Configured subsequent stop p9363/p9563 for SLS. - Subsequent response of STOP F.	Immediate input of speed setpoint = 0 and start of timer t_B . Once t_B or $n_{act} < n_{shutdown}$, STOP A is triggered.	STOP B with subsequent STOP A. The drive decelerates along the OFF3 ramp and then switches to STOP A.
STOP C	Configurable subsequent stop p9363/p9563 with SLS.	Immediate input of speed setpoint = 0 and start of timer t_C . Once t_C has elapsed, SOS is selected.	The drive decelerates along the OFF3 ramp; SOS is then selected.
STOP D	Configurable subsequent stop p9363/p9563 with SLS.	Timer t_D starts. No drive-integrated response. SOS is activated on expiration of t_D .	The drive must be decelerated by the higher-level control (within the drive group)! Once t_D has elapsed, SOS is selected. An automatic response is only triggered if the standstill tolerance window is violated in SOS.
STOP F	If a fault occurs in the crosswise data comparison. Follow-up response STOP B.	Timer t_{F1} (Basic Functions) or t_{F2} (Extended Functions) No drive response	If a safety function (SOS, SLS) has been selected or if SSM with hysteresis has been enabled, transition to STOP A after t_{F1} (Basic Functions) has elapsed or STOP B after t_{F2} (Extended Functions) has elapsed.

Note

A delay time between STOP F and STOP B should only be set if an additional response is initiated during this time when the "Internal Event" (r9722.7) message signal is evaluated.

Further, when using the delay time, a monitoring function should always be selected (e.g. SLS with a high limit speed) or the hysteresis of SSM should be configured.

When hysteresis is activated for SSM, then this should be considered to be an activated monitoring function.

On delays at the stop response transitions

- t_B : p9356/p9556
- t_C : p9352/p9552
- t_D : p9353/p9553
- t_{F1} : p9658/p9858
- t_{F2} : p9355/p9555
- $n_{shutdown}$: p9360/p9560

Stop response priorities

Table 5- 3 Stop response priorities

Priority classes	Stop response
Highest priority	STOP A
.....	STOP B
...	STOP C
..	STOP D
Lowest priority	STOP F

Priorities of stop responses and Extended Functions

Table 5- 4 Priorities of stop responses and Extended Functions

Stop response / Extended Function		Highest priority	...	...	...	Lowest priority
		STOP A	STOP B	STOP C	STOP D	STOP F
Highest priority	STO	STOP A / STO	STO	STO	STO	STO
.....	SS1	STOP A	STOP B / SS1	SS1	SS1	SS1
...	SS2	STOP A	STOP B	STOP C / SS2	SS2	SS2 / STOP B ²⁾
..	SOS	STOP A ¹⁾	STOP B ¹⁾	SOS	SOS	STOP B ²⁾
Lowest priority	SLS	STOP A ³⁾	STOP B ³⁾	STOP C ⁴⁾	STOP D ⁴⁾	STOP B ²⁾

1) The SOS monitoring function remains active, although the fault response in the event of a fault can no longer be triggered because it is already present.
2) STOP B is the subsequent stop of STOP F, which is activated after a parameterizable time. STOP F alone does not have any effect; the active safety function is still present.
3) The SLS monitoring function remains active, although the fault response in the event of a fault can no longer be triggered because it is already present.
4) SLS remains active during the braking phase, after which the system switches to SOS.

The table above specifies which stop response or safety function is set when a STOP is triggered when a safety function is active. The STOPS are arranged here from left to right in descending order of priority (STOP A-F).

No overall priority is assigned in the individual safety functions. SOS remains active, for example, even if STO is requested. The safety functions that cause the drive to decelerate (STO, SS1, SS2) are specified from top to bottom in descending order of priority.

If a field contains two entries, the stop responses and safety functions have the same priority. Explanation:

- STOP A corresponds to STO
- STOP B corresponds to SS1
- STOP C corresponds to SS2
- When the SS2 function is active, STOP F results in subsequent stop B. SS2 remains active.

Examples for illustrating the information in the table:

1. Safety function SS1 has just been selected. STOP A remains active; a STOP B operation that is currently in progress is not interrupted by this. Any remaining STOP C-F would be replaced by SS1.
2. The SLS safety function is selected. This selection does not modify the function of STOP A-D. A STOP F now triggers a STOP B because a safety function has been activated.
3. Stop response, STOP C is selected. If the STO or SS1 safety functions are active, this does not have any effect. If SS2 is active, this brake ramp is retained. If SOS is active, SOS remains effective, which is also the end status of STOP C. When SLS is selected, the drive is decelerated with STOP C.

Acknowledging the safety faults**General****NOTICE**

Safety faults can also be acknowledged (as with all other faults) by switching the drive unit off and then on again (POWER ON).
If this action has not eliminated the fault cause, the fault is displayed again immediately after power up.

Acknowledgement via TM54F

Parameter p10006 "SI acknowledgement internal event input terminal" allows faults to be acknowledged in the safety drives and in TM54F itself.

The **"safe fault acknowledgement"** mechanism functions as follows:

The safe input F-DI on the TM54F parameterized with the function p10006 "Safety Integrated acknowledgement internal event input terminal" is energized. This allows faults that occurred in the firmware installed in the Control Unit or Motor Module to be acknowledged by means of a safe input signal. The falling edge at this input resets the status "Internal Event" in the drives and, if used, in the TM54F.

To prevent safety faults from being acknowledged unintentionally or incorrectly, the signal at the TM54F F DI terminal, which was parameterized for acknowledgement purposes, must be at level "0" in the idle state. To trigger the acknowledgement (falling edge at F DI), the signal must first be set to "1" and then back to "0". If the required idle state is not reached, an alarm is output.

After "safe fault acknowledgement", an acknowledgement must be issued on the Control Unit in order to:

- Delete the TM54F faults from the fault buffer
- Reset the pending, red "Ready" LED on the TM54F

Acknowledgement via PROFIsafe

The higher-level controller sets the signal "Internal Event ACK" via the PROFIsafe telegram (STW bit 7) separately for each drive object. A falling edge in this signal sets the status "Internal Event" in the relevant drive, which acknowledges the fault.

Faults in the drive objects (DOs) cannot be acknowledged by the higher-level controller in the line-up but must instead be acknowledged separately for each individual drive object.

Description of faults and alarms

Note

The faults and alarms for SINAMICS Safety Integrated are described in the following documentation:

Reference: /LH1/ SINAMICS S120/S150 List Manual

5.11 Message buffer

In addition to the fault buffer for F... faults and the alarm buffer for A... alarms (see the corresponding chapter in: /IH1/ SINAMICS S120 Commissioning Manual), specifically for Safety Integrated Extended Functions, there is also a message buffer for C... safety messages.

The fault messages for the Safety Integrated Basic Functions are stored in the standard fault buffer (see chapter "Buffer for faults and alarms" in /IH1/ SINAMICS S120 Commissioning Manual).

The message buffer for safety messages is similar to the fault buffer for fault messages. The message buffer comprises the message code, message value, and message time (received/resolved). The following diagram shows how the message buffer is structured:

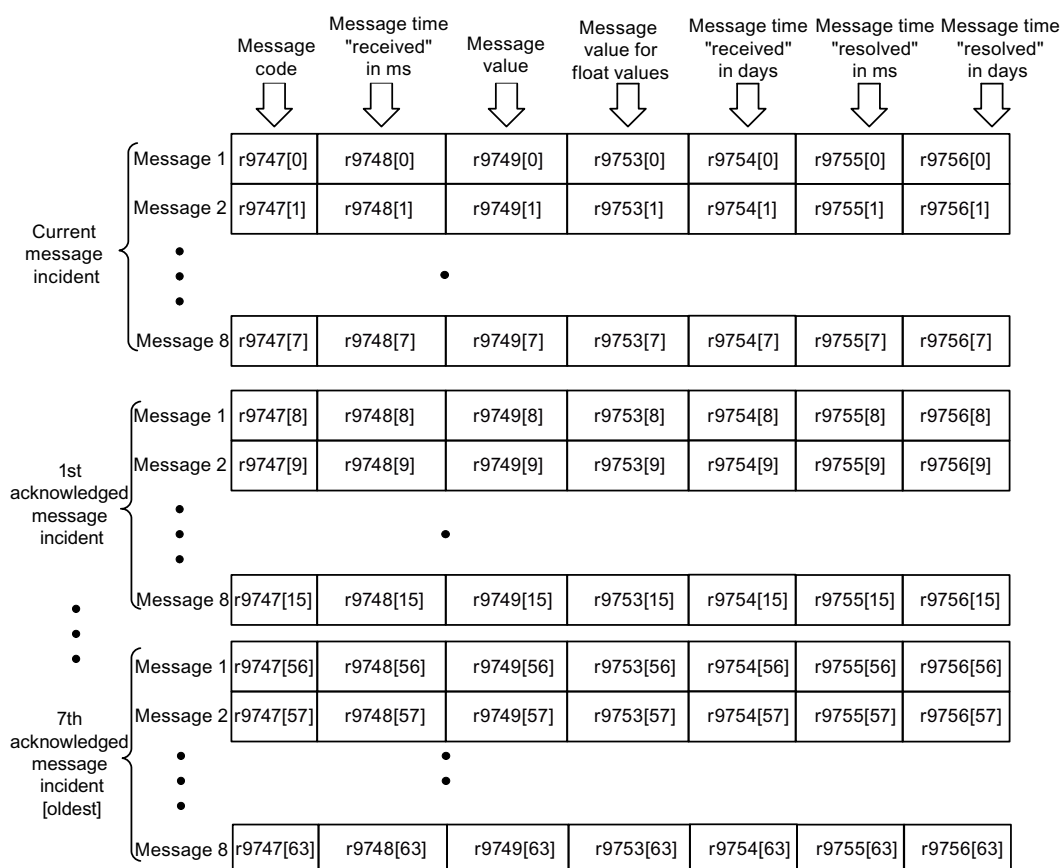


Figure 5-10 Structure of the message buffer

When a safety message is present, the bit 2139.5 = 1 ("Safety message present") is set. The entry in the message buffer is delayed. For this reason, the message buffer should not be read until a change in the buffer (r9744) has been detected after "Safety message present" is output.

The messages must be acknowledged via the fail-safe inputs F-DI of the TM54F or via PROFIsafe.

Properties of the safety message buffer:

- The entries appear in the buffer according to the time at which they occurred.
- If a new message case occurs, the message buffer is reorganized accordingly. The history is recorded in the "Acknowledged message case" 1 to 7.
- If the cause of at least one message in "Current message case" is rectified and acknowledged, the message buffer is reorganized accordingly. Messages that have not been rectified remain in "Current message case".
- If "Current message case" contains eight messages and a new message is output, the message in the parameters in index 7 is overwritten with the new message.
- r9744 is incremented each time the message buffer changes.
- A message value (r9749, r9753) can be output for a message. The message value is used to diagnose the message more accurately (refer to the message description for more details).

Deleting the message buffer:

The message buffer can be deleted as follows: p9752 = 0. Parameter p9752 (SI message cases, counter) is also reset to 0 at POWER ON. This also clears the fault memory.

Overview of important parameters (see the SINAMICS S120/S150 List Manual)

- r2139.0...12 CO/BO: Status word, faults/alarms 1
- r9744 SI message buffer changes, counter
- p9752 SI message cases, counter
- r9747[0...63] SI message code
- r9748[0...63] SI message time received in milliseconds
- r9749[0...63] SI message value
- p9752 SI message cases, counter
- r9753[0...63] SI message value for float values
- r9754[0...63] SI message time received in days
- r9755[0...63] SI message time removed in milliseconds
- r9756[0...63] SI message time removed in days

5.12 Safe actual value acquisition

Supported encoder systems

The Safety functions used to monitor motion (e.g. SS1, SS2, SOS, SLS and SSM) require safe actual value acquisition.

For safe speed/position sensing...

- Single-encoder systems or
- Two-encoder systems

...can be used.

Single-encoder system

In a single-encoder system, only the motor encoder is used to safely acquire the drive actual values. This motor encoder must be appropriately qualified (see encoder types). The safety-relevant actual values are generated either directly in the encoder or in the Sensor Module and are transferred to the Control Unit by way of fail-safe communication via DRIVE-CLiQ.

For motors without a DRIVE-CLiQ interface, the connection is established by means of additional Sensor Modules (SMC or SME).

Even if the drive is operating in the closed-loop torque controlled mode, motion monitoring functions may be selected as long as it is guaranteed that the encoder signals can be evaluated.

Special feature in the case of linear motors

The motor encoder (linear scale) of linear motors also acts as load measuring system. Only one measuring system is required for this reason. The system is connected by means of a Sensor Module or directly via DRIVE-CLiQ.

NOTICE
When specifying the standstill tolerance window, you must remember that fail-safe position monitoring takes place with a maximum of the precision displayed in r9731.

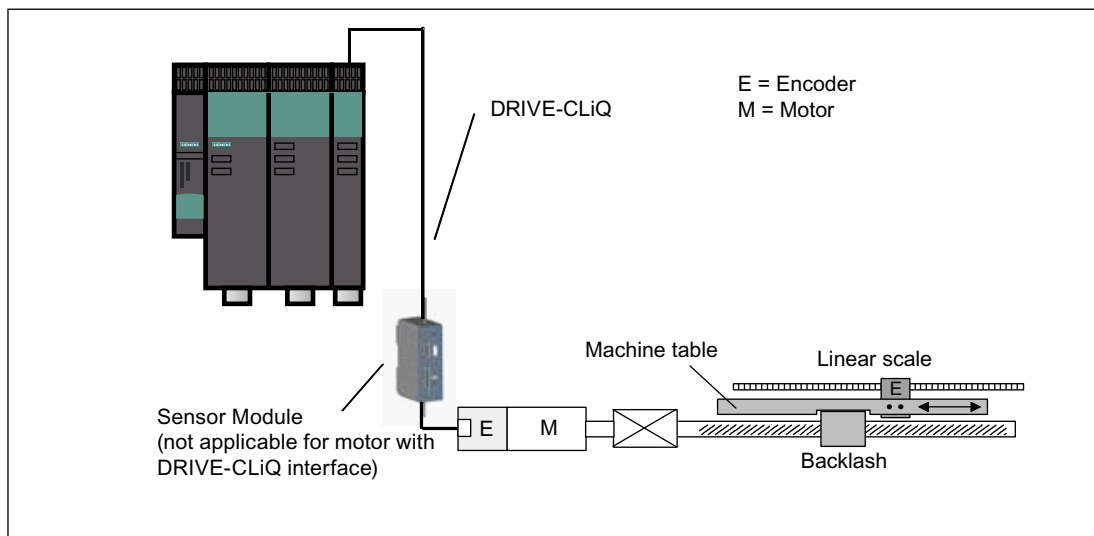


Figure 5-11 Example of an S120 single-encoder system

Two-encoder system

The fail-safe actual values for a drive are provided by two separate encoders. The actual values are transferred to the Control Unit by means of fail-safe communication via DRIVE-CLiQ.

For motors without a DRIVE-CLiQ interface, the connection is established by means of additional Sensor Modules (SMC or SME).

Each measuring system requires a separate connection or a separate Sensor Module.

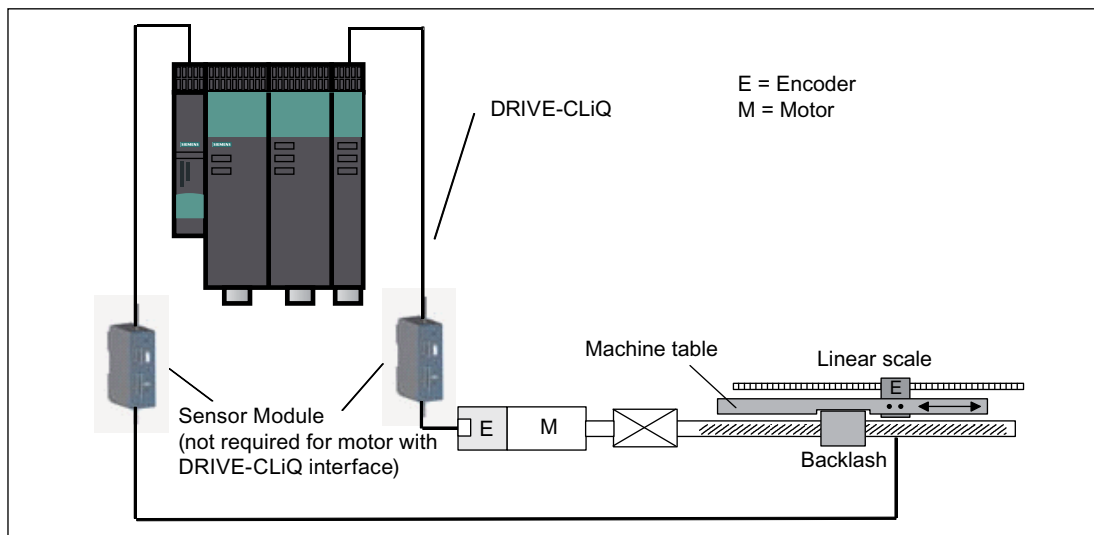


Figure 5-12 Example of an S120 two-encoder system on a linear axis via a ballscrew

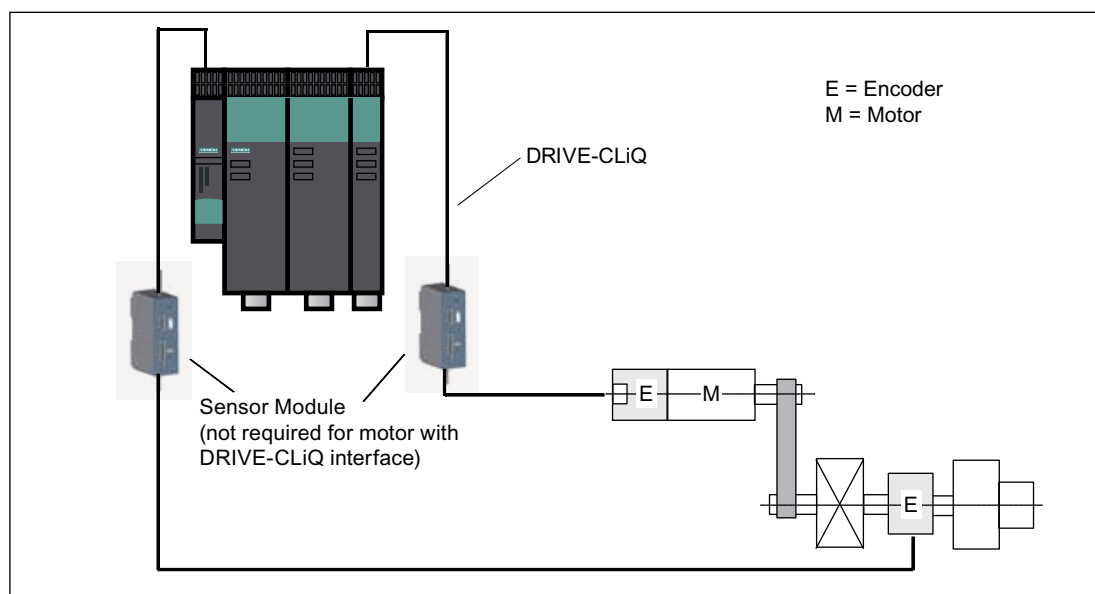


Figure 5-13 Example of an S120 two-encoder system on a rotary axis

Encoder types

Incremental encoders or absolute encoders can be used for safe acquisition of the position values on a drive.

Safe actual value acquisition relies on redundant evaluation of the incremental channels A/B that supply sin/cos signals of 1 Vpp.

The absolute position values can be transferred via the serial EnDat interface or an SSI interface to the controller.

Encoder types for 1 and 2 encoder systems

In systems with encoders with SINAMICS Safety Integrated (1 and 2 encoder systems), for safe actual value acquisition only encoders with sin/cos 1 Vss signals to the SINAMICS Sensor Modules SME20/25, SME120/125 and SMC20 are permitted which fulfill the following conditions:

1. The encoders must contain purely analog signal processing and creation. This is necessary to be able to prevent the A/B track signals with valid levels from becoming static ("freezing").
2. A failure modes and effects analysis (FMEA) must be carried out for the fixation of the encoder onto the motor shaft or the linear drive. The result should incorporate exclusion of the failure "Loosening of encoder fixation so that encoder no longer reports movement correctly" (see IEC 61800-5-2, 2008, Table D.16).

It should be noted that the machine manufacturer has sole responsibility for the fulfillment of the above-described requirements. Information on the internal realization of the encoder must come from the encoder manufacturer. The FMEA must be created by the machine manufacturer.

Certain Siemens motors with and without DRIVE-CLiQ connection can also be used for Safety Integrated functions, see:

<http://support.automation.siemens.com/WW/view/de/33512621>

Note

Basic absolute encoders (e.g. ECI, EQI) that offer an EnDat interface with additional sin/cos tracks, but operate according to an inductive measuring principle internally, are not permitted until their suitability for SINAMICS Safety Integrated has been determined.

Actual value synchronization

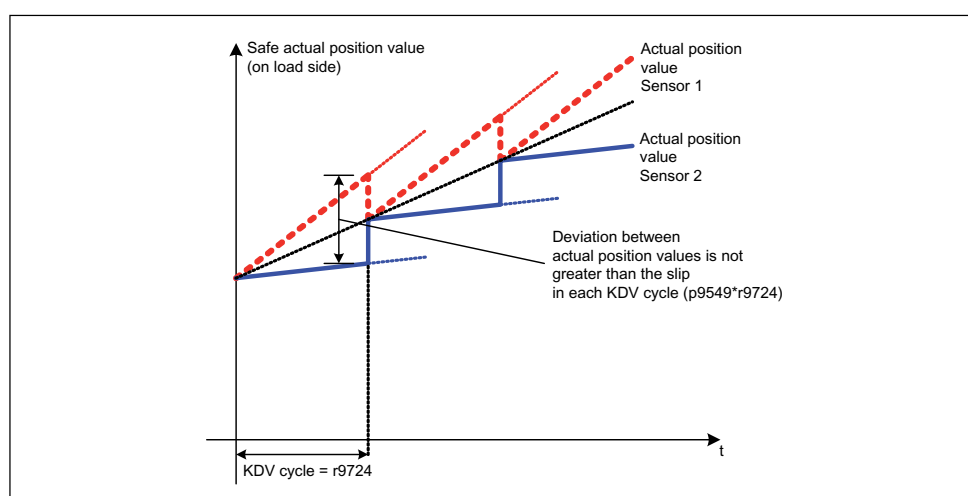


Figure 5-14 Example diagram of actual value synchronization

The mean value of the actual values of both encoders is calculated cyclically after actual value synchronization ($p9301.3 = p9501.3 = 1$) was activated. The maximum slip defined in $p9349/p9549$ is monitored within the crosswise comparison clock cycle ($r9724$). If "actual value synchronization" is not enabled, the value parameterized in $p9342/p9542$ is used as tolerance value for the crosswise comparison.

Two read parameters are available for safe motion monitoring:

r9730: SI Motion safe maximum velocity

Displays the maximum speed (load side) permissible due to the acquisition of actual values for safe motion monitoring functions. The maximum velocity for actual value acquisition depends on the actual value update clock cycle ($p9311/p9511$). Parameters $p9311/p9511$ are used to set the clock cycle time of the actual value acquisition for safe motion monitoring.

A slower clock cycle reduces the maximum permissible velocity, but also reduces the load on the Control Unit for safe actual value acquisition.

The maximum permissible velocity which, if overshoot, can trigger faults in safe actual value acquisition, is displayed in parameter $r9730$.

With a default value of $p9311/p9511$ (0 ms), the isochronous PROFIBUS clock cycle is used (or 1 ms in non-isochronous mode).

r9731: SI Motion safe position accuracy

Displays the greatest position accuracy (load side) that can be ensured due to the acquisition of the actual value for the safe motion monitoring functions.

Both parameters (r9730/r9731) depend on the relevant encoder type.

Overview of important parameters (see the SINAMICS S120/S150 List Manual)

- p9301.3 SI Motion enable safety functions (Motor Module), enable actual value synchronization
- p9501.3 SI Motion enable safety functions (Control Unit), enable actual value synchronization
- p9302 SI Motion axis type (Motor Module)
- p9502 SI Motion axis type (Control Unit)
- p9311 SI Motion actual value sensing clock cycle (Motor Module)
- p9511 SI Motion actual value sensing clock cycle (Control Unit)
- p9315 SI Motion encoder coarse position value configuration (Motor Module)
- p9515 SI Motion encoder coarse position value configuration (Control Unit)
- p9316 SI Motion motor encoder configuration safety functions (Motor Module)
- p9516 SI Motion motor encoder configuration safety functions (Control Unit)
- p9317 SI Motion linear scale grid division (Motor Module)
- p9517 SI Motion linear scale grid division (Control Unit)
- p9318 SI Motion encoder pulses per revolution (Motor Module)
- p9518 SI Motion encoder pulses per revolution (Control Unit)
- p9319 SI Motion fine resolution Gn_XIST1 (Motor Module)
- p9519 SI Motion fine resolution G1_XIST1 (Control Unit)
- p9320 SI Motion spindle pitch (Motor Module)
- p9520 SI Motion spindle pitch (Control Unit)
- p9321[0...7] SI Motion gearbox encoder/load denominator (Motor Module)
- p9521[0...7] SI Motion gearbox encoder/load denominator (Control Unit)
- p9322[0...7] SI Motion gearbox encoder/load numerator (Motor Module)
- p9522[0...7] SI Motion gearbox encoder/load numerator (Control Unit)
- p9323 SI Motion redundant coarse position value valid bits (Motor Module)
- p9324 SI Motion redundant coarse position value fine resolution (Motor Module)
- p9325 SI Motion redundant coarse position value relevant bits (Motor Module)
- p9523 SI Motion redundant coarse position value valid bits (Control Unit)
- p9524 SI Motion redundant coarse position value fine resolution (Control Unit)
- p9525 SI Motion redundant coarse position value relevant bits (Control Unit)
- p9326 SI Motion encoder assignment (Motor Module)

- p9526 SI Motion encoder assignment second channel
- p9342 SI Motion actual value comparison tolerance (crosswise) (Motor Module)
- p9542 SI Motion actual value comparison tolerance (crosswise) (Control Unit)
- p9349 SI Motion slip velocity tolerance (Motor Module)
- p9549 SI Motion slip velocity tolerance (Control Unit)
- r9713[0...3] SI Motion diagnostics position action value load side
- r9714[0...1] SI Motion diagnostics speed
- r9724 SI Motion crosswise comparison clock cycle
- r9730 SI Motion safe maximum velocity
- r9731 SI Motion safe position accuracy

5.13 Forced dormant error detection

Forced dormant error detection and function test through test stop

The functions and switch-off signal paths must be tested at least once within a defined period to establish whether they are working properly in order to meet the requirements of EN ISO 13849-1:2006 and IEC 61508 in terms of timely error detection.

The maximum permissible interval for forced dormant error detection with the Basic and Extended Functions is 9000 hours or once a year.

This functionality must be implemented by means of test stop triggering either in cyclic manual mode or by the automated process.

The test stop cycle is monitored. On expiration of the parameterized timer (also after POWER ON / warm restart), the alarm A01697: "SI Motion: Test of motion monitoring required" is generated and a status bit is set which can be transferred to an output or to a PZD bit via BICO. This alarm does not affect machine operation.

The test stop must be initiated application-specific and be executed at a time which suits application requirements. This functionality is implemented by means of a single-channel parameter p9705 which can be wired via BICO either to an input terminal on the drive unit (CU), or to an IO-PZD in the drive telegram.

- p9559 SI Motion Forced dormant error detection timer (Control Unit)
- p9705 BI: SI Motion Test stop signal source
- r9723.0 CO/BO: SI Motion diagnostics signals integrated in the drive

A test stop does not require POWER ON. The acknowledgment is set by canceling the test stop request.

When the appropriate safety devices are implemented (e.g. protective doors), it can be assumed that running machinery will not pose any risk to personnel. For this reason, only an alarm is output to inform the user that a forced dormant error detection run is due and to request that this be carried out at the next available opportunity.

Examples of when to carry out forced dormant error detection:

- When the drives are at a standstill after the system has been switched on.
- Before the protective door is opened.
- At defined intervals (e.g. every 8 hours).
- In automatic mode (time and event dependent)

Note

STO is triggered when a test stop is carried out for the Safety functions. It is not permissible to select STO before selecting the test stop and the axis must not be in operation. When using blocksize Power Modules, the test stop must be initiated at closed-loop controlled standstill (speed setpoint of 0, current is flowing through the motor).

Forced dormant error detection F-DI/F-DO of TM54F through test stop

An automatic test stop function is available for forced dormant error detection within the F-DIs/DOs test.

To ensure that the test stop function of the TM54F can be used, the F-DIs that are used must be interconnected in accordance with the following wiring example. The digital inputs of F-DI0 to F-DI4 must be connected to the "L1+" power supply. The digital inputs of F-DI5 to F-DI9 must be connected to the "L2+" power supply.

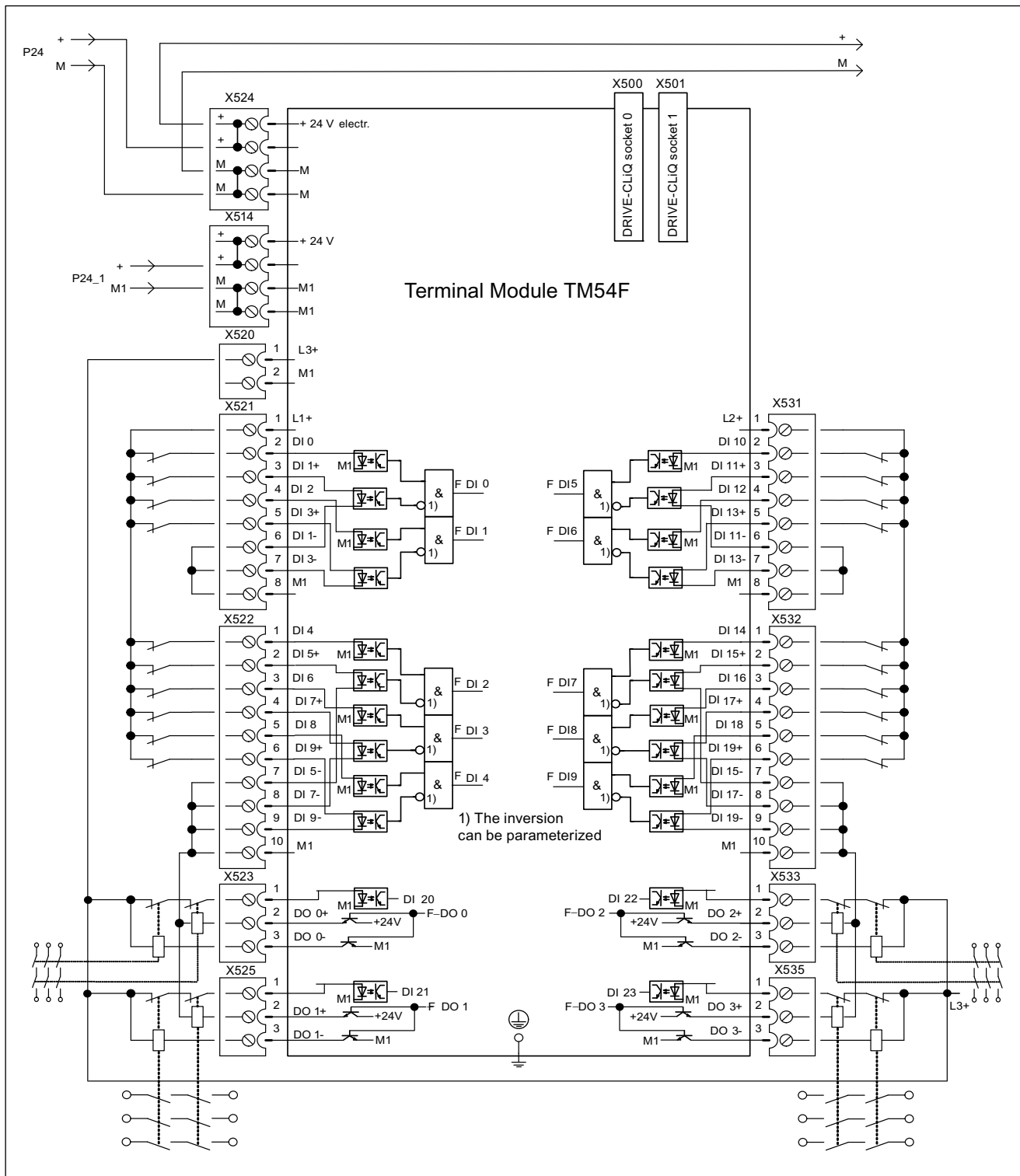


Figure 5-15 Example of the TM54F wiring

The F-DIs must be registered for the test stop by means of p10041.

CAUTION

The F-DI states are frozen for the duration of the test (approx. 100 ms)!

In order to be able to use the test stop function, the F-DOs being used must be interconnected in accordance with the connection example shown above and the forced feedback signals of the two relays must be connected to the corresponding digital input (DI 20 to DI 23).

The corresponding F-DOs must be registered for the test stop by means of p10046.

NOTICE

F-DOs which are not registered for evaluation by means of p10046 are set to "0" for the duration of the test stop ("fail-safe values").

Maximum test stop period: $19 * p10000 + 2 * 20 \text{ ms} + 6 * p10001$

WARNING

If the connected devices do not support the test stop function for specific F-DIs or F-DOs, the relevant F-DIs/F-DOs must be operated dynamically, e.g. by means of switch operation, or through specific machine functions.

The test stop must be executed at a suitable time. That is, it must be initiated application-specific. This functionality is implemented by means of a parameter p10007 which can be wired via BICO either to an input terminal on the drive unit (CU), or to an IO-PZD in the drive telegram.

The test stop cycle is monitored. On expiration of the parameterized timer (also after POWER ON / warm restart), the alarm A35014: "TM54F: Test stop required" is output.

- p10001 SI wait time for test stop at F-DO 0 ... 3
- p10003 SI forced dormant error detection timer
- p10007 BI: SI input terminal forced dormant error detection F-DO 0 ... 3
- p10041 SI F-DI test enable
- p10046 SI test sensor feedback input DI 20 ... 23

A test stop does not require POWER ON. The acknowledgment is set by canceling the test stop request.

Additional instructions for performing the test stops are provided in Chapter "Commissioning TM54F using STARTER/Scout → Test stop".

Control of the safety functions

The safety-oriented input and output terminals (F-DI and F-DO) act as an interface between the internal Safety Integrated functionality and the process.

A dual-channel signal applied to an F-DI (Fail-safe Digital Input, safety-oriented digital input = safe input terminal pair) controls the active monitoring of the activation/deactivation of safety functions. This function also depends on the status of sensors (e.g. switches).

An F-DO (Fail-safe Digital Output, safety-oriented digital output = safe output terminal pair) delivers a dual-channel signal representing feedback from the safety functions. It is suitable, for example, for the safety-oriented control of actuators (e.g. line contactor). See also the figures "F-DI 0 ... 4 overview", "F-DI 5 ... 9 overview" and "F-DO overview (without showing the main contacts on the contactors)".

Dual-channel processing of I/O signals

A dual-channel structure is implemented for data input/output and for processing safety-oriented I/O signals. All requests and feedback signals for safety-oriented functions should be entered or tapped using both channels.

The following options are available for controlling Safety Integrated functions:

- Control via terminals on the Control Unit and Motor Module (only STO, SS1 (time controlled) and SBC).
- Control via PROFIsafe
- Control by means of TM54F terminals

Control by means of terminals on the Control Unit and Motor Module can be activated alongside with one of the other two options. Only one of the two control modes can be selected for each drive object; either PROFIsafe or TM54F.

NOTICE
Per single Control Unit, either control via PROFIsafe or TM54F is permitted. Mixed operation is not permitted.

6.1 Control signals by way of terminals on the Control Unit and Motor/Power Module

Features

- Only for the STO, SS1 (time-controlled) and SBC functions
- Dual-channel structure via two digital inputs (Control Unit/power unit)
- A debounce function can be applied to the terminals of the Control Unit and the Motor Module to prevent incorrect trips due to signal disturbances or asymmetrical test signals. The filter times are set using parameters p9651 and p9851.
- Different terminal blocks depending on the format
- Automatic ANDing of up to 8 digital inputs (p9620[0...7]) on the Control Unit for chassis format power units connected in parallel

Overview of the safety function terminals for SINAMICS S120

The different power unit formats of SINAMICS S120 have different terminal designations for the inputs of the safety functions. These are shown in the following table.

Table 6- 1 Inputs for safety functions

Module	1. Switch-off signal path (p9620[0])	2. Switch-off signal path	EP terminals
Control Unit CU320-2 DP	X122.1....6 / X132.1...6 DI 0...7/16/17/20/21		
Single Motor Module Booksize/Booksize Compact	(see CU320-2 DP)	X21.3 and X21.4 (on the Motor Module)	X21.3 X21.4
Single Motor Module/ Power Module Chassis	(see CU320-2 DP)	X41.1 and X41.2	X41.1 X41.2
Double Motor Module Booksize/Booksize Compact	(see CU320-2 DP)	X21.3 and X21.4 (motor connection X1)/X22.3 and X22.4 (motor connection X2) (on the Motor Module)	X21.3 X21.4 X22.3 X22.4
Power Module Blocksize with CUA31/CUA32	(see CU320-2 DP)	X210.3 and X210.4 (on the CUA31/CUA32)	X210.3 X210.4
For further information about the terminals, see the Equipment Manuals.			

Terminals for STO, SS1 (time-controlled), SBC

The functions are separately selected/deselected for each drive using two terminals.

1. Switch-off signal path, Control Unit

The desired input terminal is selected via BICO interconnection (BI: p9620[0]).

2. Switch-off signal path Motor Module/Power Module

The input terminal is the "EP" terminal ("Enable Pulses")

The EP terminal is periodically interrogated with a sampling time, which is rounded off to an integer multiple of the current controller cycle; however, it is a minimum of 1 ms.
(example: $t_i = 400 \mu s$, $t_{EP} \Rightarrow 3 \times t_i = 1.2 ms$)

Both terminals must be operated simultaneously, otherwise a fault will be issued.

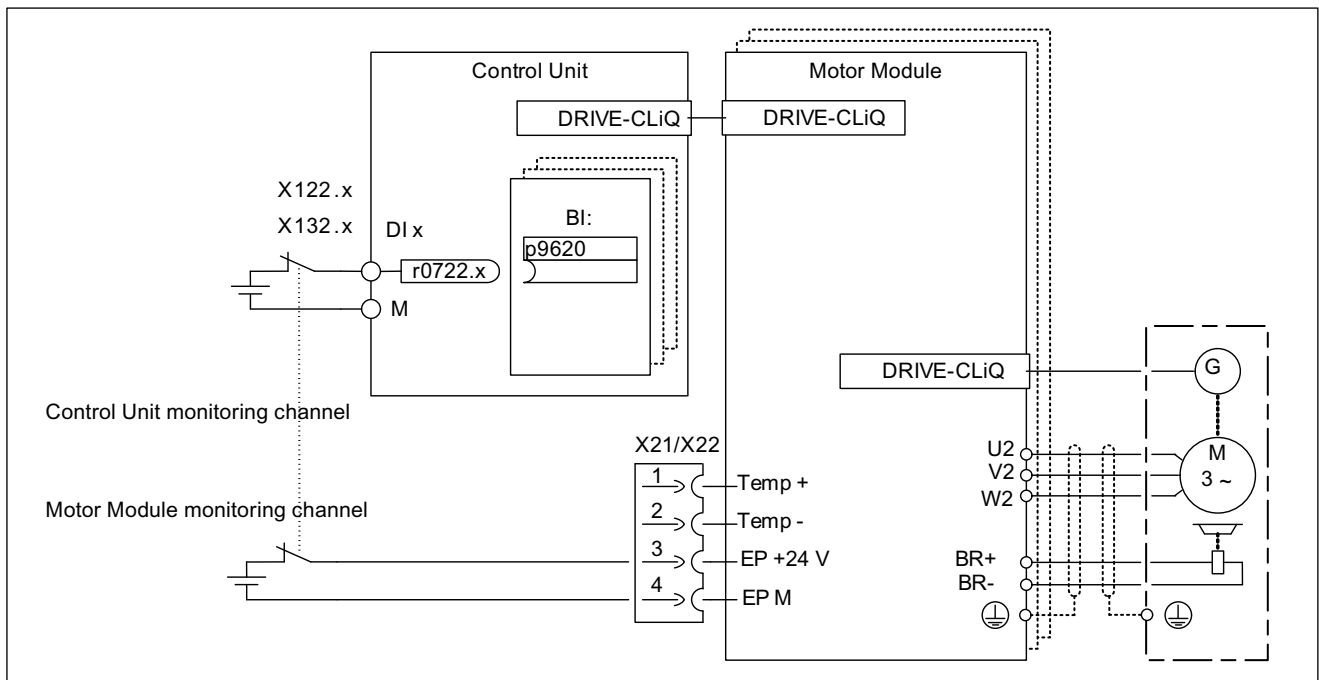


Figure 6-1 Example: Terminals for "Safe Torque Off", example for Motor Modules booksize and CU320-2 DP

Grouping drives

To ensure that the function works for more than one drive at the same time, the terminals for the corresponding drives must be grouped together as follows:

1. Switch-off signal path

By connecting the binector input to the joint input terminal on the drives in one group.

2. Switch-off signal path (Motor Module/Power Module with CUA3x)

By appropriately wiring the terminals for the individual Motor Modules/Power Modules with CUA31/CUA32 assigned to the group.

Note

The grouping must be identical in both monitoring channels.

If a fault in a drive results in a "Safe Torque Off" (STO), this does not automatically mean that the other drives in the same group also switch to "Safe Torque Off" (STO).

Example: Terminal groups

For this purpose, the same grouping for "Safe Torque Off" must be performed on both the Control Unit and the Motor Modules.

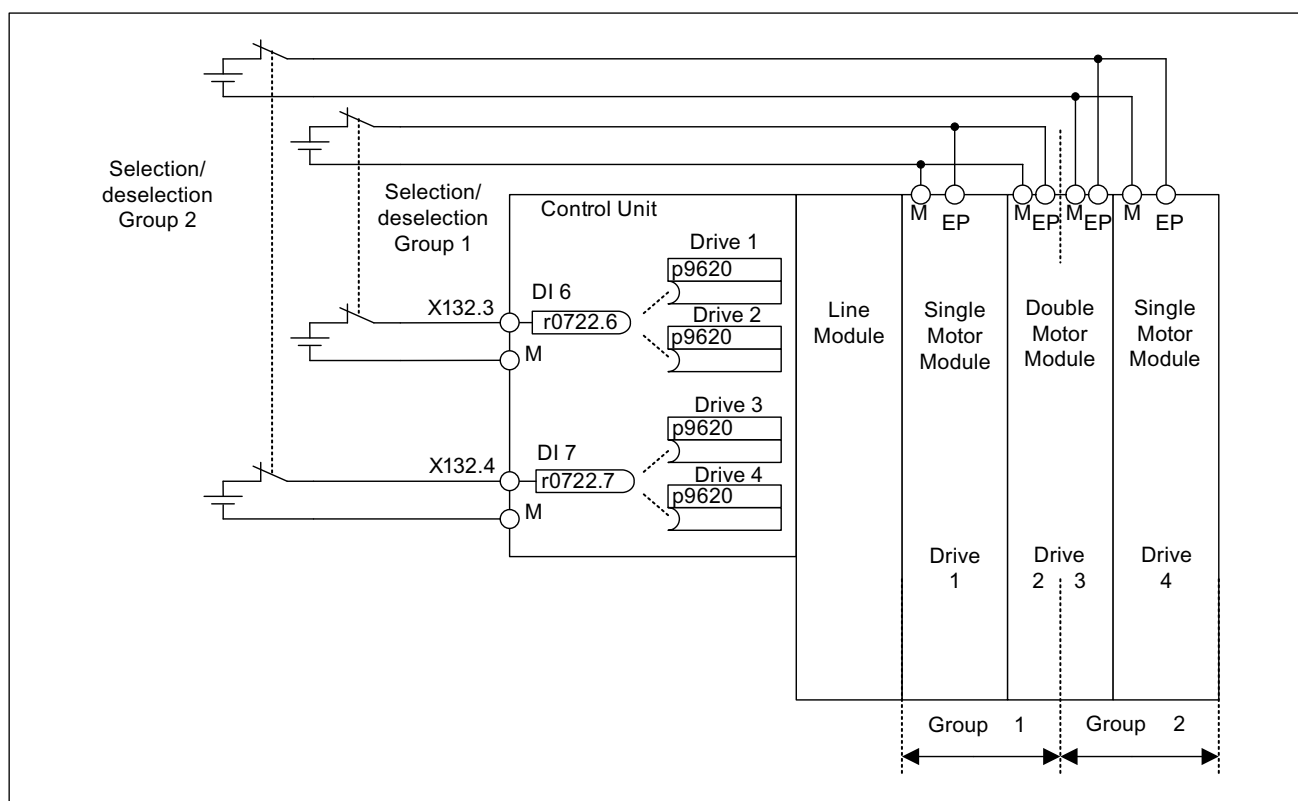


Figure 6-2 Example: Grouping terminals with Motor Modules booksize and CU320-2 DP

Information on the parallel connection of chassis type Motor Modules

When chassis type Motor Modules are connected in parallel, a safe AND element is created on the parallel drive object. The number of indexes in p9620 corresponds to the number of parallel chassis components in p0120.

6.1.1 Simultaneity and tolerance time of the two monitoring channels

Simultaneity and tolerance time of the two monitoring channels

The "Safe Torque Off" function must be selected/deselected simultaneously in both monitoring channels using the input terminals and is only effective for the associated drive.

1 signal: Deselecting the function

0 signal: Selecting the function

"Simultaneously" means:

The changeover must be complete in both monitoring channels within the parameterized tolerance time.

- p9650 SI SGE changeover tolerance time (Control Unit)
- p9850 SI SGE changeover tolerance time (Motor Module)

Note

To avoid incorrect triggering of fault messages, on these inputs the tolerance time must always be set to be smaller than the shortest time between two switching events (ON/OFF, OFF/ON).

If the "Safe Torque Off" function is not selected/deselected within the tolerance time, this is detected by the cross-comparison, and fault F01611 or F30611 (STOP F) is output. In this case, the pulses have already been canceled as a result of the selection of "Safe Torque Off" on one channel.

6.1.2 Bit pattern test

Bit pattern test of fail-safe outputs

The inverter normally responds immediately to signal changes in its fail-safe inputs. This is not desired in the following case: Several control modules test their fail-safe outputs using bit pattern tests (on/off tests) to identify faults due to either short or cross circuiting. When you interconnect a fail-safe input of the inverter with a fail-safe output of a control module, the inverter responds to these test signals.

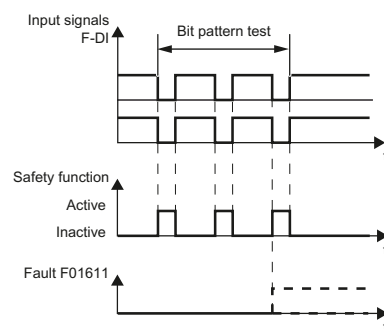


Figure 6-3 Inverter response to a bit pattern test

The dark test pulses (temporary deactivation with logical 1 on the F-DO) with a duration of <1 ms do not present a problem. The light tests (temporary activation with logical 0 on the F-DO) can take up to 4 ms. The EP terminal of the Motor Module is scanned every 1 ms and detects this test pulse. In the default setting, the digital inputs on the CU320 are scanned every 4 ms; i.e. this pulse is not detected there.

This does not present a problem when only the STO function is being used (with or without SBC), as usually the associated discrepancy time is significantly greater. However, if the SS1 function is used, discrepancy errors occur. This is because a pulse starts the SS1 timer which may then no longer be stopped (requirement of stop category 1 in accordance with EN 60204-1). The timers are compared when the discrepancy time has expired. As the timer has only been started on one channel, a discrepancy error occurs. To prevent this, you need to make suitable parameter settings to ensure the test pulse is detected on both channels. The sampling time of the digital inputs on the CU320 must therefore be reduced. The following setting must be made:

p0799 (on CU) = 2 ms

Also note that the test pulses on the different channels (P and M) of the F-DOs are switched at different times. The time difference may be up to 0.8 s. For this reason, the discrepancy time must be increased as follows:

p9650 = p9850 $\geq$ 900 ms

Overview of important parameters (see the SINAMICS S120/S150 List Manual)

- p0799[0...2] CU inputs/outputs sampling time
- p9650 SI SGE changeover tolerance time (Control Unit)
- p9850 SI SGE changeover tolerance time (Motor Module)

6.2 Activation via PROFIsafe

As an alternative to controlling Safety Integrated functions via terminals or TM54F, they can also be controlled via PROFIsafe. PROFIsafe telegram 30 is used for communication using PROFIBUS and PROFINET.

Control via PROFIsafe is available for both Safety Integrated Basic Functions and Safety Integrated Extended Functions.

The structure of the associated control and status words is described further below (see chapter "Description of telegram 30").

6.2.1 Enabling of the control via PROFIsafe

For PROFIsafe communication, SINAMICS devices require a PROFIBUS or a PROFINET interface.

Every drive with configured PROFIsafe in the drive unit represents a PROFIsafe slave (F slave) with a fail-safe communication to the F host via PROFIBUS or PROFINET and is assigned its own PROFIsafe telegram.

In so doing, a PROFIsafe safety channel, a so-called safety slot is created using the HW Config tool from SIMATIC Manager Step 7. It is then possible to also control the Basic Functions using PROFIsafe telegram 30. The structure of the associated control and status words is described further below (see Tables "PROFIsafe-STW" and "PROFIsafe-ZSW"). The PROFIsafe telegram 30 is placed in front of the standard telegram for communication (e.g. telegram 2).

Enabling PROFIsafe

The Safety Integrated Functions are enabled via PROFIsafe using parameters p9601 and p9801:

- Basic Functions: p9601.2 = 0, p9801.2 = 0
Extended Functions: p9601.2 = 1, p9801.2 = 1
- p9601.3 = 1, p9801.3 = 1

Note

Licensing for Safety Integrated Basic Functions via PROFIsafe

No license is required to use Basic Functions. This also applies to control via PROFIsafe. However, for Extended Functions, you require an appropriate license that will be charged for.

All parameters involved in PROFIsafe communication are password protected against undesirable changes and secured using a checksum. The telegrams are configured using a configuration tool (e.g. HW Config + F-Configuration Pack or SCOUT) on the F host.

Safety Integrated Basic Functions via PROFIsafe and terminals

Control of the Basic Functions via terminals on the Control Unit and on the Motor/Power Module (parameters p9601.0 = p9801.0 = 1) may be enabled in parallel. In this way, the STO and SS1 functions (time controlled) can be selected via PROFIsafe telegram 30 as well as in parallel via the onboard terminals of the Control Unit and Motor Module/Power Module.

STO has priority over SS1, i.e. if SS1 and STO are simultaneously initiated, then STO is executed.

6.2.2 Structure of telegram 30

6.2.2.1 Structure of telegram 30 (Basic Functions)

PROFIsafe control word (STW)

S_STW1, PZD1 in telegram 30, output signals
See function diagram [2840].

Table 6- 2 Description of the PROFIsafe STW

Bit	Meaning	Comments	
0	STO	1	Deselect STO
		0	Select STO
1	SS1	1	Deselect SS1
		0	Select SS1
2	SS2	0	— ¹
3	SOS	0	— ¹
4	SLS	0	— ¹
5	Reserved	-	—
6	Reserved	-	—
7	Internal Event ACK	1/0	Acknowledgement
		0	No acknowledgement
8	Reserved	-	—
9	Select SLS bit 0	-	— ²
10	Select SLS bit 1	-	
11...15	Reserved	-	—

¹ Inactive signals for Basic Functions; are set to 0.

² A static zero signal must be continuously present

PROFIsafe status word (ZSW)

S_ZSW1, PZD1 in telegram 30, input signals
See function diagram [2840].

Table 6- 3 Description of the PROFIsafe status word (ZSW)

Bit	Meaning	Comments	
0	STO active	1	STO active
		0	STO not active
1	SS1 active	1	SS1 active
		0	SS1 not active
2	SS2 active	0	— ¹
3	SOS active	0	— ¹
4	SLS active	0	— ¹

Bit	Meaning	Comments	
5	Reserved	-	–
6	Reserved	-	–
7	Internal Event	1	Internal event
		0	No internal event
8	Reserved	-	–
9	Active SLS level bit 0	-	– ¹
10	Active SLS level bit 1	-	
11	SOS selected	0	– ¹
12...14	Reserved	-	–
15	SSM (speed)	0	– ¹

¹ Inactive signals for Basic Functions; are set to 0.

6.2.2.2 Structure of telegram 30 (Extended Functions)

PROFIsafe control word (STW)

S_STW1, PZD1 in telegram 30, output signals
See function diagram [2840].

Table 6- 4 Description of the PROFIsafe STW

Bit	Meaning	Comments	
0	STO	1	Deselect STO
		0	Select STO
1	SS1	1	Deselect SS1
		0	Select SS1
2	SS2	1	Deselect SS2
		0	Select SS2
3	SOS	1	Deselect SOS
		0	Select SOS
4	SLS	1	Deselect SLS
		0	Select SLS
5	Reserved	-	-
6	Reserved	-	-
7	Internal Event ACK	1/0	Acknowledgement
		0	No acknowledgement
8	Reserved	-	-
		-	-
9	Select SLS bit 0	-	Select speed limit for SLS (2 bits) ¹
10	Select SLS bit 1	-	
11...15	Reserved	-	-

¹ A static zero signal must be continuously available

PROFIsafe status word (ZSW)

S_ZSW1, PZD1 in telegram 30, input signals

See function diagram [2840].

Table 6- 5 Description of the PROFIsafe status word (ZSW)

Bit	Meaning	Comments	
0	STO active	1	STO active
		0	STO not active
1	SS1 active	1	SS1 active
		0	SS1 not active
2	SS2 active	1	SS2 active
		0	SS2 not active
3	SOS active	1	SOS active
		0	SOS not active
4	SLS active	1	SLS active
		0	SLS not active
5	Reserved	-	-
6	Reserved	-	-
7	Internal event	1	Internal event
		0	No internal event
8	Reserved	-	-
		-	-
9	Active SLS level bit 0	-	Display of the speed limit for SLS (2 bits)
10	Active SLS level bit 1	-	
11	SOS selected	1	SOS selected
		0	SOS deactivated
12...14	Reserved	-	-
15	SSM (speed)	1	SSM (speed below limit value)
		0	SSM (speed higher than/equal to limit)

6.3 Control via TM54F

6.3.1 TM54F design

Terminal Module TM54F is a terminal expansion module for snap-on rail mounting to DIN EN 60715. The TM54F features fail-safe digital I/O for controlling the Safety Integrated Extended Functions.

Each Control Unit can be assigned only one TM54F which is connected via DRIVE-CLiQ.

NOTICE

The TM54F may not be interconnected in series with the Motor Modules and must be operated on a separate DRIVE-CLiQ line (separate port on the Control Unit). Other Terminal and Sensor Modules can be connected to this DRIVE-CLiQ line.

TM54F features the following terminals:

Table 6- 6 Overview of the TM54F interfaces

Type	Number
Fail-safe digital outputs (F-DO)	4
Fail-safe digital inputs (F-DI)	10
Sensor ¹⁾ power supplies, dynamic response supported ²⁾	2
Sensor ¹⁾ power supply, no dynamic response	1
Digital inputs for checking the F-DO with activated forced dormant error detection	4

¹⁾ Sensors: Fail-safe devices for command operations and status logging (e.g. Emergency Stop pushbuttons, safety door locks, position switches, and light arrays / light curtains).

²⁾ Dynamic response: The sensor power supply is cycled on and off by the TM54F when the forced dormant error detection is active for the sensors, cable routing, and the evaluation electronics.

The TM54F provides 4 fail-safe digital outputs and 10 fail-safe digital inputs. A fail-safe digital output consists of a 24 V DC switching output, an output switching to ground and a digital input for reading back the switching state. A fail-safe digital input consists of two digital inputs.

Note

You have the following options of acknowledging TM54F faults after troubleshooting:

- POWER ON
- Falling edge in signal "Internal Event ACK" with subsequent acknowledgement on the Control Unit.

For different signal states within a fail-safe F-DI of the TM54F, the signal states of the two digital inputs of the F-DI are frozen at logical 0 (safety function selected) until a safe acknowledgement has been carried out using an F-DI via parameter p10006 (SI acknowledgement, internal event input terminal).

The monitoring time (p10002) for the discrepancy of the two digital inputs of an F-DI must, under certain circumstances, be selected high enough so that switching operations do not initiate an undesirable response and then require a safe acknowledgement. Therefore, the signal states at the two related digital inputs (F-DI) must have the same state within this monitoring time, otherwise the following fault will be output F35151 TM54F: Discrepancy error. This requires safe acknowledgement.

Note

The discrepancy time must be set so that it is always smaller than the smallest expected switching interval of the signal to this F-DI.

6.3.2 F-DI function

Description

Fail-safe digital inputs (F-DI) consist of two digital inputs. The cathode (M) of the optocoupler is routed to the second digital input to allow the connection of an M-switching F-DO output (the anode must be connected to 24 V DC).

Parameter p10040 is used to determine whether an F-DI is operated as NC/NC or NC/NO contact. The status of DI can be read at parameter r10051 for the drive objects TM54F_MA and TM54F_SL. The same bits of both drive objects are logically linked by AND operation and return the status of the relevant F-DI.

Test signals from controls can be filtered out using parameter p10017, so that faults are not incorrectly interpreted.

Explanation of terms:

NC contact / NC contact: to select the safety function, a "zero level" must be present on both inputs.

NC contact / NO contact: to select the safety function, a "zero level" at input 1 and a "1 level" at input 2 must be present.

The signal states at the two associated digital inputs (F-DI) must assume the same status configured in p10040 within the monitoring time set in p10002.

In order to enable forced dormant error detection, connect the digital inputs of F-DI 0 ... 4 with the dynamic voltage supply L1+ and the digital inputs with F-DI 5 ... 9 to L2+ (for additional information on forced dormant error detection, see the corresponding function description in the chapter "Extended Functions").

In the SINAMICS S120/150 List Manual, function diagrams 2850 or 2851 show an overview of the fail-safe inputs F-DI 0 4 or F-DI 5 ... 9.

F-DI features

- Fail-safe configuration with two digital inputs per F-DI
- Input filter for test signals with an adjustable gating time (p10017)
- Configurable connection of NC/NC or NC/NO contacts by means of parameter p10040
- Status parameter r10051
- Adjustable time window for monitoring discrepancy at both digital inputs by means of parameter p10002 for all F-DIs (details in Chapter: Input/output interconnections of a safety switching device with TM54F)

Note

To avoid incorrect triggering of fault messages, on these inputs the tolerance time must always be set to be smaller than the shortest time between two switching events (ON/OFF, OFF/ON).

- Second digital input with additional tap of the optocoupler cathode for connecting an M-switching output of a fail-safe controller.

WARNING

In contrast to mechanical switching contacts (e.g. Emergency Stop switches), leakage currents can still flow through semiconductor switches even when they are switched off. This can lead to false switching states if digital inputs are not connected correctly.

The conditions for digital inputs/outputs specified in the relevant manufacturer documentation must be observed.

WARNING

In accordance with IEC 61131 Part 2, Chapter 5.2 (2008), only outputs that have a maximum residual current of 0.5 mA when "OFF" can be used to connect TM54F digital inputs with digital semiconductor outputs.

The inclusion of additional load resistors makes it possible to use digital outputs with larger residual currents to connect TM54F inputs.

Overview of important parameters (see the SINAMICS S120/S150 List Manual)

- p9651 SI STO/SBC/SS1 debounce time (Control Unit)
- p9851 SI STO/SBC/SS1 debounce time (Control Unit)
- p10002 SI discrepancy monitoring time
- p10017 SI digital inputs, debounce time
- p10040 SI F-DI input mode
- r10051.0...9 CO/BO: SI status of digital inputs

6.3.3 Function of the F-DO

Description

Fail-safe digital outputs (F-DO) consist of two digital outputs plus one digital input that checks the switching state for forced dormant error detection. The first digital input switches 24 V DC, and the second switches M of the X514 voltage supply.

The status of each F-DO can be read at parameter r10052. The status of the associated DIs can be read at parameter r10053 for the drive objects of the slave (TM54F_SL).

In order to enable forced dormant error detection, connect the corresponding digital input for the forced feedback signals of the relays (additional information on forced dormant error detection is provided in the chapter "Extended functions").

In the SINAMICS S120/150 List Manual, function diagram 2853 provides an overview of the fail-safe outputs F-DO 0...3, and the associated checking inputs F-DI 20...23.

F-DO signal sources

A drive group contains several drives with similar characteristics. The groups are parameterized at the p10010 and p10011 parameters.

The following signals are available for interconnecting (p10042 to p10045) each one of the four drive groups with the F-DO:

- STO active (power removed)
- SS1 active
- SS2 active
- SOS active
- SLS active
- SSM feedback active
- SOS selected
- Internal event (no active safety fault)
- Safe state

The following signals can be requested by means of p10039[0...3] for each drive group (index 0 corresponds with drive group 1 etc.):

- STO active (power removed)
- SS1 active
- SS2 active

- SOS active
- SLS active

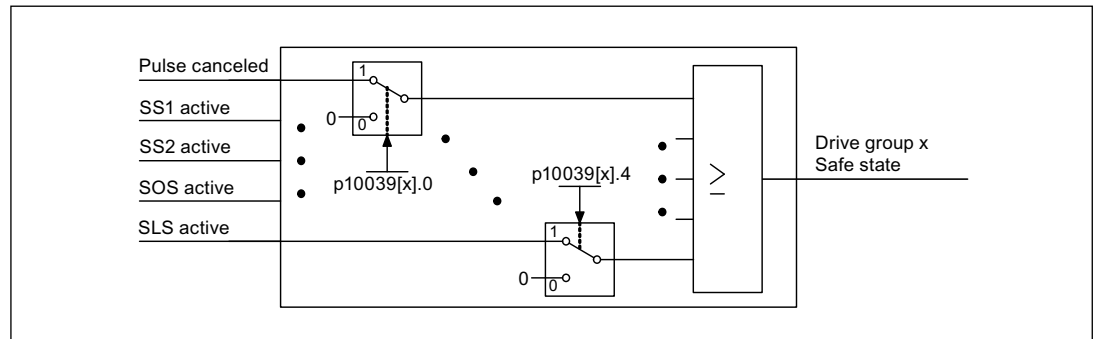


Figure 6-4 Safe state selection

The same signals (high-active) of each drive or drive group are logically linked by means of AND operation. The different signals selected through p10039 are logically linked by means of OR operation. Result of these logic operations is the "Safe State" for each drive group.

Each F-DO supports the interconnection of up to 6 signals by way of indexing (p10042[0...5] to p10045[0...5]) and their output as logical AND operation.

F-DO features

- Each F-DO with fail-safe configuration consisting of two digital outputs plus one digital input for checking the switching state for forced dormant error detection
- Status parameters r10052/r10053

Function diagrams (see SINAMICS S120/S150 List Manual)

- 2853 TM54F (F-DO 0 ... F-DO 3, DI 20 ... DI 23)
- 2856 TM54F Safe State selection
- 2857 TM54F assignment (F-DO 0 ... F-DO 3)

Overview of important parameters (see the SINAMICS S120/S150 List Manual)

- p10039 SI Safe State signal selection
- p10042[0..5] SI F-DO 0 signal sources
- p10045[0..5] SI F-DO 3 signal sources
- r10052.0...3 CO/BO: SI status of digital outputs
- r10053.0...3 CO/BO: SI digital inputs 20 ... 23

Commissioning

7.1 Safety Integrated firmware versions

Firmware versions for Safety Integrated

The safety firmware installed on the Control Unit and the safety firmware installed on the Motor Module each have separate version IDs. The parameters listed below can be used to read the version IDs from the relevant hardware.

Read the overall firmware version via:

- r0018 Control Unit firmware version

The following firmware data can be read for the basic functions:

- r9770[0...3] SI version, drive-autonomous safety functions (Control Unit)
- r9870[0...3] SI version, drive-autonomous safety functions (Motor Module)

The following firmware data can be read for the extended functions:

- r9590[0...3] SI Motion version safety motion monitoring (Control Unit)
- r9390[0...3] SI Motion version safety motion monitoring (Motor Module)
- r9890[0...2] SI version (Sensor Module)
- r10090[0...3] SI TM54F version

Basic Functions and Extended Functions

Basic and/or Extended Functions that have been enabled are checked to determine whether the parameter for the automatic firmware update is set (p7826 = 1).

This means that at each boot, the firmware version of the DRIVE-CLiQ components involved is checked in comparison to the firmware version of the Control Unit and, if required, updated.

During the acceptance test for the Safety Integrated basic functions, the safety firmware versions of the Motor Modules must be read, logged, and checked against the list below.

During the acceptance test for the Safety Integrated extended functions, the safety firmware versions of the Motor Modules, Sensor Modules and, if necessary, the Terminal Module TM54F required for the safety functions are read, logged, and checked against the list below.

When the extended functions are used, the firmware requirements for the basic functions must also be fulfilled at all times.

The list of permissible safety firmware version combinations, which must be used as a reference during the test, can be found under "Product Support" at the following address: <http://support.automation.siemens.com/WW/view/de/28554461>
The testing procedure is described at the end of the chapter.

Procedure for checking the safety firmware version combinations

The document in the link provided contains tables listing the permissible safety firmware version combinations for the different safety function classes (SINAMICS basic functions, SINAMICS extended functions, SINUMERIK Safety Integrated).

The Safety firmware version relevant for the Safety function can be read from the Control Unit. The row containing this version number specifies the associated, permissible safety firmware versions of the relevant drive components. These versions must be compatible with the versions installed on your system.

7.2 Commissioning Safety Integrated functions

The Safety functions are commissioned using the screen forms in the STARTER. You will find these functions for each drive under "Functions" -> "Safety Integrated".

The password "0" is set by default.

NOTICE
For safety-relevant reasons, using the STARTER commissioning tool (or SCOUT) you can only set the safety-relevant parameters of the SINAMICS S120 Control Unit offline. In order to set the safety-relevant parameters of the Motor Module, establish an online connection to SINAMICS S120 and transfer the parameters by clicking on the "Copy parameters" button in the start screen of the configuration.

Note

Activating changed safety parameters

When exiting the commissioning mode (p0010 = 0), most of the changed parameters immediately become active. However, for some parameters, a POWER ON is required. In this case, a STARTER message will inform you about this.

When performing an acceptance test, a POWER ON is always required.

7.2.1 Prerequisites for commissioning the Safety Integrated functions

1. Commissioning of the drives must be complete.
2. A POWER ON (switching off/on) is carried out on the drives.
3. Non-safe pulse suppression must be present, e.g. via
OFF1 = "0" or OFF2 = "0".

If the motor holding brake is connected and parameterized, the holding brake is applied.
4. For operation with SBC, the following applies:

A motor with a motor holding brake must be connected to the appropriate terminal of the module.

7.2.2 Default settings for commissioning Safety Integrated functions without encoder

Additional default settings are required before commissioning Safety functions without an encoder. The ramp-function generator is automatically created if a vector drive is configured. Please continue up to the ramp-function generator configuration. If a servodrive is configured, proceed as follows to call the ramp-function generator:

1. Activate the ramp-function generator: In the configured project, call the "Drive Navigator" offline, select the device configuration and click on "Configure drive". In the next window, under the function modules, select "Extended setpoint channel". With "Continue", proceed with the configuration and when completed, exit with "Complete". The ramp-function generator is now active and can be parameterized.

2. If the project window, open the ramp-function generator by double-clicking on **Drive unit** → **Drives** → **Servo** → **Setpoint channel** → **Ramp-function generator**:

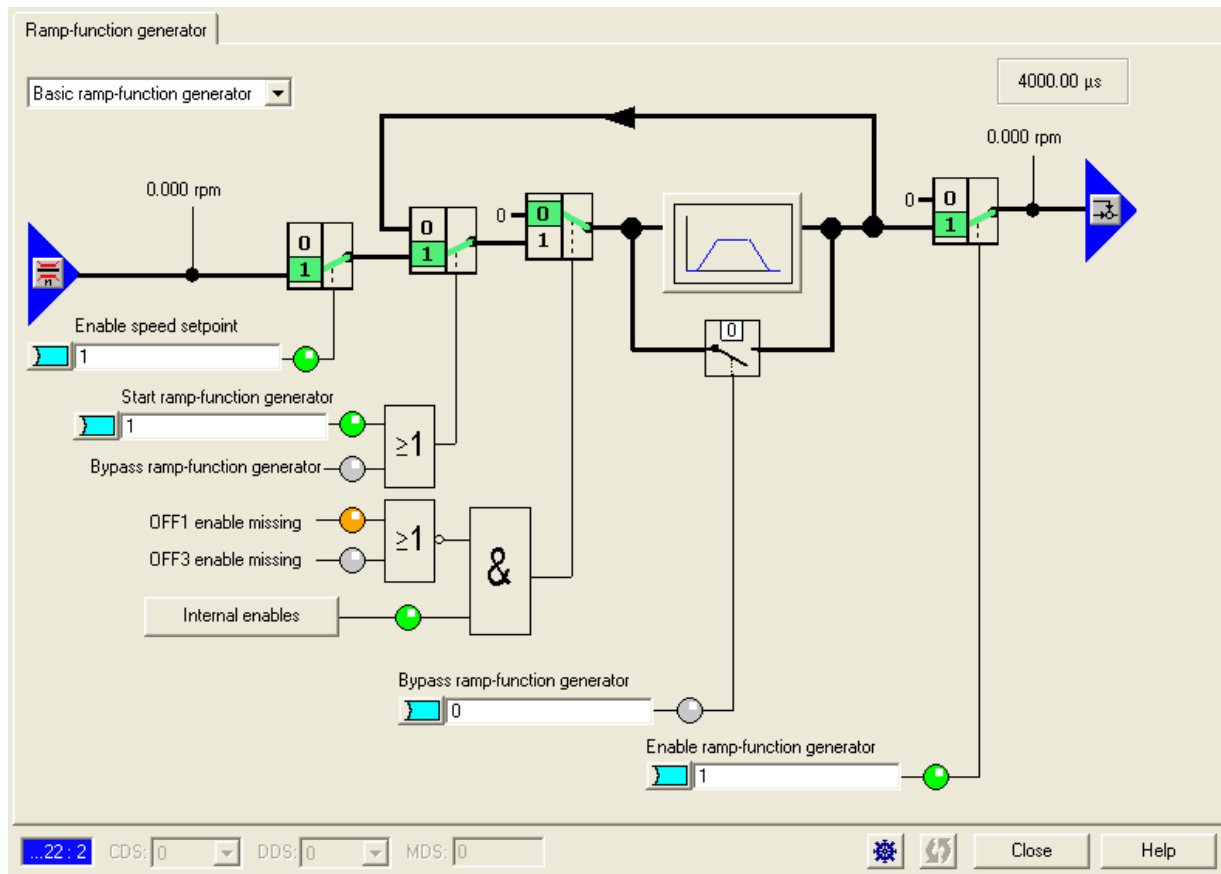


Figure 7-1 Ramp-function generator

3. Clicking on the button with the ramp opens the following window:

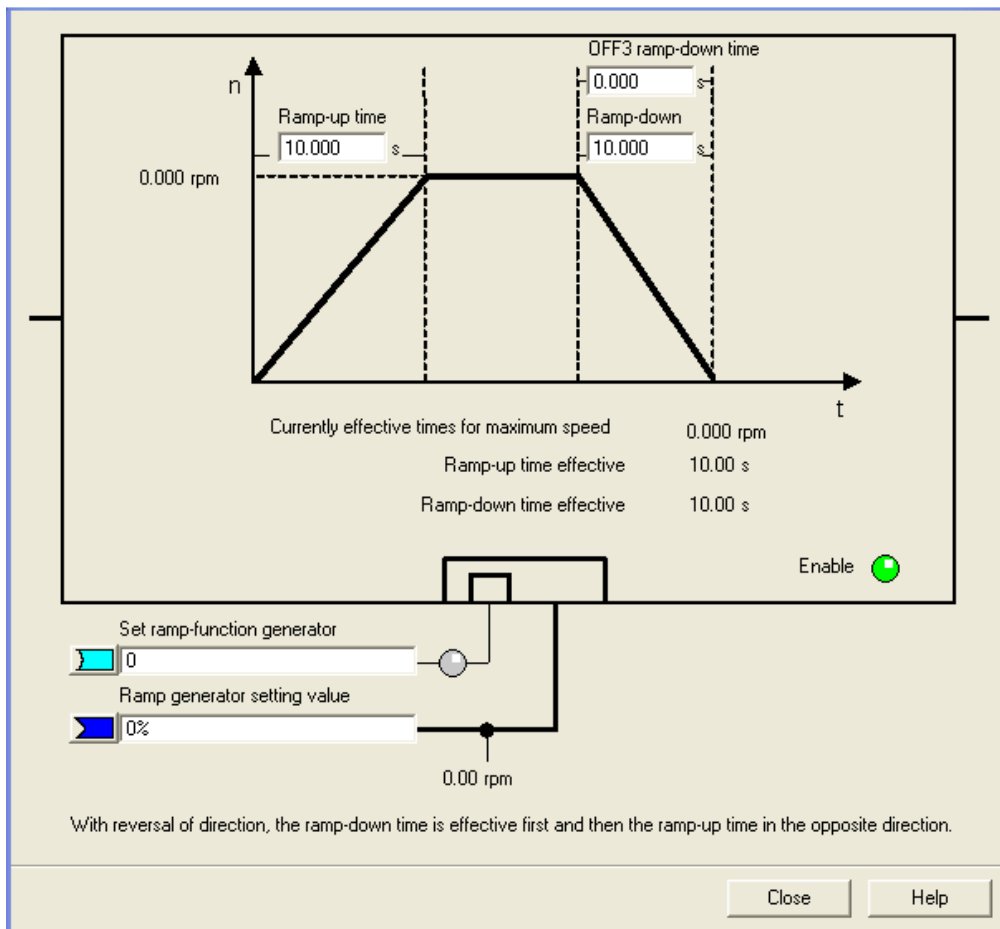


Figure 7-2 Ramp-function generator ramp

- Here, enter the data to define the ramp-function generator ramp.
- Then you must carry out the motor measurements: Start with static measurements and then take rotating measurements.

Activating Safety Integrated

- Open the Safety Integrated selection window under **Drive unit** → **Drives** → **Functions** → **Safety Integrated** and select the required Safety function:

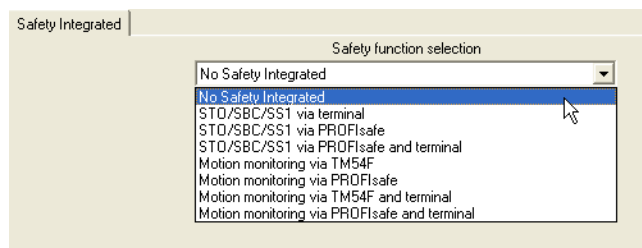


Figure 7-3 Safety Integrated selection

2. In the drop-down menu below, select "[1] Safety without encoder".
3. Then, open the configuration window and set the actual value acquisition cycle (p9511) to the value of the current controller cycle (p0115) (e.g. 125 µsec).
4. Click "Gear factor" and set the actual value tolerance (p9542) to a larger value (e.g. 10 mm/min or 10 rpm) and the number of motor revolutions to match the pole pair number (r0313).
5. Open SS1 and set the shutdown speed to >0.
6. Call the Safely-Limited Speed, change all of the stop responses to "[0]STOP A" or "[1]STOP B" and close the window.
7. The user-specific Safety settings can now be performed.
8. Click on "Copy parameters".
9. Switch off/switch on the drive to accept the changes.

Note

If, during startup, the drive outputs the message C01711, this indicates problems due to too steep ramping in connection with the precontrol. You have the following options to remedy this:

- Reduce the ramp gradient.
 - Use the extended ramp-function generator (with rounding) to set a more gentle ramp up.
 - Reduce the precontrol.
 - Change the values of parameters p9587, p9588 and p9589 (see specifications in the List Manual).
-

7.2.3 Standard commissioning of Safety Integrated functions

Standard commissioning of the safety functions

1. A commissioned project that has been uploaded to STARTER can be transferred to another drive unit keeping the existing Safety parameterization.
2. If the source and target devices have different software versions, the reference checksums may have to be adapted. This is displayed by a fault:
Reference checksums Fault
p9729[0...2] F01680(0...2)
p9399[0...1] F30680(0...1)
p9799 F01650 (fault value: 1000)
p9899 F30650 (fault value: 1000)
p10005[0...1] F01650(0...1) (TM54F)
3. Once the project has been downloaded to the target device, an acceptance must be carried out. This is indicated by fault F01650 (fault value: 2004). Additional information on the acceptance test is provided in the Function Manual "Safety Integrated" in Chapter "Acceptance test and acceptance report".

7.2.4 Setting the sampling times

Terminology

The software functions installed in the system are executed cyclically at different **sampling times** (p0115, p0799, p4099).

Safety functions are executed within the **monitoring clock cycle** (p9300/p9500) and TM54F is executed within the **sampling time** (p10000).

Communication on PROFIBUS is handled cyclically by means of the **communication clock cycle**.

During the PROFIsafe scan cycle, the PROFIsafe telegrams issued by the master are evaluated.

Rules

- The monitoring clock cycle (p9300/p9500) can be set between 500 µs to 25 ms.

Note

The monitoring clock cycle must be the same on all drives.

However, the calculation time required for the Extended Functions in the Control Unit depends on the monitoring clock cycle, that is, shorter clock cycles extend the calculation time. The availability of a specific monitoring clock cycle therefore depends on calculation time resources of the Control Unit.

Calculation time resources on the Control Unit are influenced primarily by the number of drives, the number of drives with enabled Extended Functions, the connected DRIVE-CLiQ components, the selected DRIVE-CLiQ topology, the use of a CBE20 and by the selected technological functions.

- PROFIBUS
 - The monitoring clock cycle (p9300/p9500) must be an integer multiple of the actual value update clock cycle. p9311/p9511 are generally used for the clock cycle for actual value acquisition. When p9311/p9511 = 0 in *isochronous operation* the isochronous PROFIBUS communication clock cycle is used, in *non-isochronous operation* the monitoring clock cycle in this case is 1 ms.
 - The current controller cycle must be no more than a quarter of the length of the actual value update clock cycle.
 - The sampling time of the current controller (p0115[0]) must be at least 125 µs.
- The sampling time of the TM54F must be the same as the monitoring clock cycle (p10000 = p9300/p9500).

Note

The Safety functions are carried out in the monitoring cycle (r9780/r9880 for Basic Functions or p9500/p9300 for Extended Functions). PROFIsafe telegrams are evaluated in the PROFIsafe scan cycle, which corresponds to twice the monitoring clock cycle.

Overview of important parameters (see the S120/S150 List Manual)

- p9300 SI Motion monitoring clock cycle (Motor Module) (only Extended Functions)
- p9500 SI Motion monitoring clock cycle (Control Unit) (only Extended Functions)
- p9311 SI Motion actual value sensing clock cycle (Motor Module)
- p9511 SI Motion actual value sensing clock cycle (Control Unit)
- r9780 SI monitoring clock cycle (Control Unit)
- r9880 SI monitoring clock cycle (Motor Module)
- p10000 SI sampling time (TM54F)

7.3 Commissioning TM54F by means of STARTER/SCOUT

7.3.1 Basic sequence of commissioning

The following conditions must be met before you can configure the TM54F:

- Concluded initial commissioning of all drives

Table 7- 1 Configuration sequence

Step	Execution
1	Insert the TM54F
2	Configure the TM54F and generate the drive groups
3	Configure the drive groups
4	Configure the inputs
5	Configure the outputs
6	Copy the parameters to the second drive object (TM54F_SL)
7	Change the safety password
8	Activate the configuration by selecting "Activate settings"
9	Save the project in STARTER
10	Save the project in the drive by selecting "Copy RAM to ROM"
11	Execute POWER ON
12	Acceptance test

7.3.2 Configuration start screen

Description

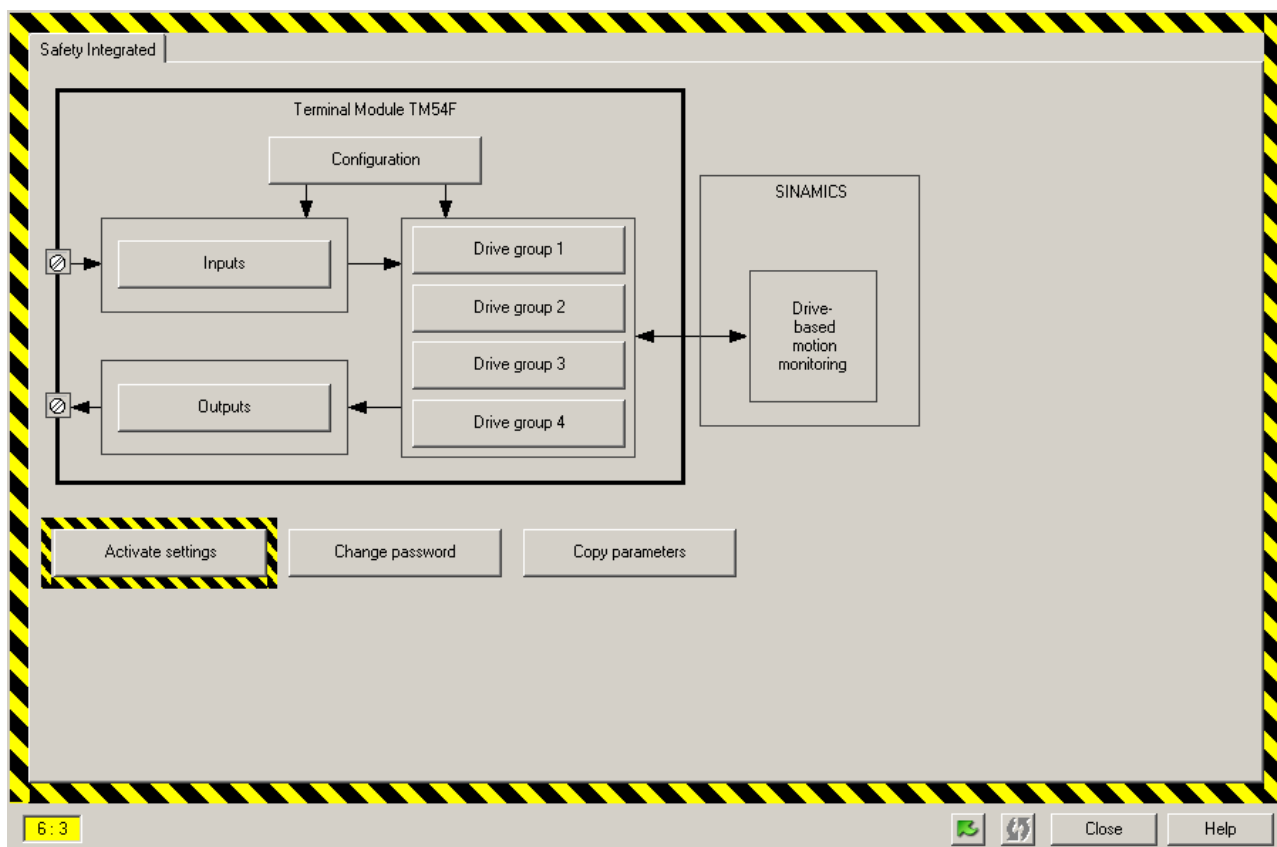


Figure 7-4 Configuration start screen TM54F

The following functions can be selected in the start screen:

- Configuration
Opens the "Configuration" screen
- Inputs
Opens the "Inputs" screen
- Outputs
Opens the "Outputs" screen
- Drive group 1 ... 4
Opens the corresponding screen of drive group 1 to 4
- Copy parameters
To copy the configuration to the second drive object (TM54F_SL), press "Copy parameters".

- Change/activate settings

- Change settings

You can select this button and enter the TM54F password in order to edit the configuration data. The button function changes to "Activate settings".

- Activate settings

This function activates your parameter settings and initiates calculation of the actual CRC and the corresponding transfer to the target CRC.

The parameters are activated after restart, and you are requested to carry out the acceptance test.

A message is output requesting you to save the project and then restart the system. It is also required to carry out an acceptance test.

- Change password (p10061 ... p10063)

In order to change the password, enter the old password (factory setting: 0) and then enter and confirm the new password.

7.3.3 TM54F configuration

Configuration screen of TM54F for Safety Integrated

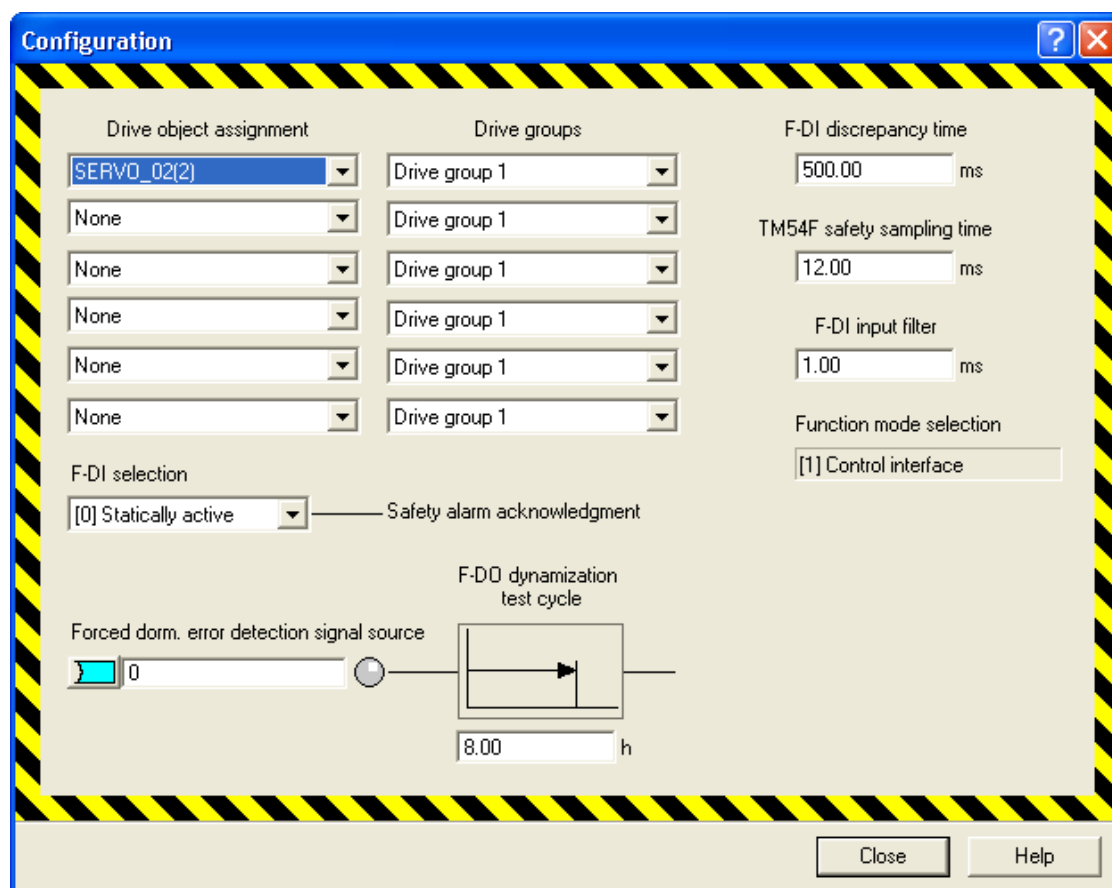


Figure 7-5 TM54F configuration

Functions of this screen:

- Assigning drive objects (p10010)
Select a drive object to be assigned to a drive group.
- Drive groups (p10011)
Each configured safety drive can be assigned to a drive group using a drop-down list box. The list box displays the drives and their names.
- Discrepancy time (p10002)
The signal states at the two terminals of an F-DI are monitored in order to determine whether these have assumed the same logical state within the discrepancy time.

Note

The discrepancy time must be set so that it is always smaller than the smallest expected switching interval of the signal to this F-DI.

- Safety sampling time (p10000)

The Safety sampling time corresponds to the sampling time of TM54F.

Note

The Safety clock cycle (p10000) of the TM54F must be set so it is the same as the monitoring clock cycle in p9300/p9500 for all the drives controlled by the TM54F.

- F-DI input filter (p10017)

Parameterizing the debounce time of the F-DIs and single-channel DIs of the TM54F. The debounce time is rounded off to whole ms and then accepted. The debounce time specifies the maximum time an interference pulse can be present at F-DIs before being interpreted as a switching operation.

- F-DI selection (p10006)

The Extended Functions enter a safety message in a special message buffer upon the detection of internal errors or violations of limits. This alarm must be acknowledged safely. You can assign an F-DI terminal pair for safe acknowledgment.

- Signal source, forced dormant error detection (p10007)

Select an input terminal to start the test stop: The test stop is started with a 0/1 signal at the input terminal and is then only possible if the drive is not in commissioning mode.

- Test cycle, dynamization F-DO (p10003)

Fail-safe I/O must be tested at defined intervals in order to validate their fail-safety (test stop, or forced dormant error detection). The TM54F module is provided with a function block which is selected by means of a BICO source to execute this forced dormant error detection (e.g. switch the L1+ and L2+ sensor power supply). Each selection triggers a timer in order to monitor the test cycle. A message is set on expiration of the monitored time.

7.3.4 Test stop of the TM54F

Test of the fail-safe inputs and outputs

Fail-safe I/O must be tested at defined intervals in order to validate their fail-safety (test stop, or forced dormant error detection). For this purpose, the TM54F contains a function block which carries out this forced dormant error detection when selected via a BICO source. To monitor the time until the next required test, a timer is started after every error-free test stop. A message is set on expiration of the monitored time and each time the Control Unit is switched on.

The failsafe digital inputs are always tested during a test stop. Three test stop modes can be selected for testing the outputs (see following section). After a time interval (p10003) has expired, the user is notified via message A01774 that a test stop must be performed for the F-DI/DO of the TM54F.

Carrying out a test stop:

Proceed as follows to parameterize the test stop:

1. Determine the appropriate test stop mode for the circuits used in your application (see diagrams in the following sections).
2. Set the test stop mode which is to be used via parameter p10047.
3. Use parameter p10046 to define which digital outputs (F-DO 0 to F-DO 3) are to be tested. Note the following:
Digital outputs which are not tested will be switched off during the test stop.
4. Use parameter p10041 to define which fail-safe digital inputs are to be checked during the test.
Inputs which do not have L1+ and L2+ power supplies may not be selected for the test.
5. Set the debounce time for the digital inputs using parameter p10017.
6. Use parameter p10001 to set the time within which the digital output signals to the corresponding digital inputs DI 20 ... DI 23 or DIAG inputs must be recognized. Select this time depending on the maximum response time of the external F-DO circuit.
7. Use parameter p10003 to set the interval within which a test stop should be carried out. After this time interval has expired, you will be notified via message A01774 that a test stop must be performed for the F-DI/DO of the TM54F.
8. Set the signal source which triggers the start of the test stop using parameter p10007. This can be, for example, a control signal or switch via a BICO switchable signal.

While the test stop is being carried out, the message A35012 (TM54F: Test stop active) appears. The messages A01774 and A35012 only disappear again after the test stop has been performed. If an error is found during the test stop, fault F35013 is output. Using the test sequence specified for each test stop mode, you can see which error has occurred from the fault value of the test step.

Duration of test stop

You can calculate the duration of the test stop with this formula:

	Comments
$T_{\text{Test stop}} = 3 \cdot p10000 + 2 \cdot (3 \text{ ms} + p10017) +$	• Test of the F-DIs • Evaluation of the inactive F-DIs
$8 \cdot p10000 + 6 \cdot (p10001 + p10017)$	• Test of the F-DOs • Evaluation of the active F-DIs

7.3.4.1 Test stop mode 1

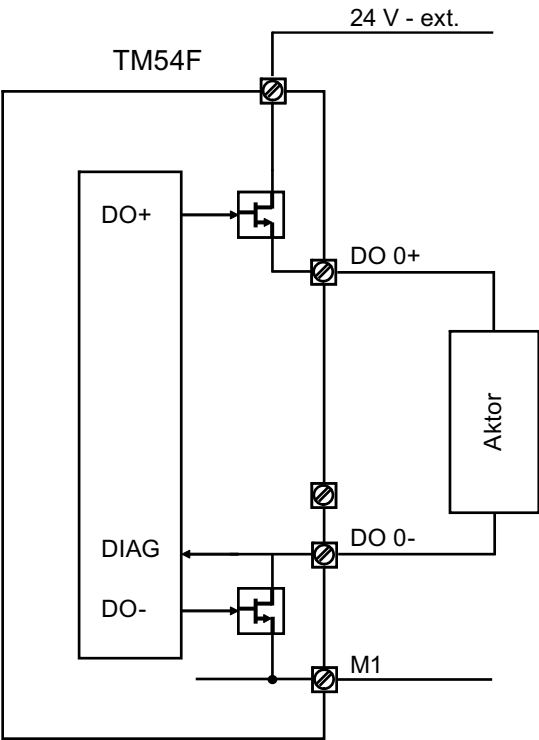


Figure 7-6 F-DO circuit, test stop mode 1

Test step ¹⁾	L1+	L2+	Comment
1	OFF	ON	Synchronization
3	OFF	OFF	F-DIs 0 ... 4 Check at 0 V
5	ON	ON	F-DIs 5 ... 9 Check at 0 V

Test step ¹⁾	DO+	DO-	Expected response, DIAG signal
6	OFF	OFF	LOW
8	ON	ON	LOW
10	OFF	ON	LOW
12	ON	OFF	HIGH
14	OFF	OFF	LOW

Test sequence for test stop mode 1

¹⁾ You can find a complete list of the steps in the SINAMICS S120/S150 List Manual /LH1/ under message F35013.

7.3.4.2 Test stop mode 2

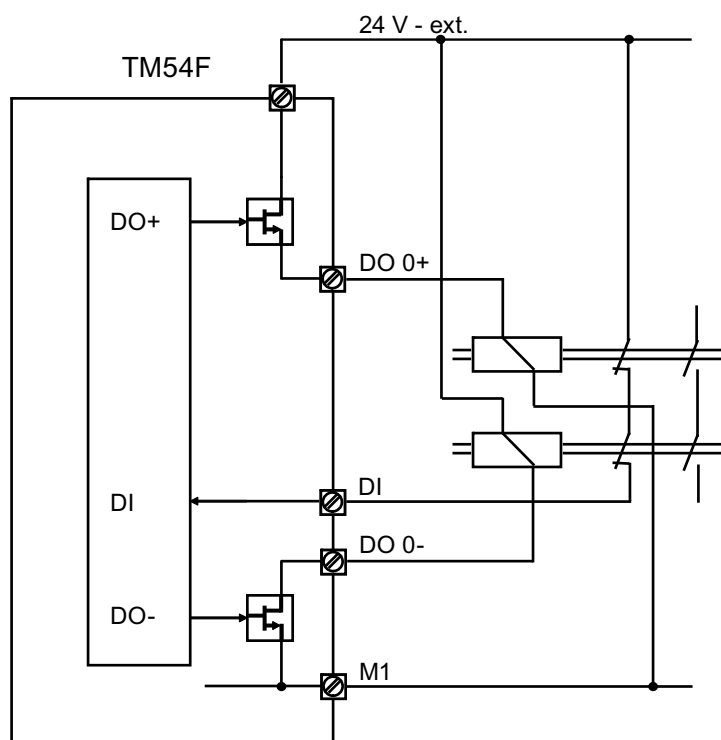


Figure 7-7 F-DO circuit, test stop mode 2

Test step ¹⁾	L1+	L2+	Comment
1	OFF	ON	Synchronization
3	OFF	OFF	F-DIs 0 ... 4 Check at 0 V
5	ON	ON	F-DIs 5 ... 9 Check at 0 V

Test step ¹⁾	DO+	DO-	Expected response, DI signal
6	OFF	OFF	HIGH
8	ON	ON	LOW
10	OFF	ON	LOW
12	ON	OFF	LOW
14	OFF	OFF	HIGH

Test sequence for test stop mode 2

¹⁾ You can find a complete list of the steps in the SINAMICS S120/S150 List Manual /LH1/ under message F35013.

7.3.4.3 Test stop mode 3

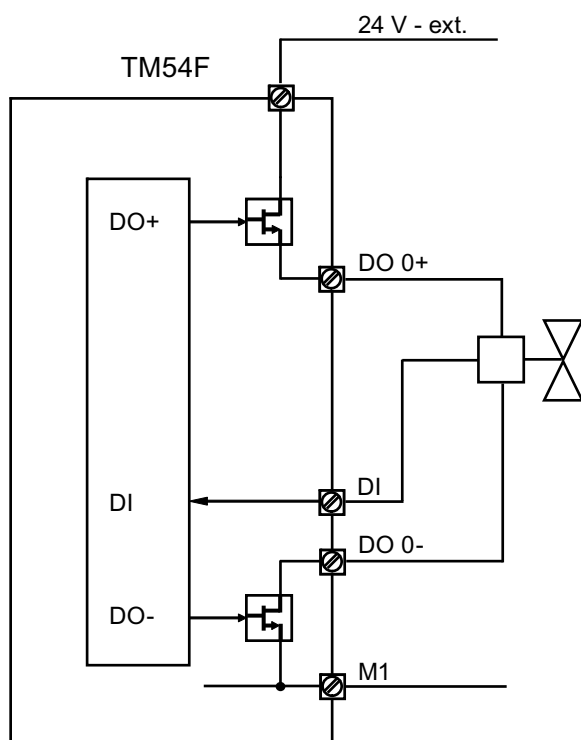


Figure 7-8 F-DO circuit, test stop mode 3

Test step ¹⁾	L1+	L2+	Comment
1	OFF	ON	Synchronization
3	OFF	ON	F-DIs 0 ... 4 Check at 0 V
5	ON	ON	F-DIs 5 ... 9 Check at 0 V

Test step ¹⁾	DO+	DO-	Expected response, DI signal
6	OFF	OFF	HIGH
8	ON	ON	LOW
10	OFF	ON	HIGH
12	ON	OFF	HIGH
14	OFF	OFF	HIGH

Test sequence for test stop mode 3

¹⁾ You can find a complete list of the steps in the SINAMICS S120/S150 List Manual /LH1/ under message F35013.

7.3.4.4 Test stop mode parameters

Overview of important parameters (see the SINAMICS S120/S150 List Manual)

- p10000 SI sampling time
- p10001 SI wait time for test stop at DO 0 ... DO 3
- p10003 SI forced dormant error detection timer
- p10007 BI: SI forced dormant error detection F-DO 0 ... 3 signal source
- p10017 SI digital inputs, debounce time
- p10046 SI test sensor feedback input DI 20 ... 23
- p10047[0...3] SI select test mode for test stop

7.3.5 F-DI/F-DO configuration

Inputs screen F-DI

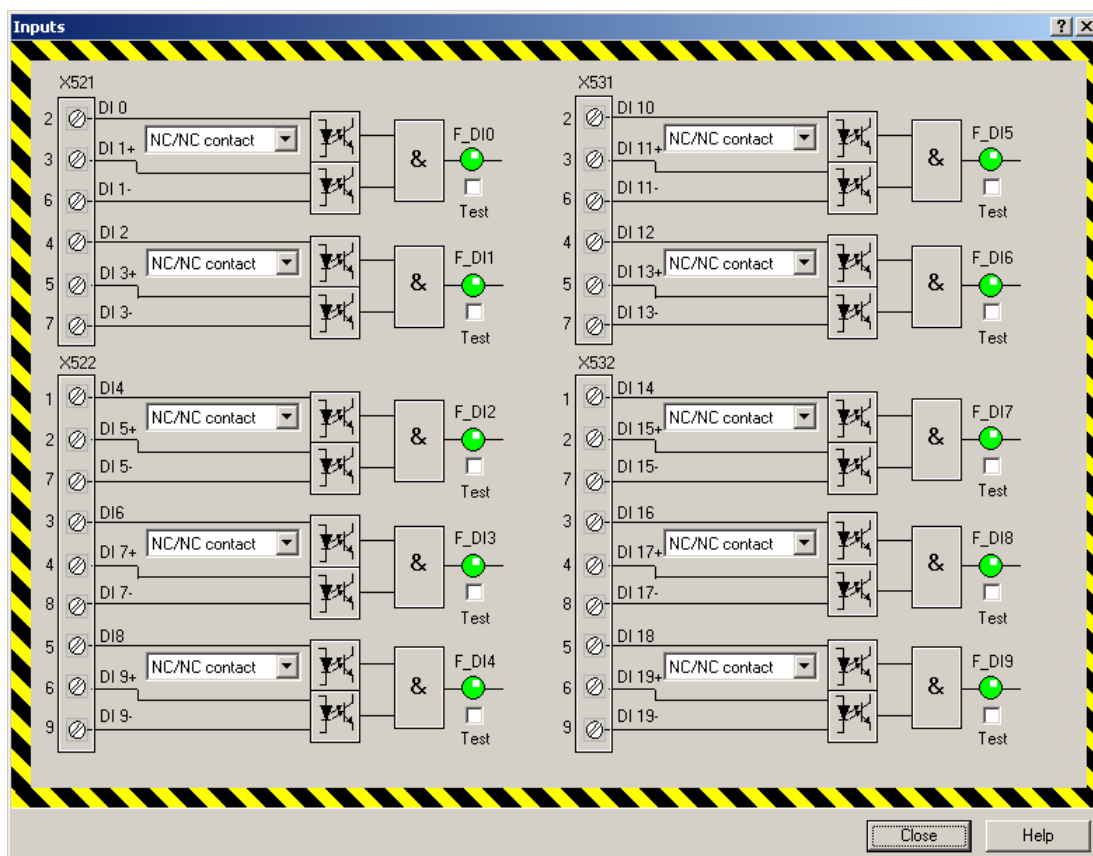


Figure 7-9 Inputs screen

NC/NO contact (p10040)

Terminal property F-DI 0-9 (p10040.0 = F-DI 0, ... p10040.9 = F-DI 9), only the property of the second (lower) digital input is set. Always connect an NC contact to digital input 1 (upper). Digital input 2 can be configured as NO contact.

Activate test mode (p10041)

A check mark at an F-DI defines whether the pair of digital inputs is to be integrated in the forced dormant error detection test of the assigned power supply (L1+ or L2+) (for additional information, see chapter "Forced dormant error detection", under Extended Functions).

LED in F-DI screen

The LED downstream of the AND element indicates the logical state (inactive: gray, active: green, discrepancy error: red).

Outputs screen F-DO

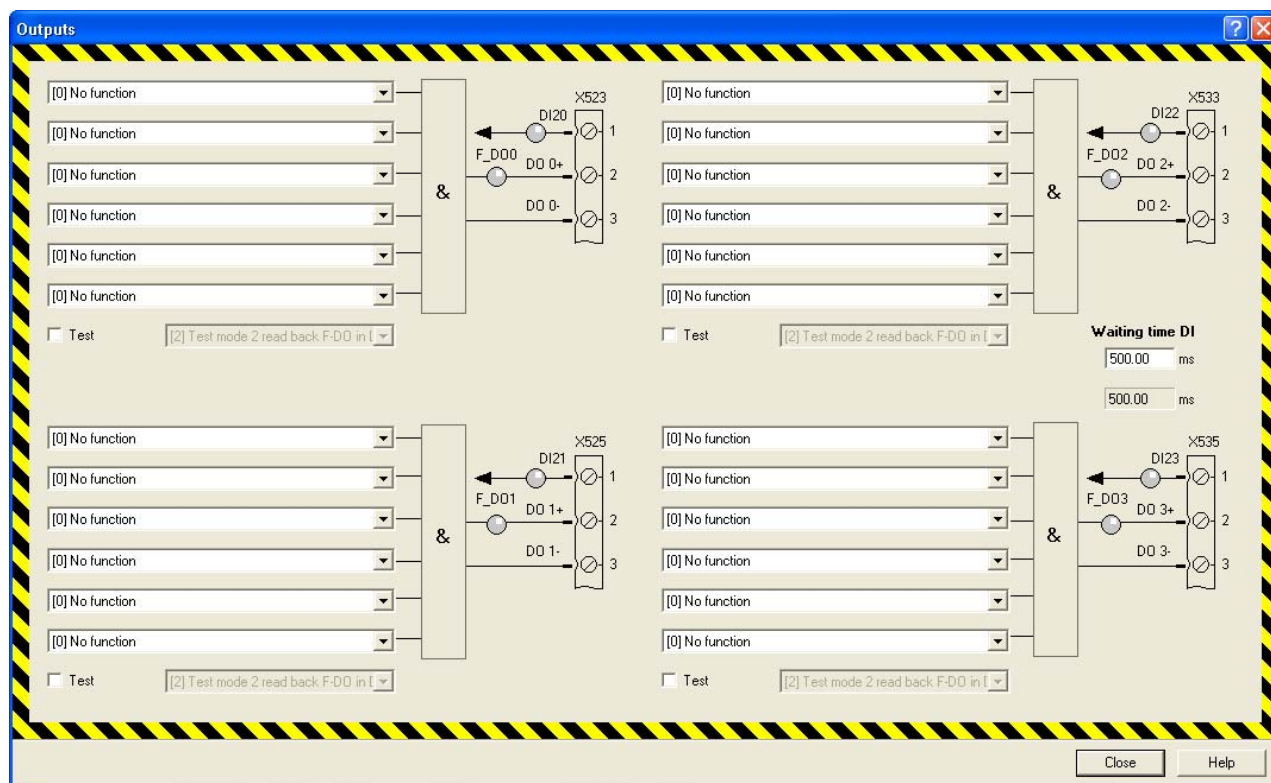


Figure 7-10 Outputs screen

Signal source for F-DO (p10042 - p10045)

An AND element with 6 inputs is interconnected with each output terminal pair of an F-DO; the signal sources for the AND inputs can be selected:

- If a signal source is not connected to an input, then the input is set to HIGH (default), exception: If a signal source is not connected at any input, then the output signal = 0
- Status signals of the drive of drive group 1 to 4

For additional information on status signals, see chapter "F-DO overview" in the "Control by means of TM54F terminals".

Select test sensor feedback signal (p10046 [0..3]) and select test mode for test stop (p10047 [0..3])

The test of the feedback line for the dynamization can be activated at each F-DO and the test mode can be selected for the test stop (for additional information, see Chapter "Forced dormant error detection" under Extended Functions).

LED in the F-DO screen

The LED downstream of the AND element indicates the logical state (inactive: gray, active: green).

The LED of the digital inputs DI20 to DI23 indicate the status of the digital input (inactive: gray, active: green).

7.3.6 Control interface of the drive group

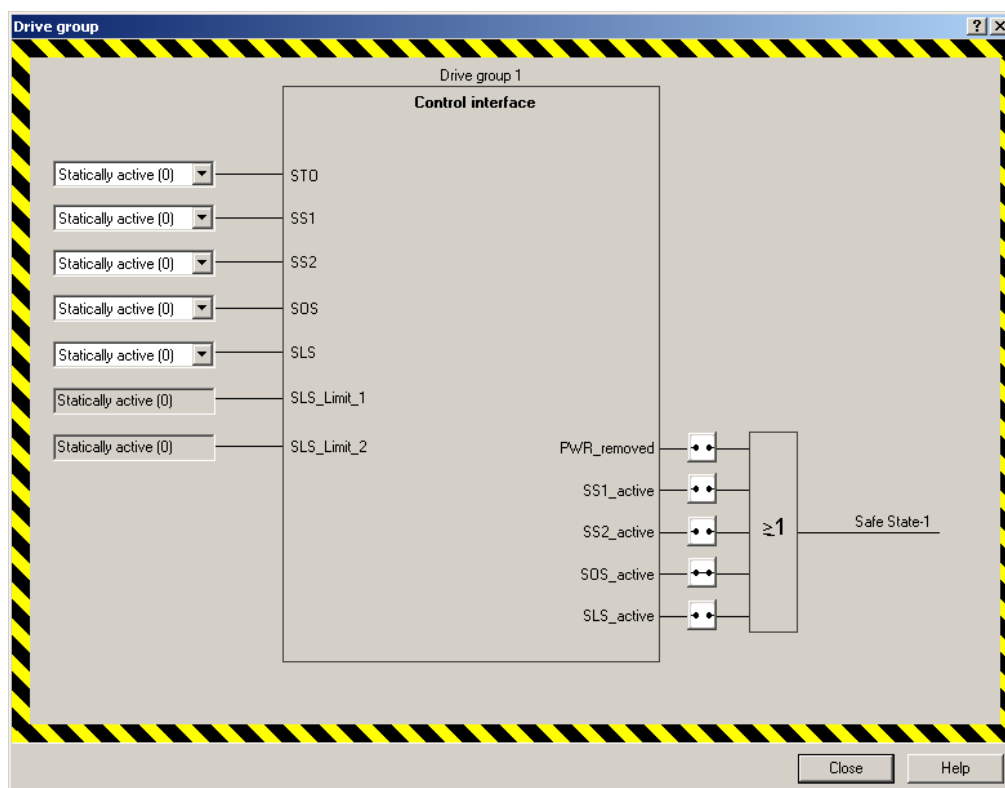


Figure 7-11 Screen, drive group

Functions of this screen:

- Selection of an F-DI for the STO, SS1, SS2, SOS and SLS functions and for SLS speed limits (bit coded) (p10022 to p10028).

A separate screen is available for each drive group. An F-DI can be assigned several functions in several drive groups.

- Configuration of the "Safe State" signal (p10039)

A safety output signal "Safe State" can be generated for each drive group from the following status signals:

- STO active (Power_removed)
- SS1 active
- SS2 active
- SOS active
- SLS active

The status signals from the same functions for different drives of a drive group are logically AND'ed. The status signals of the individual functions (STO active, SS1 active, etc.) are ORed.

The "Safe State" signals can be assigned to a an F-DO.

7.4 Procedure for configuring PROFI-safe communication

The next sections deal with a sample configuration of PROFI-safe communication between a SINAMICS S120 drive unit and higher-level SIMATIC F-CPU operating as PROFIBUS master. Here a special safety connection ("Safety-Slot") between the Master and Slave is set up automatically, which may not be changed.

HW Config can then be used to configure PROFI-safe telegram 30 (sub-module ID = 30) for the drive objects (abbreviation: DO).

Requirements for PROFI-safe communication

The following minimum software and hardware requirements apply for the configuration and operation of safety-oriented communication (F communication):

Software:

- SIMATIC Manager STEP 7 V5.4 SP4¹⁾ or higher
- S7 F Configuration Pack V5.5 SP3¹⁾ or higher
- S7 Distributed Safety Programming V5.4 SP3¹⁾ or higher
- STARTER V4.1.5 + SSP V4.3 or SIMOTION SCOUT V4.1.5 + SSP V4.3 or higher
- Drive ES Basic V5.4 SP4¹⁾ or higher

Hardware:

- A controller with safety functions (in our example, SIMATIC F-CPU 317F-2¹⁾)
- SINAMICS S120 (in our example, a CU320-2)
- Correct installation of the devices

¹⁾ When using a SIMATIC F-CPU

NOTICE
If a single software or hardware component is either older than those specified in this document or is missing, PROFI-safe cannot be configured via PROFIBUS or PROFINET.

7.4.1 Configuring PROFIsafe via PROFIBUS

Topology (network view of the project)

Components participating in F communication via PROFIBUS are basically wired as follows:

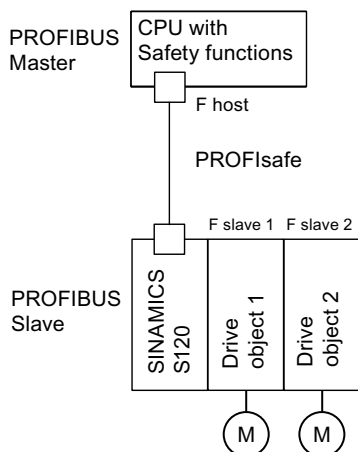


Figure 7-12 Example of a PROFIsafe topology

Configuring PROFIsafe communication using an example with a Siemens F-CU

The next sections describe a configuration of PROFIsafe communication between a SIMATIC F-CPU and a drive unit. It is helpful to regularly save intermediate states.

Creating a safety master

1. Create an F-CPU, e.g. CPU 317F-2, and a drive, e.g. SINAMICS S120 with CU320-2, in accordance with the hardware installed in HW Config.

To do this, start SIMATIC Manager and create a new project.

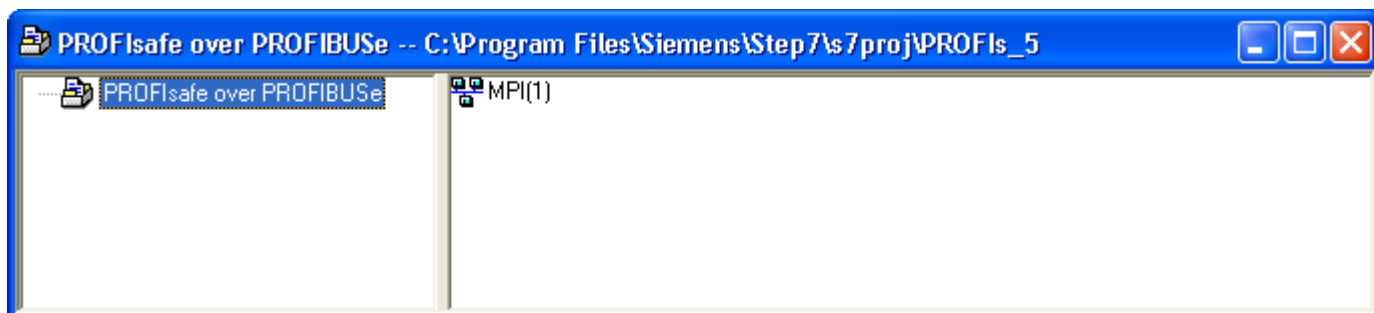


Figure 7-13 Creating a new project

2. Create a SIMATIC S300 Station under "Insert".

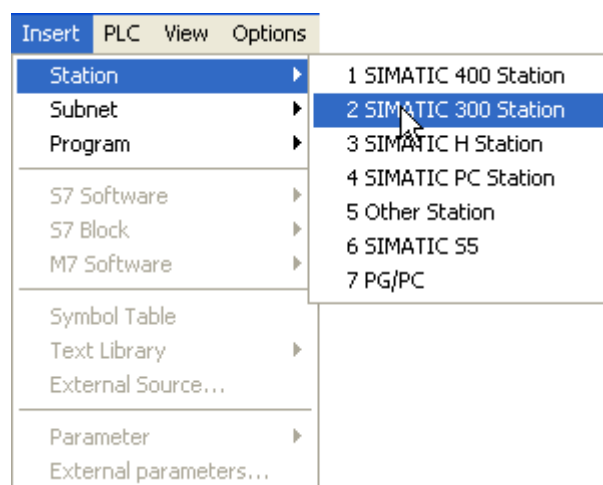


Figure 7-14 Creating a new station

3. The HW Config tool opens by double-clicking on SIMATIC S300(1), and then on "Hardware".

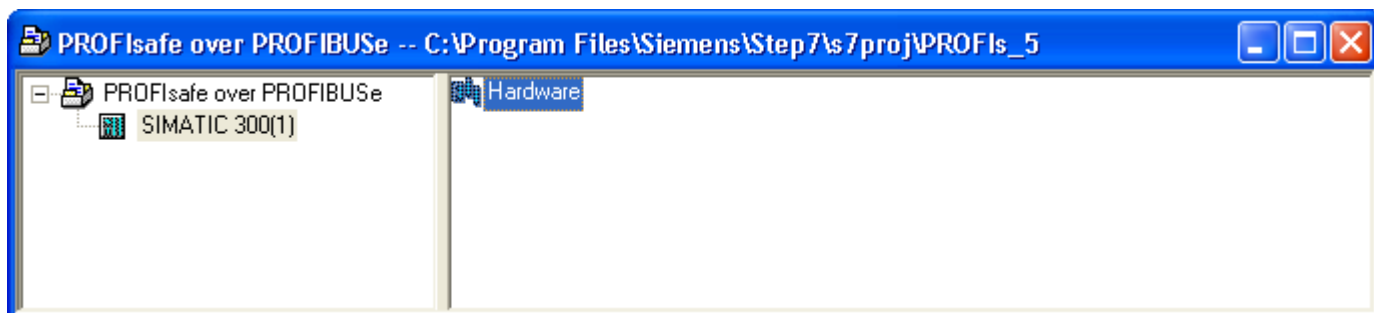


Figure 7-15 Calling HW Config

4. First create a mounting rail ((0)UR) under HW Config in the lefthand window:
From the standard catalog under SIMATIC 300/RACK-300, drag the mounting rail to the upper lefthand field (the cursor has a "+" character).

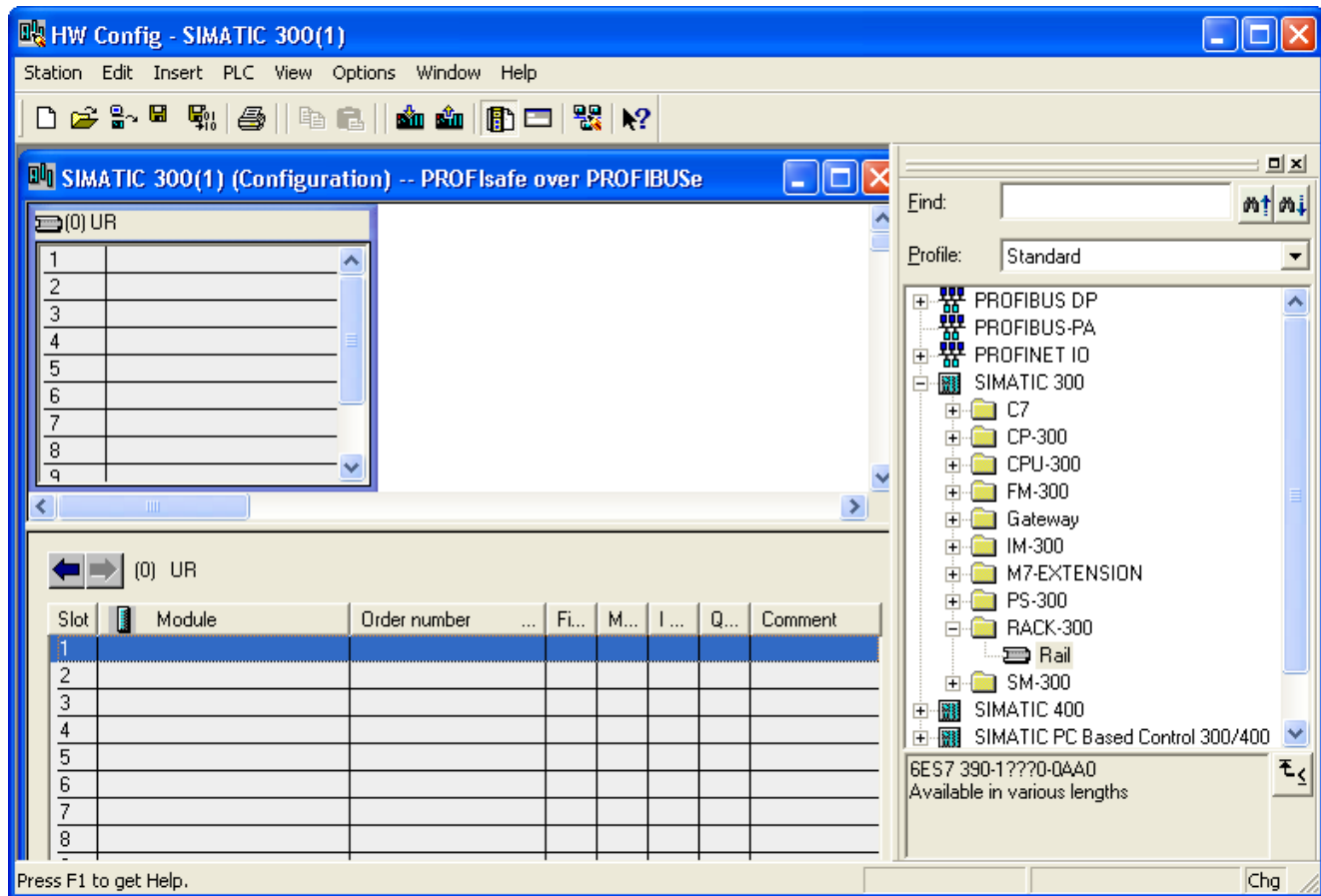


Figure 7-16 Creating a mounting rail

5. Select a safety-capable CPU under SIMATIC 300/CPU 300:
In this case, for example, drag CPU 317F-2, V2.6 into the RACK on slot 2 (highlighted).

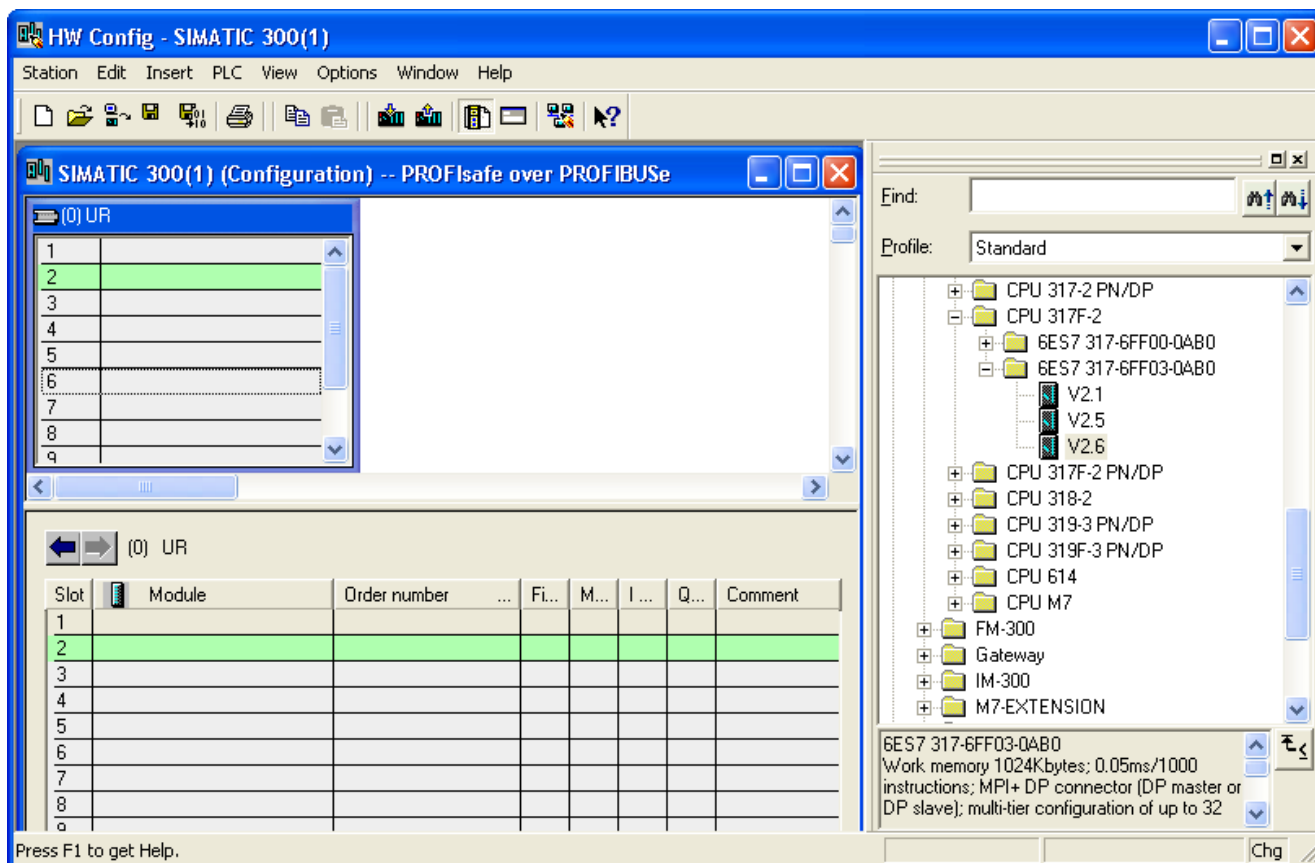


Figure 7-17 Creating an F host (master)

6. In the rack: The "Properties - PROFIBUS interface DP" window is opened by double-clicking on line X2. Under the tab "Parameter", click on "Properties..." in the interface field.

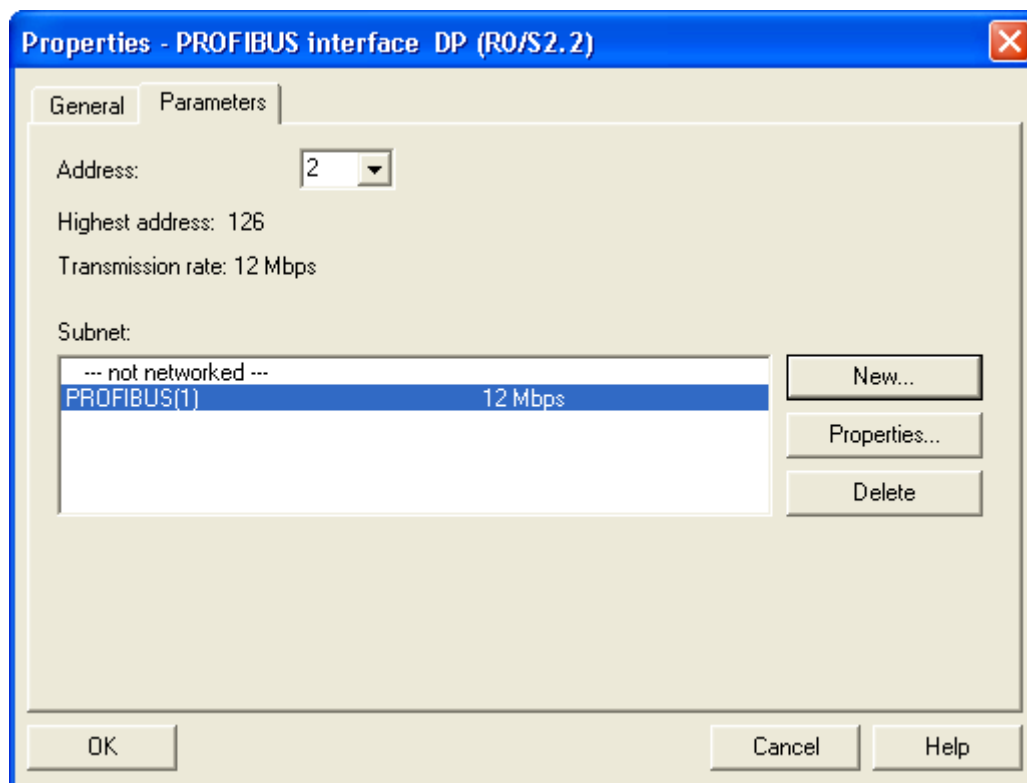


Figure 7-18 Setting the PROFIBUS interface

7. Set the PROFIBUS interface under the "Parameter" tab, set the address, and with the "Properties..." button, set the network settings, the transmission rate (e.g. 12 Mbit/s), the profile (DP) and then acknowledge with "OK". This sets up the master.

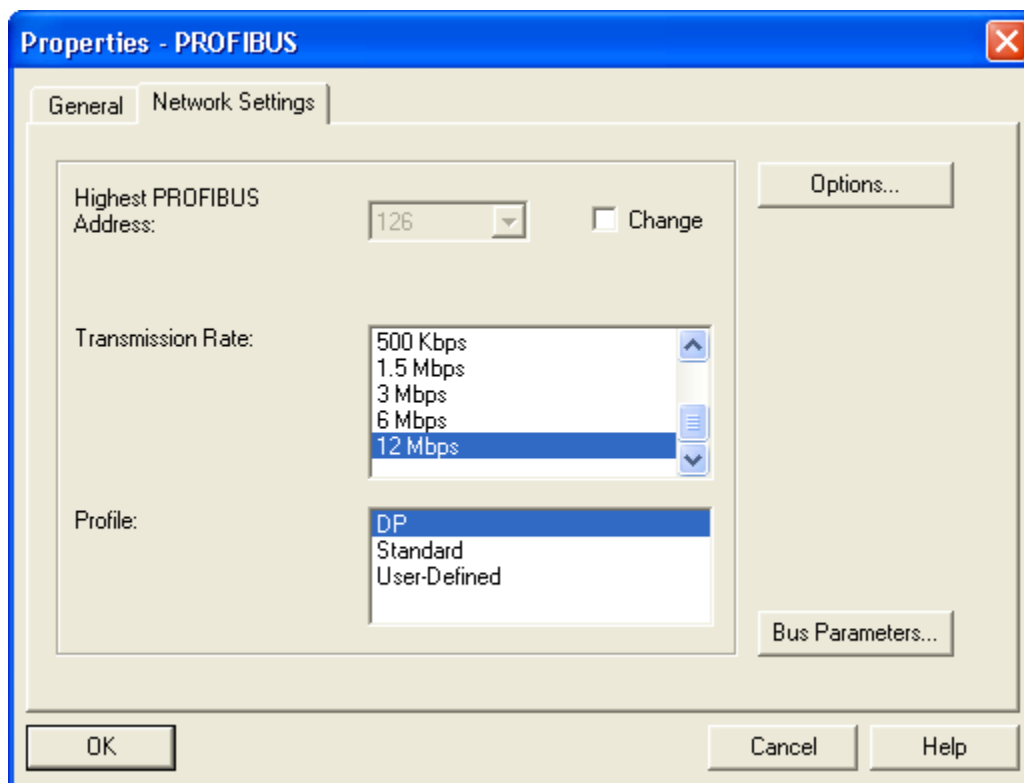


Figure 7-19 Setting the PROFIBUS profile

Creating a safety slave (drive)

1. The drive can either be selected in the catalog window under PROFIBUS-DP/SINAMICS /SINAMICS S120/SINAMICS S120 CU320 or by installing a GSD file. Using the lefthand mouse key, drag the "SINAMICS S120 CU320" drive to the PROFIBUS line in the upper lefthand window (the cursor has a + character) and release the mouse key. In the following properties window, set the PROFIBUS address of the drive and exit the following window with "OK".

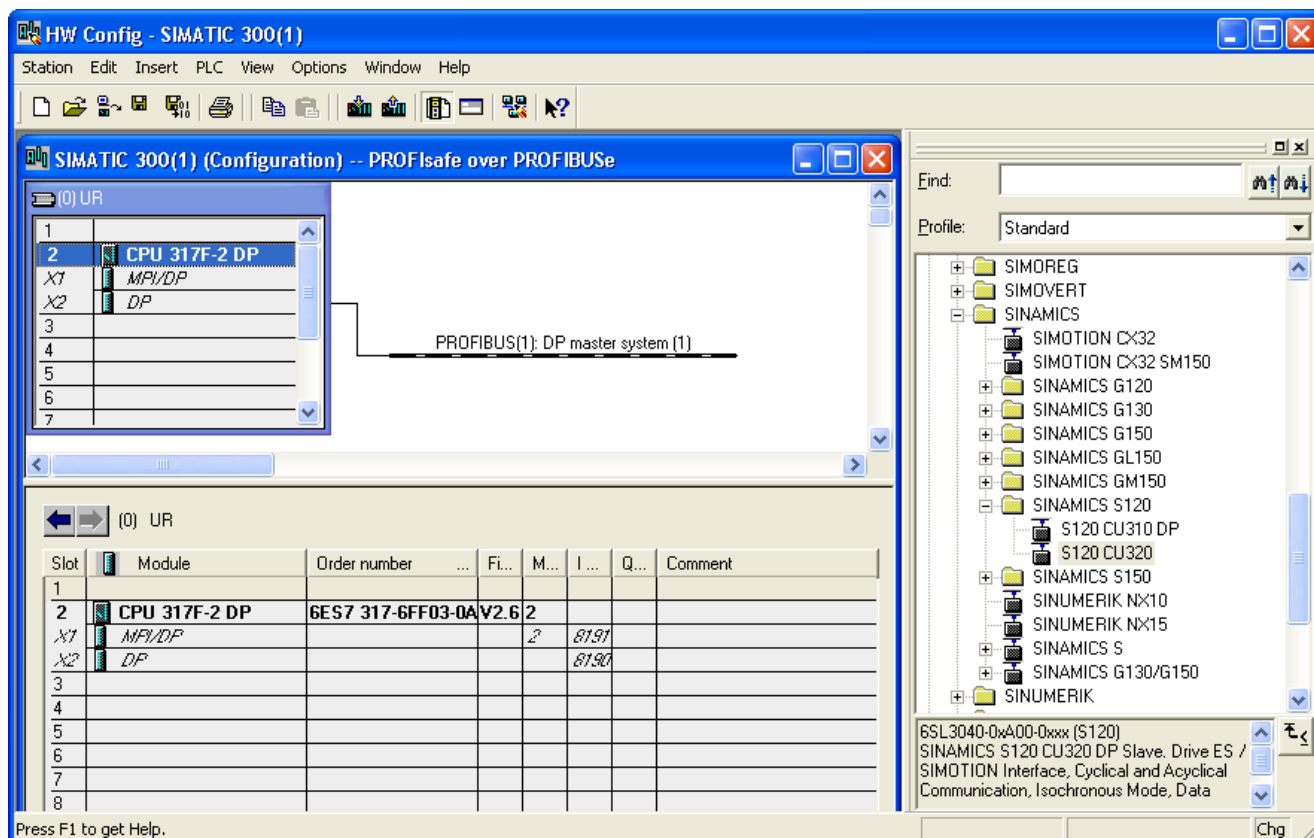


Figure 7-20 Selecting a drive

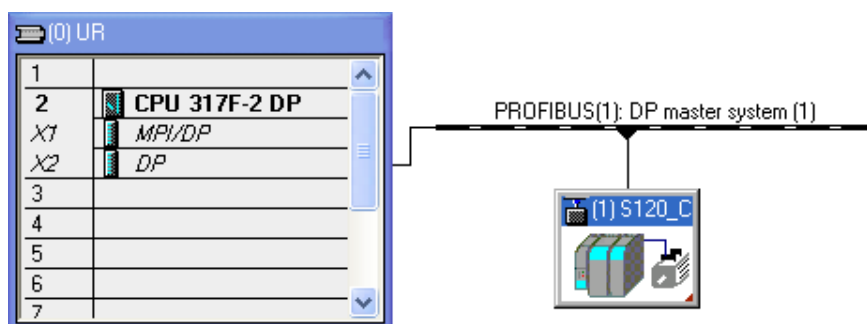


Figure 7-21 Drive created

- Double-clicking on the drive symbol opens the properties of the DP slave (here: (7)SINAMICS S120). The telegrams for F communication are selected and displayed (e.g. Siemens telegram 105) under "Configuration". Select the PROFIsafe telegram 30 under the option column. As a result, the "PROFIsafe..." button at the center left is activated.

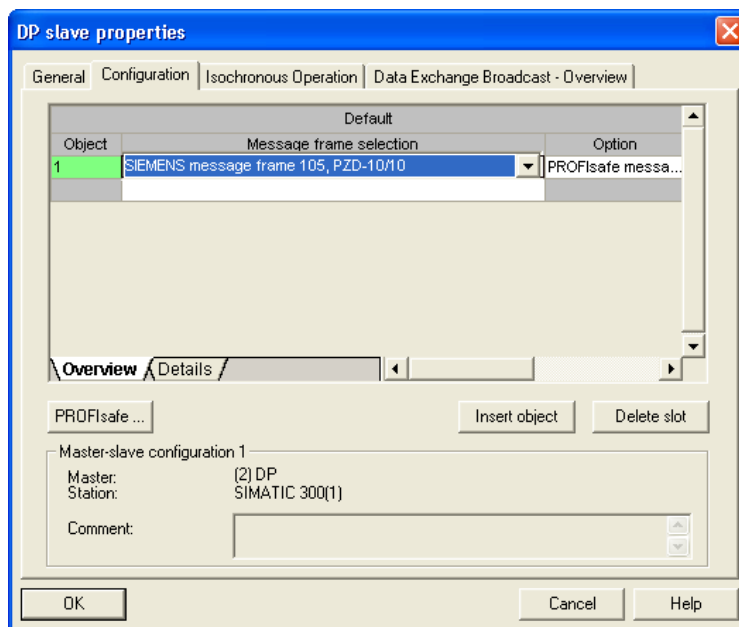


Figure 7-22 PROFIBUS DP slave properties

- The F parameters important for F communication are set using the "PROFIsafe..." button.

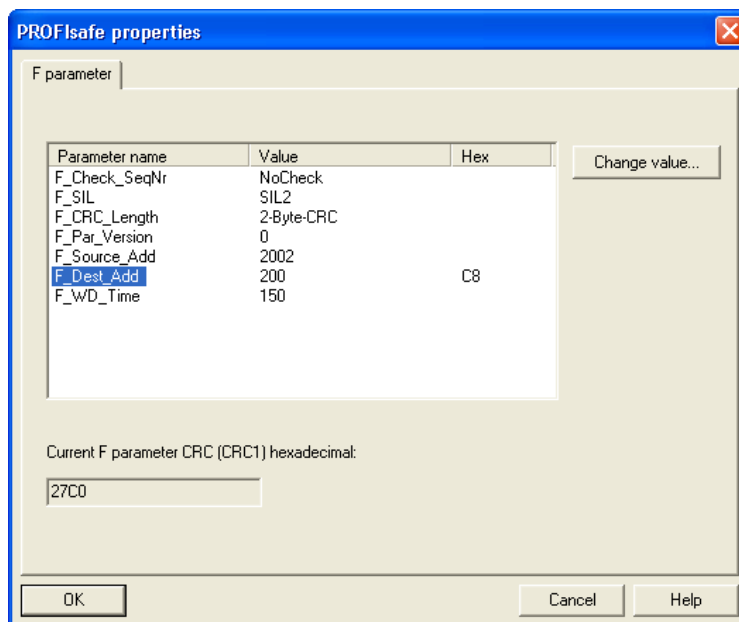


Figure 7-23 Setting the F parameters

The PROFIsafe mode is selected using parameters F_CRC_Length and P_Par_Version. The PROFIsafe address is set using parameter F_Dest_Add.

Selecting the PROFIsafe mode

The two PROFIsafe modes V1.0 and V2.0 can be selected.

- In the "F parameter" window, first click on the value that is to be changed.
- Then on the button "Change value..."
- A window will open. Select and confirm the desired value.
- Using these parameters, you can select either PROFIBUS mode version V1.0 or V2.0:
 - **PROFIsafe V1.0 mode:** F_CRC_Length = 2 byte-CRC, then P_Par_Version: 0
 - **PROFIsafe V2.0 mode:** F_CRC_Length = 3 byte-CRC, then P_Par_Version: 1

The following value ranges can be set for the last two parameters of the list:

1. PROFIsafe destination address F_Dest_Add: 1-65534

F_Dest_Add determines the PROFIsafe destination address of the drive object. Any value within the range is allowed, although it must be manually entered again in the Safety configuration of the drive in the SINAMICS drive unit. The F_Dest_Add value must be set in p9610 (Control Unit) and in p9810 (Motor Module). This can be done in a user-friendly fashion via the PROFIsafe STARTER screen form (refer to the following diagram). The PROFIsafe destination address of the F parameters must be entered here in the hexadecimal format (C8H in the example).

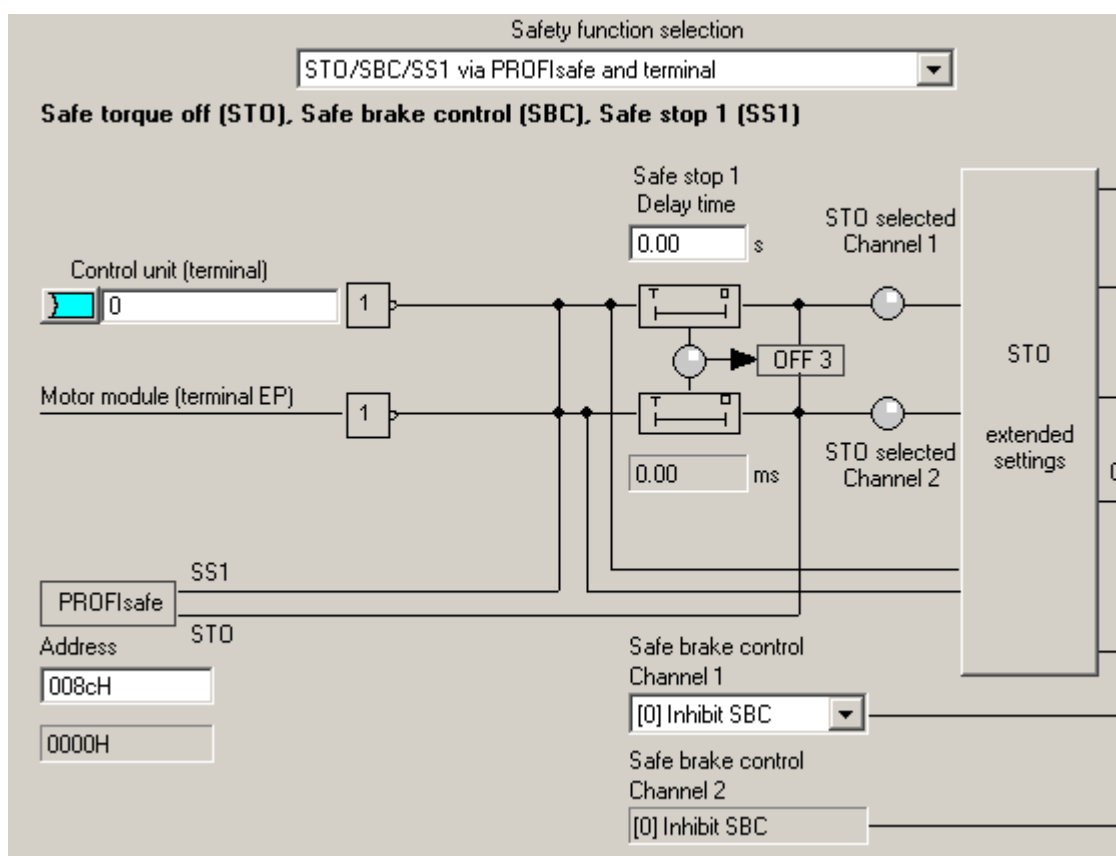


Figure 7-24 STARTER screen section from Safety Integrated: Setting the PROFIsafe address (example)

2. PROFIsafe monitoring time F_WD_Time: 10-65535

A valid current safety telegram must be received from the F-CPU within the monitoring time. The drive will otherwise switch to the safe state.

The monitoring time should be of sufficient length to ensure not only that the communication functions tolerate telegram delays, but also that the fault response is triggered quickly enough if a fault occurs (e.g. interruption of the communication connection).

For additional information on F parameters, refer to the online help of the ("Help subjects" button).

7.5 PROFIsafe via PROFINET

7.5.1 Configuring PROFIsafe via PROFINET

The next sections deal with a sample configuration of PROFIsafe communication between a SINAMICS S120 drive unit and a higher-level SIMATIC F-CPU operating as PROFINET master.

HW Config can then be used to configure PROFIsafe telegram 30 (sub-module ID = 30) for the drive objects (abbreviation: DO).

7.5.2 Requirements for PROFIsafe communication

Requirements for PROFIsafe communication

The following minimum software and hardware requirements apply for the configuration and operation of safety-oriented communication (F communication):

Software:

- SIMATIC Manager STEP 7 V5.4 SP4¹⁾ or higher
- S7 F Configuration Pack V5.5 SP3¹⁾ or higher
- S7 Distributed Safety Programming V5.4 SP3¹⁾ or higher
- STARTER V4.1.5 + SSP V4.3 or SIMOTION SCOUT V4.1.5 + SSP V4.3 or higher
- Drive ES Basic V5.4 SP4¹⁾ or higher

Hardware:

- A controller with safety functions (in our example, SIMATIC F-CPU 317F-2¹⁾)
- SINAMICS S120 (in our example, a CU320-2)
- Correct installation of the devices

¹⁾ When using a SIMATIC F-CPU

NOTICE

If a single software or hardware component is either older than those specified in this document or is missing, PROFIsafe cannot be configured via PROFIBUS or PROFINET.
--

7.5.3 Configuring PROFIsafe via PROFINET

Configuring PROFIsafe communication using SINAMICS S120 as an example

Configuring PROFIsafe via PROFINET is almost identical to configuring "PROFIsafe via PROFIBUS".

The difference is that the SINAMICS drive unit and SIMATIC F-CPU are in the same **PROFINET** subnet instead of in the same **PROFIBUS** subnet.

1. In HW Config, create a PROFINET-capable F-CPU, e.g. CPU 317F-2 PN/DP, corresponding to the hardware that has been installed. Create a **PROFINET** subnet and configure the F-CPU as an IO controller. Information about configuring an IO controller of F-CPU 317F-2 can be found in this reference:
Reference: SIMATIC PROFINET IO Getting Started: Collection
2. In the standard module catalog, under PROFINET IO, choose the module that you want to connect to the PROFINET IO subnet as an IO device, e.g. a CU320-2.
3. Drag the module to the line of the PROFINET IO subnet. The IO device is inserted. The Properties -> Ethernet Interface SINAMICS-S120-2 window opens. A suggested IP address will already be displayed here and the subnet selected. Confirm with "OK" to accept the setting.
4. Save and compile the settings in HW Config, and then load them to the target device.

This sets up a PROFINET connection between the F-CPU and the SINAMICS S120 drive.

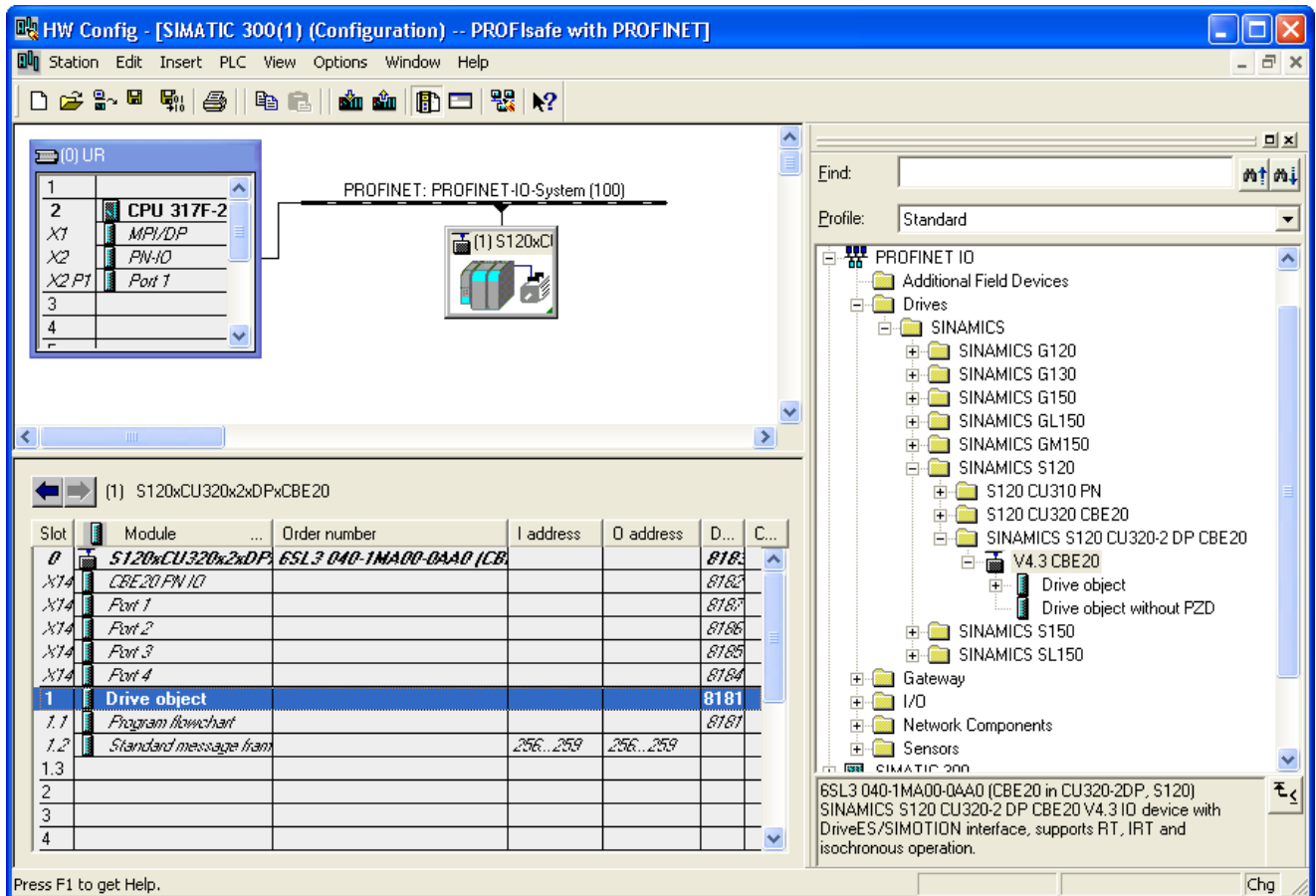


Figure 7-25 Configuration of the PROFINET connection in HW Config

1. Open the context menu of the drive object and select the command "Object properties":
The "Properties – drive object" window appears. Select the PROFIsafe telegram via PROFINET in this window. Using the "Options" tab, create the "PROFIsafe telegram 30".

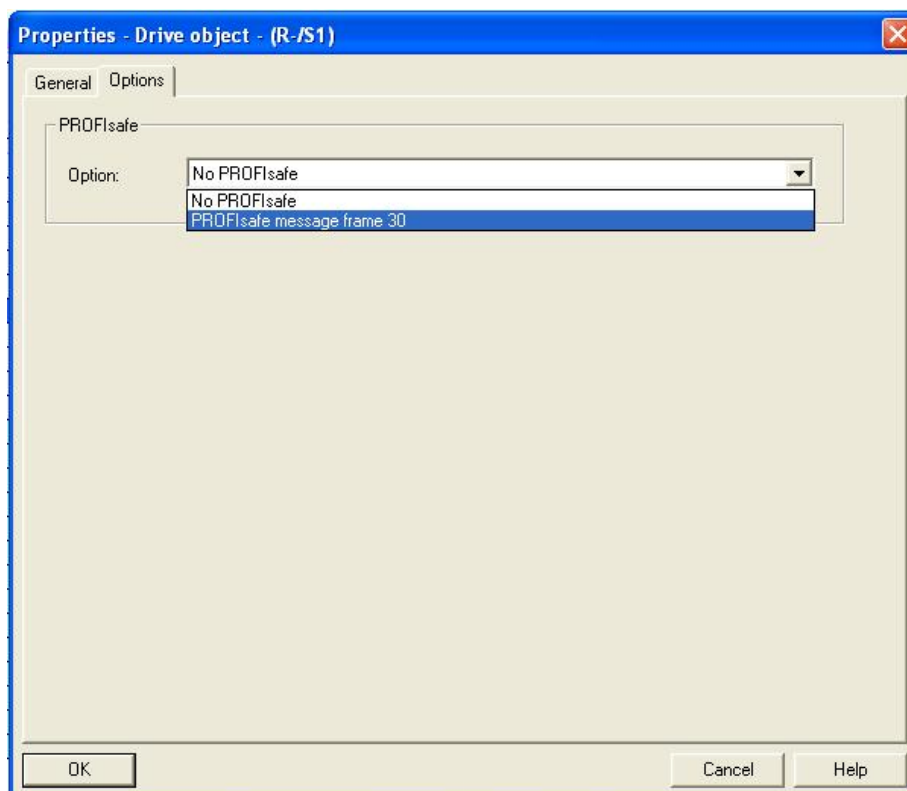


Figure 7-26 Drive object option "PROFIsafe telegram"

In the overview for the SINAMICS drive, a PROFIsafe slot that needs to be configured is displayed under "Drive object".

Slot	Module	Order number	I address	Q address	D...	C...
0	S120xCU320x2xDP	6SL3 040-1MA00-0AA0 (CB...			8183	
X14	CBE20 PN IO				8182	
X14	Port 1				8187	
X14	Port 2				8186	
X14	Port 3				8185	
X14	Port 4				8184	
1	Drive object				8181	
1.1	Program flowchart				8181	
1.2	PROFIsafe		0...5	0...5		
1.3	Standard message fram		256...259	256...259		
1.4						
2						
3						

Figure 7-27 Defining PROFIsafe for a drive

- Under the drive module, select the "PROFIsafe" line and use the right-hand mouse key to call up the properties of the PROFIsafe slot.

2. Define the address area of the PROFIsafe telegram under the "Addresses" tab. The start address for inputs and output is identical. To confirm your entries, choose "OK".

The screenshot shows the 'Properties - PROFIsafe' dialog box with the 'Addresses' tab selected. It contains two sections: 'Inputs' and 'Outputs'. Both sections have 'Start' and 'End' address fields set to 8 and 13 respectively, and a 'Process image' dropdown menu set to 'OB1 PI'. At the bottom, there are 'OK', 'Cancel', and 'Help' buttons.

Figure 7-28 Setting PROFINET addresses

3. Using the "PROFIsafe" tab, define the parameters important for Safety communication ("F parameters"). If the "PROFIsafe..." tab is inactive, then you can activate this button for control using the "Activate..." button.

The screenshot shows the 'Properties - PROFIsafe' dialog box with the 'PROFIsafe' tab selected. It features a table of F parameters and a 'Change value...' button. Below the table, there is a field for the current F parameter CRC (CRC1) hexadecimal value, which is 4E0B. At the bottom, there are 'OK', 'Cancel', and 'Help' buttons.

Parameter name	Value	Hex
F_SIL	SIL2	
F_CRC_Length	3-Byte-CRC	
F_Block_ID	0	
F_Par_Version	1	
F_Source_Add	2000	
F_Dest_Add	200	C8
F_WD_Time	150	

Current F parameter CRC (CRC1) hexadecimal:
4E0B

Figure 7-29 Setting F parameters

Setting F parameters:

The following range of values is valid for the the two last parameters of the list:

PROFIsafe destination address F_Dest_Add: 1 to 65534

F_Dest_Add determines the PROFIsafe destination address of the drive object.

Any value within the range is allowed, although it must be manually entered again in the Safety configuration of the drive in the SINAMICS drive unit. The F_Dest_Add value must be set in p9610 (Control Unit) and in p9810 (Motor Module). This can be done in a user-friendly fashion via the PROFIsafe STARTER screen form (refer to the diagram in the chapter, Commissioning PROFIsafe via PROFIBUS).

PROFIsafe monitoring time F_WD_Time: 10 to 65535

A valid current safety telegram must be received from the F-CPU within the monitoring time. The drive will otherwise switch to the safe state.

The monitoring time should be of sufficient length to ensure not only that the communication functions tolerate telegram delays, but also that the fault response is triggered quickly enough if a fault occurs (e.g. interruption of the communication connection).

Note

When you close the "PROFIsafe properties" dialog box, the fail-safe addresses (F_Dest_Add and F_Source_Add) are checked to ensure that they are unique. This function is only available, however, when the PROFINET link between SINAMICS S120 and SIMATIC F-CPU has already been established.

For additional information about creating a safety program and accessing PROFIsafe user data (e.g STW and ZSW) within the safety program, refer to the "SIMATIC, S7 Distributed Safety - Configuring and Programming" Programming and Operating Manual.

Safety configuration (online) in the SINAMICS drive

The process of configuring the SINAMICS drive via PROFINET by means of Safety Integrated screen forms is identical to that for configuration via PROFIBUS. Here, refer to the following chapter, PROFIsafe configuration with STARTER.

Acceptance inspection

Once configuring and commissioning have been successfully completed, you need to carry out an acceptance test of the drive safety functions (see Chapter "Acceptance tests and acceptance reports (Page 175)").

Note

If F parameters of the SINAMICS drive are changed in HW Config, the global signature of the safety program in the SIMATIC F-CPU changes. In this way, the global signature can be used to identify whether safety-relevant settings in the F-CPU (F parameters of the SINAMICS slave) have changed. The global signature does not, however, contain any changes to safety-relevant drive parameters set in SCOUT or STARTER.

7.5.4 Initializing the drives

Initializing the drives

In order that the master control can communicate with the drives, e.g. a CU317F-2 PN/DP with a SINAMICS S120, via PROFINET, the drives must have unique names (self-explanatory names are advantageous) and must be assigned their own IP addresses and set using STARTER or the PST initializing tool (so-called initialization).

Instructions regarding "initialization" can be found in the Commissioning Manual S120 (IH1) in Chapter "Establishing online operation - STARTER via PROFINET IO".

7.6 PROFIsafe configuration with STARTER (Basic Functions)

The Safety Integrated Basic Functions can be commissioned using STARTER in three ways.

1. STO/SS1/SBC only via terminals
2. STO/SS1/SBC only via PROFIsafe
3. STO/SS1/SBC via PROFIsafe and terminals simultaneously

The STARTER screen forms for using the Safety Integrated Basic Functions using terminals, PROFIsafe or terminals and PROFIsafe are described together here.

Safety slot

In order to use the Safety Integrated functions via PROFIBUS or PROFINET, a safety slot must first be created using the SIMATIC Manager Step 7 and HW Config. The procedure to do this was described in the previous chapters.

Expert list

The Safety Integrated Basic Functions can be individually and manually set using the expert list – but the settings using the STARTER screen forms are more user friendly and you are less prone to making mistakes.

Calling Safety Integrated in STARTER using SINAMICS S120 as example

The STARTER screen form for "Safety Integrated" is called under Drives/Functions with a double-click and can look like this (tree-type view depends on the specific project):

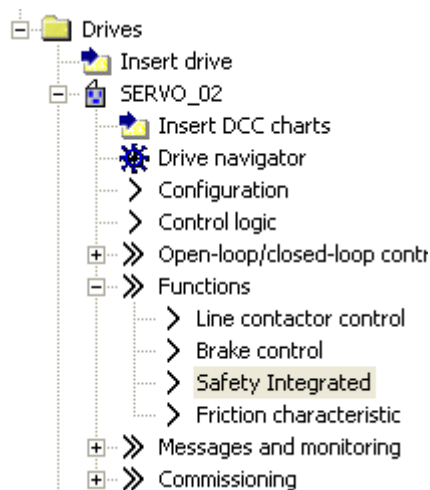


Figure 7-30 STARTER tree to call Safety Integrated

To use the full functionality of STARTER screen forms, there must be an online connection between the drives, the controller and STARTER.

Selecting using the pulldown menu:

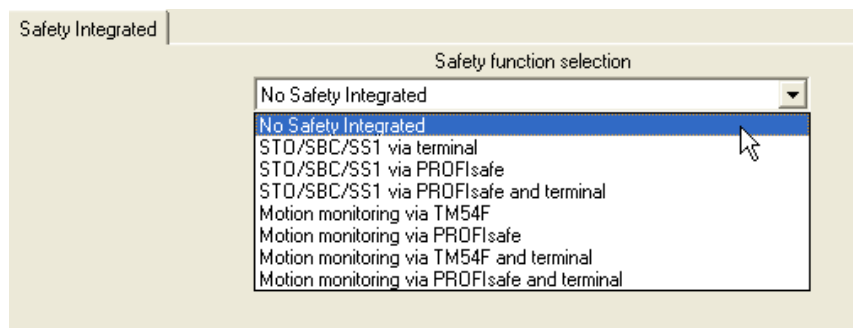


Figure 7-31 Safety_Integrated_Auswahl

Depending on the selection, different setting screen forms open:

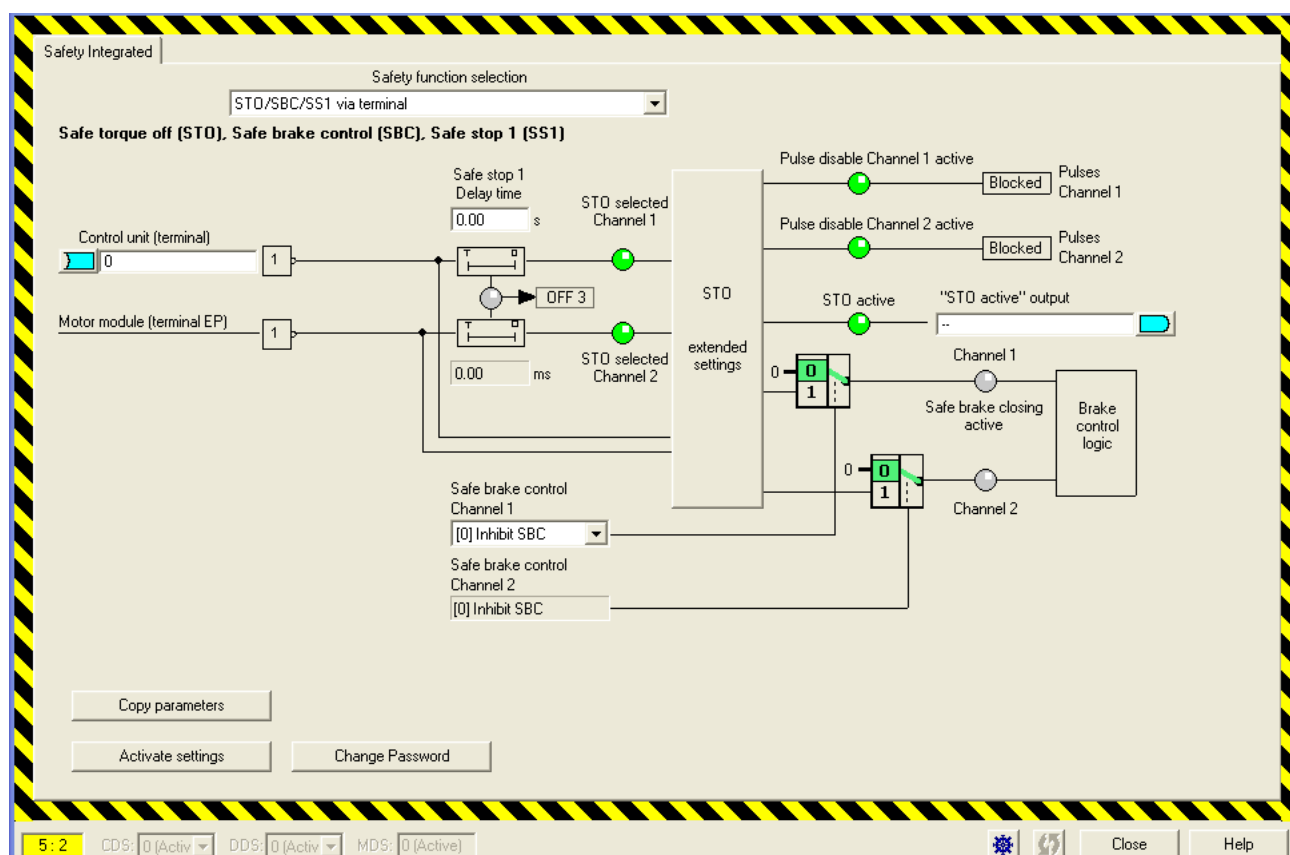


Figure 7-32 STO/SBC/SS1 via terminals

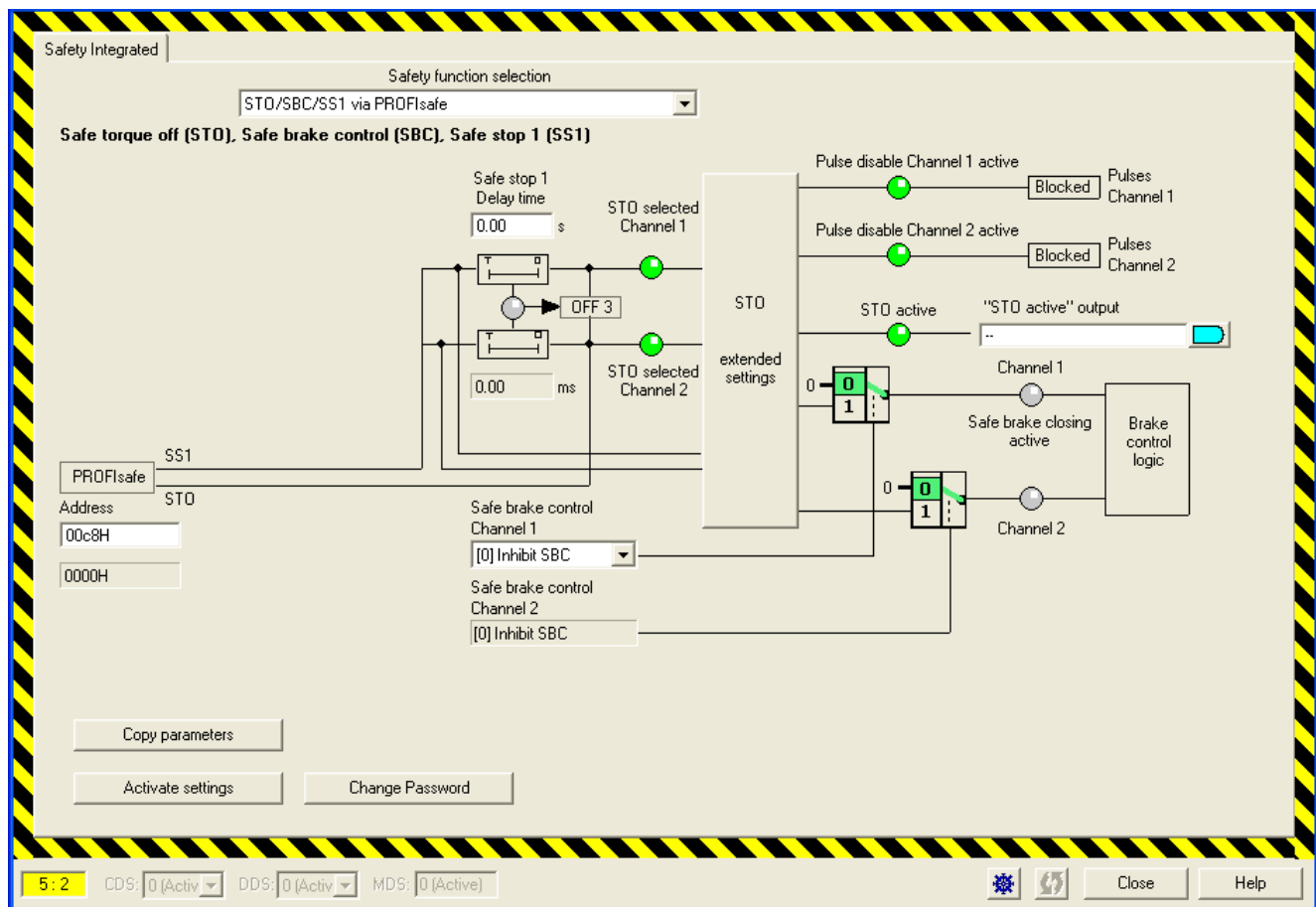


Figure 7-33 STO/SBC/SS1 via PROFIsafe

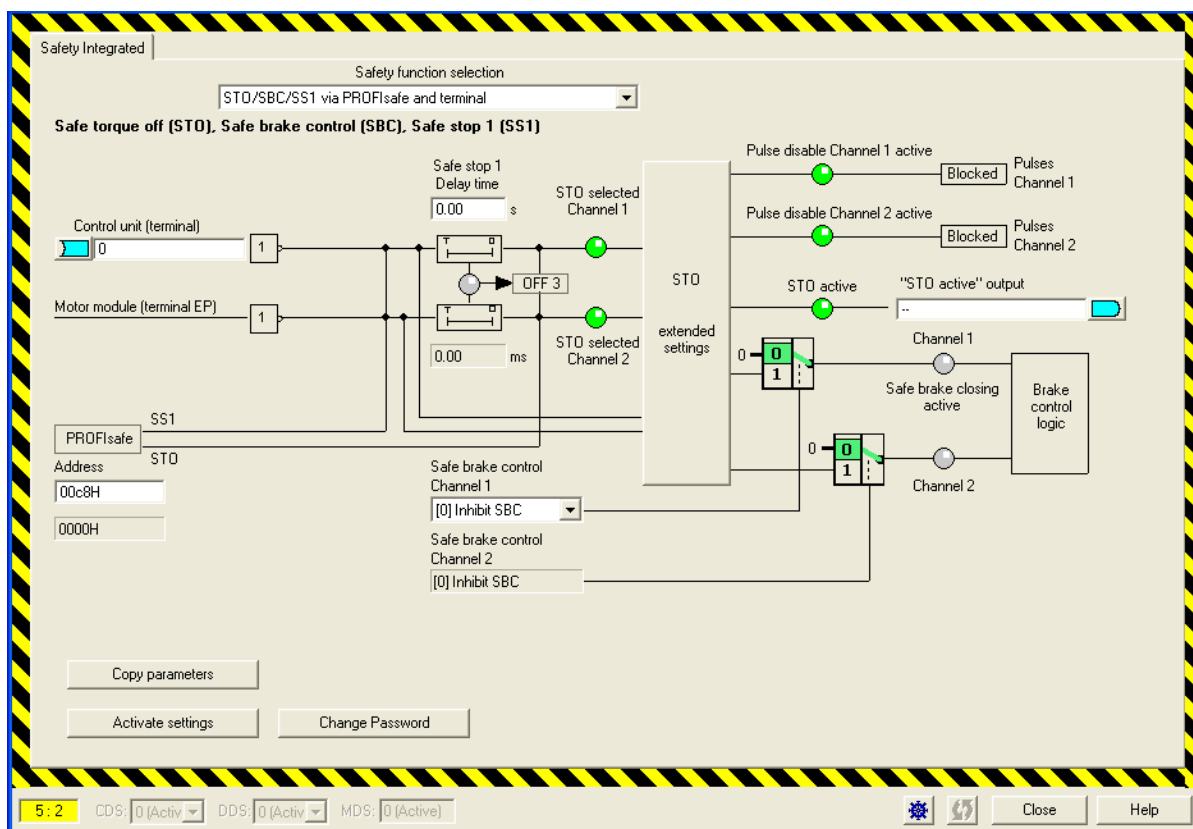


Figure 7-34 STO/SBC/SS1 via PROFIsafe and terminal

Activating PROFIsafe via the expert list

In order to activate Safety Integrated Basic Functions via PROFIsafe, in the expert list, bit 3 of p9601 and p9801 must be set to "1" and bit 2 to "0". Bit 0 must be set to either "1" or "0", depending on whether the control via terminals is to be enabled in parallel via PROFIsafe or not.

Saving and copying the Basic Function parameters

After setting the specific parameters for Safety Integrated functions (e.g. the PROFIsafe address) using the "Copy parameters" button, these must be copied from the CU into the Motor/Power Module and activated using the "Activate settings" button.

Acceptance test

An acceptance test needs to be carried out once configuration and commissioning are complete (see chapter Acceptance tests and acceptance reports (Page 175)).

Note

If F parameters of the SINAMICS drive are changed in HW Config, the global signature of the safety program in the SIMATIC F-CPU changes. This means that using the global signature it is possible to identify whether safety-relevant settings have changed in the F-CPU (F parameters of the SINAMICS slave). However, this global signature does not include the safety-relevant drive parameters so that their change cannot be checked in this way.

7.7 Commissioning a linear/rotary axis

The next section outlines the safety commissioning procedure for a linear axis/rotary axis when a TM54F is used.

1. Connect a PG to the drive and link it to the target device via STARTER.
2. In the STARTER project tree, select the required drive object and under **Functions** → **Safety Integrated** open the start screen to configure Safety Integrated.
3. Click on the **Change settings** button. The window for selecting Safety Integrated opens.
4. It is only possible to change Safety parameters after entering the valid Safety password (parameter p9761 for the drives or p10061 for the TM54F).

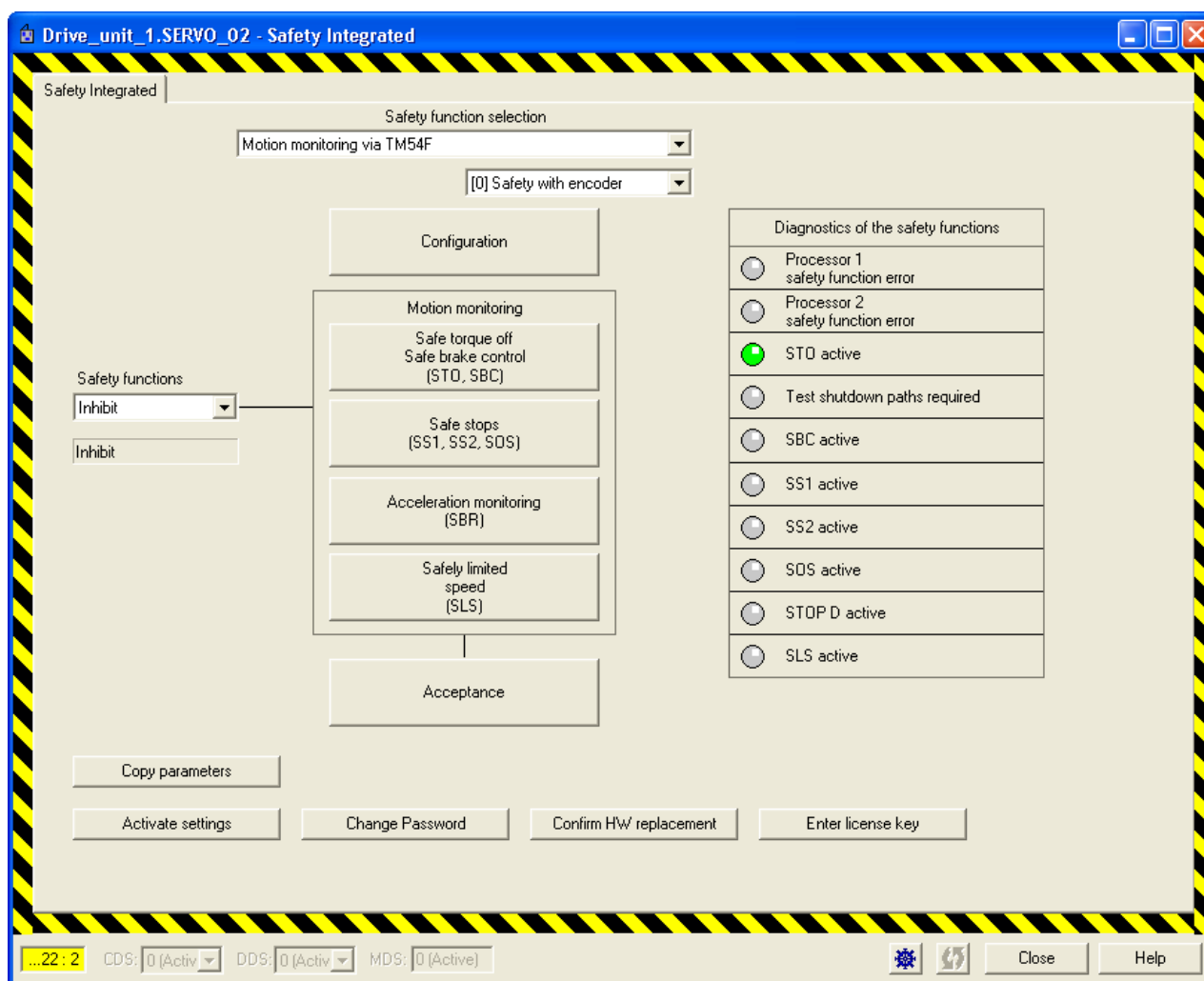


Figure 7-35 Safety Integrated commissioning of a linear/rotary axis

5. Select **Motion Monitoring via TM54F** from the list **Select Safety Function**.

6. Enable the the safety functions (p9501) via the list of **Safety functions**. Then click on the **Configuration** button.
7. The safety configuration screen of the drive opens.

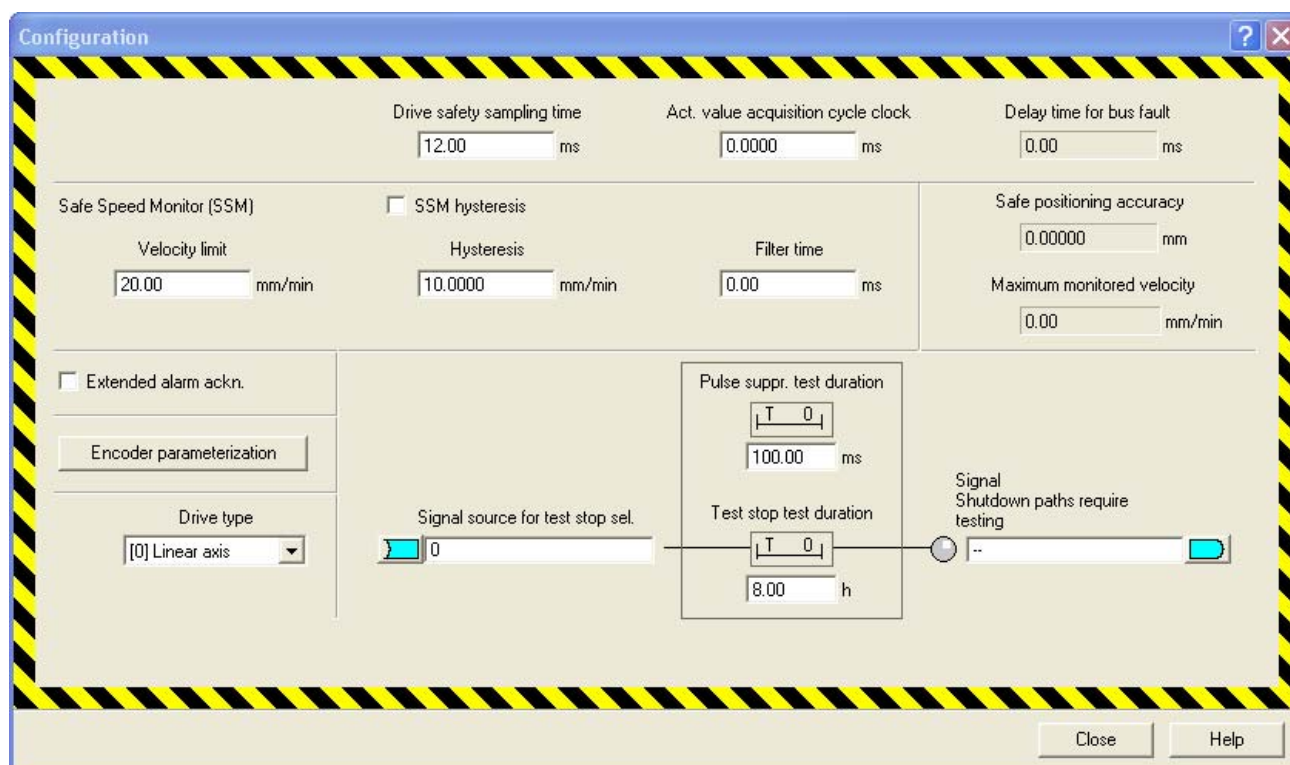


Figure 7-36 Safety configuration: Drive

8. For the drive, set the same **Monitoring clock cycle** (safety clock cycle) as for the TM54F (see "TM54F Configuration").
9. Set the required **Drive type** (linear axis / rotary axis) (p9502). Continue at item 15 if you have not changed the selected drive type.
10. Close the screen. Click on the **Copy parameters** and then click on **Activate settings** (exit commissioning mode, p0010=0).
11. Execute the "Copy RAM to ROM" function for the **Entire project** by clicking the "Entire project" button.
12. Perform a POWER ON. The new parameterization is now active.
13. Reconnect STARTER to the target device. The messages that are displayed indicate that safety commissioning was not completed (different actual and target checksum) can be ignored.
14. Load the project into the PG. The display of parameter units (rotary/linear axis) will be updated accordingly in STARTER.
15. Complete the configuration by adapting the parameterization of the required monitoring limits, timers, encoder settings, etc.

7.8 Modular machine concept Safety Integrated

The modular machine concept for Safety Integrated Basic Functions and Extended Functions provides support for commissioning modular machines. A complete machine, including all its available options, is created in a topology. Only those components that are actually implemented in the finished machine are later activated. Likewise, certain components can also be deactivated to begin with and reactivated if they are required at a later stage.

With the modular machine concept, a distinction is made between the following applications:

- Once the components with Safety functions have been activated for the first time after series commissioning, the hardware replacement needs to be confirmed (see "Information about replacing components" in this manual).
- Once all the drives (including Safety Integrated Extended Functions) have been commissioned, they are to be deactivated (p0105) without changing the hardware. They can only be activated again with a subsequent warm start or by means of POWER ON.
- The DOs of the TM54F can be deactivated by means of parameter p0105. The TM54F itself can only be deactivated when all the drives entered in p10010 "SI drive object assignment" were deactivated separately by means of p0105 beforehand.
- When spare parts are required and the drive is deactivated (p0105) during the delivery period for the required hardware component. Reactivation with subsequent warm start or POWER ON and confirmation of hardware replacement (see "Information about replacing components" in this manual).
- Component exchange on a Control Unit (e.g. to localize faults). For Safety Integrated, this is the same as a hardware replacement. After a warm start or POWER ON, the process of exchanging hardware must be confirmed in order to complete it (see "Information about replacing components" in this manual).

7.9 Information pertaining to component replacements

Replacing a component from the perspective of Safety Integrated

Note

When replacing certain components (Motor Modules when using a TM54F, Sensor Modules or motors with DRIVE-CLiQ interface), this process must be acknowledged to safeguard the communication connections to be renewed within the device. When replacing other components, no acknowledgement is required since the communication connections to be renewed are saved automatically.

For information about component replacements, see "Example of component replacements" in the SINAMICS S120 Function Manual FH1.

WARNING

Observe the instructions with regard to changing or replacing software components in the chapter "Safety instructions".

1. The faulty component was replaced in accordance with safety regulations.
2. Switch on the machine, but first ensure that there are no persons in the danger zone.
3. Only when you control the Extended Functions via TM54F:
 - Fault F35150 is output if a TM54F communication error is recognized after exchanging a Motor Module.
 - Fault C30711 with a fault value 1031 is output if a defect in a monitoring channel is recognized after exchanging a Sensor Module.
 - With STARTER/SCOUT:
 - Click on "Acknowledge hardware replacement" in the start screen of the Safety functions.
 - If you are working without STARTER with SINAMICS with BOP or with SIMOTION with HMI:
 - Start the copying function for Node Identifier (p9700 = 1D hex).
 - Confirm the hardware CRC on the drive object (p9701 = EC hex).

Carry out these two tasks after having replaced a Sensor Module at drive object servo or vector, and after having replaced a Motor Module at drive object TM54F_MA (if installed).
4. Back up all parameters on the memory card:
 - With BOP: Set p0977 = 1.
 - With STARTER: "Copy RAM to ROM" function.
5. Carry out a POWER ON (power off/on) for all components.

Note

In this case, the system will not notify you with a flashing LED that a POWER ON is required.

6. The faults F01650/F30650 (acceptance test required) are output and must be acknowledged.
7. Carry out an acceptance test and acceptance report according to Chapter "Acceptance test and acceptance report" and table "Effect of the acceptance test for certain measures".

**WARNING**

Before re-entering the danger area and before resuming operation, a simplified function test must be carried out for all the drives affected by the component exchange (see chapter "acceptance test").

7.10 Information pertaining to series commissioning

A commissioned project that has been uploaded to STARTER can be transferred to another drive unit keeping the existing safety parameterization.

1. Load the STARTER project into the drive unit.
2. Switch on the machine, but first ensure that there are no persons in the danger zone.
3. Fault F35150 (communication error after replacing a Motor Module) or C30711 with fault value 1031 (data transfer error after replacing a Sensor Module) is only output if you control Extended Functions via TM54F.
4. With STARTER/SCOUT:
 - Click on **Acknowledge hardware replacement** in the start screen of the Safety functions.
5. If you are using SINAMICS with a BOP or SIMOTION with HMI, then you must perform the following steps:
 - Start the copy function for Node Identifier (p9700 = 1D hex)
 - Confirm the hardware CRC on the drive object (p9701 = EC hex)

Carry out these two tasks after having replaced a Sensor Module at drive object servo or vector, and after having replaced a Motor Module at drive object TM54F_MA (if installed).
6. Back up all parameters on the memory card (p0977 = 1).
7. Carry out a POWER ON (power off/on) for all components.
8. Faults F01650/F30650 are output (acceptance test required, see chapter "Acceptance test and acceptance report", table "Effect of the acceptance tests for specific measures").

**WARNING**

Before re-entering the danger area and before resuming operation, a simplified function test must be carried out for all the drives affected by the component exchange (see chapter "acceptance test").

Safety message for standard commissioning with Safety Integrated Extended Functions

If third-party motors with absolute encoders are being used, a situation may arise where a Safety message prevents commissioning.

One reason for this may be that a different serial number of the absolute encoder is saved on the memory card than that in the Control Unit which is to be commissioned. The Safety message can only be acknowledged once the serial number for the absolute encoder has been corrected manually (e.g. with STARTER). Instructions on this are provided in the chapter "Information about replacing components". You can then carry on with the commissioning.

Application examples

8.1 Input/output interconnections for a safety switching device with TM54F

TM54F: interconnecting F-DO with safe input on safety switching device

Note

These typical circuit diagrams are only valid for version B of TM54F devices.

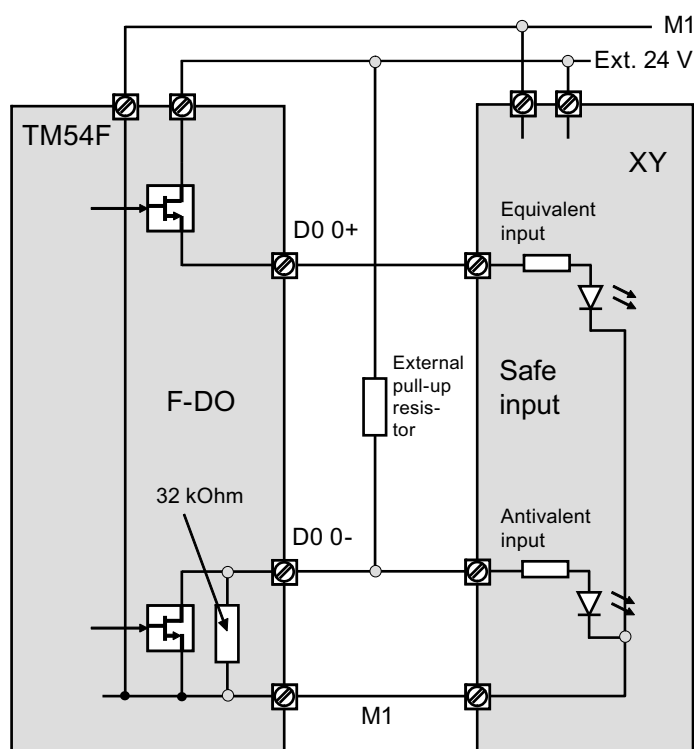


Figure 8-1 TM54F F-DO at equivalent/antivalent safe input on safety switching device (e.g. safety PLC)

The external pull-up resistor is only required in exceptional cases, see below.

TM54F: Interconnecting F-DI with a plus-minus switching output on a safety switching device



WARNING

In contrast to mechanical switching contacts (e.g. Emergency Stop switches), leakage currents can still flow in semiconductor switches such as those usually used at digital outputs even when they have been switched off. This can lead to false switching states if digital inputs are not connected correctly.

The conditions for digital inputs/outputs specified in the relevant manufacturer documentation must be observed.

Note

Test pulses from F-DOs

There are safety modules, whose F-DOs send test pulses for self-testing and for checking the circuit (transmission route). These test pulses can trigger incorrect alarms, which then require safe acknowledgement. In order to avoid these incorrect alarms, the discrepancy time p10002 should be set long enough so that a fault of the safety function itself is excluded. According to the experience that we have gained, a setting of approx. 150 ms has proven itself in practice; however, it is necessary to take into account the function description of the test pulses from the F-DOs of the safety control.



WARNING

In accordance with IEC 61131 Part 2, Chapter 5.2 (2008), when interconnecting the digital inputs of the TM54F with digital semiconductor outputs, only outputs that have a maximum residual current of 0.5 mA when in the "OFF" state can be used.

Debounce

Test signals from the controls can be filtered out using parameter p10017 (SI digital inputs, debounce time) so that faults are not misinterpreted.

F-DI = safety-oriented dual-channel digital input

F-DO = safety-oriented dual-channel digital output

If digital outputs from another device (e.g. F-DOs on a safety PLC) with a residual current greater than 0.5 mA in the "OFF" state are connected to the F-DIs of the TM54F, then F-DI load resistors should be connected up in the channel involved.

The maximum permissible voltage for a TM54F F-DI when "OFF" is 5 V (in accordance with IEC 61131-2, 2008).

The following two diagrams show exactly how the protective circuits for F-DIs with additional load resistors are wired.

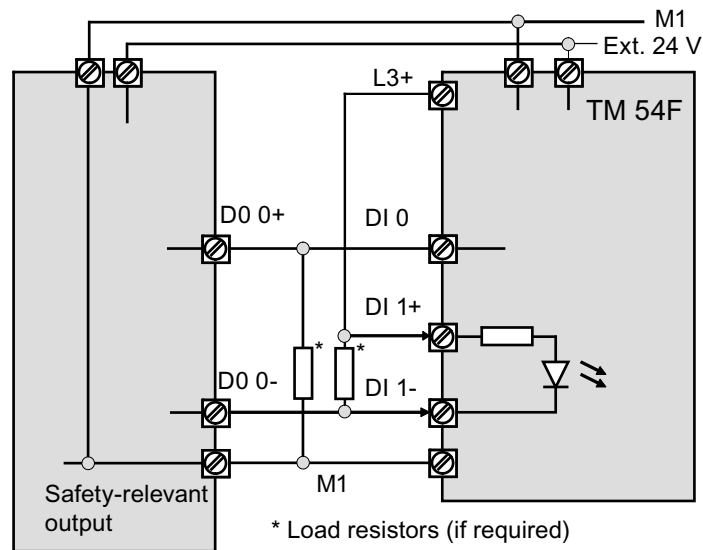


Figure 8-2 TM54F F-DI at plus-minus switching safe output on safety switching device (e.g. safety PLC)

TM54F: interconnecting F-DI with plus-plus switching output on safety switching device

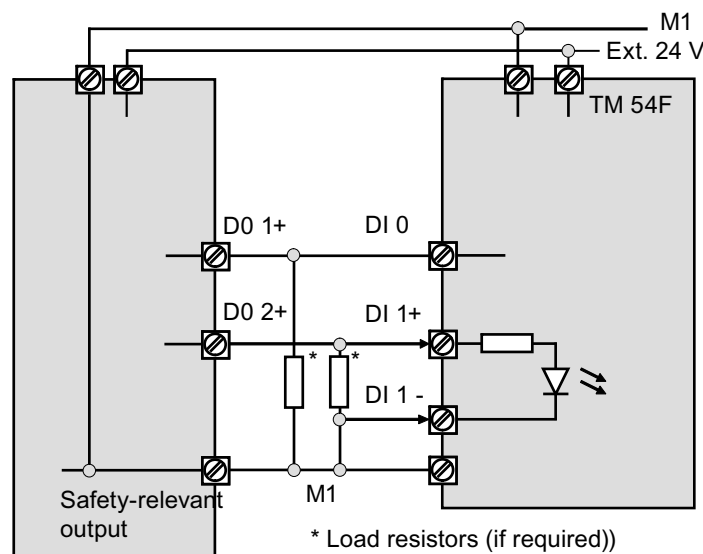


Figure 8-3 TM54F F-DI at plus-plus-switching safe output on a safety switching device (e.g. safety PLC).

Dimensioning the load resistors - example 1:

According to the manufacturer's documentation, the leakage current of an F-DO of a safety PLC for the P and F channels is 1 mA; in other words, it is around 0.5 mA higher than is permissible for the F-DI.

The necessary load resistance is therefore $R = 5 \text{ V} / 0.5 \text{ mA} = 10 \text{ k}\Omega$.

At the maximum supply voltage, the power loss for this resistor is:

$P = (28.8 \text{ V})^2 / R = 83 \text{ mW}$. The resistor is to be permanently dimensioned for this power loss.

Dimensioning the load resistors - example 2:

If additional conditions for the digital output (e.g. a minimum load or a maximum load resistance) are specified in the manufacturer's documentation, then these must be taken into account.

For example, a load between 12Ω and $1 \text{ k}\Omega$ is specified for the SIMATIC ET200S I/O module 4 F-DO (6ES7138-4FB02-0AB0).

Therefore, two additional load resistors with a continuous load capacity of at least $P = (28.8 \text{ V})^2 / R = 830 \text{ mW}$ are required to connect an F-DO of this kind to a TM54F F-DI.

When using a regulated 24 V power supply (e.g. SITOP) a resistor with a significantly lower power loss is sufficient.

Note

Open-circuit detection for a pull-up resistor

If the pull-up resistor is higher than $1 \text{ k}\Omega$, then the open-circuit detection no longer reliably functions and must be disabled.

8.2 Application examples

Application examples can be found at the following Siemens website:

<http://support.automation.siemens.com/WW/view/en/20810941/136000t>

Acceptance tests and acceptance reports

The acceptance test requirements (configuration check) for electrical drive safety functions emanate from DIN EN 61800-5-2:2008, Chapter 7.1 Point f). The acceptance test "configuration check" is named in this standard.

- Description of the application including a picture
- Description of the safety relevant components (including software versions) which are used in the application
- List of the PDS(SR) [Power Drive System(Safety Related)] safety functions used
- Results of all tests of these safety functions, using the specified testing procedure
- List of all safety relevant parameters and their values in the PDS(SR)
- Checksum, test date and confirmation by testing personnel

The acceptance test for systems with Safety Integrated functions (SI functions) is focused on validating the functionality of Safety Integrated monitoring and stop functions implemented in the drive system. The test objective is to verify proper implementation of the defined safety functions and of test mechanisms (forced dormant error detection measures) and to examine the response of specific monitoring functions to the explicit input of values outside tolerance limits. The test must cover all drive-specific Safety Integrated motion monitoring functions and global Safety Integrated functionality of Terminal Module TM54F (if used).

WARNING

A new acceptance test must be carried out if any changes were made to SI function parameters and must be logged in the acceptance report.

Note

The acceptance test is designed to ensure that the safety functions are correctly parameterized. The measured values (e.g. distance, time) and the system behavior identified (e.g. initiation of a specific stop) can be used for checking the plausibility of the configured safety functions. The objective of an acceptance test is to identify potential configuration errors and/or to document the correct function of the configuration. The measured values are typical values (not worst case values). They represent the behavior of the machine at the time of measurement. These measurements cannot be used, for example, to derive maximum values for over-travel.

9.1 Acceptance test structure

Authorized person, acceptance report

The test of each SI function must be carried out by an authorized person and logged in the acceptance report. The report must be signed by the person who carried out the acceptance test. The acceptance report must be kept in the logbook of the relevant machine. Access rights for SI parameters must be password protected and be documented accordingly in the acceptance report. Authorized in this sense refers to a person who has the necessary technical training and knowledge of the safety functions and is authorized by the machine manufacturer to carry out the acceptance test.

Note

- Observe the information in the chapter "Procedures for initial commissioning".
 - The acceptance report presented below is both an example and recommendation.
 - An acceptance report template in electronic format is available at your local Siemens sales office.
-

Necessity of an acceptance test

A complete acceptance test (as described in this chapter) is required after initial commissioning of Safety Integrated functionality on a machine. Safety-related function expansions, transfer of the commissioning settings to other series machines, hardware changes, software upgrades or similar, permit the acceptance test to be performed with a reduced scope if necessary. A summary of conditions which determine the necessary test scope or proposals in this context is provided below.

In order to define a partial acceptance test, it is necessary in the first instance to specify the acceptance test objects, and in the second instance to define logical groups which represent the elements of the acceptance test. The acceptance test must be carried out separately for each individual drive (as far as the machine allows).

Prerequisites for the acceptance test

- The machine is properly wired.
- All safety equipment such as protective door monitoring devices, light barriers or emergency limit switches are connected and ready for operation.
- Commissioning of the open-loop and closed-loop control should be completed, as e.g. the over-travel distance may otherwise change as a result of a changed dynamic response of the drive control. These include, for example:
 - Configuration of the setpoint channel
 - Position control in the higher-level controller
 - Drive control

Information about the acceptance tests

Note

As far as possible, the acceptance tests are to be carried out at the maximum possible machine speed and acceleration rates to determine the maximum braking distances and braking times that can be expected.

Note on the acceptance test mode

The acceptance test mode can be activated for a definable period (p9358/p9558) by setting the appropriate parameters (p9370/p9570). It tolerates specific limit violations during the acceptance test. For instance, the setpoint speed limits are no longer active in the acceptance test mode. To ensure that this state is not accidentally kept, the acceptance test mode is automatically exited after the time set in p9358/p9558.

It is only worth activating acceptance test mode during the acceptance test for functions SS2, SOS and SLS. It has no effect on other functions.

Normally, SOS can be selected directly or via SS2. In order to also be able to initiate violation of the standstill limits in state SS2 when the acceptance test mode is active, the acceptance test mode deactivates the brake ramp of SS2 so that the motor can be moved. When an SOS violation is acknowledged in the active acceptance test mode, the current position is adopted as the new stop position so that an SOS violation is not immediately identified again.

WARNING

If a speed setpoint other than zero is present, the active stop function SS2 is set, and the motor is at a standstill (active SOS), the axis starts to move as soon as the acceptance test is activated.

9.1.1 Content of the complete acceptance test

A) Documentation

Documentation of the machine and of safety functions

1. Machine description (with overview)
2. Specification of the controller (if this exists)
3. Configuration diagram
4. Function table:
 - Active monitoring functions depending on the operating mode and the protective door,
 - Other sensors with protective functions,
 - The table is part or is the result of the configuring work.
5. SI functions for each drive
6. Information about safety equipment

B) Functional testing of safety functions

Detailed function test and evaluation of SI functions used. For some functions this contains trace recordings of individual parameters. The procedure is described in detail in section Acceptance tests (Page 193).

1. Test of the SI function "Safe Torque Off" (STO)
 - Required when used in Basic and/or Extended Functions
 - This test is also required if you are not explicitly using STO, but just one function for which STOP A occurs as an error response.

Alternatively, you can also carry out the quantitative test using STOP A yourself, if you use the tables from sections Acceptance test Safe Torque Off (Basic Functions) (Page 194) , Acceptance test Safe Torque Off with encoder (Extended Functions) (Page 199) or Acceptance test Safe Torque Off without encoder (Extended Functions) (Page 219).
 - You do not need to prepare trace recording for this test.
2. Test of the SI function "Safe Stop 1" (SS1)
 - Required when used in Basic and/or Extended Functions
 - This test is also required if you are not explicitly using SS1, but just one function for which STOP B occurs as an error response.

Alternatively, you can also carry out the quantitative test using STOP B yourself, if you use the tables from sections Acceptance test for Safe Stop 1 with encoder (Extended Functions) (Page 200) or Acceptance test for Safe Stop 1 without encoder (Extended Functions) (Page 220).
 - Trace recording only required if Extended Functions are used
3. Test of the SI function "Safe Brake Control (SBC)"
 - Required when using Basic and/or Extended Functions
 - You do not need to prepare trace recording for this test.

4. Test of the SI function "Safe Stop 2" (SS2)

- Only required when used in Extended Functions
- This test is also required if you are not explicitly using SS2 but just one function for which STOP C occurs as an error response.

Alternatively, you can also carry out the quantitative test using STOP C yourself, if you use the table from section Acceptance test for Safe Stop 2 (Extended Functions) (Page 203).

- Trace recording is required

5. Test of the SI function "Safe Operating Stop" (SOS)

- Only required when used in Extended Functions
- This test is also required if you are not explicitly using SOS, but just one function for which STOP D occurs as an error response.

Alternatively, you can also carry out the quantitative test using STOP D yourself, if you use the table from section Acceptance test for Safe Operating Stop (Extended Functions) (Page 205).

- Trace recording is required

6. Test of the SI function "Safely Limited Speed" (SLS)

- Only required when used in Extended Functions
- Trace recordings are required for each SLS limit

7. Test of the SI function "Safe Speed Monitor" (SSM)

- Only required when used in Extended Functions
- Trace recording is required

C) Functional testing of forced dormant error detection

Test of the forced dormant error detection of the safety functions on each drive (for each control mode) and the TM54F (if used).

1. Test of the forced dormant error detection of the safety function on the drive

- If you are using Basic Functions, you need to activate and then deactivate STO once again.
- If you are using Extended Functions, you need to carry out a test stop.

2. Test of the forced dormant error detection of the TM54F (if present)

- Only if Extended Functions are used
- Carry out test stop of the TM54F

D) Conclusion of the report

Report of the commissioning status tested and countersignatures

1. Inspection of SI parameters
2. Logging of checksums (for each drive)
3. Issuing of the Safety password and documenting this process (do not specify the Safety password in the report!)
4. RAM to ROM backup, upload of project data to STARTER, and backup of the project
5. Countersignature

9.1.2 Content of the partial acceptance test

A) Documentation

Documentation of the machine and of safety functions

1. Extending/changing the hardware data
2. Extending/changing the software data (specify version)
3. Extending/changing the configuration diagram
4. Extending/changing the function table:
 - Active monitoring functions depending on the operating mode and the protective door
 - Other sensors with protective functions
 - The table is part or is the result of the configuring work
5. Extending/changing the SI functions per drive
6. Extending/changing the specifications of the safety equipment

B) Functional testing of safety functions

Detailed function test and evaluation of SI functions used. For some functions this contains trace recordings of individual parameters. The procedure is described in detail in section Acceptance tests (Page 193).

The function test can be left out if no parameters of the individual safety functions have been changed. In the case that only parameters of individual functions have been changed, only these functions need to be tested anew.

1. Test of the SI function "Safe Torque Off" (STO)
 - Required when used in Basic and/or Extended Functions
 - This test is also required if you are not explicitly using STO, but just one function for which STOP A occurs as an error response.

Alternatively, you can also carry out the quantitative test using STOP A yourself, if you use the tables from sections Acceptance test Safe Torque Off (Basic Functions) (Page 194) , Acceptance test Safe Torque Off with encoder (Extended Functions) (Page 199) or Acceptance test Safe Torque Off without encoder (Extended Functions) (Page 219).

- You do not need to prepare trace recording for this test.

2. Test of the SI function "Safe Stop 1" (SS1)

- Required when used in Basic and/or Extended Functions
- This test is also required if you are not explicitly using SS1, but just one function for which STOP B occurs as an error response.

Alternatively, you can also carry out the quantitative test using STOP B yourself, if you use the tables from Acceptance test for Safe Stop 1 with encoder (Extended Functions) (Page 200) or Acceptance test for Safe Stop 1 without encoder (Extended Functions) (Page 220).

- Trace recording only required if Extended Functions are used

3. Test of the SI function "Safe Brake Control (SBC)"

- Required when using Basic and/or Extended Functions
- You do not need to prepare trace recording for this test.

4. Test of the SI function "Safe Stop 2" (SS2)

- Only required when used in Extended Functions
- This test is also required if you are not explicitly using SS2 but just one function for which STOP C occurs as an error response.

Alternatively, you can also carry out the quantitative test using STOP C yourself, if you use the table from section Acceptance test for Safe Stop 2 (Extended Functions) (Page 203).

- Trace recording is required

5. Test of the SI function "Safe Operating Stop" (SOS)

- Only required when used in Extended Functions
- This test is also required if you are not explicitly using SOS, but just one function for which STOP D occurs as an error response.

Alternatively, you can also carry out the quantitative test using STOP D yourself, if you use the table from section Acceptance test for Safe Operating Stop (Extended Functions) (Page 205).

- Trace recording is required

6. Test of the SI function "Safely Limited Speed" (SLS)

- Only required when used in Extended Functions
- A trace recording is required for each SLS limit

7. Test of the SI function "Safe Speed Monitor" (SSM)

- Only required when used in Extended Functions
- Trace recording is required

C) Functional testing of forced dormant error detection

Test of the forced dormant error detection of the safety functions on each drive (for each control mode) and the TM54F (if used).

1. Test of the forced dormant error detection of the safety function on the drive
 - If you are using Basic Functions, you need to activate and then deactivate STO once again.
 - If you are using Extended Functions, you need to carry out a test stop.
2. Test of the forced dormant error detection of the TM54F (if present)
 - Only if Extended Functions are used
 - Carry out test stop of the TM54F

D) Functional testing of actual value acquisition

1. General testing of actual value acquisition
 - After exchanging the component, initial activation and brief operation in both directions.



WARNING

During this process, all personnel must keep out of the danger area.

2. Test of failsafe actual value acquisition
 - Only necessary if Extended Functions are used
 - If the motion monitoring functions are activated (e.g. SLS or SSM with hysteresis), briefly operate the drive in both directions.

E) Conclusion of the report

Report of the commissioning status tested and countersignatures

1. Extension of checksums (for each drive)
2. Countersignature

9.1.3 Test scope for specific measures

Scope of partial acceptance tests for specific measures

Table 9- 1 Scope of partial acceptance tests for specific measures

Measure	A) Documentation	B) Functional testing of safety functions	C) Functional testing of forced dormant error detection	D) Functional testing of actual value acquisition	E) Conclusion of the report
Replacement of the encoder system	No	No	No	Yes	Yes
Replacement of an SMC/SME	Yes, Points 1 and 2	No	No	Yes	Yes
Replacement of a motor with DRIVE-CLiQ	Yes, Points 1 and 2	No	No	Yes	Yes
Replacement of the Control Unit / power unit hardware	Yes, Points 1 and 2	No	Yes, only Point 1	Yes, only Point 1	Yes
Replacement of the Power Module or Safe Brake Relay	Yes, Points 1 and 2	Yes, Points 1 and 3	Yes, only Point 1	Yes, only Point 1	Yes
Replacing the TM54F	Yes, Points 1 and 2	Yes, but only testing of the selection of the safety functions	Yes	Yes, only Point 1	Yes
Firmware - upgrade(CU/power unit/ Sensor Modules)	Yes, only Point 2	Yes, if new safety functions are to be used	Yes	Yes, only Point 1	Yes
Change to a single parameter of a safety function (e.g. SLS limit)	Yes, Points 4 and 5.	Yes, test the appropriate function	No	Yes	Yes
Transfer of project data to other machines (series commissioning)	Yes	Yes, but only testing of the selection of the safety functions	Yes	Yes	Yes

9.2 Safety logbook

Description

The "Safety Logbook" function is used to detect changes to safety parameters that affect the associated CRC sums. CRCs are only generated when p9601/p9801 (SI enable, functions integrated in the drive CU/Motor Module) is > 0.

Data changes are detected when the CRCs of the SI parameters change. Each SI parameter change that is to become active requires the reference CRC to be changed so that the drive can be operated without SI fault messages. In addition to functional safety changes, safety changes as a result of hardware being replaced can be detected when the CRC has changed.

The following changes are recorded by the safety logbook:

- Functional changes are recorded in the checksum r9781[0]:
 - Functional CRCs of the motion monitoring functions (p9729[0]), axial (Extended Functions)
 - Functional CRCs of the Basic safety functions integrated in the drive (p9799, SI setpoint checksum SI parameters CU), axial
 - Functional CRCs of the TM54F (p10005[0]), global (Extended Functions)
 - Enabling of functions integrated in the drive (p9601), axial (Basic and Extended Functions)
- Hardware-dependent changes are recorded in the checksum r9781[1]:
 - Hardware-dependent CRC of the motion monitoring functions (p9729[2]), axial (Extended Functions)
 - Hardware-dependent CRC of the TM54F (p10005[1]), global (Extended Functions)

9.3 Acceptance reports

9.3.1 Plant description - Documentation part 1

Table 9- 2 Machine description and overview diagram

Designation	
Type	
Serial number	
Manufacturer	
End customer	
Electrical drives	
Other drives	
Overview diagram of machine	

Table 9- 3 Values of relevant parameters

Versions of the firmware and of Safety Integrated			
Component	DO number	Firmware version	SI version
Parameters Control Unit		r0018 =	r9590 = r9770 = Note: Parameters can be found in the drive.
Parameters Motor Modules	DO number	Firmware version	SI version
		r0128 =	r9390 = r9870 =
		r0128 =	r9390 = r9870 =
		r0128 =	r9390 = r9870 =
		r0128 =	r9390 = r9870 =
		r0128 =	r9390 = r9870 =

9.3 Acceptance reports

Parameters Sensor Modules	DO number	Firmware version	SI version
		r0148 =	r9890 =
		r0148 =	r9890 =
		r0148 =	r9890 =
		r0148 =	r9890 =
		r0148 =	r9890 =
		r0148 =	r9890 =
TM54F parameters	DO number	Firmware version	SI version
		r0158 =	r10090 =
		r0158 =	r10090 =
Monitoring clock cycles of Safety Integrated			
	DO number	SI monitoring clock cycle Control Unit	SI monitoring clock cycle Motor Module
Basic Functions		r9780 =	r9880 =
		r9780 =	r9880 =
		r9780 =	r9880 =
		r9780 =	r9880 =
		r9780 =	r9880 =
		r9780 =	r9880 =
	DO number	SI monitoring clock cycle Motor Module	SI monitoring clock cycle Control Unit
Extended Functions		p9300 =	p9500 =
		p9300 =	p9500 =
		p9300 =	p9500 =
		p9300 =	p9500 =
		p9300 =	p9500 =
		p9300 =	p9500 =
TM54F parameters	DO number	SI monitoring clock cycle TM54F	
		p10000 =	

9.3.2 Description of safety functions - Documentation Part 2

Note

This description of a system is for illustration purposes only. In each case, the actual settings for the system concerned will need to be modified as required.

9.3.2.1 Function table

Table 9- 4 Example table: Active monitoring functions depending on the operating mode, the protective doors or other sensors

Mode of operation	Protective door	Drive	Status of monitoring functions
Production	closed and locked	1 2	deactivated SLS enabled
	unlocked	1 2	SOS STO deactivated
Setup	closed and locked	1 2	deactivated SLS 1 enabled
	unlocked	1 2	SLS 1 deselected, enabled
...	...	...	...

9.3.2.2 Safety Integrated functions used

Table 9- 5 Example: functional overview of the safety functions

Drive	SI function	Limit	Active if
1	SOS	100 mm	refer to the function table
	SLS 1	200000 mm/min	refer to the function table
2	SOS	100 mm	refer to the function table
	SLS 1	50 rpm	refer to the function table
...	...	...	...

Comments:

All drives use the SI function SS1 for the EMERGENCY STOP functionality.

Drive 2 is equipped with a holding brake which is controlled by two channels via the corresponding Motor Module output.

Drive-specific Safety parameters

Note

You need to fill out this table for each axis.

Table 9- 6 Drive-specific data

SI function	Parameter Motor Modules / CU	Motor Module value / CU value
Enable safety functions	p9301 / p9501	0000 bin
Axis type	p9302 / p9502	0
Function specification	p9306 / p9506	0
Function configuration	p9307 / p9507	0000 bin
Actual value acquisition clock cycle	p9311 / p9511	0.0 ms
Coarse position value configuration	p9315 / p9515	0000 bin
Encoder configuration, safety functions	p9316 / p9516	0000 bin
Linear scale graduations	p9317 / p9517	10 nm
Encoder pulses per revolution	p9318 / p9518	2048
Fine resolution G1_XIST1	p9319 / p9519	11
Leadscrew pitch	p9320 / p9520	10 mm
Gearbox encoder (motor) / load denominator	p9321[0] / p9521[0] p9321[1] / p9521[1] p9321[2] / p9521[2] p9321[3] / p9521[3] p9321[4] / p9521[4] p9321[5] / p9521[5] p9321[6] / p9521[6] p9321[7] / p9521[7]	1 1 1 1 1 1 1 1
Gearbox encoder (motor) / load numerator	p9322[0] / p9522[0] p9322[1] / p9522[1] p9322[2] / p9522[2] p9322[3] / p9522[3] p9322[4] / p9522[4] p9322[5] / p9522[5] p9322[6] / p9522[6] p9322[7] / p9522[7]	1 1 1 1 1 1 1 1
Redundant coarse position value valid bits	p9323 / p9523	9
Redundant coarse position value fine resolution bits	p9324 / p9524	-2
Redundant coarse position value relevant bits	p9325 / p9525	16
Encoder assignment	p9326 / p9526	1

SI function	Parameter Motor Modules / CU	Motor Module value / CU value
Sensor Module node identifier	p9328[0] p9328[1] p9328[2] p9328[3] p9328[4] p9328[5] p9328[6] p9328[7] p9328[8] p9328[9] p9328[10] p9328[11]	0000 hex 0000 hex 0000 hex 0000 hex 0000 hex 0000 hex 0000 hex 0000 hex 0000 hex 0000 hex 0000 hex 0000 hex
SI Motion Gx_XIST1 coarse position safety most significant bit	p9329 / p9529	14
SOS standstill tolerance	p9330 / p9530	1.000°
PLC limit values	p9331[0] / p9531[0] p9331[1] / p9531[1] p9331[2] / p9531[2] p9331[3] / p9531[3]	2000.00 mm/min 2000.00 mm/min 2000.00 mm/min 2000.00 mm/min
Actual value comparison tolerance	p9342 / p9542	0.1000°
SSM Filter time	p9345 / p9545	0.0 ms
SSM speed limit	p9346 / p9546	20.00 mm/min
SSM speed hysteresis	p9347 / p9547	10 mm/min
SBR actual speed tolerance	p9348 / p9548	300.00 1/min
Slip speed tolerance	p9349 / p9549	6.0 rpm
SLS changeover delay time	p9351 / p9551	100.00 ms
STOP C -> SOS delay time	p9352 / p9552	100.00 ms
STOP D -> SOS delay time	p9353 / p9553	100.00 ms
STOP F -> STOP A delay time	p9355 / p9555	0.00 ms
Pulse cancelation delay time	p9356 / p9556	100.00 ms
Pulse suppression test time	p9357 / p9557	100.00 ms
Acceptance test mode time limit	p9358 / p9558	40000.00 ms
Pulse suppression shutdown speed	p9360/p9560	0.0 rpm
PLC stop response	p9363[0] / p9563[0] p9363[1] / p9563[1] p9363[2] / p9563[2] p9363[3] / p9563[3]	2 2 2 2
SBR speed limit	p9368 / p9568	0.0 mm/min
Acceptance test mode	p9370 / p9570	0000 hex
Forced dormant error detection timer	p9559	8.00 h
Brakeramp reference value	p9381 / p9581	1500 rpm
Brake ramp delay time	p9382 / p9582	250 ms
Brake ramp monitoring time	p9383 / p9583	10.00 s

SI function	Parameter Motor Modules / CU	Motor Module value / CU value
Minimum current actual value acquisition without encoder	p9388 / p9588	10.00 %
Acceleration voltage tolerance	p9389 / p9589	100.00 %
Enable drive-integrated functions	p9801 / p9601	0000 bin
Enable safe brake control	p9802 / p9602	0
PROFIsafe address	p9810 / p9610	0000 hex
Signal source for STO/SBC/SS1	p9620[0] p9620[1] p9620[2] p9620[3] p9620[4] p9620[5] p9620[6] p9620[7]	0 0 0 0 0 0 0 0
Signal source for SBA	p9821 / p9621	0
SBA relay wait times	p9822[0] / p9622[0] p9822[1] / p9622[1]	100.00 ms 65.00 ms
SGE changeover tolerance time	p9850 / p9650	500.00 ms
STO/SBC/SS1 debounce time	p9851 / p9651	0.00 ms
Safe Stop 1 delay time	p9852 / p9652	0.00 s
STOP F -> STOP A delay time	p9858 / p9658	0.00 µs
Forced dormant error detection timer	p9659	8.00 h

9.3.2.3 Safety parameters of the TM54F

Table 9- 7 Parameters for control via the TM54F (excerpt)

SI function	Parameters	Value
Sampling time	p10000	12.00 ms
Wait time for test stop on DO	p10001	500.00 ms
Monitoring time discrepancy	p10002	12.00 ms
Forced dormant error detection timer	p10003	8.00 h
Acknowledging internal event input terminal	p10006	0
Input terminal forced dormant error detection	p10007	0
Assignment of drive objects	p10010[0] p10010[1] p10010[2] p10010[3] p10010[4] p10010[5]	0 0 0 0 0 0

SI function	Parameters	Value
Assignment of drive groups	p10011[0] p10011[1] p10011[2] p10011[3] p10011[4] p10011[5]	1 1 1 1 1 1
Digital inputs debounce time	p10017	1.00 ms
STO input terminal	p10022[0] p10022[1] p10022[2] p10022[3]	0 0 0 0
SS1 input terminal	p10023[0] p10023[1] p10023[2] p10023[3]	0 0 0 0
SS2 input terminal	p10024[0] p10024[1] p10024[2] p10024[3]	0 0 0 0
SOS input terminal	p10025[0] p10025[1] p10025[2] p10025[3]	0 0 0 0
PLC input terminal	p10026[0] p10026[1] p10026[2] p10026[3]	0 0 0 0
SLS_Limit(1) input terminal	p10027[0] p10027[1] p10027[2] p10027[3]	0 0 0 0
SLS_Limit(2) input terminal	p10028[0] p10028[1] p10028[2] p10028[3]	0 0 0 0
Safe state signal selection	p10039[0] p10039[1] p10039[2] p10039[3]	1 hex 1 hex 1 hex 1 hex
F-DI input mode	p10040	0 hex
F-DI test enable	p10041	0 hex
F-DO 0 signal sources	p10042[0] p10042[1] p10042[2] p10042[3] p10042[4] p10042[5]	0 0 0 0 0 0

SI function	Parameters	Value
F-DO 1 signal sources	p10043[0]	0
	p10043[1]	0
	p10043[2]	0
	p10043[3]	0
	p10043[4]	0
	p10043[5]	0
F-DO 2 signal sources	p10044[0]	0
	p10044[1]	0
	p10044[2]	0
	p10044[3]	0
	p10044[4]	0
	p10044[5]	0
F-DO 3 signal sources	p10045[0]	0
	p10045[1]	0
	p10045[2]	0
	p10045[3]	0
	p10045[4]	0
	p10045[5]	0
Test Sensor feedback signal	p10046	0 hex
Selection of test mode for test stop	p10047[0]	2
	p10047[0]	2
	p10047[0]	2
	p10047[0]	2

9.3.2.4 Safety equipment

Protective door The protective door is unlocked by means of single-channel request key
Protective door switch The protective door is equipped with a safety door switch. The safety door switch returns the dual-channel signal "Door closed and locked". Changeover and selection of safety functions in accordance with the table shown above.
Mode selector switch The "Production" and "Setup" modes are set by means of a mode selector switch. The key switch features two contact levels. Changeover and selection of safety functions in accordance with the table shown above.
EMERGENCY-STOP pushbutton The dual-channel EMERGENCY-STOP pushbuttons are wired in series. The EMERGENCY STOP signal activates SS1 for all drives. Then the external brakes and STO are activated.
Test stop Activation by means of: - Machine power on - Unlocking the protective door

9.4 Acceptance tests

Note

As far as possible, the acceptance tests are to be carried out at the maximum possible machine speed and acceleration rates to determine the maximum braking distances and braking times that can be expected.

Note

If Basic Functions and Extended Functions are combined, the acceptance test for both types must be carried out for the functions used.

Note

The trace recordings for the Extended Functions aid evaluation of the more complex functionality compared to the Basic Functions (for which trace recording is not required). You may also use other recording options (e.g. via HMI) if need be.

Note**Non-critical alarms**

When evaluating the alarm buffer you can tolerate the following alarms:

- A01697 SI Motion: Motion monitoring test required
- A01796 SI Motion CU: Waiting for communication

These alarms occur after every system startup and can be evaluated as non-critical. You do not need to include these alarms in the acceptance report.

9.4.1 Acceptance tests – Basic Functions

9.4.1.1 Acceptance test Safe Torque Off (Basic Functions)

Table 9- 8 "Safe Torque Off" acceptance test

No.	Description	Status
Note: The acceptance test must be individually conducted for each configured control. The control can be realized via terminals or via PROFIsafe.		
1.	Initial state	
	• Drive in "Ready" state (p0010 = 0)	
	• STO function enabled (on-board terminals / PROFIsafe p9601.1 = 1 or p9601 = 8)	
	• No Safety faults and alarms (r0945[0...7], r2122[0...7]); see note "non-critical alarms" in section Acceptance tests (Page 193)	
	• r9772.17 = r9872.17 = 0 (STO deselection via terminals - DI CU / EP terminal Motor Module); only relevant for STO via terminal	
	• r9772.20 = r9872.20 = 0 (STO deselection via PROFIsafe); only relevant for STO via PROFIsafe	
	• r9772.0 = r9772.1 = 0 (STO deselected and inactive – Control Unit)	
	• r9872.0 = r9872.1 = 0 (STO deselected and inactive – Motor Module)	
	• r9773.0 = r9773.1 = 0 (STO deselected and inactive – drive)	
	• r9774.0 = r9774.1 = 0 (STO deselected and inactive - group); only relevant for grouping	
2.	Run the drive	
	• Ensure that the correct drive is running	
	Select STO when you issue the traversing command and check the following:	
	• The drive coasts to a standstill or is braked and stopped by the mechanical brake (if available and configured (p1215, p9602, p9802)).	
	• No Safety faults and alarms (r0945[0..7], r2122[0..7])	
	• r9772.17 = r9872.17 = 1 (STO selection via terminal - DI CU / EP terminal Motor Module); only relevant for STO via terminal	
	• r9772.20 = r9872.20 = 1 (STO selection via PROFIsafe); only relevant for STO via PROFIsafe	
	• r9772.0 = r9772.1 = 1 (STO selected and active – Control Unit)	
	• r9872.0 = r9872.1 = 1 (STO selected and active – Motor Module)	
	• r9773.0 = r9773.1 = 1 (STO selected and active – drive)	
	• r9774.0 = r9774.1 = 1 (STO selected and active - group); only relevant for grouping	

No.	Description	Status
3.	Deselect STO and check the following:	
	No Safety faults and alarms (r0945[0..7], r2122[0..7])	
	r9772.17 = r9872.17 = 0 (STO deselection via terminals - DI CU / EP terminal Motor Module); only relevant for STO via terminal	
	r9772.20 = r9872.20 = 0 (STO deselection via PROFIsafe); only relevant for STO via PROFIsafe	
	r9772.0 = r9772.1 = 0 (STO deselected and inactive – Control Unit)	
	r9872.0 = r9872.1 = 0 (STO deselected and inactive – Motor Module)	
	r9773.0 = r9773.1 = 0 (STO deselected and inactive – drive)	
	r9774.0 = r9774.1 = 0 (STO deselected and inactive - group); only relevant for grouping	
	r0046.0 = 1 (drive in "switch-on inhibited" state)	
4.	Acknowledge "switch-on inhibit" and run the drive. Ensure that the correct drive is running.	
	The following is tested: - Correct DRIVE-CLiQ wiring between Control Unit and Motor Modules - Correct assignment of drive No. – Motor Module – motor - The hardware is functioning properly - Correct wiring of the switch-off signal path (only via terminal) - Correct assignment of the terminals for STO on the Control Unit - Correct STO grouping (if available) - Correct parameterization of the STO function - Routine for forced dormant error detection of the switch-off signal paths	

9.4.1.2 Acceptance test for Safe Stop 1 (Basic Functions)

Table 9- 9 "Safe Stop 1" function

No.	Description	Status
Note: The acceptance test must be individually conducted for each configured control. The control can be realized via terminals or via PROFIsafe.		
1.	Initial state	
	Drive in "Ready" state (p0010 = 0)	
	STO function enabled (on-board terminals / PROFIsafe p9601.1 = 1 or p9601 = 8)	
	Enable SS1 function (p9652 > 0, p9852 > 0)	
	No Safety faults and alarms (r0945[0...7], r2122[0...7]); see note "non-critical alarms" in section Acceptance tests (Page 193)	
	r9772.22 = r9872.22 = 0 (SS1 deselection via terminals – DI CU / EP terminal Motor Module); only relevant for SS1 via terminal	
	r9772.23 = r9872.23 = 0 (SS1 deselection via PROFIsafe); only relevant for SS1 via PROFIsafe	
	r9772.0 = r9772.1 = 0 (STO deselected and inactive – CU)	
	r9772.5 = r9772.6 = 0 (SS1 deselected and inactive – CU)	

9.4 Acceptance tests

No.	Description	Status
	• r9872.0 = r9872.1 = 0 (STO deselected and inactive – MM)	
	• r9872.5 = r9872.6 = 0 (SS1 deselected and inactive – MM)	
	• r9773.0 = r9773.1 = 0 (STO deselected and inactive – drive)	
	• r9773.5 = r9773.6 = 0 (SS1 deselected and inactive – drive)	
	• r9774.0 = r9774.1 = 0 (STO deselected and inactive - group); only relevant for grouping	
	• r9774.5 = r9774.6 = 0 (SS1 deselected and inactive - group); only relevant for grouping	
2.	Run the drive	
	Check whether the correct drive is operational	
	Select SS1 when you issue the traversing command and check the following:	
	• The drive is braked along the OFF3 ramp (p1135)	
	Before the SS1 delay time (p9652, p9852) expires, the following applies:	
	• r9772.22 = r9872.22 = 1 (SS1 selection via terminals – DI CU / EP terminal Motor Module); only relevant for SS1 via terminal	
	• r9772.23 = r9872.23 = 1 (SS1 selection via PROFIsafe); only relevant for SS1 via PROFIsafe	
	• r9772.0 = r9772.1 = 0 (STO deselected and inactive - CU)	
	• r9772.5 = r9772.6 = 1 (SS1 selected and active – CU)	
	• r9872.0 = r9872.1 = 0 (STO deselected and inactive – MM)	
	• r9872.5 = r9872.6 = 1 (SS1 selected and active – MM)	
	• r9773.0 = r9773.1 = 0 (STO deselected and inactive)	
	• r9773.5 = r9773.6 = 1 (SS1 selected and active – drive)	
	• r9774.0 = r9774.1 = 0 (STO deselected and inactive - group); only relevant for grouping	
	• r9774.5 = r9774.6 = 1 (SS1 selected and active - group); only relevant for grouping	
	STO is initiated after the SS1 delay time expires (p9652, p9852).	
	• No Safety faults and alarms (r0945[0...7], r2122[0...7])	
	• r9772.0 = r9772.1 = 1 (STO selected and active - CU)	
	• r9772.5 = r9772.6 = 1 (SS1 selected and active – CU)	
	• r9872.0 = r9872.1 = 1 (STO selected and active – MM)	
	• r9872.5 = r9872.6 = 1 (SS1 selected and active – MM)	
	• r9773.0 = r9773.1 = 1 (STO selected and active)	
	• r9773.5 = r9773.6 = 1 (SS1 selected and active – drive)	
	• r9774.0 = r9774.1 = 1 (STO selected and active - group); only relevant for grouping	
	• r9774.5 = r9774.6 = 1 (SS1 selected and active - group); only relevant for grouping	
3.	Canceling SS1	
	• No Safety faults and alarms (r0945[0...7], r2122[0...7])	
	• r9772.22 = r9872.22 = 0 (SS1 deselection via terminals – DI CU / EP terminal Motor Module); only relevant for SS1 via terminal	
	• r9772.23 = r9872.23 = 0 (SS1 deselection via PROFIsafe); only relevant for SS1 via PROFIsafe	

No.	Description	Status
	• r9772.0 = r9772.1 = 0 (STO deselected and inactive – CU)	
	• r9772.5 = r9772.6 = 0 (SS1 deselected and inactive – CU)	
	• r9872.0 = r9872.1 = 0 (STO deselected and inactive – MM)	
	• r9872.5 = r9872.6 = 0 (SS1 deselected and inactive – MM)	
	• r9773.0 = r9773.1 = 0 (STO deselected and inactive – drive)	
	• r9773.5 = r9773.6 = 0 (SS1 deselected and inactive – drive)	
	• r9774.0 = r9774.1 = 0 (STO deselected and inactive - group); only relevant for grouping	
	• r9774.5 = r9774.6 = 0 (SS1 deselected and inactive - group); only relevant for grouping	
	• r0046.0 = 1 (drive in "switch-on inhibited" state)	
4.	Acknowledge "switch-on inhibit" and run the drive. Ensure that the correct drive is running.	
	The following is tested:	
	• Correct parameterization of the SS1 function	

9.4.1.3 Acceptance test for Safe Brake Control (Basic Functions)

Table 9- 10 "Safe Brake Control" function

No.	Description	Status
Note: The acceptance test must be individually conducted for each configured control. The control can be realized via terminals or via PROFIsafe.		
1.	Initial state	
	• Drive in "Ready" state (p0010 = 0)	
	• STO function enabled (on-board terminals / PROFIsafe p9601.1 = 1 or p9601 = 8)	
	• Enable SBC function (p9602 = 1, p9802 = 1)	
	• Brake as in sequence control or brake always released (p1215 = 1 or p1215 = 2)	
	• No Safety faults and alarms (r0945, r2122) ; see note "non-critical alarms" in section Acceptance tests (Page 193)	
	• r9772.4 = r9872.4 = 0 (SBC not requested)	
	• r9772.0 = r9772.1 = 0 (STO deselected and inactive – CU)	
	• r9872.0 = r9872.1 = 0 (STO deselected and inactive – MM)	
	• r9773.0 = r9773.1 = 0 (STO deselected and inactive – drive)	
	• r9774.0 = r9774.1 = 0 (STO deselected and inactive - group); only relevant for grouping	

9.4 Acceptance tests

No.	Description	Status
2.	Run drive (if applied, brake is released)	
	Check whether the correct drive is operational	
	Select STO/SS1 when you issue the traversing command and check the following:	
	The brake is applied (for SS1 the drive is previously decelerated along the OFF3 ramp)	
	No Safety faults and alarms (r0945[0...7], r2122[0...7])	
	r9772.4 = r9872.4 = 1 (SBC requested)	
	r9772.0 = r9772.1 = 1 (STO selected and active – CU)	
	r9872.0 = r9872.1 = 1 (STO selected and active – MM)	
	r9773.0 = r9773.1 = 1 (STO selected and active – drive)	
	r9774.0 = r9774.1 = 1 (STO selected and active - group); only relevant for grouping	
3.	Deselect STO and check the following:	
	No Safety faults and alarms (r0945[0...7], r2122[0...7])	
	r9772.4 = r9872.4 = 0 (deselect SBC)	
	r9772.0 = r9772.1 = 0 (STO deselected and inactive – CU)	
	r9872.0 = r9872.1 = 0 (STO deselected and inactive – MM)	
	r9773.0 = r9773.1 = 0 (STO deselected and inactive – drive)	
	r9774.0 = r9774.1 = 0 (STO deselected and inactive - group); only relevant for grouping	
	r0046.0 = 1 (drive in "switch-on inhibited" state)	
4.	Acknowledge "switch-on inhibit" and run the drive. Check whether the correct drive is operational.	
	The following is tested: - The brake is connected properly - The hardware is functioning properly - The SBC is parameterized correctly - Routine for the forced dormant error detection of the brake control	

9.4.2 Acceptance tests for Extended Functions (with encoder)

9.4.2.1 Acceptance test Safe Torque Off with encoder (Extended Functions)

Table 9- 11 "Safe Torque Off" function

No.	Description	Status
Notes:		
The acceptance test must be individually conducted for each configured control. The control can be realized via TM54F or via PROFIsafe.		
1.	Initial state	
	• Drive in "Ready" state (p0010 = 0)	
	• Safety Integrated Extended Functions enabled (p9601.2 = 1)	
	• Safety functions enabled (p9501.0 = 1)	
	• No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7]); see note "non-critical alarms" in section Acceptance tests (Page 193)	
	• r9772.18 = r9872.18 = 0 (STO deselected via Safe Motion Monitoring)	
	• r9772.0 = r9772.1 = 0 (STO deselected and inactive – Control Unit)	
	• r9872.0 = r9872.1 = 0 (STO deselected and inactive – Motor Module)	
	• r9773.0 = r9773.1 = 0 (STO deselected and inactive – drive)	
	• r9720.0 = 1 (STO deselected)	
	• r9722.0 = 0 (STO deselected)	
2.	Run the drive	
	• Check whether the correct drive is operational	
	Select STO when you issue the traversing command and check the following:	
	• The drive coasts to a standstill or is braked and stopped by the mechanical brake (if available and configured (p1215, p9602, p9802)).	
	• No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7])	
	• r9772.18 = r9872.18 = 1 (STO selected via Safe Motion Monitoring)	
	• r9772.0 = r9772.1 = 1 (STO selected and active – Control Unit)	
	• r9872.0 = r9872.1 = 1 (STO selected and active – Motor Module)	
	• r9773.0 = r9773.1 = 1 (STO selected and active – drive)	
	• r9720.0 = 0 (STO selected)	
	• r9722.0 = 1 (STO selected)	
3.	Deselect STO and check the following:	
	• No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7])	
	• r9772.18 = r9872.18 = 0 (STO deselected via Safe Motion Monitoring)	
	• r9772.0 = r9772.1 = 0 (STO deselected and inactive – Control Unit)	
	• r9872.0 = r9872.1 = 0 (STO deselected and inactive – Motor Module)	
	• r9773.0 = r9773.1 = 0 (STO deselected and inactive – drive)	

9.4 Acceptance tests

No.	Description	Status
	• r9720.0 = 1 (STO deselected)	
	• r9722.0 = 0 (STO deselected)	
	• r0046.0 = 1 (drive in "switch-on inhibited" state)	
4.	Acknowledge "switch-on inhibit" and run the drive. Ensure that the correct drive is running. The following is tested: - Correct DRIVE-CLiQ wiring between Control Unit and Motor Modules - Correct assignment of drive No. – Motor Module – motor - The hardware is functioning properly - Correct parameterization of the STO function - Routine for forced dormant error detection of the switch-off signal paths	

9.4.2.2 Acceptance test for Safe Stop 1 with encoder (Extended Functions)

Table 9- 12 "Safe Stop 1" function

No.	Description	Status
Note: The acceptance test must be individually conducted for each configured control. The control can be realized via TM54F or via PROFIsafe.		
1.	Initial state - Drive in "Ready" state (p0010 = 0) - Safety Integrated Extended Functions enabled (p9601.2 = 1) - Safety functions enabled (p9501.0 = 1) - No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7]); see note "non-critical alarms" in section Acceptance tests (Page 193)	
2.	Run the drive Check whether the correct drive is operational	
3.	Configure and activate trace recording. - Trigger: Trigger on variable - bit pattern (r9720.1 = 0) - Record the following values: r9714[0], r9714[1], r9720, r9722 - Select the time interval and pretrigger so you can recognize the selection of SS1 and the transition into the subsequent STO state Select SS1 while the drive is moving - The drive decelerates along the OFF3 ramp - Subsequent state STO is activated For better analysis, display the following bit values: - r9720.1 (deactivation SS1) - r9722.0 (STO active) - r9722.1 (SS1 active)	

No.	Description	Status
4.	Analyze trace:	
	• The STO is triggered after the SS1 timer (p9356/9556) has expired or if the speed drops below the shutdown speed (p9360/9560)	
	• r9714[0] is displayed in the unit [$\mu\text{m}/\text{Safety clock cycle}$ or $\text{m}^\circ/\text{Safety clock cycle}$]	
	For better analysis, display the following bit values:	
	• r9720.1 (deactivation SS1)	
	• r9722.0 (STO active)	
	• r9722.1 (SS1 active)	
5.	Save/print the trace and add it to the acceptance report (refer to the example below)	
6.	Deselect SS1	
	• No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7])	
	Acknowledge "switch-on inhibit" and run the drive	
	• Ensure that the correct drive is running	

Example Trace SS1 with encoder

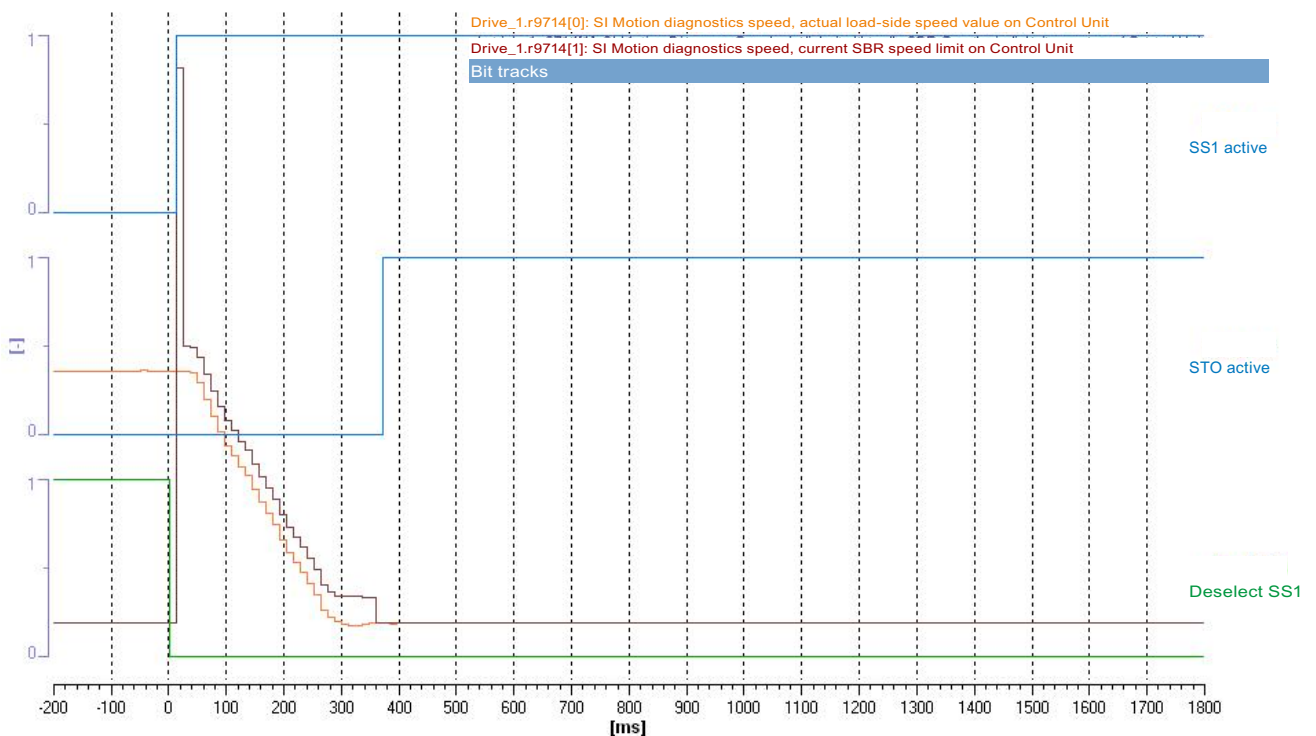


Figure 9-1 Example Trace SS1 with encoder

Trace evaluation:

- SS1 function is selected (time axis 0 ms; see bit "deselection SS1")
- Response bit "SS1 active" is set (time axis approx 20 ms)
- The drive decelerates along the configured OFF3 ramp (p1135)
- Recording of r9714[0] (orange curve) shows whether the OFF3 ramp is active
- The STO is activated (time axis approx. 370 ms; see bit "STO active"); at this point the speed falls below the shutdown speed SS1 (p9560/p9360) (drops below the shutdown speed SS1 before SS1 timer p9556/p9356 has expired)
- Brown curve: Envelope curve of function SBR (r9714[1]); a fault is generated if this is exceeded by the actual speed (r9714[0])

Note

Small time differences (of the order of 2 to 3 Safety clock cycles (here up to 36 ms)) are caused by internal calculations and do not present a problem.

9.4.2.3 Acceptance test for Safe Brake Control with encoder (Extended Functions)

Table 9- 13 "Safe Brake Control" function

No.	Description	Status
Note: The acceptance test must be individually conducted for each configured control. The control can be realized via TM54F or via PROFIsafe.		
1.	Initial state	
	• Drive in "Ready" state (p0010 = 0)	
	• Safety Integrated Extended Functions enabled (p9601.2 = 1)	
	• Safety functions enabled (p9501.0 = 1)	
	• Enable SBC function (p9602 = 1, p9802 = 1)	
	• Brake as in sequence control or brake always released (p1215 = 1 or p1215 = 2)	
	• No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7]); see note "non-critical alarms" in section Acceptance tests (Page 193)	
	• r9772.18 = r9872.18 = 0 (STO deselected via Safe Motion Monitoring)	
	• r9772.0 = r9772.1 = 0 (STO deselected and inactive – CU)	
	• r9720.0 = 1 (STO deselected)	
	• r9722.0 = 0 (STO deselected)	
2.	Run drive (if applied, brake is released)	
	• Check whether the correct drive is operational	
	Select STO when you issue the traversing command and check the following:	
	• Brake is applied	
	• No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7])	
	• r9772.4 = r9872.4 = 1 (SBC requested)	

No.	Description	Status
	r9772.18 = r9872.18 = 1 (STO selected via Safe Motion Monitoring)	
	r9720.0 = 0 (STO selected)	
	r9722.0 = 1 (STO selected)	
3.	Deselect STO and check the following:	
	No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7])	
	r9772.4 = r9872.4 = 0 (deselect SBC)	
	r9772.18 = r9872.18 = 0 (STO deselected via Safe Motion Monitoring)	
	r9720.0 = 1 (STO deselected)	
	r9722.0 = 1 (STO deselected)	
	r0046.0 = 1 (drive in "switch-on inhibited" state)	
4.	Acknowledge "switch-on inhibit" and run the drive. Check whether the correct drive is operational.	
	The following is tested: - The brake is connected properly - The hardware is functioning properly - The SBC is parameterized correctly - Routine for the forced dormant error detection of the brake control	

9.4.2.4 Acceptance test for Safe Stop 2 (Extended Functions)

Table 9- 14 "Safe Stop 2" function

No.	Description	Status
Note: The acceptance test must be individually performed for each configured control. Control may be via TM54F or PROFIsafe.		
1.	Initial state	
	Drive in "Ready" state (p0010 = 0)	
	Safety Integrated Extended Functions enabled (p9601.2 = 1)	
	Safety functions enabled (p9501.0 = 1)	
	No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7]); see note "non-critical alarms" in section Acceptance tests (Page 193)	
2.	Run the drive	
	Ensure that the correct drive is running	
3.	Configure and activate trace recording.	
	Trigger: Trigger on variable - bit pattern (r9720.2 = 0)	
	Record the following values: r9714[0], r9714[1], r9720, r9722	
	Select the time interval and pretrigger so you can recognize the selection of SS2 and the transition into the subsequent SOS state	
	Select SS2 while the drive is moving	
	The drive decelerates along the OFF3 ramp	

9.4 Acceptance tests

No.	Description	Status
	Subsequent SOS state is activated	
	No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7])	
4.	Analyze trace:	
	SOS is triggered after the SS2 timer (p9352/9552) has expired.	
	r9714[0] is displayed in the unit [$\mu\text{m}/\text{Safety clock cycle}$ or $\text{m}^\circ/\text{Safety clock cycle}$]	
	For better analysis, display the following bit values:	
	r9720.2 (deselection SS2)	
	r9722.2 (SS2 active)	
	r9722.3 (SOS active)	
5.	Save/print the trace and add it to the acceptance report (refer to the example below)	
6.	Deselect SS2	
	Ensure that the drive is running with the setpoint again	
	No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7])	

Example Trace SS2

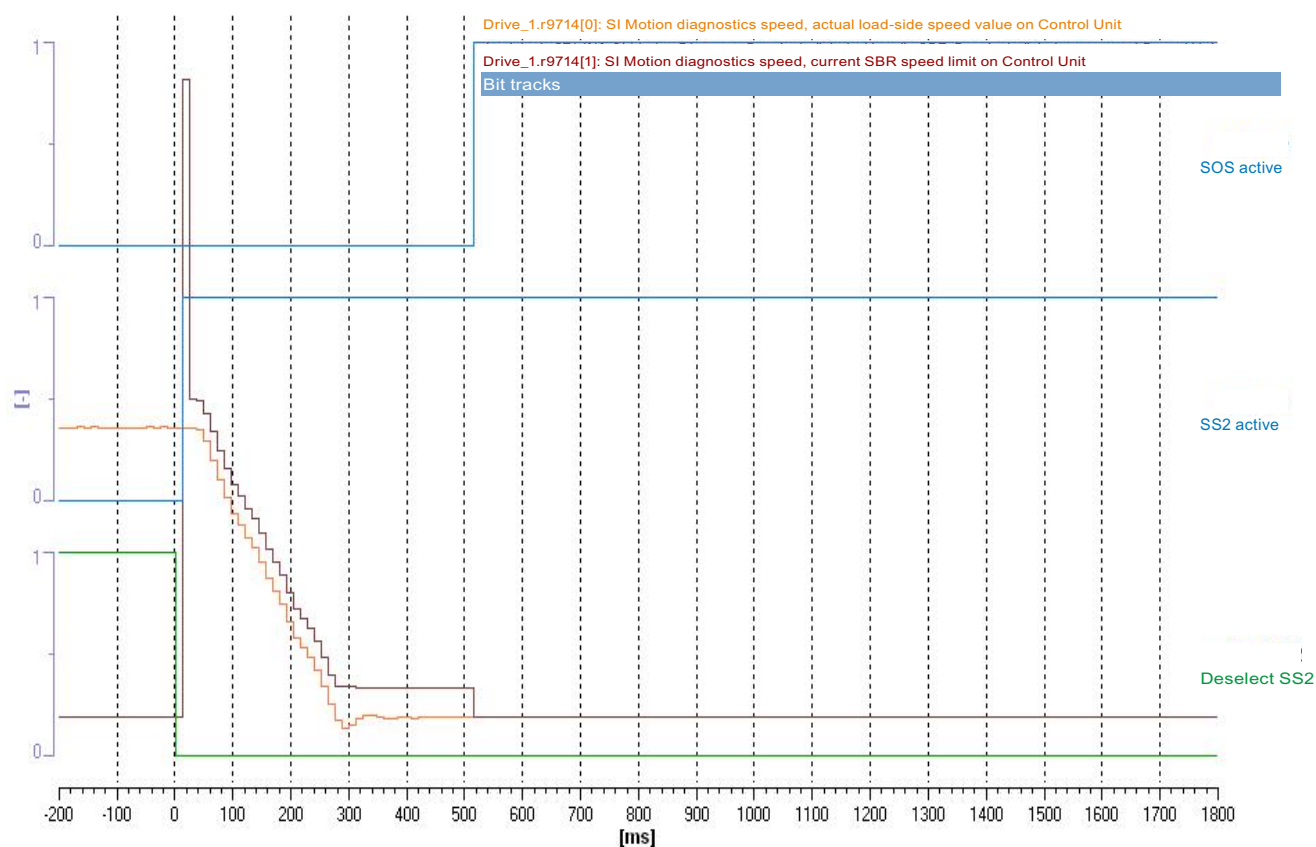


Figure 9-2 Example Trace SS2

Trace evaluation:

- SS2 function is selected (time axis 0 ms; see bit "deselection SS2")
- Response bit "SS2 active" is set (time axis approx 20 ms)
- The drive decelerates along the configured OFF3 ramp (p1135)
- Recording of r9714[0] (orange curve) shows whether the OFF3 ramp is active
- SOS is activated (time axis approx. 500 ms; see bit "SOS active"); at this point the SS2 timer (p9552/p9352) has expired
- Brown curve: Envelope curve of function SBR (r9714[1]); a fault is generated if this is exceeded by the actual speed (r9714[0])

Note

Small time differences (of the order of 2 to 3 Safety clock cycles (here up to 36 ms)) are caused by internal calculations and do not present a problem.

9.4.2.5 Acceptance test for Safe Operating Stop (Extended Functions)

Table 9- 15 "Safe Operating Stop" function

No.	Description	Status
Note: The acceptance test must be individually conducted for each configured control. The control can be realized via TM54F or via PROFIsafe.		
1.	Initial state	
	• Drive in "Ready" state (p0010 = 0)	
	• Safety Integrated Extended Functions enabled (p9601.2 = 1)	
	• Safety functions enabled (p9501.0 = 1)	
	• No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7]); see note "non-critical alarms" in section Acceptance tests (Page 193)	
2.	It may be necessary to take measures in the higher-level control to be able to run the drive with activated SOS.	
3.	Configure and activate trace recording.	
	• Trigger: Trigger on variable - bit pattern (r9722.7 = 0)	
	• Record the following values: r9713[0], r9720, r9721, r9722	
	• Select the time interval and pretrigger so you can recognize start-up of the drive and violation of the SOS tolerance window (p9330/0530)	
	Select SOS	
	Run the drive beyond the standstill limit set in p9330/p9530	
	• Check that the drive moves briefly and then decelerates back to a standstill	
	Check whether the following Safety messages are pending:	
	• C01707, C30707 (Tolerance for safe operating stop exceeded)	
	• C01701, C30701 (STOP B initiated)	

9.4 Acceptance tests

No.	Description	Status
	C01700, C30700 (STOP A initiated)	
4.	Analyze trace:	
	As soon as r9713[0] (unit μm or m°) leaves the tolerance window, a Safety message (r9722.7 = 0) becomes active	
	As a consequence, the drive is brought to a standstill with STOP B and STOP A	
	For better analysis, display the following bit values:	
	r9720.3 (deselect SOS)	
	r9721.12 (STOP A or B active)	
	r9722.0 (STO active; set for STOP A)	
	r9722.1 (SS1 active; set for STOP B)	
	r9722.3 (SOS active)	
	r9722.7 (internal event; set on the occurrence of the first Safety message)	
5.	Save/print the trace and add it to the acceptance report (refer to the example below)	
6.	Deselect SOS and acknowledge Safety messages	
	No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7])	
	r0046.0 = 1 (drive in "switch-on inhibited" state)	
	Acknowledge "switch-on inhibit" and run the drive	
	Check the drive is moving	

Example trace

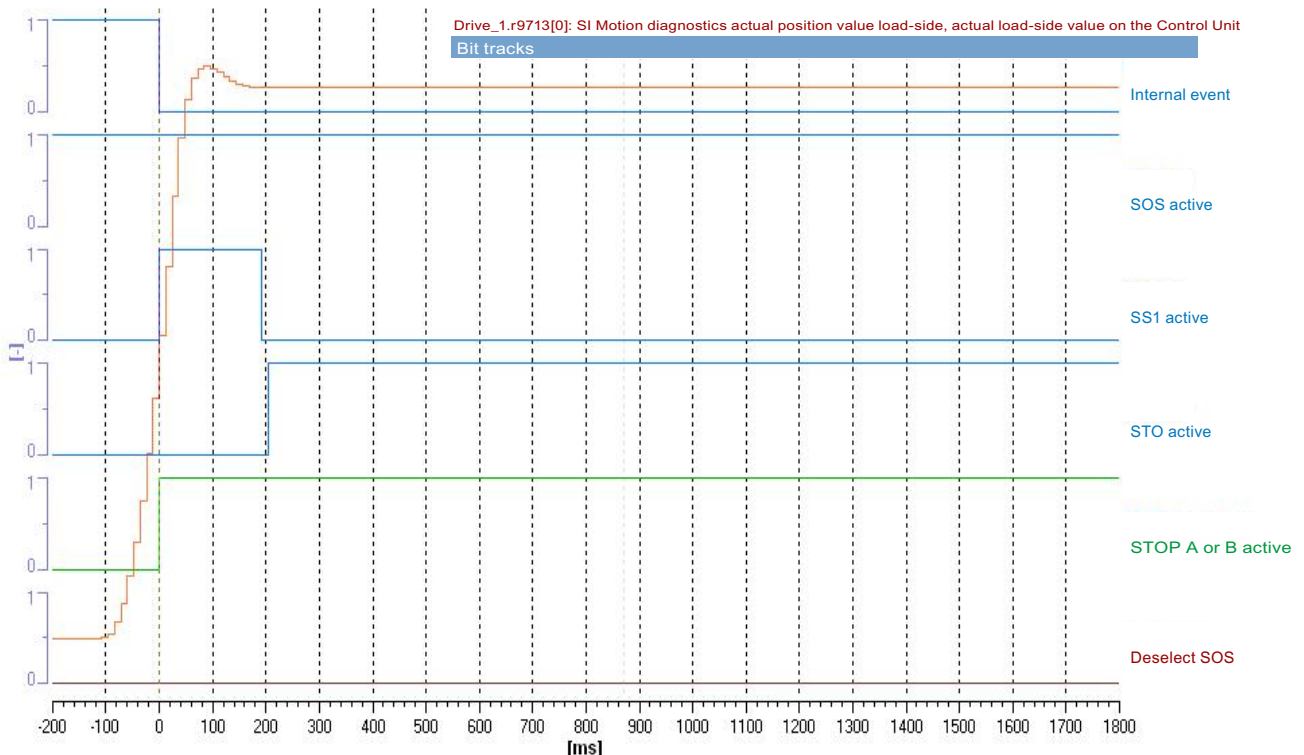


Figure 9-3 Example trace SOS

Trace evaluation:

- SOS function is activated (see bits "deselect SOS" and "SOS active")
- The drive starts moving (time axis approx -100 ms)
- Exiting the SOS tolerance window is recognized (time axis approx 0 ms)
- Safety fault is initiated (time axis approx. 0 ms; bit "internal event" is set to 0)
- Fault response STOP B is initiated (see bit "STOP A or B active" and "SS1 active")
- The drive is decelerated to a standstill
- Standstill reached (time axis approx. 200 ms)
- STOP A (as follow-up response to STOP B) is activated (see bit "STO active"); at this point the speed falls below the shutdown speed SS1 (p9560/p9360) before the SS1 timer (p9556/p9356) has expired (drops below the shutdown speed SS1 before SS1 timer (p9556/p9356) has expired)

Note

Small time differences (of the order of 2 to 3 Safety clock cycles (here up to 36 ms)) are caused by internal calculations and do not present a problem.

9.4.2.6 Acceptance tests for Safely Limited Speed with encoder

SLS with stop response "STOP A"

Table 9- 16 Function "Safely Limited Speed with encoder" with STOP A

No.	Description	Status
Note: The acceptance test must be carried out separately for each configured control and each SLS speed limit used. Control may be via TM54F or PROFIsafe.		
1.	Initial state	
	Drive in "Ready" state (p0010 = 0)	
	Safety Integrated Extended Functions enabled (p9601.2 = 1)	
	Safety functions enabled (p9501.0 = 1)	
	No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7]); see note "non-critical alarms" in section Acceptance tests (Page 193)	
2.	It may be necessary to take measures in the higher-level control to be able to exceed the active speed limit.	
3.	Configure and activate trace recording	
	Trigger: Trigger on variable - bit pattern (r9722.7 = 0)	
	Record the following values: r9714[0], r9714[1], r9720, r9721, r9722	
	Select the time interval and pretrigger so you can recognize when the active SLS limit has been exceeded and the subsequent drive responses	
	Select SLS with level x	
	Switch on the drive and specify the setpoint above the SLS limit	
	Check that the drive is moving, and after the SLS limit (p9331[x]/9531[x]) has been exceeded that it is coasting or a configured holding brake is applied	
	Check whether the following Safety messages are pending:	
	C01714 (x00), C30714 (x00); x = 1...4 depending on the SLS level (safely limited speed exceeded)	
	C01700, C30700 (STOP A initiated)	
4.	Analyze trace:	
	If r9714[0] (unit [µm/Safety clock cycle] or [m°/Safety clock cycle]) exceeds the active SLS limit, a Safety message (r9722.7 = 0) becomes active	
	STOP A is initiated as a consequence	
	For better analysis, display the following bit values:	
	r9720.4 (deselection SLS) and r9720.9/.10 (selection SLS level)	
	r9721.12 (STOP A or B active)	
	r9722.0 (STO active; set for STOP A)	
	r9722.4 (SLS active) and r9722.9/.10 (active SLS level)	
	r9722.7 (internal event; set on the occurrence of the first Safety message)	
5.	Save/print the trace and add it to the acceptance report (refer to the example below)	

No.	Description	Status
6.	Deselect SLS and acknowledge Safety messages	
	• No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7])	
	• r0046.0 = 1 (drive in "switch-on inhibited" state)	
	Acknowledge "switch-on inhibit" and run the drive	
	Check the drive is moving	

Example trace SLS with STOP A

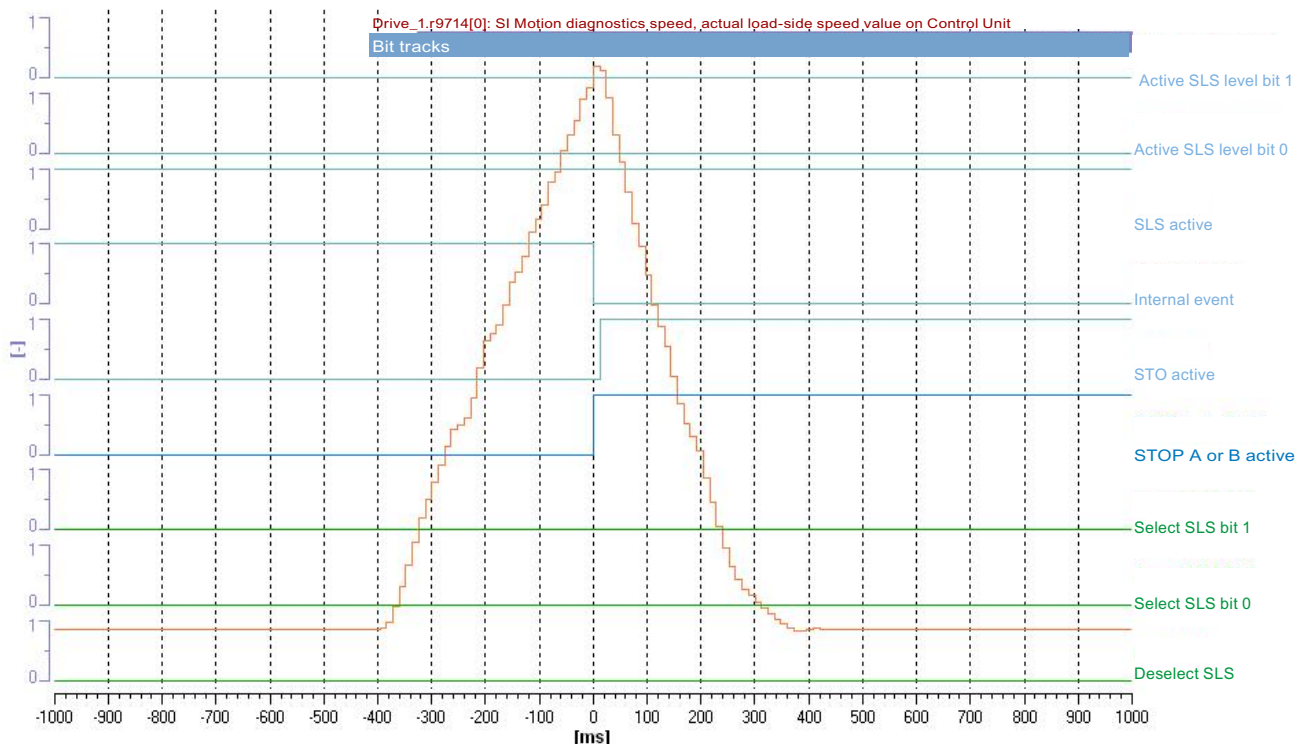


Figure 9-4 Example trace: SLS with STOP A

Trace evaluation:

- SLS function with SLS level 1 is active (see bits "deselection SLS", "selection SLS bit 0", "selection SLS bit 1" and "SLS active", "active SLS level bit 0" and "active SLS level bit 1")
- Drive is accelerated beyond the SLS limit (time axis from approx. -400 ms)
- Exceeding of the limit is recognized (time axis 0 ms)
- Safety fault is initiated (time axis 0 ms; bit "internal event" is set to 0)

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- Fault response STOP A is initiated (time axis 0 ms; see bit "STOP A or B active" and "STO active")
- Drive coasts (see orange curve of r9714[0])

Note

Small time differences (of the order of 2 to 3 Safety clock cycles (here up to 36 ms)) are caused by internal calculations and do not present a problem.

SLS with stop response "STOP B"

Table 9- 17 Function "Safely Limited Speed with encoder" with STOP B

No.	Description	Status
Note: The acceptance test must be carried out separately for each configured control and each SLS speed limit used. Control may be via TM54F or PROFIsafe.		
1.	Initial state	
	• Drive in "Ready" state (p0010 = 0)	
	• Safety Integrated Extended Functions enabled (p9601.2 = 1)	
	• Safety functions enabled (p9501.0 = 1)	
	• No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7]); see note "non-critical alarms" in section Acceptance tests (Page 193)	
2.	It may be necessary to take measures in the higher-level control to be able to exceed the active speed limit.	
3.	Configure and activate trace recording.	
	• Trigger: Trigger on variable - bit pattern (r9722.7 = 0)	
	• Record the following values: r9714[0], r9714[1], r9720, r9721, r9722	
	• Select the time interval and pretrigger so you can recognize when the active SLS limit has been exceeded and the subsequent drive responses	
	Select SLS with level x	
	Switch on the drive and specify the setpoint above the SLS limit	
	• Check that the drive is moving, and after the SLS limit (p9331[x]/9531[x]) has been exceeded that it is decelerated along the OFF3 ramp before STOP A becomes active	
	Check whether the following Safety messages are pending:	
	• C01714 (x00), C30714 (x00); x = 1...4 depending on the SLS level (safely limited speed exceeded)	
	• C01701, C30701 (STOP B initiated)	
	• C01700, C30700 (STOP A initiated)	

No.	Description	Status
4.	Analyze trace:	
	• If r9714[0] (unit [$\mu\text{m}/\text{Safety clock cycle}$] or [$\text{m}^\circ/\text{Safety clock cycle}$]) exceeds the active SLS limit, a Safety message (r9722.7 = 0) becomes active	
	• A STOP B is initiated as a consequence (with subsequent stop STOP A)	
	For better analysis, display the following bit values:	
	• r9720.4 (deselection SLS) and r9720.9/.10 (selection SLS level)	
	• r9721.12 (STOP A or B active)	
	• r9722.0 (STO active; set for STOP A)	
	• r9722.1 (SS1 active; set for STOP B)	
	• r9722.4 (SLS active) and r9722.9/.10 (active SLS level)	
	• r9722.7 (internal event; set on the occurrence of the first Safety message)	
5.	Save/print the trace and add it to the acceptance report (refer to the example below)	
6.	Deselect SLS and acknowledge Safety messages	
	• No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7])	
	• r0046.0 = 1 (drive in "switch-on inhibited" state)	
	Acknowledge "switch-on inhibit" and run the drive	
	• Check the drive is moving.	

Example trace SLS with STOP B

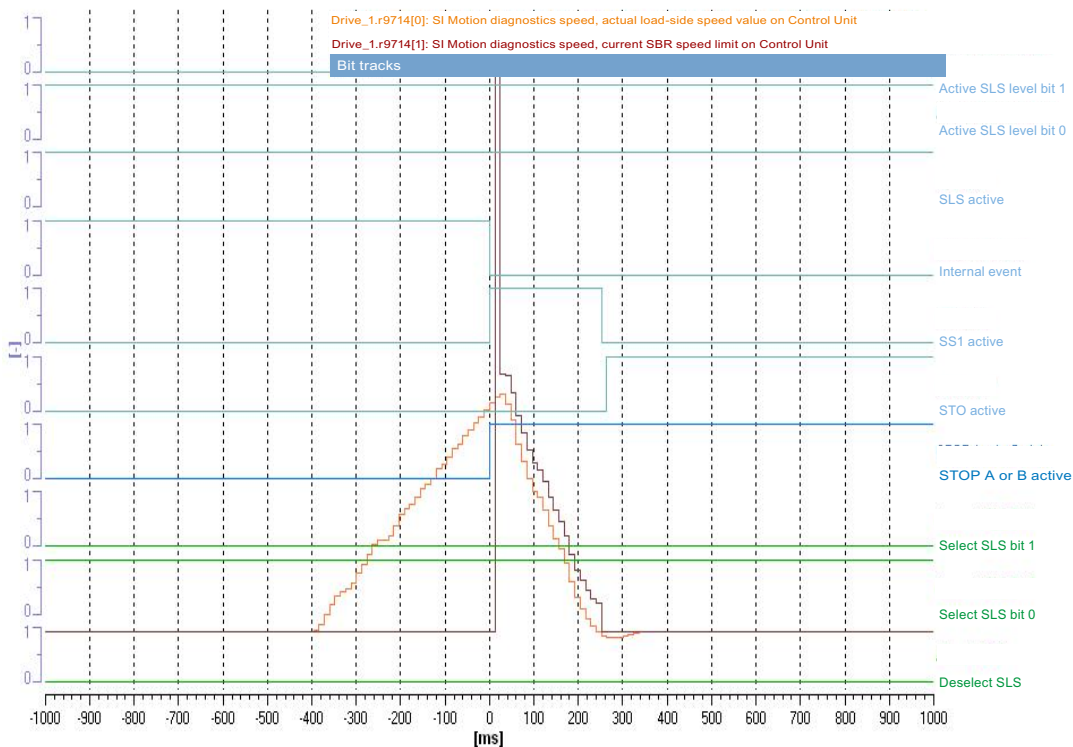


Figure 9-5 Example trace: SLS with STOP B

Trace evaluation:

- SLS function with SLS level 2 is active (see bits "deselection SLS", "selection SLS bit 0", "selection SLS bit 1" and "SLS active", "active SLS level bit 0" and "active SLS level bit 1")
- Drive is accelerated beyond the SLS limit (time axis from approx. -400 ms)
- Exceeding of the limit is recognized (time axis 0 ms)
- Safety fault is initiated (time axis 0 ms; bit "internal event" is set to 0)
- Fault response STOP B is initiated (time axis 0 ms; see bit "STOP A or B active" and "SS1 active")
- Drive is decelerated to a standstill (see orange curve of r9714[0])
- Standstill reached (time axis from approx. 250 ms)
- STOP A (as follow-up response to STOP B) is activated (see bit "STO active"); at this point the speed falls below the shutdown speed SS1 (p9560/p9360) (drops below the shutdown speed SS1 before SS1 timer p9556/p9356 has expired)

Note

Small time differences (of the order of 2 to 3 Safety clock cycles (here up to 36 ms)) are caused by internal calculations and do not present a problem.

SLS with stop response "STOP C"

Table 9- 18 Function "Safely Limited Speed with encoder" with STOP C

No.	Description	Status
Note: The acceptance test must be carried out separately for each configured control and each SLS speed limit used. Control may be via TM54F or PROFIsafe.		
1.	Initial state	
	• Drive in "Ready" state (p0010 = 0)	
	• Safety Integrated Extended Functions enabled (p9601.2 = 1)	
	• Safety functions enabled (p9501.0 = 1)	
	• No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7]); see note "non-critical alarms" in section Acceptance tests (Page 193)	
2.	It may be necessary to take measures in the higher-level control to be able to exceed the active speed limit	
3.	Configure and activate trace recording.	
	• Trigger: Trigger on variable - bit pattern (r9722.7 = 0)	
	• Record the following values: r9714[0], r9714[1], r9720, r9721, r9722	
	• Select the time interval and pretrigger so you can recognize when the active SLS limit has been exceeded and the subsequent drive responses	
	Select SLS with level x	
	Switch on the drive and specify the setpoint above the SLS limit	
	• Check that the drive is moving, and after the SLS limit (p9331[x]/9531[x]) has been exceeded that it is decelerated to a standstill along the OFF3 ramp	

No.	Description	Status
	Check whether the following Safety messages are pending:	
	C01714 (x00), C30714 (x00); x = 1...4 depending on the SLS level (safely limited speed exceeded)	
	C01708, C30708 (STOP C initiated)	
4.	Analyze trace:	
	If r9714[0] (unit [μm/Safety clock cycle] or [m°/Safety clock cycle]) exceeds the active SLS limit, a Safety message (r9722.7 = 0) becomes active	
	STOP C is initiated as a consequence	
	For better analysis, display the following bit values:	
	r9720.4 (deselection SLS) and r9720.9/.10 (selection SLS level)	
	r9721.13 (STOP C active)	
	r9722.2 (STO active; set for STOP C)	
	r9722.3 (SOS active)	
	r9722.4 (SLS active) and r9722.9/.10 (active SLS level)	
	r9722.7 (internal event; set on the occurrence of the first Safety message)	
5.	Save/print the trace and add it to the acceptance report (refer to the example below)	
6.	Deselect SLS and acknowledge Safety messages	
	Ensure that the drive is running with the setpoint again	
	No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7])	

Example trace SLS with STOP C

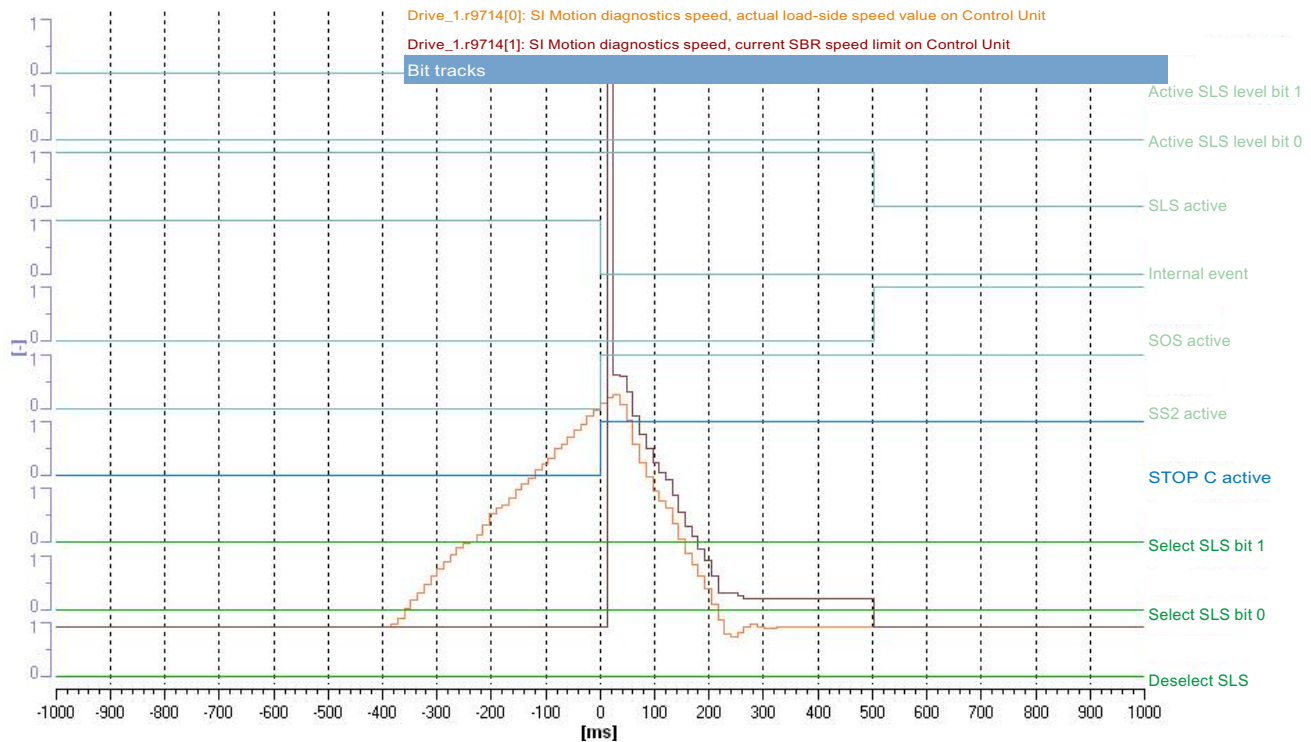


Figure 9-6 Example trace: SLS with STOP C

Trace evaluation:

- SLS function with SLS level 1 is active (see bits "deselection SLS", "selection SLS bit 0", "selection SLS bit 1" and "SLS active", "active SLS level bit 0" and "active SLS level bit 1")
- Drive is accelerated beyond the SLS limit (time axis from approx. -400 ms)
- Exceeding of the limit is recognized (time axis 0 ms)
- Safety fault is initiated (time axis 0 ms; bit "internal event" is set to 0)
- Fault response STOP C is initiated (see bit "STOP C active" and "SS2 active")
- Drive is decelerated to a standstill (see orange curve of r9714[0])
- After the SS2 timer has expired the follow-up function SOS is activated (time axis 500 ms)
- The "SOS active" bit is set and "SLS active" is reset

Note

Small time differences (of the order of 2 to 3 Safety clock cycles (here up to 36 ms)) are caused by internal calculations and do not present a problem.

SLS with stop response "STOP D"

Table 9- 19 Function "Safely Limited Speed with encoder" with STOP D

No.	Description	Status
Note: The acceptance test must be carried out separately for each configured control and each SLS speed limit used. Control may be via TM54F or PROFIsafe.		
1.	Initial state	
	Drive in "Ready" state (p0010 = 0)	
	Safety Integrated Extended Functions enabled (p9601.2 = 1)	
	Safety functions enabled (p9501.0 = 1)	
	No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7]); see note "non-critical alarms" in section Acceptance tests (Page 193)	
2.	It may be necessary to take measures in the higher-level control to be able to exceed the active speed limit	
3.	Configure and activate trace recording	
	Trigger: Trigger on variable - bit pattern (r9722.7 = 0)	
	Record the following values: r9714[0], r9714[1], r9720, r9721, r9722	
	Select the time interval and pretrigger so that the exceeding of the active SLS limit and the subsequent drive responses can be recognized	
	Select SLS with level x	
	Switch on the drive and specify the setpoint above the SLS limit	
	Check that the drive is moving, and after the SLS limit (p9331[x]/9531[x]) and the SOS standstill tolerance window have been exceeded, that it is decelerated along the OFF3 ramp before STOP A becomes active as a consequence	
	Check whether the following Safety messages are pending:	
	C01714 (x00), C30714 (x00); x = 1...4 depending on the SLS level (safely limited speed exceeded)	
	C01709, C30709 (STOP D initiated)	
	C01707, C30707 (Tolerance for safe operating stop exceeded)	
	C01701, C30701 (STOP B initiated)	
	C01700, C30700 (STOP A initiated)	
4.	Analyze trace:	
	If r9714[0] (unit [µm/Safety clock cycle] or [m°/Safety clock cycle]) exceeds the active SLS limit, a Safety message (r9722.7 = 0) becomes active	
	STOP D is initiated as a consequence.	
	As a consequence of STOP D (selection SOS) the above-described responses will be triggered if the drive is not stopped by the higher-level control on activation of STOP D	
	For better analysis, display the following bit values:	
	r9720.4 (deselection SLS) and r9720.9/.10 (selection SLS level)	
	r9721.14 (STOP D active)	
	r9722.3 (STO active; set for STOP D)	
	r9722.4 (SLS active) and r9722.9/.10 (active SLS level)	

9.4 Acceptance tests

No.	Description	Status
	r9722.7 (internal event; set on the occurrence of the first Safety message)	
5.	Save/print the trace and add it to the acceptance report (refer to the example below)	
6.	Deselect SLS and acknowledge Safety messages	
	No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7])	
	r0046.0 = 1 (drive in "switch-on inhibited" state)	
	Acknowledge "switch-on inhibit" and run the drive	
	Check the drive is moving	

Example trace SLS with STOP D

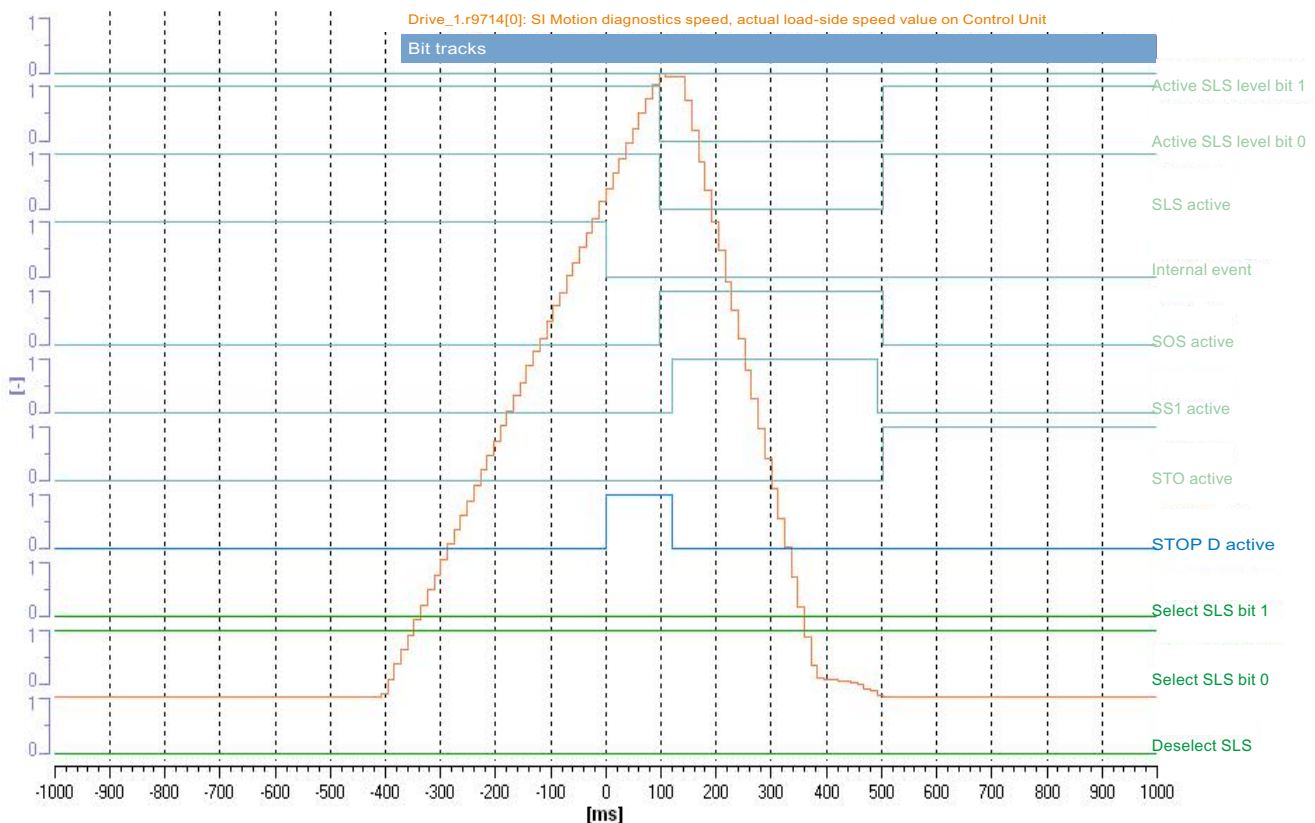


Figure 9-7 Example trace: SLS with STOP D

Trace evaluation:

- SLS function with SLS level 2 is active (see bits "deselection SLS", "selection SLS bit 0", "selection SLS bit 1" and "SLS active", "active SLS level bit 0" and "active SLS level bit 1")
- Drive is accelerated beyond the SLS limit (time axis from approx. -400 ms)
- Exceeding of the limit is recognized (time axis 0 ms)
- Safety fault is initiated (time axis 0 ms; bit "internal event" is set to 0)
- Fault response STOP D (corresponds to selection SOS) is initiated (see bit "STOP D active")

- Only after the delay time between selection and activation SOS (p9551/p9351) has expired is the standstill position safely monitored (time axis 100 ms; see bit "SOS active")
- But as the axis continues to turn, the standstill tolerance window is violated (time axis approx. 120 ms)
- STOP B is initiated (see bit "SS1 active")
- The drive is decelerated to a standstill
- Standstill is reached (time axis approx. 500 ms)
- STOP A (as follow-up response to STOP B) is activated (see bit "STO active"); at this point the speed falls below the shutdown speed SS1 (p9560/p9360) (drops below the shutdown speed SS1 before SS1 timer p9556/p9356 has expired).

Note

Small time differences (of the order of 2 to 3 Safety clock cycles (here up to 36 ms)) are caused by internal calculations and do not present a problem.

9.4.2.7 Acceptance test for Safe Speed Monitor (Extended Functions)

Table 9- 20 "Safe Speed Monitor" function

No.	Description	Status
1.	Initial state	
	• Drive in "Ready" state (p0010 = 0)	
	• Safety Integrated Extended Functions enabled (p9601.2 = 1)	
	• Safety functions enabled (p9501.0 = 1)	
	• No Safety message (r0945, r2122, r9747); see note "non-critical alarms" in section Acceptance tests (Page 193)	
2.	Switch off the drive or specify speed setpoint = 0	
	Configure and activate trace recording	
	• Trigger: Trigger on variable - bit pattern (r9722.15 = 1)	
	• Record the following values: r9714[0], r9722	
	• Select the time interval and pretrigger so you can recognize when the level exceeds the SSM limit (p9346/9546) and subsequently falls below it again	
	Switch on the drive and specify the setpoint so that the level briefly exceeds the SSM limit and then drops below it once more	
	• Check the drive is turning	
3.	Analyze trace:	
	• If r9714[0] (unit [μm /Safety clock cycle] or [m° /Safety clock cycle]) exceeds the SSM limit p9346/9546, r9722.15 = 0 is valid	
	• Once the level is lower than the limit r9722.15 = 1 is valid	
	• If hysteresis is active then r9722.15 only becomes 1 once more if r9714[0] falls below the limit p9346/9546 minus the hysteresis value p9347/9547	

No.	Description	Status
	For better analysis, display the following bit values:	
	r9722.15 (SSM speed under the limit value)	
4.	Save/print the trace and add it to the acceptance report (refer to the example below)	

Example Trace SSM (with hysteresis)

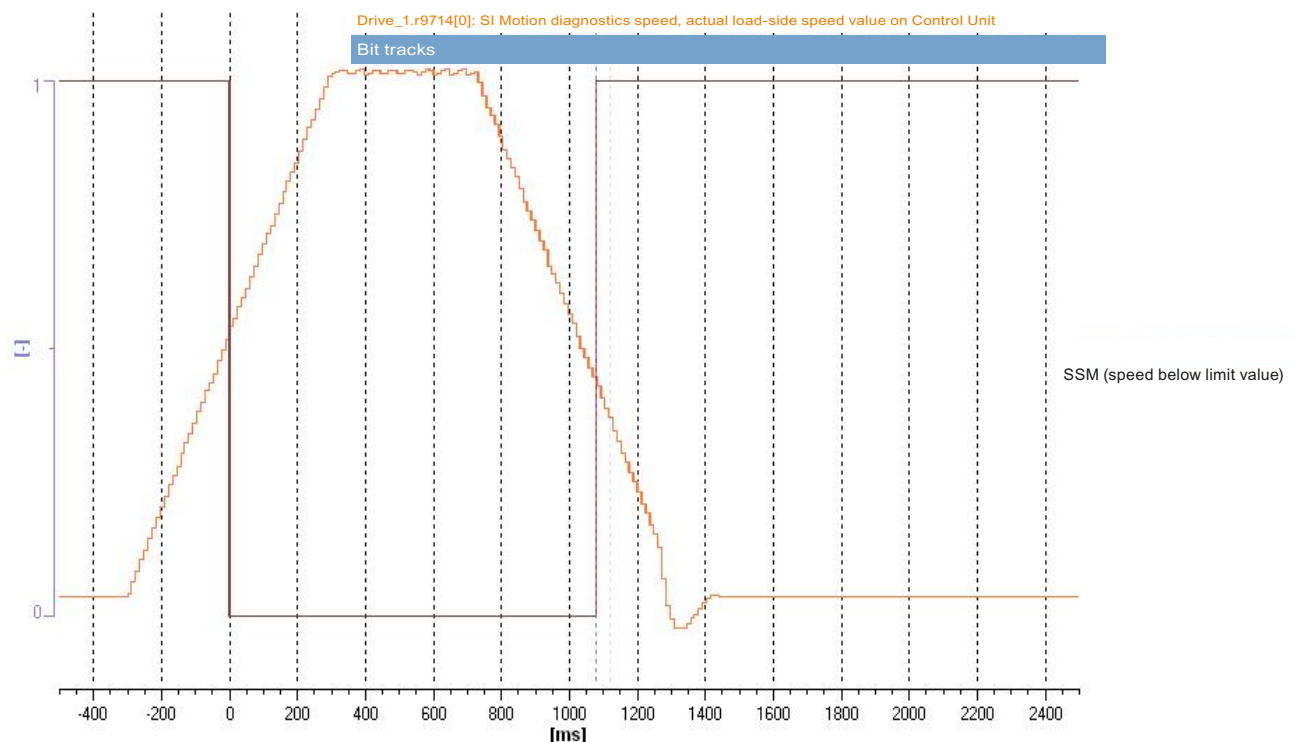


Figure 9-8 Example Trace SSM (with hysteresis)

Trace evaluation:

- Drive is accelerated (time axis from approx. -300 ms)
- SSM limit value (p9546/p9346) is exceeded (time axis 0 ms)
- Bit "SSM (speed below limit value)" is set to 0 (time axis 0 ms)
- Drive is decelerated again (time axis approx. 750 ms)
- Hysteresis active: The above named bit is only set to 1 once more if the speed falls below the SSM limit value minus the hysteresis value (p9547/p9347) (time axis approx. 1080 ms)

Note

Small time differences (of the order of 2 to 3 Safety clock cycles (here up to 36 ms)) are caused by internal calculations and do not present a problem.

9.4.3 Acceptance tests for Extended Functions (without encoder)

9.4.3.1 Acceptance test Safe Torque Off without encoder (Extended Functions)

Table 9- 21 Function "Safe Torque Off without encoder"

No.	Description	Status
Notes: The acceptance test must be individually conducted for each configured control. The control can be realized via TM54F or via PROFIsafe.		
1.	Initial state	
	• Drive in "Ready" state (p0010 = 0)	
	• Safety Integrated Extended Functions enabled (p9601.2 = 1)	
	• Safety functions enabled (p9501.0 = 1)	
	• Safety configured without encoder (p9506 = 1)	
	• No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7]); see note "non-critical alarms" in section Acceptance tests (Page 193)	
	• r9772.18 = r9872.18 = 0 (STO deselected via Safe Motion Monitoring)	
	• r9772.0 = r9772.1 = 0 (STO deselected and inactive – Control Unit)	
	• r9872.0 = r9872.1 = 0 (STO deselected and inactive – Motor Module)	
	• r9773.0 = r9773.1 = 0 (STO deselected and inactive – drive)	
	• r9720.0 = 1 (STO deselected)	
	• r9722.0 = 0 (STO deselected)	
2.	Run the drive	
	• Check whether the correct drive is operational	
	Select STO when you issue the traversing command and check the following:	
	• The drive coasts to a standstill or is braked and stopped by the mechanical brake (if available and configured (p1215, p9602, p9802)).	
	• No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7])	
	• r9772.18 = r9872.18 = 1 (STO selected via Safe Motion Monitoring)	
	• r9772.0 = r9772.1 = 1 (STO selected and active – Control Unit)	
	• r9872.0 = r9872.1 = 1 (STO selected and active – Motor Module)	
	• r9773.0 = r9773.1 = 1 (STO selected and active – drive)	
	• r9720.0 = 0 (STO selected)	
	• r9722.0 = 1 (STO selected)	
3.	Deselect STO and check the following:	
	• No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7])	
	• r9772.18 = r9872.18 = 0 (STO deselected via Safe Motion Monitoring)	
	• r9772.0 = r9772.1 = 0 (STO deselected and inactive – Control Unit)	
	• r9872.0 = r9872.1 = 0 (STO deselected and inactive – Motor Module)	

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No.	Description	Status
	- r9773.0 = r9773.1 = 0 (STO deselected and inactive – drive) - r9720.0 = 1 (STO deselected) - r9722.0 = 0 (STO deselected) - r0046.0 = 1 (drive in "switch-on inhibited" state)	
4.	Acknowledge "switch-on inhibit" and run the drive. Ensure that the correct drive is running. The following is tested: - Correct DRIVE-CLiQ wiring between Control Unit and Motor Modules - Correct assignment of drive No. – Motor Module – motor - The hardware is functioning properly - Correct parameterization of the STO function - Routine for forced dormant error detection of the switch-off signal paths	

9.4.3.2 Acceptance test for Safe Stop 1 without encoder (Extended Functions)

Table 9- 22 Function "Safe Stop 1 without encoder"

No.	Description	Status
Note: The acceptance test must be individually conducted for each configured control. The control can be realized via TM54F or via PROFIsafe.		
1.	Initial state - Drive in "Ready" state (p0010 = 0) - Safety Integrated Extended Functions enabled (p9601.2 = 1) - Safety functions enabled (p9501.0 = 1) - Safety configured without encoder (p9506 = 1) - No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7]); see note "non-critical alarms" in section Acceptance tests (Page 193)	
2.	Run the drive Check whether the correct drive is operational	
3.	Configure and activate trace recording - Trigger: Trigger on variable - bit pattern (r9720.1 = 0) - Record the following values: r9714[0], r9714[1], r9720, r9722 - Select the time interval and pretrigger so you can recognize the selection of SS1 and the transition into the subsequent STO state Select SS1 while the drive is moving - The drive decelerates along the OFF3 ramp - Subsequent state STO is activated For better analysis, display the following bit values: - r9720.1 (deactivation SS1) - r9722.0 (STO active) - r9722.1 (SS1 active)	

No.	Description	Status
4.	Analyze trace:	
	- STO is triggered if the speed drops below the shutdown speed (p9360/9560) - r9714[0] is displayed in the unit [$\mu\text{m}/\text{Safety clock cycle}$ or $\text{m}^\circ/\text{Safety clock cycle}$]	
	For better analysis, display the following bit values:	
	- r9720.1 (deactivation SS1) - r9722.0 (STO active) - r9722.1 (SS1 active)	
5.	Save/print the trace and add it to the acceptance report (refer to the example below)	
6.	Canceling SS1	
	No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7])	
	Acknowledge "Switch-on inhibit" and run the drive	
	Ensure that the correct drive is running	

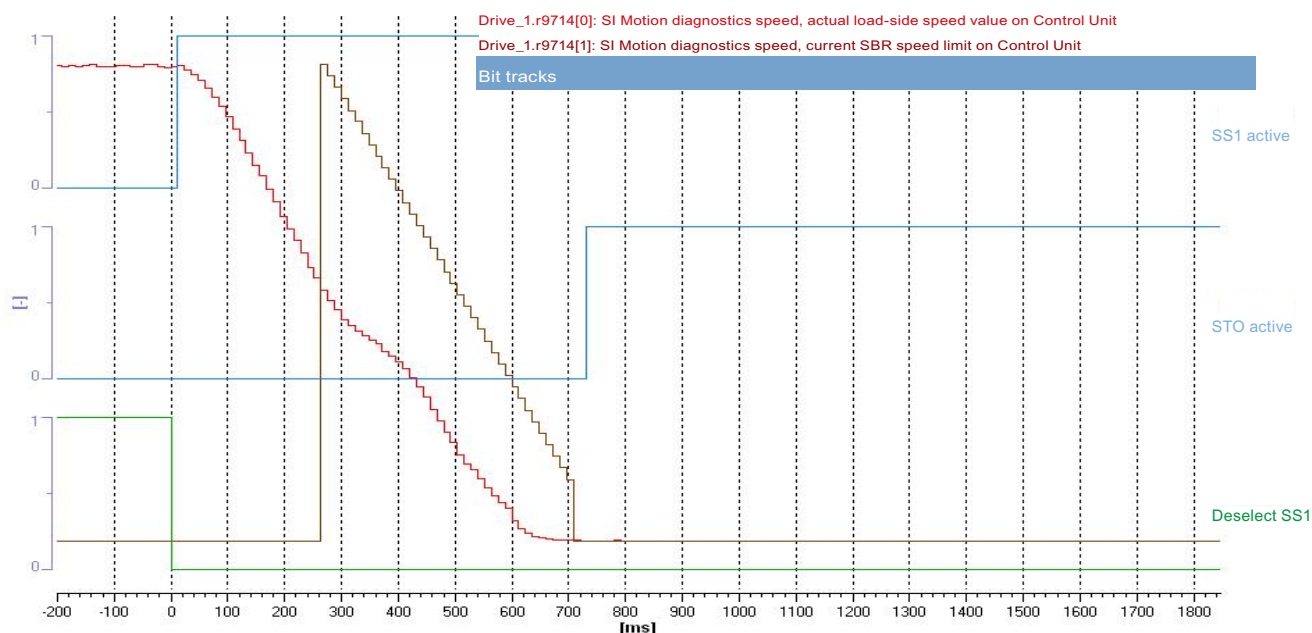


Figure 9-9 Example Trace SS1 without encoder

Trace evaluation:

- SS1 function is selected (time axis 0 ms; see bit "deselection SS1")
- Response bit "SS1 active" is set (time axis approx 20 ms)
- The drive decelerates along the configured OFF3 ramp (p1135)
- Recording of r9714[0] (orange curve) shows whether the OFF3 ramp is active

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- STO is activated (time axis approx. 720 ms; see bit "STO active"); at this point the speed falls below the shutdown speed SS1 (p9560/p9360)
- Brown curve: Envelope curve of function SBR (r9714[1]); faults are generated if this is exceeded by the actual speed (r9714[0])

In contrast to the SBR for Safety with encoder, this curve is not tracked according to actual speed but calculated using Safety parameters. Furthermore, this monitoring only becomes active after a configurable time has elapsed (in this case the time is 250 ms).

Note

Small time differences (of the order of 2 to 3 Safety clock cycles (here up to 36 ms)) are caused by internal calculations and do not present a problem.

9.4.3.3 Acceptance test for Safe Brake Control without encoder (Extended Functions)

Table 9- 23 Acceptance test "Safe Brake Control without encoder"

No.	Description	Status
Note: The acceptance test must be individually conducted for each configured control. The control can be realized via TM54F or via PROFIsafe.		
1.	Initial state	
	• Drive in "Ready" state (p0010 = 0)	
	• Safety Integrated Extended Functions enabled (p9601.2 = 1)	
	• Safety functions enabled (p9501.0 = 1)	
	• Safety configured without encoder (p9506 = 1)	
	• Enable SBC function (p9602 = 1, p9802 = 1)	
	• Brake as in sequence control or brake always released (p1215 = 1 or p1215 = 2)	
	• No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7]); see note "non-critical alarms" in section Acceptance tests (Page 193)	
	• r9772.18 = r9872.18 = 0 (STO deselected via Safe Motion Monitoring)	
	• r9772.0 = r9772.1 = 0 (STO deselected and inactive – CU)	
	• r9720.0 = 1 (STO deselected)	
	• r9722.0 = 0 (STO deselected)	
2.	Run drive (if applied, brake is released)	
	• Check whether the correct drive is operational	
	Select STO when you issue the traversing command and check the following:	
	• Brake is applied	
	• No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7])	
	• r9772.4 = r9872.4 = 1 (SBC requested)	
	• r9772.18 = r9872.18 = 1 (STO selected via Safe Motion Monitoring)	
	• r9720.0 = 0 (STO selected)	
	• r9722.0 = 1 (STO selected)	

No.	Description	Status
3.	Deselect STO and check the following:	
	• No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7])	
	• r9772.4 = r9872.4 = 0 (deselect SBC)	
	• r9772.18 = r9872.18 = 0 (STO deselected via Safe Motion Monitoring)	
	• r9720.0 = 1 (STO deselected)	
	• r9722.0 = 0 (STO deselected)	
	• r0046.0 = 1 (drive in "switch-on inhibited" state)	
4.	Acknowledge "switch-on inhibit" and run the drive. Check whether the correct drive is operational.	
	The following is tested: • The brake is connected properly • The hardware is functioning properly • The SBC is parameterized correctly • Routine for the forced dormant error detection of the brake control	

9.4.3.4 Acceptance tests for Safely Limited Speed without encoder (Extended Functions)

SLS with stop response "STOP A"

Table 9- 24 Function "Safely Limited Speed without encoder" with "STOP A"

No.	Description	Status
Note: The acceptance test must be carried out separately for each configured control and each SLS speed limit used. Control may be via TM54F or PROFIsafe.		
1.	Initial state	
	• Drive in "Ready" state (p0010 = 0)	
	• Safety Integrated Extended Functions enabled (p9601.2 = 1)	
	• Safety functions enabled (p9501.0 = 1)	
	• Safety configured without encoder (p9506 = 1)	
	• No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7]); see note "non-critical alarms" in section Acceptance tests (Page 193)	
2.	It may be necessary to take measures in the higher-level control to be able to exceed the active speed limit	
3.	Configure and activate trace recording	
	• Trigger: Trigger on variable - bit pattern (r9722.7 = 0)	
	• Record the following values: r9714[0], r9720, r9721, r9722	
	• Select the time interval and pretrigger so you can recognize when the active SLS limit has been exceeded and the subsequent drive responses	
	Select SLS with level x	
	Switch on the drive and specify the setpoint above the SLS limit	
	• Check that the drive is moving, and after the SLS limit (p9331[x]/9531[x]) has been exceeded that it is coasting or a configured holding brake is applied	

9.4 Acceptance tests

No.	Description	Status
	Check whether the following Safety messages are pending:	
	C01714 (x00), C30714 (x00); x = 1...4 depending on the SLS level (safely limited speed exceeded)	
	C01700, C30700 (STOP A initiated)	
4.	Analyze trace:	
	If r9714[0] (unit [µm/Safety clock cycle] or [m°/Safety clock cycle]) exceeds the active SLS limit, a Safety message (r9722.7 = 0) becomes active	
	STOP A is initiated as a consequence	
	For better analysis, display the following bit values:	
	r9720.4 (deselection SLS) and r9720.9/.10 (selection SLS level)	
	r9721.12 (STOP A or B active)	
	r9722.0 (STO active; set for STOP A)	
	r9722.4 (SLS active) and r9722.9/.10 (active SLS level)	
	r9722.7 (internal event; set on the occurrence of the first Safety message)	
5.	Save/print the trace and add it to the acceptance report (refer to the example below)	
6.	Deselect SLS and acknowledge Safety messages.	
	No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7])	
	r0046.0 = 1 (drive in "switch-on inhibited" state)	
	Acknowledge "switch-on inhibit" and run the drive	
	Check the drive is moving	

Example trace SLS sensorless (STOP A)

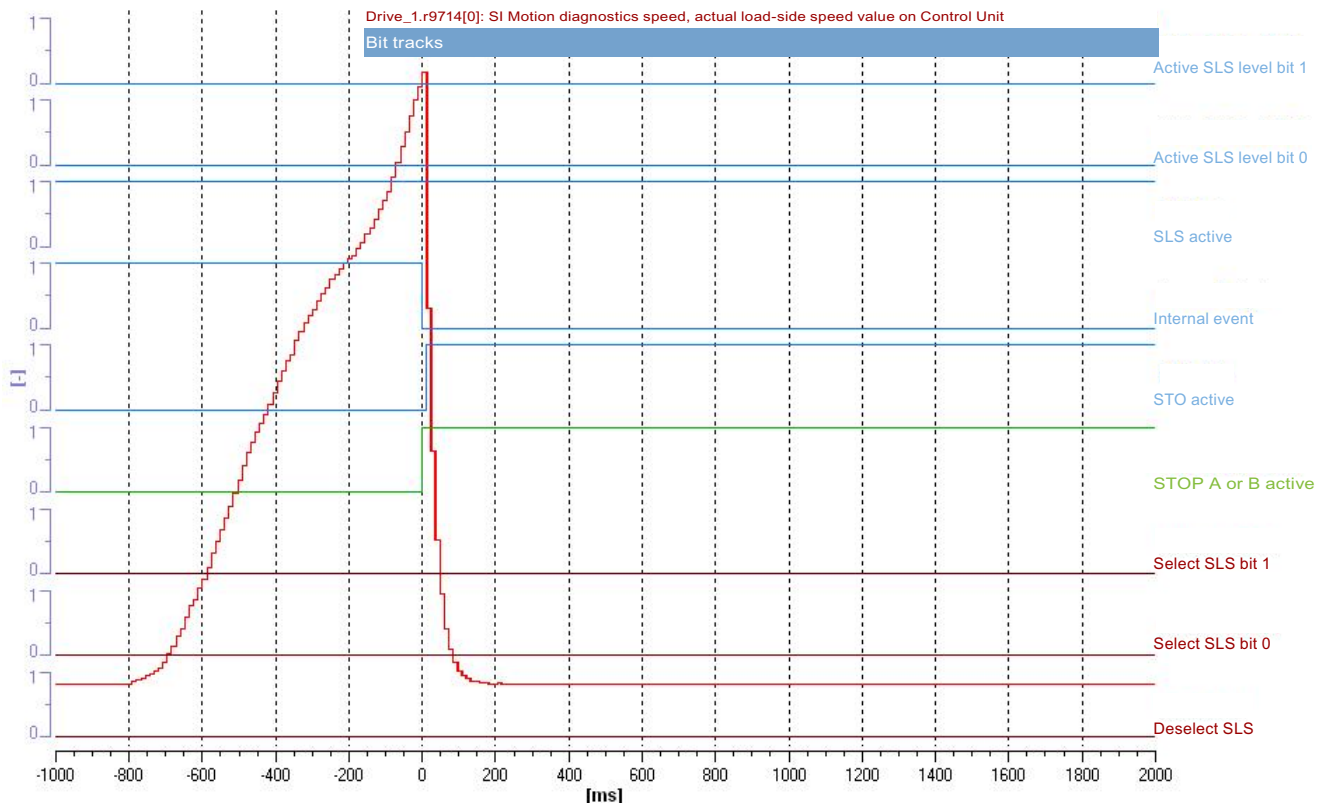


Figure 9-10 Example trace SLS without encoder with STOP A

Trace evaluation:

- SLS function with SLS level 1 is active (see bits "deselection SLS", "selection SLS bit 1" and "selection SLS bit 0", "SLS active", "active SLS level bit 0" and "active SLS level bit 1")
- Drive is accelerated beyond the SLS limit (time axis from approx. -800 ms)
- Exceeding of the limit is recognized (time axis 0 ms)
- Safety fault is initiated (time axis 0 ms; bit "internal event" is set to 0)
- Fault response STOP A is initiated (time axis 0 ms; see bit "STOP A or B active" and "STO active")
- Drive coasts (see red curve of r9714[0])

Note

Small time differences (of the order of 2 to 3 Safety clock cycles (here up to 36 ms)) are caused by internal calculations and do not present a problem.

SLS with stop response "STOP B"

Table 9- 25 Function "Safely Limited Speed without encoder" with "STOP B"

No.	Description	Status
Note: The acceptance test must be carried out separately for each configured control and each SLS speed limit used. Control may be via TM54F or PROFIsafe.		
1.	Initial state	
	Drive in "Ready" state (p0010 = 0)	
	Safety Integrated Extended Functions enabled (p9601.2 = 1)	
	Safety functions enabled (p9501.0 = 1)	
	Safety configured without encoder (p9506 = 1)	
	No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7]); see note "non-critical alarms" in section Acceptance tests (Page 193)	
2.	It may be necessary to take measures in the higher-level control to be able to exceed the active speed limit	
3.	Configure and activate trace recording	
	Trigger: Trigger on variable - bit pattern (r9722.7 = 0)	
	Record the following values: r9714[0], r9714[1], r9720, r9721, r9722	
	Select the time interval and pretrigger so you can recognize when the active SLS limit has been exceeded and the subsequent drive responses	
	Select SLS with level x	
	Switch on the drive and specify the setpoint above the SLS limit	
	Check that the drive is moving, and after the SLS limit (p9331[x]/9531[x]) has been exceeded that it is decelerated along the OFF3 ramp before STOP A becomes active	
	Check whether the following Safety messages are pending:	
	C01714 (x00), C30714 (x00); x = 1...4 depending on the SLS level (safely limited speed exceeded)	
	C01701, C30701 (STOP B initiated)	
	C01700, C30700 (STOP A initiated)	
4.	Analyze trace:	
	If r9714[0] (unit [µm/Safety clock cycle] or [m°/Safety clock cycle]) exceeds the active SLS limit, a Safety message (r9722.7 = 0) becomes active	
	A STOP B is initiated as a consequence (with subsequent stop STOP A)	
	For better analysis, display the following bit values:	
	r9720.4 (deselection SLS) and r9720.9/.10 (selection SLS level)	
	r9721.12 (STOP A or B active)	
	r9722.0 (STO active; set for STOP A)	
	r9722.1 (SS1 active; set for STOP B)	
	r9722.4 (SLS active) and r9722.9/.10 (active SLS level)	
	r9722.7 (internal event; set on the occurrence of the first Safety message)	
5.	Save/print the trace and add it to the acceptance report (refer to the example below)	

No.	Description	Status
6.	Deselect SLS and acknowledge Safety messages	
	No Safety faults and alarms (r0945[0...7], r2122[0...7], r9747[0...7])	
	r0046.0 = 1 (drive in "switch-on inhibited" state)	
	Acknowledge "switch-on inhibit" and run the drive	
	Check the drive is moving	

Example trace SLS without encoder (STOP B)

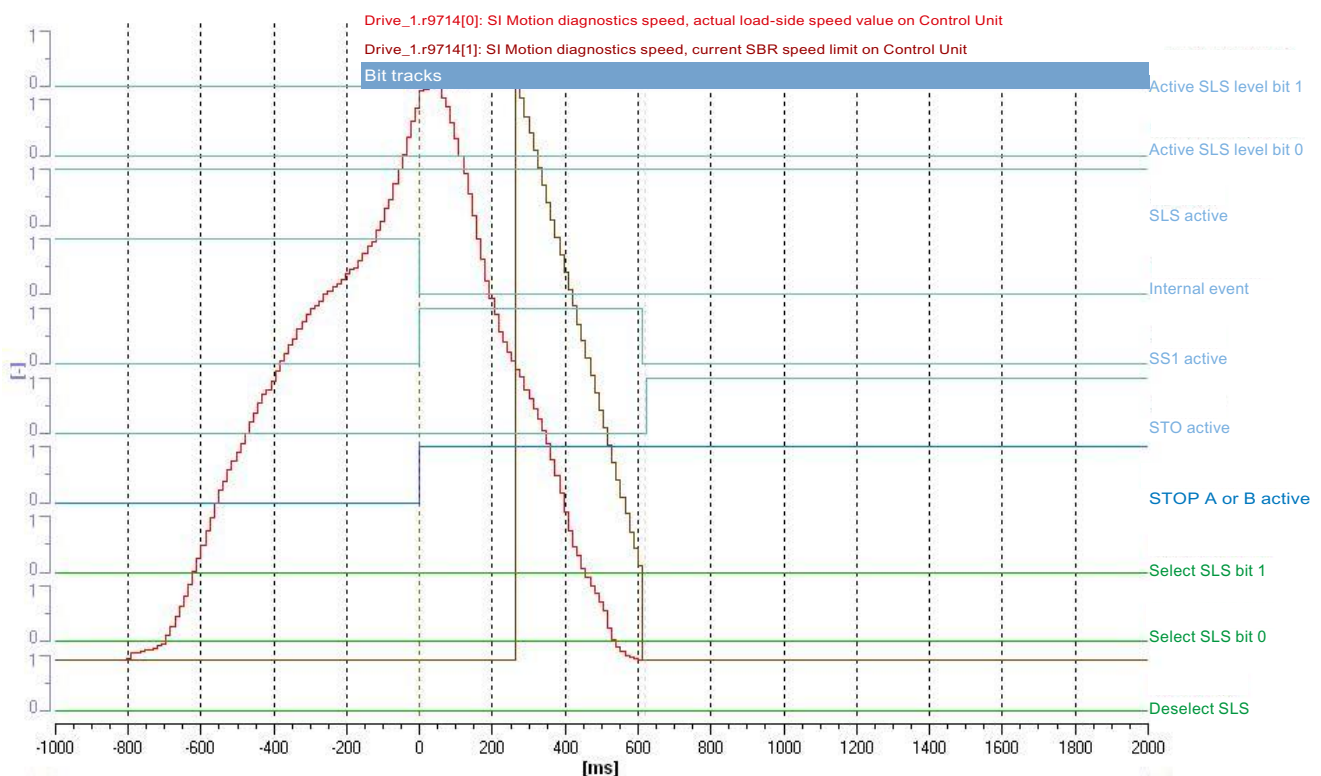


Figure 9-11 Example trace SLS without encoder with STOP B

Trace evaluation:

- SLS function with SLS level 1 is active (see bits "deselection SLS", "selection SLS bit 0", "selection SLS bit 1" and "SLS active", "active SLS level bit 0" and "active SLS level bit 1")
- Drive is accelerated beyond the SLS limit (time axis from approx. -800 ms)
- Exceeding of the limit is recognized (time axis 0 ms)
- Safety fault is initiated (time axis 0 ms; bit "internal event" is set to 0)
- Fault response STOP B is initiated (time axis 0 ms; see bit "STOP A or B active" and "SS1 active")
- Drive is decelerated to a standstill (see orange curve of r9714[0])
- Standstill reached (time axis from approx. 600 ms)

- STOP A (as follow-up response to STOP B) is activated (see bit "STO active"); at this point the speed falls below the shutdown speed SS1 (p9560/p9360)
- SBR monitoring is activated after 250 ms

Note

Small time differences (of the order of 2 to 3 Safety clock cycles (here up to 36 ms)) are caused by internal calculations and do not present a problem.

9.5 Completion of certificate

SI parameters

	Specified values checked?	
	Yes	No
Control Unit		
Motor Module		

Checksums

Basic Functions + Extended Functions			
Drive name	Drive number	SI reference checksum SI parameters (Control Unit)	SI reference checksum SI parameters (Motor Module)
		p9799 =	p9899 =
		p9799 =	p9899 =
		p9799 =	p9899 =
		p9799 =	p9899 =
		p9799 =	p9899 =
		p9799 =	p9899 =
only Extended Functions			

Drive name	Drive number	SI reference checksum SI parameters (Control Unit)	SI reference checksum SI parameters (Motor Module)
		p9399[0] = p9399[1] =	p9729[0] = p9729[1] = p9729[2] =
		p9399[0] = p9399[1] =	p9729[0] = p9729[1] = p9729[2] =
		p9399[0] = p9399[1] =	p9729[0] = p9729[1] = p9729[2] =

Drive name	Drive number	SI reference checksum SI parameters (Control Unit)	SI reference checksum SI parameters (Motor Module)
		p9399[0] = p9399[1] =	p9729[0] = p9729[1] = p9729[2] =
		p9399[0] = p9399[1] =	p9729[0] = p9729[1] = p9729[2] =
		p9399[0] = p9399[1] =	p9729[0] = p9729[1] = p9729[2] =
TM54F			
	Reference checksum:	p10005[0] =	p10005[1] =

Safety logbook

	Functional ¹⁾
Checksum for functional tracking of changes	r9781[0] =
Checksum for hardware dependent tracking of changes	r9781[1] =
Time stamp for functional tracking of changes	r9782[0] =
Time stamp for hardware dependent tracking of changes	r9782[1] =

1) These parameters can be found in the expert list of the Control Unit.

Data backup

	Storage medium			Storage location
	Type	Designation	Date	
Parameter				
PLC program				
Circuit diagrams				

Countersignatures

Commissioning engineer

This confirms that the tests and checks have been carried out properly.

Date	Name	Company/dept.	Signature

Machine manufacturer

This confirms that the parameters recorded above are correct.

Date	Name	Company/dept.	Signature

Appendix A

A.1 List of abbreviations

Note:

The following list of abbreviations includes all abbreviations and their meanings used in the entire SINAMICS user documentation.

Abbreviation	Source of abbreviation	Meaning
A		
A...	Alarm	Alarm
AC	Alternating Current	Alternating current
ADC	Analog Digital Converter	Analog digital converter
AI	Analog Input	Analog input
AIM	Active Interface Module	Active Interface Module
ALM	Active Line Module	Active Line Module
AO	Analog Output	Analog output
AOP	Advanced Operator Panel	Advanced Operator Panel
APC	Advanced Positioning Control	Advanced Positioning Control
AR	Automatic Restart	Automatic restart
ASC	Armature Short Circuit	Armature short circuit
ASCII	American Standard Code for Information Interchange	American standard code for information interchange
ASM	Asynchronmotor	Induction motor
B		
BERO	-	Contactless proximity switch
BI	Binector Input	Binector input
BIA	Berufsgenossenschaftliches Institut für Arbeitssicherheit	Germany's Institute for Occupational Safety and Health
BICO	Binector Connector Technology	Binector connector technology
BLM	Basic Line Module	Basic Line Module
BO	Binector Output	Binector output
BOP	Basic Operator Panel	Basic Operator Panel

Abbreviation	Source of abbreviation	Meaning
C		
C	Capacitance	Capacitance
C...	-	Safety message
CAN	Controller Area Network	Serial bus system
CBC	Communication Board CAN	Communication board CAN
CD	Compact Disc	Compact Disc
CDC	Crosswise data comparison	Crosswise data comparison
CDS	Command Data Set	Command data set
CF Card	CompactFlash Card	CompactFlash Card
CI	Connector Input	Connector input
CLC	Clearance Control	Clearance control
CNC	Computer Numerical Control	Computer numerical control
CO	Connector Output	Connector output
CO/BO	Connector Output/Binector Output	Connector/binector output
COB ID	CAN Object Identification	CAN Object identification
COM	Common contact of a changeover relay	Center contact of a changeover contact
COMM	Commissioning	Commissioning
CP	Communication Processor	Communication processor
CPU	Central Processing Unit	Central processing unit
CRC	Cyclic Redundancy Check	Cyclic redundancy check
CSM	Control Supply Module	Control Supply Module
CU	Control Unit	Control Unit
CUA	Control Unit Adapter	Control Unit Adapter
CUD	Control Unit DC MASTER	Control Unit DC MASTER
D		
DAC	Digital Analog Converter	Digital analog converter
DC	Direct Current	DC current
DC link	DC link	DC link
DCB	Drive Control Block	Drive Control Block
DCC	Drive Control Chart	Drive Control Chart
DCC	Data Cross Check	Crosswise data comparison
DCN	Direct Current Negative	DC current negative
DCP	Direct Current Positive	DC current positive
DDS	Drive Data Set	Drive data set
DI	Digital Input	Digital input
DI/DO	Digital Input/Digital Output	Digital input/output bidirectional
DMC	DRIVE-CLiQ Hub Module Cabinet	DRIVE-CLiQ Hub Module Cabinet
DME	DRIVE-CLiQ Hub Module External	DRIVE-CLiQ Hub Module External
DO	Digital Output	Digital output
DO	Drive Object	Drive object

Abbreviation	Source of abbreviation	Meaning
DP	Decentralized Peripherals	Distributed IOs
DPRAM	Dual Ported Random Access Memory	Memory with dual access ports
DRAM	Dynamic Random Access Memory	Dynamic memory
DRIVE-CLiQ	Drive Component Link with IQ	Drive Component Link with IQ
DSC	Dynamic Servo Control	Dynamic Servo Control
E		
EASC	External Armature Short Circuit	External armature short circuit
EDS	Encoder Data Set	Encoder data set
ESD	Electrostatic Sensitive Devices	Electrostatic sensitive devices
ELCB	Earth Leakage Circuit Breaker	Earth leakage circuit breaker
ELP	Earth Leakage Protection	Earth leakage protection
EMC	Electromagnetic Compatibility	Electromagnetic compatibility
EMF	Electromagnetic Force	Electromagnetic force
EMC	Electromagnetic compatibility	Electromagnetic compatibility
EN	European standard	European standard
EnDat	Encoder Data Interface	Encoder interface
EP	Enable Pulses	Pulse enable
EPOS	Einfachpositionierer	Basic positioner
ES	Engineering System	Engineering System
ESB	Equivalent circuit diagram	Equivalent circuit diagram
ESD	Electrostatic Sensitive Devices	Electrostatic sensitive devices
ESR	Extended Stop and Retract	Extended stop and retract
F		
F...	Fault	Fault
FAQs	Frequently Asked Questions	Frequently asked questions
FBL	Free Blocks	Free function blocks
FCC	Function Control Chart	Function Control Chart
FCC	Flux Current Control	Flux current control
FD	Function Diagram	Function diagram
F-DI	Failsafe Digital Input	Fail-safe digital input
F-DO	Failsafe Digital Output	Fail-safe digital output
FEM	Fremderregter Synchronmotor	Separately excited synchronous motor
FEPROM	Flash EPROM	Non volatile read and write memory
FG	Function Generator	Function generator
FI	-	Fault current
FOC	Fiber-Optic Cable	Fiber-optic cable
FP	Function diagram	Function diagram
FPGA	Field Programmable Gate Array	Field Programmable Gate Array

Abbreviation	Source of abbreviation	Meaning
FW	Firmware	Firmware
G		
GB	Gigabyte	Gigabyte
GC	Global Control	Global Control Telegram (Broadcast Telegramm)
GND	Ground	Reference potential for all signal and operating voltages, usually defined as 0 V (also referred to as G)
GSD	Generic Station Description	Generic station description: Describes the characteristics of a PROFIBUS slave
GSV	Gate Supply Voltage	Gate Supply Voltage
GUID	Globally Unique Identifier	Globally unique identifier
H		
HF	High Frequency	High frequency
HFD	Hochfrequenzdrossel	High-frequency reactor
HMI	Human Machine Interface	Human machine interface
HTL	High-Threshold Logic	Logic with a high fault threshold
HW	Hardware	Hardware
I		
I/O	Input/Output	Input/output
I2C	Inter-Integrated Circuit	Internal serial data bus
IASC	Internal Armature Short Circuit	Internal armature short circuit
IBN	Inbetriebnahme	Commissioning
ID	Identifier	Identification
IE	Industrial Ethernet	Industrial Ethernet
IEC	International Electrotechnical Commission	International Electrotechnical Commission
IF	Interface	Interface
IGBT	Insulated Gate Bipolar Transistor	Insulated gate bipolar transistor
IGCT	Integrated Gate-Controlled Thyristor	Semiconductor power switch with integrated control electrode
IL	Impulslöschung	Pulse cancelation
IP	Internet Protocol	Internet Protocol
IPO	Interpolator	Interpolator
IT	Isolé Terré	Non-grounded three-phase power supply
IVP	Internal Voltage Protection	Internal voltage protection
J		
JOG	Jogging	Jogging

Abbreviation	Source of abbreviation	Meaning
K		
KIP	Kinetische Pufferung	Kinetic buffering
Kp	-	Proportional gain
KTY	-	Special temperature sensor
L		
L	-	Formula symbol for inductance
LED	Light Emitting Diode	Light Emitting Diode
LIN	Linear motor	Linear motor
LSB	Least Significant Bit	Least significant bit
LSC	Line-Side Converter	Line-side converter
LSS	Line Side Switch	Line side switch
LU	Length Unit	Length unit
M		
M	-	Formula symbol for torque
M	Masse	Reference potential for all signal and operating voltages, usually defined as 0 V (also referred to as GND)
MB	Megabyte	Megabyte
MCC	Motion Control Chart	Motion Control Chart
MDS	Motor Data Set	Motor data set
MLFB	Maschinenlesbare Fabrikatebezeichnung	Machine-Readable Product Code
MMC	Man-Machine Communication	Man-machine communication
MMC	Micro Memory Card	Micro memory card
MSB	Most Significant Bit	Most significant bit
MSC	Motor-Side Converter	Motor-side converter
MSCY_C1	Master Slave Cycle Class 1	Cyclic communication between master (Class 1) and slave
MSR	Motorstromrichter	Motor-side converter
MT	Machine Tool	Machine tool
N		
N. C.	Not Connected	Not connected
N...	No Report	No message or internal message
NAMUR	Normenarbeitsgemeinschaft für Mess- und Regeltechnik in der chemischen Industrie	Standardization association for measurement and control in the chemical industry
NC	Normally Closed (contact)	NC contact
NC	Numerical Control	Numerical control
NEMA	National Electrical Manufacturers Association	Standardization body in the US
NM	Nullmarke	Zero mark
NO	Normally Open (contact)	NO contact

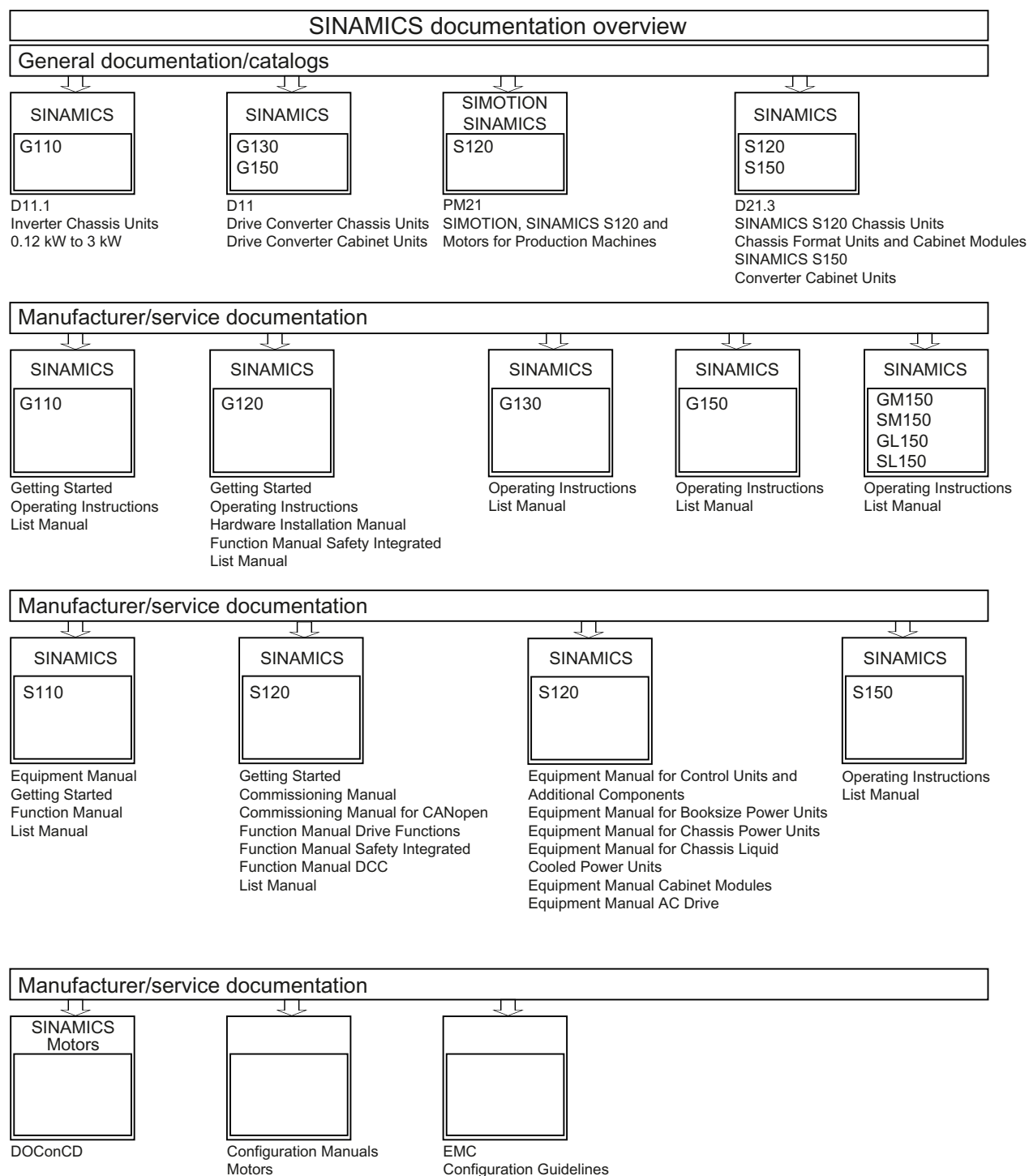
Abbreviation	Source of abbreviation	Meaning
NSR	Netzstromrichter	Line-side converter
NVRAM	Non-Volatile Random Access Memory	Non-volatile read/write memory
O		
OA	Open Architecture	Open Architecture
OC	Operating Condition	Operating condition
OEM	Original Equipment Manufacturer	Original Equipment Manufacturer
OLP	Optical Link Plug	Fiber-optic bus connector
OMI	Option Module Interface	Option module interface
P		
p...	-	Adjustable parameters
PB	PROFIBUS	PROFIBUS
PC	Position Controller	Position Controller
PcCtrl	PC Control	Control for master
PD	PROFIdrive	PROFIdrive
PDS	Power unit Data Set	Power unit data set
PE	Protective Earth	Protective earth
PELV	Protective Extra Low Voltage	Protective extra low voltage
PEM	Permanenterregter Synchronmotor	Permanent-magnet synchronous motor
PG	Programmiergerät	Programming device
PI	Proportional Integral	Proportional integral
PID	Proportional Integral Differential	Proportional integral differential
PLC	Programmable Logic Controller	Programmable logic controller
PLL	Phase-Locked Loop	Phase-locked loop
PN	PROFINET	PROFINET
PNO	PROFIBUS Nutzerorganisation	PROFIBUS user organization
PPI	Point-to-Point Interface	Point-to-point interface
PRBS	Pseudo Random Binary Signal	White noise
PROFIBUS	Process Field Bus	Serial data bus
PS	Power Supply	Power supply
PSA	Power Stack Adapter	Power Stack Adapter
PTC	Positive Temperature Coefficient	Positive temperature coefficient
PTP	Point-To-Point	Point-to-Point
PWM	Pulse Width Modulation	Pulse width modulation
PZD	Prozessdaten	Process data
R		
r...	-	Display parameters (read-only)
RAM	Random Access Memory	Read/write memory
RCCB	Residual Current Circuit Breaker	Residual current operated circuit breaker
RCD	Residual Current Device	Residual current operated circuit breaker
RCM	Residual Current Monitor	Residual current monitor

Abbreviation	Source of abbreviation	Meaning
RFG	Ramp-Function Generator	Ramp-function generator
RJ45	Registered Jack 45	Term for an 8-pin socket system for data transmission with shielded or non-shielded multi-wire copper cables
RKA	Rückkühlanlage	Cooling unit
RO	Read Only	Read only
RPDO	Receive Process Data Object	Receive process data object
RS232	Recommended Standard 232	Interface standard for cable-connected serial data transmission between a sender and receiver (also known under EIA232)
RS485	Recommended Standard 485	Interface standard for a cable-connected differential, parallel, and/or serial bus system (data transmission between a number of senders and receivers, also known under EIA485)
RTC	Real Time Clock	Real time clock
RZA	Raumzeigerapproximation	Space vector approximation
S		
S1	-	Uninterrupted duty
S3	-	Intermittent duty
SBC	Safe Brake Control	Safe brake control
SBH	Sicherer Betriebshalt	Safe operating stop
SBR	-	Safe acceleration monitoring
SCA	Safe Cam	Safe cam
SD Card	SecureDigital Card	Secure digital memory card
SE	Sicherer Software-Endschalter	Safe software limit switch
SG	Sicher reduzierte Geschwindigkeit	Safely reduced speed
SGA	Sicherheitsgerichteter Ausgang	Safety-related output
SGE	Sicherheitsgerichteter Eingang	Safety-related input
SH	Sicherer Halt	Safe standstill
SI	Safety Integrated	Safety Integrated
SIL	Safety Integrity Level	Safety Integrity Level
SLM	Smart Line Module	Smart Line Module
SLP	Safely-Limited Position	Safely-limited position
SLS	Safely Limited Speed	Safely limited speed
SLVC	Sensorless Vector Control	Vector control without encoder
SM	Sensor Module	Sensor Module
SMC	Sensor Module Cabinet	Sensor Module Cabinet
SME	Sensor Module External	Sensor Module External
SN	Sicherer Software-Nocken	Safe software cam
SOS	Safe Operating Stop	Safe operating stop

Abbreviation	Source of abbreviation	Meaning
SP	Service Pack	Service pack
SPC	Setpoint Channel	Setpoint channel
SPI	Serial Peripheral Interface	Serial interface for connecting peripherals
SS1	Safe Stop 1	Safe stop 1 (monitored for time and ramping up)
SS2	Safe Stop 2	Safe stop 2
SSI	Synchronous Serial Interface	Synchronous serial interface
SSM	Safe Speed Monitor	Safe feedback for speed monitoring (n < nx)
SSP	SINAMICS Support Package	SINAMICS support package
STO	Safe Torque Off	Safe torque off
STW	Steuerwort	Control word
T		
TB	Terminal Board	Terminal Board
TIA	Totally Integrated Automation	Totally Integrated Automation
TM	Terminal Module	Terminal module
TN	Terre Neutre	Grounded three-phase supply network
Tn	-	Integral time
TPDO	Transmit Process Data Object	Transmit process data object
TT	Terre Terre	Grounded three-phase supply network
TTL	Transistor-Transistor Logic	Transistor-transistor logic
Tv	-	Rate time
U		
u.d.	under development	Under development: This feature is not currently available
UL	Underwriters Laboratories Inc.	Underwriters Laboratories Inc.
UPS	Uninterruptible Power Supply	Uninterruptible power supply
UTC	Universal Time Coordinated	Universal time coordinated
V		
VC	Vector Control	Vector control
Vdc	-	DC link voltage
VdcN	-	Partial DC link voltage negative
VdcP	-	Partial DC link voltage positive
VDE	Verband Deutscher Elektrotechniker	Association of German electrical engineers
VDI	Verein Deutscher Ingenieure	Association of German Engineers
VPM	Voltage Protection Module	Voltage Protection Module
Vpp	Volt peak-to-peak	Volt peak-to-peak
VSM	Voltage Sensing Module	Voltage Sensing Module

Abbreviation	Source of abbreviation	Meaning
X		
XML	Extensible Markup Language	Standard language for Web publishing and document management
Z		
ZM	Zero Mark	Zero mark
ZSW	Zustandswort	Status word

A.2 Document structure



Suggested improvements

If you come across any misprints in this document, please let us know using this form. We would also be grateful for any suggestions and recommendations for improvement.

To: SIEMENS AG I DT MC MS1 P.O. Box 3180 D-91050 Erlangen, Federal Republic of Germany Fax: +49 (0) 9131 / 98 - 2176 (documentation) mailto:docu.motioncontrol@siemens.com http://www.siemens.com/automation/service&support	From Name:
	Address of your Company/Dept.
	Street:
	Postal code: Location:
	Phone: /
	Fax: /

Suggestions and/or corrections

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